Lecture notes on quantum information technology

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Part I

Mathematics, Basics, and Tools

In this part, we introduce the mathematical foundations related to quantum information theory, as well as the basic concepts of quantum mechanics and quantum information. We first introduce the mathematical background of linear algebra, including the definitions and important properties of vectors and matrices. Next, we introduce the basic framework of quantum mechanics, including the vector representation of quantum states, operators, evolution of quantum systems, and projective measurements. Furthermore, we discuss the description of quantum mechanics using density matrices, quantum channels, and POVM measurements. Finally, we introduce important concepts in quantum information, including the impossibility of perfectly distinguishing quantum states, the no-cloning theorem, and the uncertainty principle.

Chapter 1

Mathematics: Linear algebra

We first introduce the basics of linear algebra related to quantum information.

1.1 Vectors and Hilbert Space

1.1.1 Hilbert Space

Mathematically, a ***complex vector*** $\boldsymbol{x}$ can be viewed as a linear mapping from the alphabet set Σ to the set of complex numbers $\mathbb{C}$. For example, if we choose $\Sigma = \{0, 1, \ldots, d-1\}$, and denote $x_i = \boldsymbol{x}(i)$, $\forall i \in \{0, 1, \ldots, d-1\}$, we can represent a d-dimensional vector $\boldsymbol{x}$ as

$$\boldsymbol{x} := \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{d-1} \end{pmatrix}. \tag{1.1}$$

We define the scalar multiplication of a vector as

$$a\boldsymbol{x} := \begin{pmatrix} ax_0 \\ ax_1 \\ \vdots \\ ax_{d-1} \end{pmatrix}. \tag{1.2}$$

Given vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, the vector addition is defined as

$$\boldsymbol{x}+\boldsymbol{y} := \begin{pmatrix} x_0+y_0 \\ x_1+y_1 \\ \vdots \\ x_{d-1}+y_{d-1} \end{pmatrix}, \tag{1.3}$$

Thus, all d-dimensional complex vectors form a linear space, which is called the ***Hilbert space***, denoted as $\mathcal{H}_d$. Since quantum states are defined on Hilbert spaces, we also use capital letters A, B, C, ... to represent quantum systems. Therefore, for a quantum system A, the corresponding Hilbert space is denoted as $\mathcal{H}_A$.

Given two d-dimensional vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, the inner product of the vectors is defined as

$$\langle \boldsymbol{x},\ \boldsymbol{y}\rangle := \sum_j x_j^* \cdot y_j, \tag{1.4}$$

where x_j^* is the complex conjugate of x_j. The inner product satisfies the linearity property

$$\langle \boldsymbol{x},\ a\boldsymbol{y}+b\boldsymbol{z}\rangle = a\,\langle \boldsymbol{x},\ \boldsymbol{y}\rangle + b\,\langle \boldsymbol{x},\ \boldsymbol{z}\rangle, \tag{1.5}$$

and the complex conjugate property

$$\langle \boldsymbol{x},\ \boldsymbol{y}\rangle = \langle \boldsymbol{y},\ \boldsymbol{x}\rangle^*. \tag{1.6}$$

The corresponding ***Euclidean norm*** of the vector is defined as

$$\|\boldsymbol{x}\| := \sqrt{\langle \boldsymbol{x},\ \boldsymbol{x}\rangle}, \tag{1.7}$$

which satisfies the following properties:

- Non-negativity: $\|\boldsymbol{x}\| \geq 0$.
- Linearity: $\|a\boldsymbol{x}\| = |a|\|\boldsymbol{x}\|$.
- Triangle inequality: $\|\boldsymbol{x}\| + \|\boldsymbol{y}\| \geq \|\boldsymbol{x}+\boldsymbol{y}\|$.
- Cauchy-Schwarz inequality: $\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| \geq |\langle \boldsymbol{x},\ \boldsymbol{y}\rangle|$.

Problem 1.1. *Use the Cauchy inequality to prove the Cauchy-Schwarz inequality*

$$|\langle \boldsymbol{x},\boldsymbol{y}\rangle| \leq \|\boldsymbol{x}\|\|\boldsymbol{y}\| \tag{1.8}$$

1.1.2 Dual Vectors

A ***dual vector*** corresponds to a linear mapping from Hilbert spaces to complex numbers. Specifically, for a d-dimensional vector $\boldsymbol{x}$, we can define a corresponding dual vector as

$$\boldsymbol{x}^\dagger = (x_0^*,\ x_1^*,\ \ldots,\ x_{d-1}^*), \tag{1.9}$$

and its satisfies scalar multiplication and addition of dual vectors as

$$(a\boldsymbol{x})^\dagger = (a^* x_0^*,\ a^* x_1^*,\ \ldots,\ a^* x_{d-1}^*), \tag{1.10}$$

and

$$(\boldsymbol{x}+\boldsymbol{y})^\dagger = \boldsymbol{x}^\dagger + \boldsymbol{y}^\dagger = (x_0^* + y_0^*,\ x_1^* + y_1^*,\ \ldots,\ x_{d-1}^* + y_{d-1}^*). \tag{1.11}$$

Following the rules of matrix multiplication, the multiplication of a dual vector $\boldsymbol{x}^\dagger$ and a vector $\boldsymbol{y}$ is defined as

$$\boldsymbol{x}^\dagger \cdot \boldsymbol{y} = \sum_j x_j^* \cdot y_j = \langle \boldsymbol{x},\ \boldsymbol{y} \rangle, \tag{1.12}$$

which therefore can be seen as linear mappings from the Hilbert space to complex numbers.

Problem 1.2. *Prove that all linear mappings from the Hilbert space to complex numbers are isomorphic to the space of dual vectors.*

Problem 1.3. *Prove the duality relation*

$$(\boldsymbol{x}^\dagger \cdot \boldsymbol{y})^* = \boldsymbol{y}^\dagger \cdot \boldsymbol{x} \tag{1.13}$$

1.1.3 Basis Vectors of Vectors

Consider a d-dimensional Hilbert space. We define a set of vectors $\{\boldsymbol{e}_j\}$ as

$$\boldsymbol{e}_0 := \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},\ \boldsymbol{e}_1 := \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix},\ \ldots,\ \boldsymbol{e}_{d-1} := \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \tag{1.14}$$

This set of vectors $\{\boldsymbol{e}_j\}$ satisfies the orthonormal property,

$$\|\boldsymbol{e}_j\| = 1,\ \langle \boldsymbol{e}_j,\ \boldsymbol{e}_{j'} \rangle = \delta_{j,j'}. \tag{1.15}$$

where $\delta_{j,j'} = 1$ if $j = j'$, and $\delta_{j,j'} = 0$ if $j \neq j'$. Thus, $\{\boldsymbol{e}_j\}$ forms an ***orthonormal basis*** for the Hilbert space. Any d-dimensional vector can be represented as a linear combination of these basis vectors:

$$\boldsymbol{x} = \sum_j x_j \boldsymbol{e}_j. \tag{1.16}$$

The corresponding dual basis vectors are

$$e_0^\dagger = (1,\ 0,\ \ldots,\ 0),\ e_1^\dagger = (0,\ 1,\ \ldots,\ 0),\ \ldots,\ e_{d-1}^\dagger = (0,\ 0,\ \ldots,\ 1). \tag{1.17}$$

Any dual vector $\boldsymbol{x}^\dagger$ can be represented as

$$\boldsymbol{x}^\dagger = \sum_j x_j^* \boldsymbol{e}_j^\dagger. \tag{1.18}$$

Using the above decomposition, the product of a dual vector and a vector is

$$\boldsymbol{x}^\dagger \cdot \boldsymbol{y} = \sum_{j,j'} x_j^* y_{j'} \boldsymbol{e}_j^\dagger \cdot \boldsymbol{e}_{j'} = \sum_j x_j^* y_j. \tag{1.19}$$

Here, we used $\boldsymbol{e}_j^\dagger \cdot \boldsymbol{e}_{j'} = \delta_{j,j'}$.

Problem 1.4. *Given* $\boldsymbol{x} = \sum_j x_j \boldsymbol{e}_j$ *and* $\boldsymbol{e}_j = \sum_k U_{j,k} \boldsymbol{e}'_k$, *find the* x'_k *that satisfies* $\boldsymbol{x} = \sum_k x'_k \boldsymbol{e}'_k$.

1.1.4 Direct Sum and Tensor Product

For a vector $\boldsymbol{x}$ of dimension d_x and a vector $\boldsymbol{y}$ of dimension d_y, the ***direct sum*** of the vectors is defined as

$$\boldsymbol{x} \oplus \boldsymbol{y} = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{d_x-1} \\ y_0 \\ y_1 \\ \vdots \\ y_{d_y-1} \end{pmatrix}. \tag{1.20}$$

The ***tensor product*** of the vectors is defined as

$$\boldsymbol{x} \otimes \boldsymbol{y} = \begin{pmatrix} x_0 y_0 \\ x_0 y_1 \\ \vdots \\ x_0 y_{d_y-1} \\ x_1 y_0 \\ x_1 y_1 \\ \vdots \\ x_1 y_{d_y-1} \\ \vdots \\ x_{d_x-1} y_0 \\ x_{d_x-1} y_1 \\ \vdots \\ x_{d_x-1} y_{d_y-1} \end{pmatrix}. \tag{1.21}$$

We note that $\boldsymbol{x} \oplus \boldsymbol{y}$ corresponds to a vector of dimension $d_x + d_y$, while $\boldsymbol{x} \otimes \boldsymbol{y}$ corresponds to a vector of dimension $d_x \cdot d_y$.

Problem 1.5. *Prove that* $\langle \boldsymbol{x} \oplus \boldsymbol{y}, \boldsymbol{x}' \oplus \boldsymbol{y}' \rangle = \langle \boldsymbol{x}, \boldsymbol{x}' \rangle + \langle \boldsymbol{y}, \boldsymbol{y}' \rangle$ *and* $\langle \boldsymbol{x} \otimes \boldsymbol{y}, \boldsymbol{x}' \otimes \boldsymbol{y}' \rangle = \langle \boldsymbol{x}, \boldsymbol{x}' \rangle \times \langle \boldsymbol{y}, \boldsymbol{y}' \rangle$.

1.1.5 Braket Representation of Vectors

In quantum mechanics, a commonly used vector representation is the braket notation. Simply put, the vector $\boldsymbol{x}$ and its dual vector $\boldsymbol{x}^\dagger$ are represented as

$$\boldsymbol{x} := |x\rangle\,,\ \boldsymbol{x}^\dagger := \langle x|\,. \tag{1.22}$$

Similarly, the orthonormal basis vectors for $\boldsymbol{x}$ and its dual vector are represented as

$$\boldsymbol{e}_j := |e_j\rangle\,,\ \boldsymbol{e}_j^\dagger := \langle e_j|\,. \tag{1.23}$$

Here, we further abbreviate $|e_j\rangle$ as $|j\rangle$ and $\langle e_j|$ as $\langle j|$, and the set of basis vectors $\{|j\rangle\}$ is often called the ***computational basis***. According to the multiplication rules for vectors and dual vectors, we have

$$\begin{aligned} \langle j|x\rangle &= \boldsymbol{e}_j^\dagger \cdot \boldsymbol{x} = x_j, \\ \langle x|j\rangle &= \boldsymbol{x}^\dagger \cdot \boldsymbol{e}_j = x_j^*. \end{aligned} \tag{1.24}$$

Thus, the ket-bra representation of a vector is

$$\begin{aligned} |x\rangle &= \sum_j x_j \, |j\rangle \,, \\ \langle x| &= \sum_j \langle j| \, x_j^* . \end{aligned} \tag{1.25}$$

The inner product of two vectors is given by

$$\langle x|y\rangle = \sum_{j,j'} x_j^* y_j \, \langle j|j'\rangle = \sum_j x_j^* y_j \,. \tag{1.26}$$

The direct sum and tensor product representations of the vectors $|x\rangle$ and $|y\rangle$ are

$$|x\rangle \oplus |y\rangle \,, \;\; |x\rangle \otimes |y\rangle \,. \tag{1.27}$$

In many cases, the tensor product of vectors $|x\rangle \otimes |y\rangle$ is also abbreviated as $|x\rangle \, |y\rangle$ or even $|x, y\rangle$ or $|xy\rangle$.

1.2 Matrices

1.2.1 Definitions

A ***matrix*** can be considered as a linear map from one vector space to another. Specifically, consider a linear map M from a d_1-dimensional vector $|x\rangle$ in $\mathcal{H}_{d_1}$ to a d_2-dimensional vector $|y\rangle$ in $\mathcal{H}_{d_2}$. We have

$$|y\rangle = M(|x\rangle) := M \, |x\rangle \,. \tag{1.28}$$

Using the expansions $|y\rangle = \sum_j y_j \, |j\rangle$ and $|x\rangle = \sum_k x_k \, |k\rangle$, we get

$$\sum_j y_j \, |j\rangle = \sum_k x_k M \, |k\rangle \,, \tag{1.29}$$

which implies

$$y_j = \sum_k M_{jk} x_k \,, \tag{1.30}$$

where $M_{jk} = \langle j| \, M \, |k\rangle$. The information of the matrix M is completely encoded in the components M_{jk}. Therefore, a common matrix representation is

$$M = \sum_{jk} M_{jk} \, |j\rangle \, \langle k| \,. \tag{1.31}$$

Note that the definition of M_{jk} depends on the choice of basis vectors $\{|j\rangle\}$ and $\{|k\rangle\}$. If we choose another set of basis vectors $\{|\tilde{j}\rangle\}$ and $\{|\tilde{k}\rangle\}$, we similarly have

$$M = \sum_{jk} \tilde{M}_{jk} |\tilde{j}\rangle \langle\tilde{k}|, \tag{1.32}$$

where $\tilde{M}_{jk} = \langle\tilde{j}|M|\tilde{k}\rangle$. It is important to note that although the specific representation M_{jk} of the matrix changes with the choice of basis vectors, the matrix M itself is independent of the choice of basis. That is, the matrix as a linear map on the Hilbert space is independent of the choice of basis vectors.

Suppose matrix M represents a linear map from a d_1-dimensional vector $|x\rangle$ to a d_2-dimensional vector $|y\rangle$, and matrix N represents a linear map from a d_2-dimensional vector $|y\rangle$ to a d_3-dimensional vector $|z\rangle$. We can define the matrix multiplication as the successive application of the linear maps, i.e.,

$$N \cdot M(|x\rangle) = N(M(|x\rangle)). \tag{1.33}$$

If the matrices M and N are represented as $M = \sum_{jk} M_{jk} |j\rangle \langle k|$ and $N = \sum_{ij'} N_{ij'} |i\rangle \langle j'|$, then the matrix multiplication $N \cdot M$ is represented as

$$\begin{aligned} N \cdot M &= \left(\sum_{ij'} N_{ij'} |i\rangle \langle j'|\right) \cdot \left(\sum_{jk} M_{jk} |j\rangle \langle k|\right), \\ &= \sum_{ij'} \sum_{jk} N_{ij'} \cdot M_{jk} |i\rangle \langle j'| \cdot |j\rangle \langle k|, \\ &= \sum_{ik} \left(\sum_{j} N_{ij} \cdot M_{jk}\right) |i\rangle \langle k|, \end{aligned} \tag{1.34}$$

Thus, the matrix representation of $N \cdot M$ is given by

$$(N \cdot M)_{ik} = \sum_{j} M_{jk} \cdot N_{ij}. \tag{1.35}$$

Note that matrix multiplication is essentially the composition of quantum linear maps, and linear maps themselves are independent of the choice of basis. Therefore, the definition of matrix multiplication is also independent of the basis vectors.

On the other hand, when we have $N \cdot M = M \cdot N$, we say that M and N ***commute***. However, since the order of linear maps is generally not interchangeable,

we typically have $N \cdot M \neq M \cdot N$, which means that matrices N and M generally do not commute. The non-commutativity of matrices makes matrix operations more complex compared to scalar operations. In quantum information theory, operators may also satisfy the ***anti-commutative*** relation, i.e., $MN = -NM$. We can further define ***commutator*** and ***anti-commutator*** operations as

$$[M, N] = MN - NM, \quad \{M, N\} = MN + NM, \tag{1.36}$$

Thus, the commutation and anti-commutation relations are represented as

$$[M, N] = 0, \quad \{M, N\} = 0. \tag{1.37}$$

For a matrix from $\mathcal{H}_A$ to $\mathcal{H}_B$, we denote its space as $\mathcal{B}(\mathcal{H}_A \to \mathcal{H}_B)$ or shorthand as $\mathcal{B}(A \to B)$. When $H_A \equiv H_B$, we further abbreviate it as $\mathcal{B}(\mathcal{H}_A)$ or $\mathcal{B}(A)$.

For any Hilbert space, a special matrix is the identity matrix:

$$\mathbb{I} = \sum_j |j\rangle \langle j|, \tag{1.38}$$

Some commonly used equalities are

$$\mathbb{I} \cdot |x\rangle = |x\rangle, \ \langle x| \cdot \mathbb{I} = \langle x|. \tag{1.39}$$

This is the identity map from a vector to itself. Moreover, it is easy to verify that the identity matrix commutes with any other matrix, i.e., $M \cdot I = I \cdot M = M$.

1.2.2 Direct Sum and Tensor Product of Matrices

The direct sum and tensor product of matrices are defined similarly to the direct sum and tensor product of vectors. Given matrices $M = \sum_{j_1 k_1} M_{j_1 k_1} |j_1\rangle \langle k_1|$ and $N = \sum_{j_2 k_2} N_{j_2 k_2} |j_2\rangle \langle k_2|$, the direct sum and tensor product of M and N are

$$\begin{aligned} M \oplus N &= \begin{pmatrix} M & 0 \\ 0 & N \end{pmatrix}, \\ M \otimes N &= \sum_{j_1,\, j_2,\, k_1,\, k_2} M_{j_1 k_1} \cdot N_{j_2 k_2} |j_1 \otimes j_2\rangle \langle k_1 \otimes k_2|. \end{aligned} \tag{1.40}$$

For example, given a 2×2 matrix M and a 2×3 matrix N,

$$M = \begin{pmatrix} M_{00} & M_{01} \\ M_{10} & M_{11} \end{pmatrix}, \ N = \begin{pmatrix} N_{00} & N_{01} & N_{02} \\ N_{10} & N_{11} & N_{12} \end{pmatrix}, \tag{1.41}$$

The direct sum and tensor product of M and N are

$$
\begin{aligned}
M \oplus N &= \begin{pmatrix} M_{00} & M_{01} & 0 & 0 & 0 \\ M_{10} & M_{11} & 0 & 0 & 0 \\ 0 & 0 & N_{00} & N_{01} & N_{02} \\ 0 & 0 & N_{10} & N_{11} & N_{12} \end{pmatrix}, \\
M \otimes N &= \begin{pmatrix} M_{00}N_{00} & M_{00}N_{01} & M_{00}N_{02} & M_{01}N_{00} & M_{01}N_{01} & M_{01}N_{02} \\ M_{00}N_{10} & M_{00}N_{11} & M_{00}N_{12} & M_{01}N_{10} & M_{01}N_{11} & M_{01}N_{12} \\ M_{10}N_{00} & M_{10}N_{01} & M_{10}N_{02} & M_{11}N_{00} & M_{11}N_{01} & M_{11}N_{02} \\ M_{10}N_{10} & M_{10}N_{11} & M_{10}N_{12} & M_{11}N_{10} & M_{11}N_{11} & M_{11}N_{12} \end{pmatrix}.
\end{aligned} \tag{1.42}
$$

For the same matrix M, we generally use the following notation:

$$
\begin{aligned}
M^{\oplus n} &\equiv \underbrace{M \oplus M \oplus \cdots \oplus M}_{n}, \\
M^{\otimes n} &\equiv \underbrace{M \otimes M \otimes \cdots \otimes M}_{n}
\end{aligned} \tag{1.43}
$$

Problem 1.6. *Prove the following equalities:*

$$
\begin{aligned}
(xM_1 + yM_2) \oplus (xN_1 + yN_2) &= x\,(M_1 \oplus N_1) + y\,(M_2 \oplus N_2)\,, && (1.44) \\
(M_1 \oplus N_1)\,(M_2 \oplus N_2) &= (M_1 M_2) \oplus (N_1 N_2)\,, && (1.45) \\
(M_1 \otimes N_1)\,(M_2 \otimes N_2) &= M_1 M_2 \otimes N_1 N_2 && (1.46) \\
I_n \otimes M &= M^{\oplus n}, && (1.47) \\
(xM_1 + yM_2) \otimes N &= x\,(M_1 \otimes N) + y\,(M_2 \otimes N)\,, && (1.48) \\
M \otimes (xN_1 + yN_2) &= x\,(M \otimes N_1) + y\,(M \otimes N_2) && (1.49)
\end{aligned}
$$

1.2.3 Complex Conjugate, Transpose, and Conjugate Transpose of Matrices

Given a matrix $M = \sum_{jk} M_{jk} |j\rangle \langle k|$, we define the ***complex conjugate***, ***transpose***, and ***conjugate transpose*** of the matrix as follows:

$$\begin{aligned} M^* &= \sum_{jk} M^*_{jk} |j^*\rangle \langle k^*|, \\ M^\top &= \sum_{jk} M_{jk} |k^*\rangle \langle j^*|, \\ M^\dagger &= \sum_{jk} M^*_{jk} |k\rangle \langle j|. \end{aligned} \tag{1.50}$$

It is important to note that, for the definitions of complex conjugate and transpose, we must apply the complex conjugate to the basis vectors to ensure that they are independent of the choice of basis. Without applying the Hermitian conjugate to the basis vectors, different choices of basis could result in different values for M^* or $M^\top$. However, the definition of the conjugate transpose is basis-independent when the Hermitian conjugate is not applied to the basis vectors. Some common equalities are:

$$\begin{aligned} (aM)^* = a^* M^*,\ (aM)^\top = aM^\top,\ (aM)^\dagger = a^* M^\dagger, \\ (M \cdot N)^* = M^* \cdot N^*,\ (M \cdot N)^\top = N^\top \cdot M^\top,\ (M \cdot N)^\dagger = N^\dagger \cdot M^\dagger, \\ M^\dagger = (M^*)^\top = (M^\top)^*. \end{aligned} \tag{1.51}$$

Problem 1.7. *Prove the following equalities:*

$$(M \oplus N)^\dagger = M^\dagger \oplus N^\dagger, \tag{1.52}$$

$$(M \otimes N)^\dagger = M^\dagger \otimes N^\dagger. \tag{1.53}$$

Problem 1.8. *Prove the relationship between matrix operations and basis vectors.*

- *Prove that the complex conjugate, transpose, and conjugate transpose operation is basis-independent.*

  Hint: A direct proof is to show that for different bases $\{|j\rangle\}$ and $\{|\tilde{j}\rangle\}$, assume $M = \sum_{jk} M_{jk} |j\rangle \langle k| = \sum_{jk} \tilde{M}_{jk} |\tilde{j}\rangle \langle \tilde{k}|$, and prove that

$$\sum_{jk} M^*_{jk} |k\rangle \langle j| = \sum_{jk} \tilde{M}^*_{jk} |\tilde{k}\rangle \langle \tilde{j}|.$$

- *Prove that the complex conjugate and transpose operations defined below depend on the choice of basis.*

$$\begin{aligned} M^* &= \sum_{jk} M^*_{jk} \left|j\right\rangle \left\langle k\right|, \\ M^\top &= \sum_{jk} M_{jk} \left|k\right\rangle \left\langle j\right|. \end{aligned} \tag{1.54}$$

1.2.4 Trace, Partial Trace, and Determinant of Matrices

The ***trace*** of a square matrix M is defined as the average inner product of input and output vectors (multiplied by the dimension of the matrix), i.e.,

$$\mathrm{Tr}[M] = \sum_j \left\langle j\right|M\left|j\right\rangle. \tag{1.55}$$

Although the trace of a matrix is the sum of its diagonal elements, it is basis-independent, and satisfies the property

$$\mathrm{Tr}[M \cdot N] = \mathrm{Tr}[N \cdot M], \tag{1.56}$$

For example, $\mathrm{Tr}[\left|x\right\rangle \left\langle y\right|] = \left\langle y|x\right\rangle$. Using the properties of the identity matrix and the trace, we can also derive the definition of the trace, i.e.,

$$\mathrm{Tr}[M \cdot I] = \mathrm{Tr}[M \sum_j \left|j\right\rangle \left\langle j\right|] = \sum_j \left\langle j\right|M\left|j\right\rangle. \tag{1.57}$$

Based on the trace of a matrix, the Hilbert-Schmidt inner product of two matrices is defined as

$$\left\langle M, N\right\rangle = \mathrm{Tr}[M^\dagger \cdot N]. \tag{1.58}$$

Problem 1.9. *Prove the following results:*

- *The trace of a matrix is independent of the choice of basis.*
- *Prove that* $\mathrm{Tr}[M \cdot N] = \mathrm{Tr}[N \cdot M]$.
- *For V satisfying $V^\dagger V = I$, we have* $\mathrm{Tr}[VMV^\dagger] = \mathrm{Tr}[M]$.
- *Prove that* $\mathrm{Tr}[M \oplus N] = \mathrm{Tr}[M] + \mathrm{Tr}[N]$ *and* $\mathrm{Tr}[M \otimes N] = \mathrm{Tr}[M]\mathrm{Tr}[N]$.

1.2.5 Partial Trace of Matrices in Quantum Information

In quantum information, a common operation is the ***partial trace*** of a matrix. We use subscripts A and B to represent different systems, which correspond to different Hilbert spaces. Suppose the matrix M is a square matrix of size $(d_A \times d_B) \times (d_A \times d_B)$, and the matrix is represented in the bases $\{|j\rangle_A\}$ and $\{|k\rangle_B\}$ as follows:

$$M = \sum_{j_A j'_A k_B k'_B} M_{j_A j'_A k_B k'_B} |j\rangle_A |k\rangle_B \langle j'|_A \langle k'|_B . \tag{1.59}$$

We can view the matrix M as a map from the vectors of systems A and B to the vectors of systems A and B.

The partial trace of matrix M over system B is defined as the average matrix of M over different inputs of the B system, i.e.,

$$\mathrm{Tr}_B[M] = \sum_{k_B} \langle k_B|M|k_B\rangle = \sum_{j_A j'_A} M_{j_A j'_A} |j\rangle_A \langle j'|_A . \tag{1.60}$$

The matrix after performing the partial trace over system B, denoted as $\mathrm{Tr}_B[M]$, can be expressed as $M_{j_A j'_A} = \sum_{k_B} M_{j_A j'_A k_B k_B}$. Similarly, the partial trace operation is basis-independent.

Problem 1.10. *Prove the following results:*

$$\mathrm{Tr}_A[M_A \otimes N_B] = \mathrm{Tr}_A[M_A] N_B, \tag{1.61}$$

$$\mathrm{Tr}_B[M_A \otimes N_B] = \mathrm{Tr}_B[M_B] M_A. \tag{1.62}$$

The ***determinant*** of a matrix M is defined as

$$\det(M) = \sum_{\pi \in \mathrm{Sym}(n)} \mathrm{sign}(\pi) \prod M_{j\pi(j)}, \tag{1.63}$$

where $\mathrm{Sym}(n)$ denotes the permutation group of a set with n elements. For example, the permutation group of the set $\{0,\ 1,\ 2\}$ is

$$\begin{aligned} \mathrm{Sym}(n) = \Big\{ & (0 \to 0,\ 1 \to 1,\ 2 \to 2),\ (0 \to 1,\ 1 \to 0,\ 2 \to 2), \\ & (0 \to 2,\ 1 \to 1,\ 2 \to 0),\ (0 \to 0,\ 1 \to 2,\ 2 \to 1), \\ & (0 \to 1,\ 1 \to 2,\ 2 \to 0),\ (0 \to 2,\ 1 \to 0,\ 2 \to 1) \Big\}. \end{aligned} \tag{1.64}$$

Here, $\mathrm{sign}(\pi)$ equals $+1$ (-1) depending on whether the permutation π is an even (odd) permutation. Although the determinant of a matrix is defined by its matrix elements, it is also basis-independent.

1.2.6 Singular Value Decomposition and Spectral Decomposition

The ***singular value decomposition (SVD)*** of a matrix: For any matrix M, there exists a set of orthonormal vectors $\{|u_j\rangle\}$ and $\{|v_j\rangle\}$, and a set of non-negative real numbers $\{\lambda_j\}$, such that

$$M = \sum_j \lambda_j |u_j\rangle \langle v_j|, \tag{1.65}$$

where $\{\lambda_j\}$ are also called the singular values of M.

Problem 1.11. *Prove that the absolute value of the determinant of a matrix is the product of its singular values, i.e.,*

$$|\det(M)| = \prod_j \lambda_j. \tag{1.66}$$

Assume that M has r non-zero singular values, then we can define a set of basis vectors $\{|j\rangle : j = 1, 2, \ldots, r\}$, and find matrices U and V such that

$$|u_j\rangle = U|j\rangle, \ |v_j\rangle = V|j\rangle. \tag{1.67}$$

Here, U and V satisfy $U^\dagger U = V^\dagger V = \mathbb{I}$, which are the isometric transformations we will mention later. Define the matrix $D = \sum_j \lambda_j |j\rangle \langle j|$, then the matrix M can be written as

$$M = U \cdot D \cdot V^\dagger. \tag{1.68}$$

Thus, any matrix can be written as a diagonal matrix multiplied by isometric transformations on both sides.

For any matrix M and its singular value decomposition $M = U \cdot D \cdot V^\dagger$, we have

$$M = (U \cdot D \cdot U^\dagger) \cdot (U \cdot V^\dagger) = (U \cdot V^\dagger) \cdot (V \cdot D \cdot V^\dagger). \tag{1.69}$$

We define $P = U \cdot D \cdot U^\dagger$, $\tilde{P} = V \cdot D \cdot V^\dagger$, and $W = U \cdot V^\dagger$, so we get

$$M = P \cdot W = W \cdot \tilde{P}. \tag{1.70}$$

Here, P and $\tilde{P}$ are ***positive definite*** matrices with a spectral decomposition of positive real numbers, and W is an isometric transformation. This decomposition is also called the ***polar decomposition***.

Problem 1.12. *Compute the singular value decomposition and polar decomposition of the matrix*

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{1.71}$$

Consider $N_1 = M \cdot M^\dagger$ and $N_2 = M^\dagger \cdot M$, the singular value decompositions of the matrices N_1 and N_2 are

$$N_1 = \sum_j \lambda_j^2 \left|u_j\right\rangle \left\langle u_j\right|, \; N_2 = \sum_j \lambda_j^2 \left|v_j\right\rangle \left\langle v_j\right|. \tag{1.72}$$

Note that the ket and bra parts of the singular value decomposition of N_1 (N_2) are the same, and this type of decomposition is called the ***spectral decomposition***. In general, for any matrix M satisfying $M \cdot M^\dagger = M^\dagger \cdot M$, i.e., a ***normal matrix***, there exists a ***spectral decomposition*** or ***eigenvalue decomposition***, which can be written as

$$M = \sum_j \lambda_j \left|u_j\right\rangle \left\langle u_j\right|, \tag{1.73}$$

where $\left|u_j\right\rangle$ forms an orthonormal basis, also called the eigenvectors, and λ_j are the eigenvalues, which are generally complex numbers. The number of non-zero eigenvalues is called the ***rank*** of the matrix. It is easy to verify that the matrices N_1 and N_2 are normal matrices.

Problem 1.13. *Prove that the spectral decomposition exists if and only if the matrix is normal.*

Problem 1.14. *Given the spectral decomposition of a matrix $M = \sum_j \lambda_j \left|u_j\right\rangle \left\langle u_j\right|$, prove that the determinant and trace of the matrix are*

$$\begin{aligned} \det(M) &= \prod_j \lambda_j, \\ \mathrm{Tr}[M] &= \sum_j \lambda_j. \end{aligned} \tag{1.74}$$

1.2.7 Matrix Functions

Given the spectral decomposition of a normal matrix $M = \sum_j \lambda_j \left|u_j\right\rangle \left\langle u_j\right|$, the function of the matrix M can be expressed as

$$f(M) = \sum_j f(\lambda_j) \left|u_j\right\rangle \left\langle u_j\right|. \tag{1.75}$$

For example, some common matrix functions are

$$\begin{aligned} M^k &= \sum_j \lambda_j^k \left|u_j\right\rangle \left\langle u_j\right|, \\ e^M &= \sum_j e^{M_j} \left|u_j\right\rangle \left\langle u_j\right|, \\ \log(M) &= \sum_j \log(\lambda_j) \left|u_j\right\rangle \left\langle u_j\right|. \end{aligned} \tag{1.76}$$

Problem 1.15. *Prove the following conclusions:*

- *Given the spectral decomposition of a matrix $M = \sum_j \lambda_j \left|u_j\right\rangle \left\langle u_j\right|$, prove that*

$$e^M = \sum_k \frac{M^k}{k!}. \tag{1.77}$$

- *Assume $M^2 = \mathbb{I}$, prove that*

$$e^{i\theta M} = \cos\theta \mathbb{I} + i \sin\theta M. \tag{1.78}$$

- *For a general function, consider its Taylor expansion, and prove formula (1.75).*
- *Prove that $f(M \oplus N) = f(M) \oplus f(N)$, and consider whether a similar conclusion holds for $\otimes$.*

1.2.8 Matrix Norms

A matrix norm is a special type of matrix function. A representative class of matrix norms is the ***Schatten norm***, defined as

$$\|M\|_p = \left[\mathrm{Tr}\left[(M^\dagger \cdot M)^{p/2}\right]\right]^{1/p} = \left[\mathrm{Tr}\left[(|M|^p)\right]\right]^{1/p}, \tag{1.79}$$

where $p \in [1, \infty)$ and $|M| = \sqrt{M^\dagger \cdot M}$. The Schatten norm of a matrix is entirely determined by the singular values of the matrix, i.e.,

$$\|M\|_p = \left[\sum_j |\lambda_j|^p\right]^{1/p}, \tag{1.80}$$

where λ_j are the singular values of the matrix M. The Schatten norm satisfies the following properties:

- Monotonicity: For any $1 \leq q \leq p \leq \infty$, we have $\|M\|_p \leq \|M\|_q$.
- Invariance under unitary transformations: For any unitary transformations U and V, we have $\|M\|_p = \|U \cdot M \cdot V\|_p$.
- Invariance under conjugate, transpose, and conjugate transpose: $\|M\|_p = \|M^*\|_p = \|M^\top\|_p = \|M^\dagger\|_p$.
- Hölder's inequality: For any $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$, we have

$$|\langle M,\ N\rangle| \leq \|M\|_p \cdot \|N\|_q. \tag{1.81}$$

- Duality: For any $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$, we have

$$\|M\|_p = \sup\left\{\langle M,\ N\rangle,\ \|N\|_q = 1\right\}. \tag{1.82}$$

- Sub-multiplicativity: For any $p \in [1, \infty]$, we have

$$\|MNS\|_p \leq \|M\|_\infty \|N\|_p \|S\|_\infty \tag{1.83}$$

  Taking $S = I$ and scaling $\|M\|_\infty \leq \|N\|_p$, we get

$$\|MN\|_p \leq \|M\|_p \|N\|_p \tag{1.84}$$

Some special cases are:

- ***Trace norm***, $p = 1$: $\|M\|_1 = \sum_j |\lambda_j| = \max\{\langle M,\ U\rangle,\ U \text{ is unitary}\}$.
- ***Frobenius norm***, $p = 2$: $\|M\|_2 = \sqrt{\langle M, M\rangle} = \sqrt{\sum_j |\lambda_j|^2}$. Thus, the Frobenius norm can be understood as the Euclidean norm of the matrix in this linear space.
- ***Spectral norm***, $p = \infty$: $\|M\|_\infty = \max|\lambda_j| = \max\{\|M|u\rangle,\ \||u\rangle\| = 1\|\}$. By monotonicity, the spectral norm is also the smallest Schatten norm. In the absence of special instructions, the spectral norm is generally denoted as $\|M\|$, also referred to as the operator norm.

Problem 1.16. *Prove the following conclusions:*

- *For the absolute value of a matrix, we have*

1. $\mathrm{Tr}[|A|] = \mathrm{Tr}[|A^\dagger|]$
2. $|\mathrm{Tr}[M]| \leq \mathrm{Tr}[|M|]$
3. $|\mathrm{Tr}[A]\mathrm{Tr}[B]| \leq \mathrm{Tr}[|A||B|] \leq \mathrm{Tr}[|A|]\mathrm{Tr}[|B|]$

- *Prove that the trace norm is non-increasing under partial trace.*
- *Prove the matrix Cauchy-Schwarz inequality* $\langle M, N\rangle \leq \|M\|_2\|N\|_2$.
- *Given vectors* $|x\rangle$ *and* $|y\rangle$*, compute the trace norm, Frobenius norm, and spectral norm of* $|x\rangle\langle y|$ *and* $|x\rangle\langle x| - |y\rangle\langle y|$.

Special Matrices

Earlier, we mentioned several types of special matrices. Here, we further summarize various different special matrices and their corresponding forms in quantum information.

- ***Normal Matrix***: A matrix satisfying $M \cdot M^\dagger = M^\dagger \cdot M$; normal matrices are square matrices that have a spectral decomposition $M = \sum_j \lambda_j |u_j\rangle\langle u_j|$ or $M = U \cdot D \cdot U^\dagger$, where U is a unitary transformation (defined later) and D is a diagonal matrix.
- ***Hermitian Matrix***: A matrix satisfying $M = M^\dagger$; Hermitian matrices are special normal matrices with real eigenvalues. Hermitian matrices are often used to represent operators in quantum mechanics.
- ***Positive Semi-Definite Matrix***: A Hermitian matrix whose eigenvalues are non-negative; for any positive semi-definite matrix M, there exists a matrix N such that $M = N^\dagger \cdot N$. A matrix M is positive semi-definite if and only if $M \geq 0$, meaning all eigenvalues of the matrix are greater than or equal to zero. For Hermitian matrices M and N, if $M - N \geq 0$, we write $M \geq N$.
- ***Positive Definite Matrix***: A Hermitian matrix whose eigenvalues are strictly positive; positive definite matrices are a special case of positive semi-definite matrices with $\det(M) \neq 0$. A matrix M is positive definite if and only if $M > 0$.
- ***Density Matrix***: A positive semi-definite matrix $\rho \geq 0$ that satisfies $\mathrm{Tr}[\rho] = 1$; density matrices are generally used to represent quantum states.

- ***Projection Matrix***: A positive semi-definite matrix Π satisfying $\Pi^2 = \Pi$; the eigenvalues of a projection matrix can only be 0 or 1. A projection matrix with only one non-zero eigenvalue is also called a rank-1 projection matrix. Projection matrices are generally used to describe standard quantum measurements.

- ***Isometry Transformation***: A matrix V satisfying $V^\dagger \cdot V = \mathbb{I}$; for any vectors $|x\rangle$ and $|y\rangle$, we have $\langle x|y\rangle = \langle x|V^\dagger \cdot V|y\rangle$. This means that V acting on a vector does not change the inner product of vectors.

- ***Unitary Transformation (also called Unitary Operation)***: A square matrix U satisfying $U^\dagger \cdot U = U \cdot U^\dagger = \mathbb{I}$; for any vectors $|x\rangle$ and $|y\rangle$, we have $\langle x|y\rangle = \langle x|U^\dagger \cdot U|y\rangle = \langle x|U \cdot U^\dagger|y\rangle$. This means that both U and $U^\dagger$ acting on vectors do not change the inner product of vectors. Unitary transformations are generally used to describe the dynamical evolution of quantum systems.

Problem 1.17. *Prove the following conclusions.*

- *Prove that $M + M^\dagger$ is a Hermitian matrix.*

- *Prove that a normal matrix is Hermitian if and only if its eigenvalues are real.*

- *For a Hermitian matrix M, e^{iM} is a unitary matrix.*

- *For a unitary matrix U, $i \log U$ is a Hermitian matrix.*

1.2.9 Matrix Vectorization

Although matrices are linear transformations of vectors, matrices themselves also form a larger vector space. Consider a matrix $M = \sum_{jk} M_{jk} |j\rangle \langle k|$, a simple method of matrix vectorization is given by

$$\mathrm{vec}(M) = \sum_{jk} M_{jk} |j\rangle |k\rangle \tag{1.85}$$

Let the space of matrix M be $\mathcal{B}(A \to B)$, and this vectorization method maps $\mathcal{B}(A \to B)$ to $\mathcal{H}_B \otimes \mathcal{H}_A$. Clearly, this mapping is a one-to-one correspondence. Using this mapping, we can apply the methods and conclusions of vector analysis to matrices, and vice versa. $\mathrm{vec}(M)$ is often also denoted as $|M\rangle\rangle$.

Choosing different basis, matrices have many other methods of vectorization. Specifically, consider a set of basis matrices $\{e_j\}$ satisfying $\langle e_j, e_k\rangle = \mathrm{Tr}[e_j^\dagger \cdot e_k] = \delta_{jk}$, then we can always decompose any matrix M as

$$M = \sum_j M_j e_j, \tag{1.86}$$

which is analogous to the decomposition of a vector. Clearly, $\{|j\rangle\langle k|\}$ forms an orthogonal basis. In particular, consider Hermitian matrices and Pauli basis vectors (as in Equation (2.18)), we can vectorize Hermitian matrices in the Pauli basis.

Problem 1.18. *Prove the following properties for the vectorized form of the matrix:*

- *The inner product of matrices is the same as the inner product of their vectorized forms:*
$$\langle M, N\rangle = \langle \mathrm{vec}(M), \mathrm{vec}(N)\rangle \tag{1.87}$$
- *Consider the vectorization method in Equation* (1.85)*, prove the following conclusions:*
  - *The vectorization and inverse vectorization transformations can be written as*
$$\begin{aligned} \mathrm{vec}(M)_{BA} &= (M_{A\to B} \otimes I_A)\,|\Psi\rangle_{AA}\,, \\ M_{A\to B} &= (I_B \otimes \langle\Psi|_{AA}) \cdot (\mathrm{vec}(M)_{BA} \otimes I_A) \end{aligned} \tag{1.88}$$
    where $|\Psi\rangle_{AA} = \sum_k |kk\rangle_{AA}$, and we add the system subscripts to clearly indicate the matrix and vector multiplication relations.
  - *Consider matrices $X \in \mathcal{B}(B \to C)$ and $Y \in \mathcal{B}(A \to D)$, then we have*
$$(X \otimes Y)\mathrm{vec}(M) = \mathrm{vec}(XMY^\top) \tag{1.89}$$

Chapter 2

Basics of Quantum Information

2.1 Pure States, Unitary Evolution, and projective measurement

In this section, we will introduce the description of quantum mechanics, including the vector representation of quantum states, the matrix representation of quantum operators and evolution, and measurements based on projection operators.

2.1.1 Pure States

The state of a quantum system can be described by a vector in a Hilbert space. For example, here the quantum system could be an atom in different energy levels, a photon in different positions or paths, an electron with different spins, or a superconducting qubit in a quantum circuit.

Quantum Bit

We first consider the simplest quantum system, the ***quantum bit (qubit)***. A qubit corresponds to a system with two distinguishable states, denoted as $|0\rangle$ and $|1\rangle$. Here, $|0\rangle$ and $|1\rangle$ could represent different atomic energy levels, the position of a particle, the polarization and path of a photon, the spin of an electron, and so on. A general qubit can exist in a ***superposition*** of $|0\rangle$ and $|1\rangle$, i.e.,

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, \tag{2.1}$$

where a and b are complex numbers that satisfy the normalization condition

$$\langle\psi|\psi\rangle = |a|^2 + |b|^2 = 1. \tag{2.2}$$

Here, we consider the two-dimensional complex vector $|\psi\rangle$ as an abstract mathematical definition of a quantum state. Its specific meaning will become clearer as we explore the expectation values of operators and measurements.

According to the normalization condition, another equivalent representation of the quantum bit is

$$|\psi\rangle = e^{i\alpha}\left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\psi}|1\rangle\right), \tag{2.3}$$

where $a = e^{i\alpha}\cos\frac{\theta}{2}$ and $b = e^{i(\alpha+\psi)}\sin\frac{\theta}{2}$. Note that α is a global phase of the quantum state, and since a global phase does not affect the expectation values of operators or measurements, we can always set the global phase to zero by using $e^{-i\alpha}|\psi\rangle$. For a quantum state that ignores the global phase, we have

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\psi}|1\rangle. \tag{2.4}$$

We can represent this state with two parameters, $\theta \in [0,\pi]$ and $\psi \in [0,2\pi]$. As shown in Figure 2.1, any quantum bit can be uniquely mapped to a point on the sphere, known as the ***Bloch sphere***. The six quantum states at the two ends of the three coordinate axes are

$$|0\rangle,\ |1\rangle,\ |\pm\rangle = \frac{(|0\rangle \pm |1\rangle)}{\sqrt{2}},\ |\pm i\rangle = \frac{(|0\rangle \pm i|1\rangle)}{\sqrt{2}}. \tag{2.5}$$

Any quantum state $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\psi}|1\rangle$ corresponds to a point on the surface of the sphere, where θ and ψ are the angles between the point and the Z and X axes, respectively. Although this correspondence appears simple and direct, we will later provide a deeper connection to this relationship.

Problem 2.1. *Prove that when two qubit states are orthogonal, the corresponding vectors on the Bloch sphere are in opposite directions.*

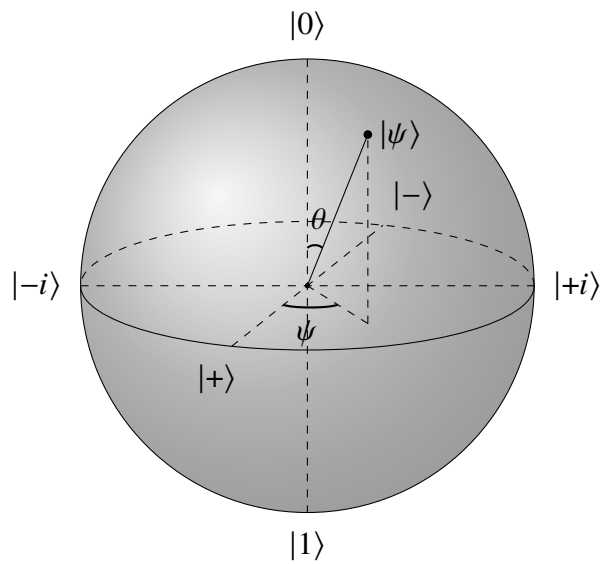


Figure 2.1: Bloch sphere of a single qubit.

Multi-Qubit Quantum States

A single quantum bit constitutes the smallest computational unit of quantum computation. A general quantum computer typically operates on multiple qubits simultaneously. Consider a system of n qubits, where the basis vectors for the k-th quantum system are $\{|0\rangle_k, |1\rangle_k\}$. A general n-qubit state $|\psi\rangle$ can be written as

$$|\psi\rangle = \sum_{j_1,\, j_2,\, \ldots,\, j_n \in \{0,1\}} \alpha_{j_1 j_2 \ldots j_n} |j_1\rangle_1 \otimes |j_2\rangle_2 \otimes \cdots \otimes |j_n\rangle_n \,. \tag{2.6}$$

where $\alpha_{j_1 j_2 \ldots j_n}$ is a complex number satisfying the normalization condition

$$\sum_{j_1 j_2 \ldots j_n} |\alpha_{j_1 j_2 \ldots j_n}|^2 = 1,$$

and $|j_1\rangle_1 \otimes |j_2\rangle_2 \otimes \cdots \otimes |j_n\rangle_n$ represents the n-qubit state, which is the basis vector of $\mathcal{H}_2^{\otimes n}$. In quantum computation, we often omit the subscripts indicating the system and combine all the kets, denoting the basis vectors as

$$|j_1\rangle_1 \otimes |j_2\rangle_2 \otimes \cdots \otimes |j_n\rangle_n \rightarrow |j_1 j_2 \ldots j_n\rangle \,. \tag{2.7}$$

Thus, the n-qubit state $|\psi\rangle$ can be simplified as

$$|\psi\rangle = \sum_{j_1,\, j_2,\, \ldots,\, j_n \in \{0,1\}} \alpha_{j_1 j_2 \ldots j_n} |j_1 j_2 \ldots j_n\rangle \,. \tag{2.8}$$

Note that the quantum state $|\psi\rangle$ generally cannot be written as $|\psi_1\rangle|\psi_2\rangle\ldots|\psi_n\rangle$, meaning there are quantum correlations between the qubits, or ***quantum entanglement***. For example, a commonly used entangled state for two qubits is the Bell state

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle),\ |\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle), \tag{2.9}$$

which are ***Bell states***. Another famous multipartite entangled state is the ***Greenberger–Horne–Zeilinger (GHZ)*** state or the cat state

$$|GHZ\rangle_n = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}) \tag{2.10}$$

Quantum entanglement plays a crucial role in both quantum information and quantum computation.

High-Dimensional Quantum States

A quantum bit is the smallest unit of a quantum system, but general quantum systems may have higher-dimensional states. For a general d-dimensional system, the quantum state is represented by a vector in $\mathcal{H}_d$, i.e.,

$$|\psi\rangle = \sum_{j=0}^{d-1} \alpha_j |j\rangle . \tag{2.11}$$

The quantum bit is the special case when $d = 2$. Common quantum systems also have $d = 3$ (qutrit) or $d = \infty$, corresponding to continuous variable systems. On the other hand, the quantum state of n qubits corresponds to a $d = 2^n$ system, so equation 2.8 can also be written as

$$|\psi\rangle = \sum_{j=0}^{2^n-1} \alpha_j |j\rangle , \tag{2.12}$$

where $j_1 j_2 \ldots j_n$ can be viewed as the binary representation of j.

Schmidt Decomposition

Similar to the spectral decomposition of matrices, we can always perform a ***Schmidt decomposition*** on a two-body quantum state. Consider a quantum state $|\psi\rangle_{AB}$ of

the AB system (note that the sizes of the A and B systems may differ). We can always find orthogonal basis vectors $\{|\varphi_j\rangle_A\}$ for the A system and $\{|\phi_j\rangle_B\}$ for the B system, such that the state $|\psi\rangle_{AB}$ can be decomposed as

$$|\psi\rangle_{AB} = \sum_j \alpha_j |\varphi_j\rangle_A \otimes |\phi_j\rangle_B, \tag{2.13}$$

where α_j are real numbers (the phase part can always be absorbed into the orthogonal basis vectors). The key to the Schmidt decomposition lies in utilizing the vectorization of matrices, or matrix representation of the two-body quantum state. For the quantum state $|\psi\rangle_{AB}$, we consider the inverse transformation of vectorization (equation (1.85)),

$$M = \mathrm{vec}^{-1}(|\psi\rangle_{AB}). \tag{2.14}$$

Since the matrix M has a singular value decomposition, we have $M = \sum_j \alpha_j |\varphi_j\rangle_A \otimes \langle\phi_j^*|_B$. Further vectorizing the matrix M leads to the Schmidt decomposition of the two-body quantum state.

Problem 2.2. *Is it possible to perform a Schmidt decomposition on a quantum state of three systems, ABC?*

2.1.2 Operators and the Expectation Values of Operators

Basic Definition of Operators

The specific meaning of a quantum state can be manifested through operators, which correspond to observables. Common operators include position $\hat{X}$, momentum $\hat{P}$, angular momentum $\hat{L}$, energy $\hat{H}$, spin $\hat{S}$, etc. For a general operator $\hat{O}$, we can obtain the expectation value of the operator by measuring the quantum state $|\psi\rangle$ (which will be defined later), i.e.,

$$\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle. \tag{2.15}$$

Note that since the measurement result is generally a real number, the operator $\hat{O}$ is Hermitian. Therefore, the specific meaning of the vector representation of the quantum state is, in part, determined by the expectation value of the operator as given by the above equation.

For example, suppose the quantum state $|\psi\rangle = a|0\rangle + b|1\rangle$ represents a superposition state of positions $|0\rangle$ and $|1\rangle$, where the position operator can be written as

$$\hat{X} = x_0 |0\rangle\langle 0| + x_1 |1\rangle\langle 1|, \tag{2.16}$$

where x_0 and x_1 represent specific positions. The expectation value of the position operator is then

$$\langle \hat{X} \rangle = x_0 |a|^2 + x_1 |b|^2. \tag{2.17}$$

Here, $|a|^2$ and $|b|^2$ can be interpreted as the probabilities of the quantum system being at positions x_0 and x_1, respectively, while $\langle \hat{X} \rangle$ represents the average position. We will further explore the meaning of expectation values of operators through quantum measurements later. The symbolˆgenerally represents an operator in quantum mechanics, but for simplicity, we will omit theˆnotation in the rest of the paper.

Qubit Operators

The ***Pauli operators*** $\{\sigma_X,\ \sigma_Y,\ \sigma_Z\}$ are commonly used operators for quantum bit systems,

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.18}$$

The Pauli operators, along with the identity matrix $\sigma_I := \mathbb{I}$, form an unnormalized basis set for all Hermitian operators, satisfying

$$\langle \sigma_i,\ \sigma_j \rangle = 2\delta_{ij}, \tag{2.19}$$

where $i,\ j \in \{I,\ x,\ y,\ z\}$. Therefore, any Hermitian operator O can be expressed as a linear combination of Pauli operators:

$$O = \alpha_0 \sigma_I + \alpha_1 \sigma_X + \alpha_2 \sigma_Y + \alpha_3 \sigma_Z. \tag{2.20}$$

From the properties of the Pauli operators, the coefficients of this linear combination are given by

$$\alpha_i = \langle \sigma_i, O \rangle / 2 = \mathrm{Tr}[\sigma_i \cdot O]/2. \tag{2.21}$$

Similarly, $\{\sigma_I,\ \sigma_X,\ \sigma_Y,\ \sigma_Z\}^{\otimes n}$ also forms a basis for n-qubit operators, i.e.,

$$O = \sum_{j_1 j_2 \cdots j_n} \alpha_{j_1 j_2 \cdots j_n} \sigma_{j_1} \otimes \sigma_{j_2} \otimes \cdots \otimes \sigma_{j_n}. \tag{2.22}$$

Problem 2.3. *Properties of Pauli matrices.*

- *Find the eigenvalues and eigenvectors of the Pauli operators.*

- *Pauli operators anti-commute to each other*

$$\{\sigma_i, \sigma_j\} = 2\delta_{i,j}. \tag{2.23}$$

- *Prove that* $\sigma_X\sigma_Y = i\sigma_Z$, $\sigma_Y\sigma_Z = i\sigma_X$, *and* $\sigma_Z\sigma_X = i\sigma_Y$.

- *Define* $\epsilon_{i,j,k} = \mathrm{sign}(\pi_{\mathrm{i,j,k}})$ *with* $(i,j,k) = \pi_{i,j,k}(1,2,3)$, *then we have*

$$\sigma_i\sigma_j = i\epsilon_{i,j,k}\sigma_k. \tag{2.24}$$

  Here, $1,2,3$ *refers to* X,Y,Z, *respectively.*

- *For two strings of Pauli operators* $\sigma_{j_1}\otimes\sigma_{j_2}\otimes\cdots\otimes\sigma_{j_n}$ *and* $\sigma_{k_1}\otimes\sigma_{k_2}\otimes\cdots\otimes\sigma_{k_n}$, *do they commute or anti-commute?*

Higher-Dimensional Quantum State Operators

For a d-dimensional quantum system, we consider the ***generalized Gell-Mann*** matrices as follows:

$$\begin{aligned}
\Lambda_s^{jk} &= |j\rangle\langle k| + |k\rangle\langle j|, \quad 1 \le j < k \le d \\
\Lambda_a^{jk} &= -i|j\rangle\langle k| + i|k\rangle\langle j|, \quad 1 \le j < k \le d \\
\Lambda^l &= \sqrt{\frac{2}{l(l+1)}}(-|l+1\rangle\langle l+1| + \sum_{j=1}^{l}|j\rangle\langle j|), \quad 1 \le l \le d-1
\end{aligned} \tag{2.25}$$

All the Λ matrices satisfy the trace condition $\mathrm{Tr}(\Lambda) = 0$ and are orthogonal. When combined with the identity matrix, they form an orthogonal, unnormalized basis. Therefore, any d-dimensional Hermitian matrix can be expressed as a linear combination of these generalized Gell-Mann matrices.

Problem 2.4. *Prove that the generalized Gell-Mann matrices form an orthogonal, unnormalized basis.*

For example, the Gell-Mann matrices in 3 dimensions are:

$$\begin{aligned}
\lambda_0 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \\
\lambda_3 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \lambda_4 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_5 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \\
\lambda_6 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \quad \lambda_7 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix}
\end{aligned} \tag{2.26}$$

We can verify that $\langle \lambda_i, \lambda_j \rangle = 3\delta_{ij}$.

2.1.3 Evolution of Quantum Systems

Schrödinger Equation

According to the ***Schrödinger equation***, the time evolution of a quantum state is given by

$$\frac{\mathrm{d}\,|\psi\rangle}{\mathrm{d}t} = -iH(t)\,|\psi\rangle\,, \tag{2.27}$$

where H is a Hermitian operator, known as the system's ***Hamiltonian***. For common systems, the Hamiltonian is usually expressed as the sum of the kinetic and potential energy terms, i.e.,

$$H(t) = -\frac{\hbar^2 \nabla^2}{2m} + V(x,\ t), \tag{2.28}$$

where m is the mass of the particle, and $\hbar$ is the reduced Planck constant. In quantum computing, since the quantum state is a multi-qubit state, the corresponding Hamiltonian can be represented by a Hermitian operator acting on multiple qubits. For example, the Hamiltonian of the Ising model with nearest-neighbor interactions in a constant field is given by

$$H = J \sum_j Z_j Z_{j+1} + h \sum_j X_j, \tag{2.29}$$

where Z_j (X_j) represents the Pauli-Z (X) operator acting on the j-th qubit. In later chapters, we will describe how to map the Hamiltonian of actual systems to

Hermitian operators acting on qubits, how to simulate the Schrödinger equation on a quantum computer, and the specific forms of physical Hamiltonians for different quantum computing platforms.

For a time-independent Hamiltonian H, the quantum state $|\psi_0\rangle$ evolves from time $t = 0$ to time t as

$$|\psi(t)\rangle = e^{-iHt}|\psi_0\rangle, \tag{2.30}$$

where e^{-iHt} is defined according to the functional rule for Hermitian matrices. Note that the transformation from quantum state $|\psi_0\rangle$ to $|\psi(t)\rangle$ is implemented by

$$U(t) = e^{-iHt}. \tag{2.31}$$

Since $U(t)$ is a unitary transformation, any transformation of quantum states in quantum mechanics is achieved by a unitary transformation[1]. Since quantum computers also follow the fundamental principles of quantum mechanics, operations in quantum computing are likewise implemented by unitary transformations. In the next chapter, we will specifically discuss how to implement unitary transformations for single, double, and multi-qubit systems.

Problem 2.5. *Prove the following results:*

- *For $U(t) = e^{-iHt}$, we have $U(t_1)U(t_2) = U(t_1 + t_2)$.*
- *When $[M, N] = 0$, it follows that $e^{-i(M+N)t} = e^{-iMt}e^{-iNt}$.*
- *Prove the Baker–Campbell–Hausdorff formula. Given Hermitian operators M, N, we have*

$$e^M e^N = e^{M+N+\frac{1}{2}[M,N]+\frac{1}{12}[M,[M,N]]-\frac{1}{12}[N,[M,N]]+\ldots} \tag{2.32}$$

Single-Qubit Pauli Rotation

Consider the case where the Hamiltonian of a single qubit is given by a Pauli operator. For a time $t = \theta/2$, the corresponding unitary transformation operations are as follows:

[1]This statement holds for pure states. For mixed states represented by density matrices, the transformation is realized by a quantum channel.

$$\mathtt{R}_X(\theta) = \exp(-i\sigma_X\theta/2) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \tag{2.33}$$

$$\mathtt{R}_Y(\theta) = \exp(-i\sigma_Y\theta/2) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \tag{2.34}$$

$$\mathtt{R}_Z(\theta) = \exp(-i\sigma_Z\theta/2) = \begin{pmatrix} \exp(-i\theta/2) & 0 \\ 0 & \exp(i\theta/2) \end{pmatrix}. \tag{2.35}$$

Based on the homomorphism mapping from SU(2) to SO(3), we can prove the following identities:

$$\mathtt{R}_X(\theta) \cdot \boldsymbol{\sigma} \cdot \mathtt{R}_X(\theta)^\dagger = R_X(\theta) \cdot \boldsymbol{\sigma}, \tag{2.36}$$

$$\mathtt{R}_Y(\theta) \cdot \boldsymbol{\sigma} \cdot \mathtt{R}_Y(\theta)^\dagger = R_Y(\theta) \cdot \boldsymbol{\sigma}, \tag{2.37}$$

$$\mathtt{R}_Z(\theta) \cdot \boldsymbol{\sigma} \cdot \mathtt{R}_Z(\theta)^\dagger = R_Z(\theta) \cdot \boldsymbol{\sigma}, \tag{2.38}$$

where $R_X(\theta)$, $R_Y(\theta)$, and $R_Z(\theta)$ correspond to counterclockwise rotation operations around the X, Y, and Z axes in three-dimensional space, respectively. Specifically,

$$R_X(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \tag{2.39}$$

$$R_Y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}, \tag{2.40}$$

$$R_Z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2.41}$$

and the vector $\boldsymbol{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$. For a unit vector $\boldsymbol{n}$, we can also consider a general Hamiltonian of the form $\boldsymbol{n} \cdot \boldsymbol{\sigma}$, leading to the following expression:

$$\mathtt{R}_{\boldsymbol{n}}(\theta) = \exp(-i\boldsymbol{n} \cdot \boldsymbol{\sigma}\theta/2) = \cos\left(\frac{\theta}{2}\right)\sigma_I - i\sin\left(\frac{\theta}{2}\right)\left(n_x\sigma_X + n_y\sigma_Y + n_z\sigma_Z\right). \tag{2.42}$$

Similarly, for a rotation matrix $R_{\boldsymbol{n}}(\theta)$ that rotates by an angle θ around the unit vector $\boldsymbol{n}$ in three-dimensional space, we have:

$$\mathtt{R}_{\boldsymbol{n}}(\theta) \cdot \boldsymbol{\sigma} \cdot \mathtt{R}_{\boldsymbol{n}}(\theta)^\dagger = R_{\boldsymbol{n}}(\theta) \cdot \boldsymbol{\sigma}. \tag{2.43}$$

Problem 2.6. *Prove the equations in this section.*

Combining with the general relation between qubits and the Bloch sphere, as discussed in (2.83), we can further prove that operations on quantum bit states can correspond to rotations on the Bloch sphere. Specifically, for the case of pure states, suppose the quantum bit state $|\psi(\boldsymbol{x})\rangle$ corresponds to a point $\boldsymbol{x}$ on the Bloch sphere, then we have

$$\mathtt{R}_{\boldsymbol{n}}(\theta)\,|\psi(\boldsymbol{x})\rangle = |\psi(R_{\boldsymbol{n}}(\theta)\cdot\boldsymbol{x})\rangle\,. \tag{2.44}$$

Problem 2.7. *For any quantum bit state $|\psi\rangle$, provide a method to implement it using single-qubit Pauli rotations.*

2.1.4 Projective measurement

Measurement and Collapse

Previously, we defined the expectation value of an operator O on a quantum state $|\psi\rangle$. Here, we will further understand the expectation value of an operator and the meaning of quantum states through projective measurements.

For a d-dimensional quantum state, for example, $|\psi\rangle = \sum_j \alpha_j\,|j\rangle$, it is a superposition of the different basis states $\{|j\rangle\}$, meaning the quantum state could be simultaneously in different states $|j\rangle$. A fundamental question is: when we perform a measurement on the quantum state $|\psi\rangle$, how will it change? According to the ***Born rule***, when we observe in the basis $\{|j\rangle\}$, the quantum state $|\psi\rangle$ will collapse randomly to $|j\rangle$, and the collapse probability is given by:

$$p_j = |\langle j|\psi\rangle|^2. \tag{2.45}$$

For example, for $|\psi\rangle = \sum_j \alpha_j\,|j\rangle$, we have $p_j = |\alpha_j|^2$. This collapse process due to measurement is referred to as the ***measurement*** of the quantum state.

Rank-1 projective measurement

Mathematically, we can abstract the measurement process as a rank-1 projective measurement on the quantum state. That is, for any set of orthogonal basis vectors $\{|u_j\rangle\}$, we have a corresponding set of projective measurements:

$$\Pi = \left\{\Pi_j = |u_j\rangle\langle u_j|\right\}. \tag{2.46}$$

For the quantum state $|\psi\rangle$, the probability of obtaining the result u_j from the projective measurement is:

$$p_j = \langle\psi|\Pi_j|\psi\rangle = |\langle u_j|\psi\rangle|^2. \tag{2.47}$$

Assuming the measurement is non-destructive, the corresponding quantum state will collapse to:

$$|\psi\rangle \to \frac{\Pi_j|\psi\rangle}{\sqrt{p_j}}. \tag{2.48}$$

We will now provide three examples to illustrate the meaning of quantum measurement:

- Given a quantum bit state $|\psi\rangle = a|0\rangle + b|1\rangle$ and the projective measurements $\{\Pi_0 = |0\rangle\langle 0|, \Pi_1 = |1\rangle\langle 1|\}$, we have

$$|\psi\rangle \to \begin{cases} |0\rangle & p_0 = |a|^2 \\ |1\rangle & p_1 = |b|^2 \end{cases} \tag{2.49}$$

- Given a quantum bit state $|\psi\rangle = a|0\rangle + b|1\rangle$ and the projective measurements $\{\Pi_+ = |+\rangle\langle +|, \Pi_- = |-\rangle\langle -|\}$, we have

$$|\psi\rangle \to \begin{cases} |+\rangle & p_0 = |a+b|^2/2 \\ |-\rangle & p_1 = |a-b|^2/2 \end{cases} \tag{2.50}$$

- We can also consider local measurements. For example, consider a two-body quantum bit state $|\psi\rangle = a|00\rangle_{AB} + b|11\rangle_{AB}$ and a projective measurement on the first system A, $\{\Pi_+ = |+\rangle\langle +|_A, \Pi_- = |-\rangle\langle -|_A\}$, we have

$$|\psi\rangle \to \begin{cases} |+\rangle_A (a|0\rangle + b|1\rangle)_B & p_0 = 1/2 \\ |-\rangle_A (a|0\rangle - b|1\rangle)_B & p_1 = 1/2 \end{cases} \tag{2.51}$$

  Here, the subscripts AB denote the two quantum systems.

Problem 2.8. *Projective measurements on single and multiple qubits.*

- *For the quantum state $|\psi\rangle = a|0\rangle + b|1\rangle$, first perform the $\{\Pi_0 = |0\rangle\langle 0|, \Pi_1 = |1\rangle\langle 1|\}$ measurement, and then perform the $\{\Pi_+ = |+\rangle\langle +|, \Pi_- = |-\rangle\langle -|\}$ measurement. Calculate the measurement results and the corresponding probabilities. If we first perform the $\{\Pi_+ = |+\rangle\langle +|, \Pi_- = |-\rangle\langle -|\}$ measurement, and then perform the $\{\Pi_0 = |0\rangle\langle 0|, \Pi_1 = |1\rangle\langle 1|\}$ measurement, are the results the same?*

- *Consider the quantum state* $|\psi\rangle = \frac{1}{\sqrt{2}}(|000\rangle_{ABC} + |111\rangle_{ABC})$. *First, perform a projective measurement on the system A,* $\{\Pi_+^A = |+\rangle\langle+|_A, \Pi_-^A = |-\rangle\langle-|_A\}$, *and then perform a projective measurement on system B,* $\{\Pi_+^B = |+\rangle\langle+|_B, \Pi_-^B = |-\rangle\langle-|_B\}$. *Calculate the probabilities and the corresponding quantum states for all four measurement results. Assume the measurement on B is performed before the measurement on A. Are the results the same?*

Any rank-1 projection measurement $\Pi = \{\Pi_j = |u_j\rangle\langle u_j|\}$ can be implemented by a unitary transformation and the standard computational basis projection measurement $\bar{\Pi} = \{\bar{\Pi}_j = |j\rangle\langle j|\}$. It is always possible to find a unitary transformation U such that $|u_j\rangle = U|j\rangle$, so we can implement any rank-1 projection measurement by first applying $U^\dagger$ to the quantum state, followed by a projection measurement in the computational basis. That is, we have:

$$\begin{aligned} p_j &= \langle\psi|\Pi_j|\psi\rangle = \langle\psi|U\cdot\bar{\Pi}_j\cdot U^\dagger|\psi\rangle, \\ |\psi\rangle &\to \frac{\Pi_j|\psi\rangle}{\sqrt{p_j}} = \frac{\bar{\Pi}_j\cdot U^\dagger|\psi\rangle}{\sqrt{p_j}}. \end{aligned} \tag{2.52}$$

For example, if we want to measure the state $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis, we can first apply the unitary rotation

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \tag{2.53}$$

and then perform a measurement on the quantum state in the computational basis, which is equivalent to the desired projection measurement.

The measurement and collapse of a quantum state can be described using ***von Neumann's measurement model***. Consider a simple example: a projection measurement of a quantum bit state $|\psi\rangle = a|0\rangle + b|1\rangle$ in the computational basis. The measurement process can be described as follows:

1. Initial state: The quantum state to be measured is $|\psi\rangle_S = a|0\rangle_S + b|1\rangle_S$, and the measurement device is in the state $|0\rangle_M$.

2. Interaction between the measurement device and the quantum state: The quantum state and the measurement device evolve into $a|0\rangle_S|0\rangle_M + b|1\rangle_S|1\rangle_M$.

3. Collapse: The measurement device is in a macroscopic distinguishable state $|0\rangle_M$ or $|1\rangle_M$, and the device couples with the environment, causing collapse. The overall quantum state collapses to $|0\rangle_S|0\rangle_M$ with probability $|a|^2$ and to $|1\rangle_S|1\rangle_M$ with probability $|b|^2$.

Of course, von Neumann's measurement model does not explain how the collapse process occurs specifically, which remains one of the few unresolved controversies in quantum mechanics.

General Projective Measurements

In quantum mechanics, a more general projective measurement Π is defined as:

$$\Pi = \left\{ \Pi_j : \ \Pi_i \Pi_j = \delta_{ij} \Pi_i, \ \sum_j \Pi_j = \mathbb{I} \right\}. \tag{2.54}$$

Rank-1 projective measurements are a special case of general projective measurements where each projection operator Π_j has rank 1. For example, in the case of local system projective measurements, we can represent the projection measurement as follows:

$$\{\Pi_+ = |+\rangle\langle+|_A \,,\ \Pi_- = |-\rangle\langle-|_A\} := \{\Pi_+ = |+\rangle\langle+|_A \otimes \mathbb{I}_B,\ \Pi_- = |-\rangle\langle-|_A \otimes \mathbb{I}_B\}, \tag{2.55}$$

where Π_+ and Π_- both have rank 2. General projective measurements can be implemented by adding auxiliary qubits, applying a global unitary transformation on the auxiliary qubits and the system to be measured, and then measuring the auxiliary qubits.

Consider another example, where we have a qutrit quantum state $|\psi\rangle = a|0\rangle + b|1\rangle + c|2\rangle$ and the projection measurement:

$$\Pi = \{\Pi_{01} = |0\rangle\langle 0| + |1\rangle\langle 1| \,,\ \Pi_2 = |2\rangle\langle 2|\}, \tag{2.56}$$

The measurement outcomes are:

$$|\psi\rangle \to \begin{cases} (a|0\rangle + b|1\rangle)/\sqrt{|a|^2+|b|^2} & p_0 = |a|^2 + |b|^2, \\ |2\rangle & p_1 = |c|^2, \end{cases} \tag{2.57}$$

Here, the projection measurement projects the quantum state onto the subspace $\{|0\rangle, |1\rangle\}$ and the state $|2\rangle$.

To implement this projection measurement Π using a trace-1 projection operator, we consider an auxiliary qubit initially in the state $|0\rangle$, and then apply the unitary transformation to the combined state of the auxiliary qubit and the quantum state $|0\rangle|\psi\rangle$:

$$U = \mathbb{I} \otimes (\Pi_{01}) + X \otimes \Pi_2, \tag{2.58}$$

where X is the Pauli-X operator. After the unitary transformation, the quantum state becomes:

$$U|0\rangle|\psi\rangle = |0\rangle(a|0\rangle + b|1\rangle) + c|1\rangle|2\rangle. \tag{2.59}$$

Thus, by performing a trace-1 projection measurement on the auxiliary qubit in the computational basis, we can equivalently implement the projection measurement Π.

Problem 2.9. *Does the measurement* $\Pi = \{\Pi_{01} = |0\rangle\langle 0| + |1\rangle\langle 1|,\ \Pi_2 = |2\rangle\langle 2|\}$ *give the same results as the measurement* $\tilde{\Pi} = \{\Pi_0 = |0\rangle\langle 0|,\ \Pi_1 = |1\rangle\langle 1|,\ \Pi_2 = |2\rangle\langle 2|\}$*?*

Projective Measurements and the Relation to Operator Averages

We previously defined the average value of an operator O in a quantum state $|\psi\rangle$ as $\langle\psi|O|\psi\rangle$. Here, we explore how to understand this average value through projective measurements.

1. First, we decompose the operator O into its spectral decomposition:

$$O = \sum_j \lambda_j |u_j\rangle\langle u_j|. \tag{2.60}$$

2. A projection measurement $\{\Pi_j = |u_j\rangle\langle u_j|\}$ is then performed on the quantum state $|\psi\rangle$, and we obtain the result u_j with probability $p_j = \langle\psi|\Pi_j|\psi\rangle$.

3. The average value of the operator O is equivalent to the weighted sum of its eigenvalues, given by:

$$\langle O\rangle = \sum_j p_j \lambda_j. \tag{2.61}$$

Thus, the average value of an operator is essentially the average of its eigenvalues, obtained by performing a projection measurement in the eigenbasis of the operator.

For example, let's consider the average value of the Pauli operator σ_X for the quantum state $|\psi\rangle = a|0\rangle + b|1\rangle$. We can perform an equivalent projection measurement on $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis, where we get probabilities $p_\pm = |a \pm b|^2/2$. The average value of σ_X is then:

$$\langle\sigma_X\rangle = p_+ - p_- = \frac{|a+b|^2}{2} - \frac{|a-b|^2}{2} = ab^* + a^*b, \tag{2.62}$$

which is consistent with the direct calculation of $\langle\psi|\sigma_X|\psi\rangle$.

Indeed, using projective measurements to obtain the average value of an operator is a way to interpret and experimentally implement operator averaging. In most cases, we can directly use the formula for the average value of an operator to compute it. For Pauli operators $\sigma_X, \sigma_Y, \sigma_Z$, the corresponding measurements are referred to as measurements in the X, Y, and Z bases, respectively.

For two operators O and O', when their commutator $[O, O'] \neq 0$, their eigenstates are generally not the same, so we cannot measure both operators simultaneously in the same basis. However, when $[O, O'] = 0$, we can choose a common eigenbasis to diagonalize both operators and measure them simultaneously.

Problem 2.10. *Consider the two-qubit quantum state $|\psi\rangle_{AB} = a|00\rangle + b|11\rangle$:*

- *Compute the average values of the XX and ZZ operators.*
- *Perform measurements on the state $|\psi\rangle_{AB}$ in the X and Z bases, and use the measurement probabilities along with the eigenvalues of the XX and ZZ operators to calculate their average values.*
- *Notice that the X and Z bases do not commute, so they cannot be measured simultaneously. However, XX and ZZ do commute, so in principle, they can be measured simultaneously. Construct a set of projection measurement bases such that XX and ZZ can be measured simultaneously.*

2.2 Density Matrices, Quantum Channels, and POVMs

Although quantum mechanics can be fully described using pure states, unitary transformations, and projective measurements, we can further extend this theoretical framework to make it easier to describe situations such as open systems, subsystems, and general theories for studying quantum information. We will see that quantum states, evolution, and measurements can be described using more general density matrices, quantum channels, and POVMs. We will also observe that these two frameworks are equivalent when calculating any physical observables.

2.2.1 Density Matrix

We will introduce the density matrix description of quantum states from three different perspectives: mathematical, quantum ensembles, and quantum subsystems.

We will see that the density matrix serves as a more systematic mathematical description of quantum states, and, on the other hand, provides a more concise method for describing more general situations.

Mathematical Perspective

In previous sections, we learned that quantum states are described by vectors, such as $|\psi\rangle$; observables are described by operators or matrices, such as O; and the expectation value of an observable is defined as $\langle O\rangle_{|\psi\rangle} = \langle\psi|O|\psi\rangle$. Physically, an observable defines different possible states, and the quantum state determines the specific state of the physical system (which may be in a superposition). Therefore, for any observable, different quantum states will yield different expectation values for that observable. From a mathematical point of view, quantum states can be considered as functions mapping observables to their expectation values.

Let the observable be O, the quantum state be ρ, and the expectation value of the observable be $\langle O\rangle_\rho$, thus we have

$$\langle O\rangle_\rho := \rho(O). \tag{2.63}$$

We note that any linear transformation f of a matrix M to a real number can be written as

$$f(M) = \mathrm{Tr}[A \cdot M]. \tag{2.64}$$

Quantum mechanics also satisfies linear properties, so equation (2.63) can be further written as

$$\langle O\rangle_\rho := \mathrm{Tr}[\rho \cdot O], \tag{2.65}$$

Here, we use ρ to represent the matrix corresponding to this linear transformation. We can further determine the conditions that the matrix ρ must satisfy:

- Since the operator O is Hermitian and the expectation value of an operator is real, the matrix ρ must also be Hermitian.
- Additionally, when we choose O as a projection operator, the expectation value $\langle O\rangle_\rho$ corresponds to the probability of the measurement outcome. Since probabilities are non-negative, ρ must satisfy the condition of being non-negative.
- Finally, when we select O as the identity matrix, representing the projection measurement over all possible measurement outcomes, we have $\langle O\rangle_\rho = \mathrm{Tr}[\rho] = 1$, which ensures normalization.

Since the matrix ρ determines the probability distribution for measurement outcomes, it is referred to as the ***density matrix***, and it must satisfy $\rho \geq 0$ and $\mathrm{Tr}[\rho] = 1$. Therefore, from a mathematical perspective, the more general description of a quantum state is the density matrix.

Problem 2.11. *For the matrix* $M = \begin{pmatrix} a & b \\ c & d, \end{pmatrix}$*, give the conditions under which* M *is a density matrix.*

Meanwhile, the vector description of a quantum state can also be converted into the density matrix form. We note that the expectation value of an operator is given by

$$\langle\psi|O|\psi\rangle = \mathrm{Tr}\big[\,|\psi\rangle\langle\psi|\,O\big], \tag{2.66}$$

Therefore, according to equation (2.65), we have

$$\rho = |\psi\rangle\langle\psi|\,. \tag{2.67}$$

Such a state is referred to as a ***pure state***, while other states are called ***mixed states***. A simple method to verify if a density matrix corresponds to a pure state is to calculate $\mathrm{Tr}[\rho^2]$, where the state is pure if and only if $\mathrm{Tr}[\rho^2] = 1$. Of course, this density matrix corresponds only to a rank-1 density matrix. But what does a general density matrix correspond to in terms of quantum states? We will address this question in the following section.

Quantum Ensembles

Consider a quantum ensemble, which is a quantum system whose possible states are $\{|\psi_j\rangle\}$, with the state $|\psi_j\rangle$ occurring with probability p_j, denoted as $\{p_j, |\psi_j\rangle\}$. Let us consider the evolution and measurement results of such a quantum ensemble. First, if the ensemble undergoes evolution under U, the quantum state $|\psi_j\rangle$ will evolve to $U|\psi_j\rangle$, while the probability p_j remains unchanged. Therefore, the ensemble evolves to $\{p_j, U|\psi_j\rangle\}$. On the other hand, if we measure the expectation value of an observable O for the ensemble, we can calculate the average value of the observable for each state $|\psi_j\rangle$, which is $\langle\psi_j|O|\psi_j\rangle$. Thus, the total average value for the ensemble is

$$\langle O\rangle_{\{p_j,|\psi_j\rangle\}} = \sum_j p_j\,\langle\psi_j|O|\psi_j\rangle\,. \tag{2.68}$$

This is the method of representing a quantum ensemble by mixing vectors and classical probabilities. But can we simplify this representation further? Can we

describe it using a density matrix? First, according to equation (2.66), equation (2.68) can be rewritten as

$$\begin{aligned}\langle O\rangle_{\{p_j,|\psi_j\rangle\}} &= \sum_j p_j \mathrm{Tr}\big[\,|\psi_j\rangle\langle\psi_j|\,O\big],\\ &= \mathrm{Tr}\left[\left(\sum_j p_j\,|\psi_j\rangle\langle\psi_j|\right)O\right].\end{aligned} \tag{2.69}$$

Thus, if we define $\rho = \sum_j p_j\,|\psi_j\rangle\langle\psi_j|$, we have

$$\langle O\rangle_{\{p_j,|\psi_j\rangle\}} = \mathrm{Tr}\left[\rho\cdot O\right], \tag{2.70}$$

which shows that the expectation value of the observable can also be equivalently obtained using the density matrix. Meanwhile, the evolution of the ensemble under U corresponds to the evolution of the density matrix as $\rho \to U\rho U^\dagger$. Therefore, we can equivalently represent the quantum ensemble using the density matrix ρ.

Problem 2.12. *Verify that* $\langle O\rangle_{\{p_j,U|\psi_j\rangle\}} = \mathrm{Tr}[\rho'\cdot O]$, *where* $\rho' = U\rho U^\dagger$.

A quantum ensemble also provides an interpretation of the density matrix, namely, it represents a quantum state that is a classical mixture. Physically, the introduction of ensembles was initially to study thermodynamic systems, i.e., macroscopic phenomena resulting from a large number of microscopic particles. Since we generally cannot directly consider the state of all microscopic particles, and macroscopic phenomena are not determined by the state of a single particle, we describe macroscopic physical phenomena using ensembles, which represent the probabilities of the system being in different states. To study thermodynamic phenomena in quantum systems, a natural idea is to extend classical ensembles to quantum ensembles. Of course, quantum ensembles are not limited to describing quantum systems in the thermodynamic limit. In physics, they can also describe thermodynamic effects in general small- to medium-scale quantum systems, such as quantum simulators. In quantum information, quantum ensembles can describe classically mixed quantum systems, for instance, a device that randomly prepares quantum states.

We note that the key point in a quantum ensemble is that the quantum state may be classically mixed with different states. This classical mixing is fundamentally different from quantum superposition. Quantum superposition can give rise to coherence and quantum entanglement, which are essential resources for quantum information and quantum computing. In contrast, classical mixing is

purely a classical operation, like randomly rolling a die, which represents the average state of the die. Classically mixing quantum states generally does not lead to further quantum effects. In experiments, quantum state mixing typically results from noise, operational errors, or information loss. On the other hand, we do not usually consider classical mixing for classical systems. For example, in the case of randomly rolling a die, although its state is unknown and random, we can always observe its state each time and record the specific state, thus avoiding classical mixing. For quantum systems, due to the no-cloning or no-distinguishability principle (discussed in Section 2.3), we often cannot know which specific state a quantum system is in, thus we inevitably need classical probability mixing. We will further explore the concept of quantum ensembles and density matrices in the context of quantum subsystems.

Quantum Subsystems

Next, we introduce the density matrix from the perspective of quantum subsystems. Consider a quantum state $|\psi\rangle_{AB}$ for the AB system. In terms of operations, we can always perform operations on the subsystems of AB. For example, we can consider the evolution or measurement operations on the A or B subsystems. Suppose the AB system is spatially separated such that the state of subsystem B and any operations on it do not affect subsystem A (according to causality). In this case, any operation or measurement on subsystem A should only depend on the state of subsystem A.

Since $|\psi\rangle_{AB}$ is the state of the entire AB system, how can we obtain the local state of subsystem A? Consider measuring an observable O_A on subsystem A, which corresponds to the global observable $O = O_A \otimes \mathbb{I}_B$ [2]. Then, the expectation value of the observable is

$$\langle O \rangle = \langle \psi | O_A \otimes \mathbb{I}_B | \psi \rangle_{AB} = \mathrm{Tr}_{AB}[(O_A \otimes \mathbb{I}_B) \cdot |\psi\rangle \langle \psi|_{AB}]. \tag{2.71}$$

Thus, we can define the density matrix for subsystem A as

$$\rho_A = \mathrm{Tr}_B[|\psi\rangle \langle \psi|_{AB}], \tag{2.72}$$

so that for any measurement on subsystem A, we have

$$\langle O_A \rangle = \mathrm{Tr}[\rho_A \cdot O_A]. \tag{2.73}$$

Similarly, the evolution of subsystem A under a unitary operation U_A corresponds to the evolution of the density matrix ρ_A as $\rho_A \to U_A \rho_A U_A^\dagger$.

[2] This can be understood from the perspective of projective measurements.

Problem 2.13. *Verify that* $\mathrm{Tr}_B[U_A|\psi\rangle\langle\psi|_{AB}U_A^\dagger] = U_A\rho_A U_A^\dagger$, *where* $\rho_A = \mathrm{Tr}_B[|\psi\rangle\langle\psi|_{AB}]$.

Quantum subsystems are closely related to quantum ensembles. Consider a joint quantum state $|\psi\rangle_{AB}$ of the AB system, where B is a qubit. This state can always be written as

$$|\psi\rangle_{AB} = \alpha_0|\psi_0\rangle_A|0\rangle_B + \alpha_1|\psi_1\rangle_A|1\rangle_B, \tag{2.74}$$

and the corresponding density matrix for subsystem A is $\rho_A = p_0|\psi_0\rangle\langle\psi_0|_A + p_1|\psi_1\rangle\langle\psi_1|_A$, where $p_0 = |\alpha_0|^2$ and $p_1 = |\alpha_1|^2$.

Suppose subsystem B is measured in the $\{|0\rangle, |1\rangle\}$ basis. The measurement outcomes correspond to obtaining $|\psi_0\rangle_A|0\rangle_B$ with probability $p_0 = |\alpha_0|^2$, and $|\psi_1\rangle_A|1\rangle_B$ with probability $p_1 = |\alpha_1|^2$. Since the measurement is random, the state of subsystem A is described by a quantum ensemble $\{p_i, |\psi_i\rangle_A\}_{i=1,2}$, and the corresponding density matrix is $\rho_A = p_0|\psi_0\rangle\langle\psi_0|_A + p_1|\psi_1\rangle\langle\psi_1|_A$. Hence, the density matrix obtained from the subsystem and the ensemble is the same.

There are a few points to note here:

- Since subsystem B can perform arbitrary basis measurements, or even POVM measurements as discussed later, subsystem A might correspond to different quantum ensembles. However, the density matrix for subsystem A remains unchanged.

- All measurements on subsystem A are completely determined by its density matrix. Therefore, we cannot distinguish between different quantum ensembles obtained from measurements on subsystem B. This is consistent with causality, meaning subsystem A cannot know any operations performed on subsystem B.

- Thus, we can assume that subsystem A first performs a measurement, with the measurement outcome determined by its corresponding density matrix. Alternatively, we can assume that subsystem B always performs a measurement, and the density matrix is obtained through the quantum ensemble of subsystem A. Unless subsystem B informs subsystem A of the measurement results, these two processes or descriptions are equivalent and indistinguishable from the perspective of subsystem A.

Therefore, when we consider quantum phenomena of quantum ensembles or subsystems, we can always unify the description using density matrices.

Problem 2.14. *For the quantum state* $|\psi\rangle_{ABC} = \alpha|000\rangle_{ABC} + \beta|111\rangle_{ABC}$*, calculate* ρ_A*,* ρ_B*,* ρ_C*, and* ρ_{AB}*.*

Problem 2.15. *A quantum state* ρ *is a pure state if and only if* $\mathrm{Tr}[\rho^2] = 1$*.*

Different Ensemble Representations of the Same Density Matrix

Given a density matrix ρ, its corresponding ensemble or decomposition is not unique. For example, for $\rho = \mathbb{I}_2/2$, we have the following different representations:

$$\rho = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}(|+\rangle\langle +| + |-\rangle\langle -|) = \frac{1}{2}(\frac{\mathbb{I}_2}{2} + \frac{\mathbb{I}_2}{2}),$$

and there are infinitely many other decompositions. For two equivalent pure-state ensemble decompositions:

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j| = \sum_k q_k |\phi_k\rangle\langle\phi_k|, \tag{2.75}$$

then and only then, there exists a unitary transformation U such that

$$\sqrt{p_j}|\psi_j\rangle = \sum_k U_{jk}\sqrt{q_k}|\phi_k\rangle. \tag{2.76}$$

If the number of $|\psi_j\rangle$ and $|\phi_k\rangle$ are not the same, we add zeros to make them have the same number of elements.

Proof. When the equation (2.76) holds, it is straightforward to verify the equivalence of the quantum decompositions. Conversely, for two equivalent decompositions, we first consider the spectral decomposition of ρ:

$$\rho = \sum_e p_e |e\rangle\langle e|. \tag{2.77}$$

Because $\{|e\rangle\}$ forms an orthogonal basis, we can express $|\psi_j\rangle$ and $|\phi_k\rangle$ in terms of this basis as:

$$|\psi_j\rangle = \sum_e \psi_{je}|e\rangle, \quad |\phi_k\rangle = \sum_e \phi_{ke}|e\rangle. \tag{2.78}$$

Therefore, we have

$$\rho = \sum_e p_e |e\rangle\langle e| = \sum_{j,e,e'} p_j \psi_{je}\psi^*_{je'}|e\rangle\langle e'| = \sum_{k,e,e'} q_k \phi_{ke}\phi^*_{ke'}|e\rangle\langle e'|, \tag{2.79}$$

and

$$\sum_j p_j \psi_{je} \psi^*_{je'} = \sum_k q_k \phi_{ke} \phi^*_{ke'} = p_e \delta_{ee'}. \tag{2.80}$$

Assuming we add zeros to ensure that the sets $\{|e\rangle\}$, $\{|\psi_j\rangle\}$, and $\{|\phi_k\rangle\}$ all have the same number of elements, we have:

$$\sqrt{p_j}\,|\psi_j\rangle = \sum_e U_{je}\sqrt{p_e}\,|e\rangle\,, \quad \sqrt{q_k}\,|\phi_k\rangle = \sum_e V_{ke}\sqrt{p_e}\,|e\rangle\,, \tag{2.81}$$

where $U_{je} = \psi_{je}$ and $V_{ke} = \phi_{ke}$ are unitary transformations. Therefore, we have:

$$\sqrt{p_j}\,|\psi_j\rangle = \sum_k (U \cdot V^\dagger)_{jk}\sqrt{q_k}\,|\phi_k\rangle\,. \tag{2.82}$$

□

Problem 2.16. *Prove the following conclusions:*

- *A quantum state ρ has different ensemble representations containing different quantum states if and only if ρ is a mixed state.*
- *Any mixed state has infinitely many ensemble representations.*
- *For any two equivalent ensemble representations $|\psi_j\rangle = \sum_e \psi_{je}|e\rangle$, and $|\phi_k\rangle = \sum_e \phi_{ke}|e\rangle$, we have $\sum_j p_j = \sum_k q_k = 1$ and*

$$\sum_{j,j'} p_j p_{j'} |\langle\psi_j|\psi_{j'}|\rangle^2 = \sum_{k,k'} q_k q_{k'} |\langle\phi_k|\phi_{k'}|\rangle^2\,.$$

Bloch Sphere

The density matrix ρ of a single qubit can be written as

$$\rho = \frac{1 + \boldsymbol{n}\cdot\boldsymbol{\sigma}}{2}, \tag{2.83}$$

where $\boldsymbol{n} = \mathrm{Tr}[\rho\cdot\boldsymbol{\sigma}]$ is a vector with a norm less than 1, and $\boldsymbol{\sigma} = (\sigma_X, \sigma_Y, \sigma_Z)$ represents the Pauli matrices. As shown in Figure 2.2, any single qubit can be

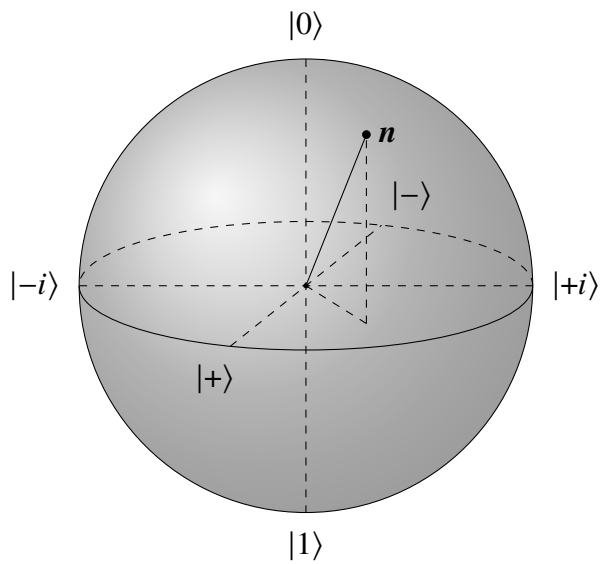


Figure 2.2: Bloch sphere and density matrices.

uniquely mapped to a point on a Bloch sphere of radius 1. This mapping is consistent with the Bloch sphere depicted in Figure 2.1. Specifically, for any pure state $\rho = |\psi\rangle\langle\psi|$ where

$$|\psi\rangle = e^{i\alpha}\left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\psi}|1\rangle\right),$$

we have the following components of the Bloch vector $\boldsymbol{n}$:

$$\begin{aligned} n_x &= \mathrm{Tr}[|\psi\rangle\langle\psi|\sigma_X] = \sin\theta\cos\psi, \\ n_y &= \mathrm{Tr}[|\psi\rangle\langle\psi|\sigma_Y] = \sin\theta\sin\psi, \\ n_z &= \mathrm{Tr}[|\psi\rangle\langle\psi|\sigma_Z] = \cos\theta. \end{aligned} \tag{2.84}$$

From earlier discussions, we know that the rotation of the Bloch sphere corresponds directly to quantum bit operations.

Problem 2.17. *Prove the formula* (2.44).

Problem 2.18. *Prove that a quantum state is pure if and only if* $\boldsymbol{n}$ *is a unit vector.*

For a system of n qubits, the density matrix ρ can be written similarly in terms of the basis $\{\mathbb{I}, \boldsymbol{\sigma}\}^{\otimes n}$, giving the expansion

$$\rho = \frac{1}{2^n}\sum_{\sigma\in\{\mathbb{I},\boldsymbol{\sigma}\}^{\otimes n}} \alpha(\sigma)\sigma, \tag{2.85}$$

where the real coefficients $\alpha(\sigma) = \mathrm{Tr}[\rho \cdot \sigma] \in [0, 1]$ and we have $\alpha(\mathbb{I}) = 1$.

Problem 2.19. *Pauli coefficients for multi-qubit states.*

- *Prove that the coefficients $\alpha(\sigma)$ of the density matrix ρ for n qubits satisfy*

$$\sum_{\sigma \in \{\mathbb{I}, \boldsymbol{\sigma}\}^{\otimes n}} \alpha(\sigma)^2 \leq 2^n,$$

  and that equality holds if and only if the quantum state is pure.

- *Is it true that for any coefficients satisfying the condition*

$$\sum_{\sigma \in \{\mathbb{I}, \boldsymbol{\sigma}\}^{\otimes n}} \alpha(\sigma)^2 \leq 2^n,$$

  there exists a corresponding density matrix?

Purification

For the AB system, we can define the density matrix of subsystem A or subsystem B. Conversely, given the density matrix of one subsystem, we can always find a larger system such that the overall system is in a pure state, while the subsystem has the given density matrix. Specifically, given the density matrix ρ_A of subsystem A, we can always find its ***purification***, which is a pure state ψ_{AB} of the AB system that satisfies $\rho_A = \mathrm{Tr}_B[\psi_{AB}]$, where we abbreviate $\psi_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|$. Some important features of purification are as follows:

- The size of the purified system B must be at least the rank of subsystem A.

- Purification is not unique. For two different purifications, ψ_{AB} and ϕ_{AC}, suppose that the dimension of B is no larger than that of C. These purifications can be related by an isometric transformation V, i.e., $\psi_{AC} = V\psi_{AB}V^\dagger$, and we have $V^\dagger V = I$.

- For any given decomposition $\rho_A = \sum_j p_j |\psi_j\rangle\langle\psi_j|_A$, one possible purification is given by

$$|\psi\rangle_{AB} = \sum_j \sqrt{p_j}\, |\psi_j\rangle_A |j\rangle_B .$$

- Define the maximally entangled state as $|\Phi^+\rangle_{AB} = \sum_j |jj\rangle_{AB}$. One possible purification is given by

$$|\psi\rangle_{AB} = \sqrt{\rho_A} \otimes \mathbb{I}_B |\Phi^+\rangle_{AB}.$$

Problem 2.20. *Prove that for* $|\psi\rangle_{AB} = \sqrt{\rho_A} \otimes \mathbb{I}_B |\Phi^+\rangle_{AB}$*, we have* $\rho_A = \mathrm{Tr}_B[\psi_{AB}]$.

For a given pure state ψ_{AB}, we have $\rho_A = \mathrm{Tr}_B[\psi_{AB}]$ and $\rho_B = \mathrm{Tr}_A[\psi_{AB}]$. Suppose that the dimension of A is no larger than the dimension of B, an important result is that there exists an isometric transformation V such that

$$\rho_B = V \rho_A V^\dagger,$$

which means that the eigenvalues of the density matrices of subsystems A and B are identical. Therefore, when the AB system is the purification of subsystem A, the state of subsystem B will be correlated with the state of subsystem A.

2.2.2 Quantum Channels

For the evolution of quantum states, we will first introduce quantum channels from both the mathematical perspective and the perspective of subsystem evolution. From the mathematical standpoint, we will start by introducing completely positive and trace-preserving linear maps, and provide concrete representations through two descriptions: the Choi–Jamiolkowski isomorphism and the Kraus representation. We will further discuss the physical realization of quantum channels, which corresponds to the evolution of subsystems, and highlight the equivalence of these different approaches. Additionally, we will introduce Pauli transfer matrices, the purification of quantum channels, and some typical quantum channels.

Mathematical Description

The evolution of quantum states essentially refers to transforming one quantum state into another (note that the dimensions of the quantum states may differ). Since quantum states are described by density matrices, the evolution of quantum states corresponds to a transformation of matrices. This transformation must satisfy the following conditions:

- ***Linearity***: Due to the linear nature of quantum mechanics, the transformation must also be linear. That is, for a linear transformation $\mathcal{E}(\rho)$ of a density matrix ρ, we must have

$$\mathcal{E}(\rho_1 + \rho_2) = \mathcal{E}(\rho_1) + \mathcal{E}(\rho_2).$$

- ***Positive Definity***: Since density matrices are positive semidefinite, the transformation must preserve this property. That is, for any positive semidefinite matrix $\rho_A \geq 0$, we must have $\mathcal{E}_A(\rho_A) \geq 0$. However, positive definiteness alone is not sufficient. For example, consider a larger system ρ_{AB}. A positive definite map $\mathcal{E}_A$ acting only on subsystem A does not guarantee that $\mathcal{E}_A(\rho_{AB}) \geq 0$. A famous counterexample is the transposition map, which is positive definite but does not guarantee $\rho_{AB}^\top \geq 0$. Therefore, we require that the map be ***completely positive*** (CP), which ensures that for any $\rho_{AB} \geq 0$, we have $\mathcal{E}_A(\rho_{AB}) \geq 0$.

- ***Trace Preservation***: Since the trace of a quantum state is 1, the transformation should not change the trace. Combining the linearity condition with the trace-preservation requirement, we have

$$\mathrm{Tr}[\mathcal{E}(\rho)] = \mathrm{Tr}[\rho].$$

Thus, a legitimate quantum state transformation must satisfy the conditions of linearity, complete positivity, and trace preservation, which together define a ***quantum channel*** (CPTP map).

For a matrix ρ, the linear map $\mathcal{E}$ can generally be represented as

$$\mathcal{E}(\rho) = \sum_{j,k} M_j \rho N_k,$$

where M_j and N_k are operators associated with the channel. The trace-preserving condition requires that

$$\mathrm{Tr}[\mathcal{E}(\rho)] = \sum_{j,k} \mathrm{Tr}[M_j \rho N_k] = \mathrm{Tr}\left[\sum_{j,k} N_k M_j \rho\right] = 1,$$

which implies that

$$\sum_{j,k} N_k M_j = \mathbb{I}.$$

On the other hand, to ensure that $\mathcal{E}$ is completely positive, we need to introduce the Choi–Jamiolkowski isomorphism.

Choi–Jamiolkowski Isomorphism

Consider a quantum channel $\mathcal{E}_{A\to B}$ that evolves the system A to system B (with potentially different dimensions for A and B). The Choi matrix of the channel $\mathcal{E}$ is defined as

$$\Phi_{\mathcal{E}_{A\to B}} = \sum_{j,k} |j\rangle\langle k|_A \otimes \mathcal{E}_{A\to B}(|j\rangle\langle k|)_B, \tag{2.86}$$

which maps the channel $\mathcal{E}_{A\to B}$ from system A to system B into a matrix acting on the combined AB system. The Choi matrix satisfies the following properties:

- ***Choi Matrix as a Quantum State***: The Choi matrix is a non-normalized quantum state:

  $$\Phi_{\mathcal{E}_{A\to B}} \geq 0, \quad \text{and} \quad \mathrm{Tr}[\Phi_{\mathcal{E}_{A\to B}}] = d_A,$$

  where d_A is the dimension of system A. Specifically, for the unnormalized maximally entangled state $|\Phi^+\rangle_{A'A} = \sum_j |jj\rangle_{A'A}$, we have

  $$\Phi_{\mathcal{E}_{A\to B}} = \mathcal{E}_{A\to B}(\Phi^+_{A'A}),$$

  where $A \equiv A'$.

- ***Choi Matrix Conditions***: The Choi matrix satisfies the conditions

  $$\Phi_{\mathcal{E}_{A\to B}} \geq 0 \quad \text{and} \quad \mathrm{Tr}_B[\Phi_{\mathcal{E}_{A\to B}}] = \mathbb{I}_A.$$

- ***Inverse Relationship***: For any Choi matrix Φ_{AB} satisfying the above conditions, it corresponds to a quantum channel $\mathcal{E}_{A\to B}$, and we can express the channel's action on a density matrix ρ_A as

  $$\mathcal{E}_{A\to B}(\rho_A) = \mathrm{Tr}_A[\Phi_{AB} \cdot \rho_A^\top]. \tag{2.87}$$

- ***Choi–Jamiolkowski Isomorphism***: The Choi–Jamiolkowski isomorphism provides a one-to-one correspondence between a quantum channel and its Choi matrix. That is, we can map a quantum channel $\mathcal{E}$ to a Choi matrix $\Phi_{\mathcal{E}}$ and vice versa.

We note that the completely positive and trace-preserving conditions for quantum channels are equivalent to the conditions $\Phi_{\mathcal{E}_{A\to B}} \geq 0$ and $\mathrm{Tr}_B[\Phi_{\mathcal{E}_{A\to B}}] = \mathbb{I}_A$ for the Choi matrix. Therefore, through the Choi–Jamiolkowski isomorphism, we can transform a relatively complex CPTP (completely positive, trace-preserving) linear map into a simpler Choi matrix representation.

Kraus Decomposition

Recall that the Choi matrix acts on the combined AB system. Consider an arbitrary decomposition (e.g., spectral decomposition) of the Choi matrix Φ_{AB}:

$$\Phi_{AB} = \sum_j |\psi_j\rangle \langle\psi_j|_{AB}, \tag{2.88}$$

where $|\psi_j\rangle$ are unnormalized states. Further, we assume that each $|\psi_j\rangle$ can be decomposed as

$$|\psi_j\rangle = \sum_k |k\rangle_A |\psi_{j,k}\rangle_B, \tag{2.89}$$

where $|\psi_{j,k}\rangle_B$ are also unnormalized states.

Using equation (2.87), we can express the quantum channel $\mathcal{E}_{A\to B}$ as

$$\begin{aligned}
\mathcal{E}_{A\to B}(\rho_A) &= \mathrm{Tr}_A\left[\sum_j |\psi_j\rangle \langle\psi_j|_{AB} \cdot \rho_A^\top\right], \\
&= \mathrm{Tr}_A\left[\sum_{j,k,k'} |k\rangle_A |\psi_{j,k}\rangle_B \langle k'|_A \langle\psi_{j,k'}|_B \cdot \rho_A^\top\right], \\
&= \sum_{j,k,k'} |\psi_{j,k}\rangle_B \langle\psi_{j,k'}|_B \cdot \langle k'|_A \rho_A^\top |k\rangle_A, \\
&= \sum_j K_j \rho_A K_j^\dagger,
\end{aligned} \tag{2.90}$$

where the Kraus operators K_j are defined as

$$K_j = \sum_k |\psi_{j,k}\rangle_B \langle k|_A. \tag{2.91}$$

Therefore, the quantum channel $\mathcal{E}_{A\to B}$ can be written as

$$\mathcal{E}_{A\to B}(\rho_A) = \sum_j K_j \rho_A K_j^\dagger, \tag{2.92}$$

where the Kraus operators satisfy the trace-preserving condition

$$K_j^\dagger K_j = \mathbb{I}. \tag{2.93}$$

This decomposition is known as the ***Kraus decomposition***, and the operators K_j are referred to as ***Kraus operators***.

Furthermore, we can prove that any Kraus decomposition satisfying $K_j^\dagger K_j = \mathbb{I}$ corresponds to a completely positive and trace-preserving (CPTP) map. Therefore, the Kraus decomposition provides an equivalent description of a quantum channel.

Problem 2.21. *Prove that $\mathcal{E}_{A\to B}(\rho_A) = \sum_j K_j \rho_A K_j^\dagger$ is completely positive*

Similar to different decompositions of quantum states, the Kraus decomposition of a given quantum channel can take various forms. Specifically, for two different Kraus decompositions $\{K_j\}$ and $\{\tilde{K}_k\}$ corresponding to the same quantum channel, there exists a unitary transformation U such that

$$K_j = \sum_k U_{jk} \tilde{K}_k. \tag{2.94}$$

If the number of elements in $\{K_j\}$ and $\{\tilde{K}_k\}$ is not the same, we can add zeros to make the two sets have the same number of elements.

Subsystem Evolution

Mathematically, the evolution of a quantum state is described by a CPTP (completely positive trace-preserving) map. What is the physical meaning of a CPTP map?

As we learned earlier, a density matrix corresponds to the state of a subsystem in a larger quantum system. Similarly, a quantum channel describes the evolution of a subsystem as part of the evolution of a larger system.

Consider two quantum systems AE, with the initial state $\rho_A \otimes |0\rangle\langle 0|_E$, and the evolution U_{AE}. The evolved quantum state is given by

$$\rho'_{AE} = U_{AE}(\rho_A \otimes |0\rangle\langle 0|_E) U_{AE}^\dagger. \tag{2.95}$$

Next, considering the density matrix of the A system after evolution, we have

$$\begin{aligned}
\rho'_A &= \mathrm{Tr}_E[\rho'_{AE}], \\
&= \mathrm{Tr}_E[U_{AE}(\rho_A \otimes |0\rangle\langle 0|_E) U_{AE}^\dagger], \\
&= \sum_j \langle j|_E U_{AE}(\rho_A \otimes |0\rangle\langle 0|_E) U_{AE}^\dagger |j\rangle_E, \\
&= \sum_j K_j \rho_A K_j^\dagger,
\end{aligned} \tag{2.96}$$

where $K_j = \langle j|_E U_{AE}|0\rangle_E$, which corresponds to the Kraus operator form of the channel that evolves ρ_A into ρ'_A. Therefore, when considering the evolution of a subsystem due to the evolution of the global system, it can be described by a quantum channel.

Physically, the evolution of a system that does not interact with other systems is called a ***closed system***, and it can be described by a unitary transformation. When the evolution of a system is entangled with the evolution of another system, this system is referred to as an ***open system***. In such cases, either the entire system's unitary evolution is considered, or the subsystem's evolution is described by a quantum channel.

Equivalence of Different Descriptions

Clearly, the evolution of any subsystem can be described by a quantum channel. Conversely, any CPTP transformation can be implemented through subsystem evolution, meaning the four descriptions are equivalent.

Specifically, consider the Kraus form of the channel $\mathcal{E}(\rho_A) = \sum_j K_j \rho_A K_j^\dagger$. We can always introduce an auxiliary system E, perform a global unitary transformation on the AE system, and thereby equivalently implement the channel transformation on the A system. Specifically, we consider the auxiliary system in the initial state $|0\rangle_E$, and the overall unitary transformation is given by

$$U_{EA} = \sum_j K_j \otimes |j\rangle_E \langle 0|_E + \cdots = \begin{bmatrix} K_1 & . & . & . \\ K_2 & . & . & . \\ K_3 & . & . & . \\ \cdots & . & . & . \end{bmatrix} \tag{2.97}$$

Here, for convenience, we have swapped the indices of AE. Note that we have only specified the first column of the matrix for U_{EA}, and other elements may still have some degree of freedom. It is easy to verify that by applying the evolution U_{EA} on the EA system, we equivalently perform the channel transformation $\mathcal{E}(\rho_A)$ on the A system. Therefore, any quantum channel can be implemented through subsystem evolution. Due to the equivalence between Kraus form, Choi matrix, and CPTP maps, we also prove the equivalence of the four descriptions mentioned above.

Problem 2.22. *Prove that there exists a unitary transformation such that formula* (2.97) *holds.*

Pauli Transfer Matrix

Another description of quantum channels is the ***Pauli transfer matrix***. First, note that matrices themselves form a linear space, so the transformation of matrices can also be described by matrices. We previously discussed several ways of vectorizing matrices. For quantum states, one common method of vectorization is to decompose the state in the Pauli basis. Let the Pauli basis be denoted by $\{\sigma_j\}$, then any quantum state can be written as

$$\rho = \sum_j \lambda_j \sigma_j. \tag{2.98}$$

For a quantum channel $\mathcal{E}$, we have

$$\mathcal{E}(\rho) = \sum_k \lambda_k \mathcal{E}(\sigma_k). \tag{2.99}$$

Similarly to matrices, we define E_{jk} as

$$E_{jk} = \mathrm{Tr}[\sigma_j \cdot \mathcal{E}(\sigma_k)], \tag{2.100}$$

so that the action of the quantum channel $\mathcal{E}$ can be equivalently expressed as

$$\mathcal{E}(\rho) = \sum_j \sum_k E_{jk} \lambda_k \sigma_j. \tag{2.101}$$

Let $\rho := |\rho\rangle\rangle$ and $\sigma := |\sigma\rangle\rangle$, then we have

$$\begin{aligned} |\rho\rangle\rangle &= \sum_k \lambda_k |\sigma_k\rangle\rangle, \\ \mathcal{E}(\rho) &= \sum_j \sum_k E_{jk} \lambda_k |\sigma_j\rangle\rangle = E |\rho\rangle\rangle. \end{aligned} \tag{2.102}$$

Thus, the ***Pauli transfer matrix*** is given by

$$E = \sum_{jk} E_{jk} |\sigma_j\rangle\rangle\langle\langle\sigma_k|. \tag{2.103}$$

Therefore, by vectorizing the quantum state, we can describe the quantum channel using the Pauli transfer matrix. The advantage of the Pauli transfer matrix is that it provides a simple matrix representation of the channel. However, the

disadvantage is that the condition for complete positivity of the channel is not as straightforwardly represented in terms of the Pauli transfer matrix.

Consider a quantum channel from n qubits to n qubits. The general form of the Pauli transfer matrix E is $E = \begin{pmatrix} 1 & 0 \\ \boldsymbol{s} & T \end{pmatrix}$, where $\boldsymbol{s}$ is a vector of dimension $N = 2^n - 1$ and T is an $N \times N$ real matrix. When the channel satisfies the unital condition, i.e., $\mathcal{E}(\mathbb{I}) = \mathbb{I}$, then $\boldsymbol{s} = 0$.

Some Typical Quantum Channels

Quantum channels provide the most general description of processes in quantum mechanics. Here, we list some typical quantum channels.

- ***State Preparation Channel***: $\mathcal{E}(\emptyset) = \rho$, which means that when the input is empty, the output is a deterministic quantum state. This can be thought of as a channel that prepares a specific quantum state in the absence of any initial input.

- ***Unitary Channel***: $\mathcal{E}(\rho) = U\rho U^\dagger$. This is the most fundamental type of quantum channel, representing a unitary evolution of the quantum state ρ under a unitary operator U.

- ***Projective Measurement***: $\mathcal{E}(\rho) = \sum_j \langle j|\rho|j\rangle \, |j\rangle \langle j|$. This represents a measurement process where the quantum state ρ is projected onto the basis states $|j\rangle$, and the outcome is a probabilistic projection of ρ onto the state $|j\rangle$ with the probability $\langle j|\rho|j\rangle$.

- ***Destructive Measurement***: $\mathcal{E}(\rho) = \sum_j \mathrm{Tr}[\rho O_j] \, |j\rangle \langle j|$, where $O_j \geq 0$ and $\sum_j O_j = \mathbb{I}$. This describes a quantum measurement process that results in a "destructive" outcome, meaning that the system collapses to one of the possible outcomes $|j\rangle$, with the measurement operators O_j satisfying the completeness relation.

- ***Non-destructive Measurement***: $\mathcal{E}(\rho) = \sum_j \mathrm{Tr}[K_j \rho K_j^\dagger] \rho_j$, where $\rho_j = K_j \rho K_j^\dagger / \mathrm{Tr}[K_j \rho K_j^\dagger]$ and $\sum_j K_j^\dagger K_j = \mathbb{I}$. This is a more generalized measurement where the system may undergo a transformation described by the Kraus operators K_j, but the state is not collapsed to a specific outcome, which contrasts with the destructive measurement described earlier.

- ***Typical Single-Qubit Channels***:

- ***Dephasing Channel*** or ***Phase Flip***: $\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z$. This channel models decoherence, where the phase information of a quantum state is lost, and it corresponds to a probabilistic flip of the phase of the state ρ.
- ***Bit Flip***: $\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$. This models a bit flip noise, where the state ρ experiences a probabilistic flip between the $|0\rangle$ and $|1\rangle$ states, represented by the Pauli X operator.
- ***Depolarizing Channel***: $\mathcal{E}(\rho) = (1-p)\rho + p(X\rho X + Y\rho Y + Z\rho Z)$. This models a random noise process where the quantum state ρ has a probability p of being replaced by a random Pauli error (bit flip, phase flip, or both).
- ***Amplitude Damping Channel***: The Kraus operators are $K_0 = \sqrt{\gamma}\,|0\rangle\langle 1|$ and $K_1 = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|$. This channel describes the energy loss of a qubit, such as in the case of spontaneous emission of photons, where the qubit transitions from the excited state $|1\rangle$ to the ground state $|0\rangle$.
- ***Erasure Channel***: $\mathcal{E}(\rho) = (1-p)\rho + p\,|2\rangle\langle 2|$. This channel represents a scenario where the quantum state is probabilistically erased and replaced by a fixed state $|2\rangle\langle 2|$, which is a kind of "reset" to a known state.

Problem 2.23. *For a single-qubit quantum state $\rho = \frac{1}{2}(I + \boldsymbol{n}\cdot\boldsymbol{\sigma})$, provide the equivalent transformation of the vector $\boldsymbol{n}$ under the above single-qubit channels.*

2.2.3 POVM

Next, we consider a more general form of measurement, namely the ***Positive Operator-Valued Measure (POVM)***. Previously, we learned that quantum measurements involve projecting a quantum state onto a basis set of vectors. This measurement process can also be described as a quantum channel. For example, measuring a quantum state in the $\{|j\rangle\}$ basis corresponds to the following channel:

$$\mathcal{E}(\rho) = \sum_j \langle j|\rho|j\rangle\,|j\rangle\langle j|\,. \tag{2.104}$$

Now, let us consider a more general scenario. Before measuring the quantum state ρ_A, we introduce an ancillary state $|0\rangle\langle 0|_E$, and apply a unitary transformation

U_{AE} on both the quantum state and the ancillary state. We then measure the E system in the $\{|j\rangle_E\}$ basis. The measurement probability is given by:

$$p_j = \langle j|_E \operatorname{Tr}_A[U_{AE}(\rho_A \otimes |0\rangle\langle 0|_E)U_{AE}^\dagger]\,|j\rangle_E\,. \tag{2.105}$$

Simplifying this expression yields:

$$p_j = \operatorname{Tr}[O_j \cdot \rho_A], \tag{2.106}$$

where $O_j = K_j^\dagger K_j$ and $K_j = \langle j|_E U_{AE}|0\rangle_E$. This process can be understood as a measurement of the positive semi-definite operators $\{O_j\}$, where the set of operators satisfies $\{O_j \geq 0\}$ and $\sum_j O_j = \mathbb{I}$, which is the definition of a POVM.

Similarly, the above measurement process can also be written as a quantum channel:

$$\begin{aligned}\mathcal{M}(\rho) &= \sum_j K_j \rho_A K_j^\dagger \otimes |j\rangle\langle j|_E\,,\\ &= \sum_j p_j \rho_{A,\,j} \otimes |j\rangle\langle j|_E\,,\end{aligned} \tag{2.107}$$

where $p_j = \operatorname{Tr}[K_j \rho_A K_j^\dagger]$ and $\rho_{A,\,j} = \frac{K_j \rho_A K_j^\dagger}{p_j}$. Note that, since we are measuring the E system, the A system will collapse to the state $\rho_{A,\,j}$ corresponding to the measurement outcome. Ignoring the E system (i.e., when the measurement result is unknown), the measurement channel can be written as:

$$\mathcal{M}(\rho) = \sum_j \operatorname{Tr}[K_j \rho_A K_j^\dagger] \frac{K_j \rho_A K_j^\dagger}{\operatorname{Tr}[K_j \rho_A K_j^\dagger]}. \tag{2.108}$$

For a given measurement result j, the quantum state $\rho_{A,\,j}$ is obtained with probability p_j. For general POVM measurements, we have the following conclusions:

- For any Kraus decomposition of a channel $\mathcal{E}(\rho) = \sum_j K_j \rho K_j^\dagger$, we can design the corresponding unitary transformation and perform a measurement on the ancillary system to obtain the quantum state $\rho_j = K_j \rho_A K_j^\dagger / p_j$ with probability $p_j = \operatorname{Tr}[K_j \rho_A K_j^\dagger]$. Therefore, not only can we implement any Kraus decomposition of a channel, but we can also obtain each Kraus component's corresponding quantum state with the appropriate probability.

- Let $O_j = K_j^\dagger K_j$. The above measurement process can be written as $\mathcal{M}(\rho) = \sum_j \operatorname{Tr}[O_j \cdot \rho]\rho_j$, which corresponds to a general non-destructive measurement.

- If we are unable to obtain the quantum state ρ_j after the measurement and only have random classical measurement outcomes, the measurement process is $\mathcal{M}(\rho) = \sum_j \mathrm{Tr}[O_j \cdot \rho] \, |j\rangle \langle j|$, which corresponds to a destructive measurement.

Problem 2.24. *Construct an ancillary system and unitary transformation to implement a POVM with the elements* $\{|0\rangle\langle 0|/3, |1\rangle\langle 1|/3, |\pm\rangle\langle\pm|/3, |\pm i\rangle\langle\pm i|/3\}$ *using a projection measurement on the ancillary system.*

2.3 Fundamental Principles of Quantum Information

In this section, we introduce some of the fundamental principles of quantum information, including the principle of the impossibility of completely distinguishing non-orthogonal quantum states, the no-cloning theorem for unknown quantum states, and the uncertainty relations for quantum measurements. We note that these principles do not have analogs in classical physics. Specifically, in classical mechanics, we can completely distinguish classical states, perfectly clone classical states, and perform measurements on classical states without uncertainty. Therefore, these principles are unique to quantum mechanics. More importantly, we will see that violating these principles leads to the destruction of causality. Thus, the validity of these principles reflects the self-consistency of quantum mechanics and causality.

2.3.1 Distinguishing Non-Orthogonal States

Consider two quantum states $|\psi_0\rangle$ and $|\psi_1\rangle$ that are randomly generated. The task of distinguishing quantum states is essentially about whether there exists a Positive Operator-Valued Measure (POVM) $\{\Pi_0, \Pi_1\}$ such that:

$$\begin{aligned} &\mathrm{Tr}[\Pi_0\psi_0] = 1, \ \mathrm{Tr}[\Pi_0\psi_1] = 0, \\ &\mathrm{Tr}[\Pi_1\psi_0] = 0, \ \mathrm{Tr}[\Pi_1\psi_1] = 1. \end{aligned} \tag{2.109}$$

This means that we can perfectly distinguish the two quantum states through $\{\Pi_0, \Pi_1\}$. We can prove that the above equation holds if and only if $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal, i.e., perfect state discrimination is only possible when the two states are orthogonal. For non-orthogonal quantum states, we cannot distinguish them perfectly using a POVM.

In contrast, in the classical case, since all states are mutually orthogonal, we can always distinguish classical states perfectly. However, because quantum states may not be orthogonal to each other, we cannot perfectly distinguish quantum states.

Problem 2.25. *Prove that the equation* (2.109) *holds if and only if* $\mathrm{Tr}[\psi_0\psi_1] = 0$. *This result also holds for mixed states.*

2.3.2 No-Cloning Theorem

In classical physics, we can perfectly distinguish between different classical states, and consequently, we can also make perfect copies of classical states. However, for any arbitrary quantum state $|\psi\rangle$, we are unable to perform a perfect cloning operation. Suppose, for the sake of argument, that we could clone arbitrary quantum states, such as $|\psi_1\rangle$ and $|\psi_2\rangle$, i.e., suppose there exists a unitary transformation U such that:

$$\begin{aligned} U\,|\psi_1\rangle\,|0\rangle &= |\psi_1\rangle\,|\psi_1\rangle\,, \\ U\,|\psi_2\rangle\,|0\rangle &= |\psi_2\rangle\,|\psi_2\rangle\,. \end{aligned} \tag{2.110}$$

Here, $|0\rangle$ represents the auxiliary system (the "blank" state) that is prepared to receive the cloned state. To see why this leads to a contradiction, let's take the inner product of the two equations. We obtain:

$$\langle\psi_2|\psi_1\rangle = \langle\psi_2|\psi_1\rangle^2\,. \tag{2.111}$$

This equation can only hold true if $\langle\psi_2|\psi_1\rangle = 0$ or $\langle\psi_2|\psi_1\rangle = 1$.

However, since $|\psi_1\rangle$ and $|\psi_2\rangle$ are arbitrary quantum states, the inner product $\langle\psi_2|\psi_1\rangle$ can take any value in the range $[0, 1]$. This implies that the unitary operation U cannot exist, because it would only work under very specific circumstances where $|\psi_1\rangle$ and $|\psi_2\rangle$ are either orthogonal ($\langle\psi_2|\psi_1\rangle = 0$) or identical ($\langle\psi_2|\psi_1\rangle = 1$).

Thus, it is impossible to perform a perfect cloning operation for arbitrary quantum states. This result is one of the cornerstones of quantum mechanics and has no equivalent in classical mechanics, where perfect cloning of states is trivially possible.

The no-cloning theorem also plays a critical role in quantum cryptography, as it ensures that eavesdropping on quantum communication is not possible without detection. If an adversary tries to clone the quantum state being transmitted, the act of cloning itself disturbs the quantum state, revealing the presence of the eavesdropper.

Problem 2.26. *Prove the no-cloning theorem using the linearity of quantum mechanics. Assume that there exists a unitary operator U that clones an arbitrary state $|\psi\rangle$, and show that this leads to a contradiction.*

2.3.3 Uncertainty Relation

Consider a quantum state $|\psi\rangle$ and measurements of two observables, A and B. The standard deviations of the measurement outcomes are given by:

$$\Delta(A) = \sqrt{\langle A^2\rangle - \langle A\rangle^2}, \quad \Delta(B) = \sqrt{\langle B^2\rangle - \langle B\rangle^2}, \tag{2.112}$$

The uncertainty relation between these two observables can be expressed as:

$$\Delta(A)\cdot\Delta(B) \geq \frac{1}{2}|\langle [A,B]\rangle|, \tag{2.113}$$

where $[A,B] = AB - BA$ is the commutator of A and B, and it encodes information about whether these operators commute. If the operators A and B do not commute, there will be an inherent uncertainty in simultaneously measuring these two observables.

We can introduce the shifted operators:

$$\tilde{A} = A - \langle A\rangle, \quad \tilde{B} = B - \langle B\rangle, \tag{2.114}$$

which represent the deviations of A and B from their expectation values. The standard deviations of $\tilde{A}$ and $\tilde{B}$ are then given by:

$$\Delta(A) = \Delta(\tilde{A}) = \sqrt{\langle \tilde{A}^2\rangle}, \quad \Delta(B) = \Delta(\tilde{B}) = \sqrt{\langle \tilde{B}^2\rangle}. \tag{2.115}$$

Therefore, the uncertainty relation in equation (2.113) can be rewritten as:

$$
\begin{aligned}
\Delta(A)\cdot\Delta(B) &= \Delta(\tilde{A})\cdot\Delta(\tilde{B}),\\
&= \sqrt{\langle\tilde{A}^2\rangle\langle\tilde{B}^2\rangle},\\
&\geq \sqrt{|\langle\tilde{A}\tilde{B}\rangle|^2},\\
&\geq \frac{1}{2}\sqrt{|\langle\tilde{A}\tilde{B}+\tilde{B}\tilde{A}+\tilde{A}\tilde{B}-\tilde{B}\tilde{A}\rangle|^2},\\
&= \frac{1}{2}\sqrt{\langle(\tilde{A}\tilde{B}+\tilde{B}\tilde{A})\rangle^2+\langle(\tilde{A}\tilde{B}-\tilde{B}\tilde{A})\rangle^2},\\
&\geq \frac{1}{2}\sqrt{|\langle[\tilde{A},\tilde{B}]\rangle|^2},\\
&= \frac{1}{2}|\langle[A,B]\rangle|.
\end{aligned}
\tag{2.116}
$$

Thus, when the operators A and B do not commute, i.e., when $[A,B]\neq 0$, the product of their standard deviations $\Delta(A)\cdot\Delta(B)$ will not be zero, implying that we cannot simultaneously measure the observables A and B with arbitrary precision.

This is a manifestation of the ***Heisenberg uncertainty principle***, which asserts that there are inherent limits to the precision with which certain pairs of physical properties (like position and momentum) can be known simultaneously. The non-commuting nature of certain quantum operators is responsible for this limitation.

The uncertainty relation is one of the key features that distinguishes quantum mechanics from classical physics, where it is always possible to measure pairs of observables with arbitrary precision (provided the measurement apparatus is ideal). The uncertainty principle places a fundamental limit on the information that can be extracted from quantum systems.

Problem 2.27. *Consider the position operator $\hat{x}$ and the momentum operator $\hat{p}$ in one dimension. Show that these operators do not commute and use the uncertainty relation to derive the Heisenberg uncertainty principle for position and momentum:*

$$\Delta x\cdot\Delta p\geq\frac{1}{2}.$$

2.3.4 Applications and Implications

These quantum information principles are not just theoretical; they have direct practical applications in the emerging field of quantum technologies. For example:

- ***Quantum Cryptography***: The no-cloning theorem ensures the security of quantum key distribution protocols like BB84, as any attempt to intercept quantum communication would disturb the system, alerting the sender and receiver to the eavesdropping attempt.

- ***Quantum Computing***: The inability to perfectly distinguish non-orthogonal states and the no-cloning theorem impose constraints on quantum algorithms and error correction techniques, influencing the design and implementation of fault-tolerant quantum computers.

- ***Quantum Measurements and Sensors***: The uncertainty relations provide a fundamental limit on the precision of quantum measurements, which is crucial in the design of quantum sensors and interferometers.

These principles also play a role in the ongoing search for a quantum theory of gravity and other advanced topics in theoretical physics.

Problem 2.28. *Discuss the implications of the no-cloning theorem in quantum communication and quantum computing. How does it affect the potential for error correction in quantum computers?*

Chapter 3

Quantum Information Tools

3.1 Quantum State Distance and Fidelity

3.1.1 Trace Distance

We begin by introducing the definition and operational meaning of the trace distance for classical probability distributions. Next, we define the trace distance for quantum states using the trace norm. We will also discuss some important properties of the trace distance for quantum states.

Distance Between Classical Probability Distributions

Consider two classical probability distributions $\mathbf{p}$ and $\mathbf{q}$ (which satisfy the conditions of being positive and normalized). The difference between these two distributions can be described by the ***trace distance*** between the two probability distributions:

$$D(\mathbf{p},\mathbf{q}) = \frac{1}{2}\sum_j |p_j - q_j|. \tag{3.1}$$

Two important properties of the trace distance are:

- ***Non-negativity***: $D(\mathbf{p},\mathbf{q}) \geq 0$, with equality if and only if $\mathbf{p} = \mathbf{q}$.
- ***Triangle inequality***: $D(\mathbf{p}_1,\mathbf{p}_2) + D(\mathbf{p}_2,\mathbf{p}_3) \geq D(\mathbf{p}_1,\mathbf{p}_3)$.

We can verify that the maximum value of the trace distance $D(\mathbf{p},\mathbf{q})$ is 1, which occurs when $\mathbf{p}$ and $\mathbf{q}$ are orthogonal. The minimum value is 0, which occurs when

$\mathbf{p} = \mathbf{q}$. Therefore, the magnitude of the trace distance characterizes the difference between these two cases: orthogonal distributions correspond to completely distinguishable random variables, and identical distributions correspond to completely indistinguishable random variables. Thus, what does the trace distance mean for any two probability distributions $\mathbf{p}$ and $\mathbf{q}$?

The most important operational meaning of the trace distance is in the task of distinguishing random variables. Consider two random variables P and Q, taking values from $\{j\}$, with probability distributions $\mathbf{p}$ and $\mathbf{q}$, respectively. Suppose P and Q are generated with uniform probability, and our task is to distinguish whether each random event is generated by P or Q. When event j occurs, the probabilities that it comes from P or Q are $p(j)/2$ and $q(j)/2$, respectively. Suppose that each time event j occurs, we determine it to be from P with probability $f(j)$ and from Q with probability $1-f(j)$, where $f(j) \in [0,1]$. Then, the average probability of success in distinguishing P and Q is:

$$P_{\text{succ}} = \frac{1}{2}\sum_j p(j)f(j) + q(j)(1-f(j)) = \frac{1+\sum_j f(j)(p(j)-q(j))}{2}. \tag{3.2}$$

We can further optimize the strategy $f(j)$ to maximize the success probability P_{succ}. In particular, an optimal strategy is to set $f(j) = 1$ when $p(j) - q(j) \geq 0$, and $f(j) = 0$ otherwise. Thus, the maximum probability of successfully distinguishing the uniformly distributed random variables P and Q is:

$$P_{\text{maxsucc}} = \frac{1+\sum_{j:p(j)\geq q(j)} |p(j)-q(j)|}{2}. \tag{3.3}$$

We notice that $0 = \sum_j p(j) - q(j) = \sum_{j:p(j)\geq q(j)} |p(j)-q(j)| - \sum_{j:p(j)<q(j)} |p(j) - q(j)|$, so we have:

$$\sum_{j:p(j)\geq q(j)} |p(j)-q(j)| = D(\mathbf{p},\mathbf{q}), \tag{3.4}$$

which implies:

$$P_{\text{maxsucc}} = \frac{1+D(\mathbf{p},\mathbf{q})}{2}. \tag{3.5}$$

Therefore, when the trace distance between $\mathbf{p}$ and $\mathbf{q}$ is larger, we can more easily distinguish them. Conversely, when the trace distance is smaller, they become harder to distinguish, and the distinguishing strategy becomes increasingly similar to random guessing.

To characterize the distance between two quantum states, i.e., density matrices, a natural idea is to use the trace distance. Consider two quantum states ρ

and σ, which are assumed to have the same eigenstates, i.e., $\rho = \sum_j p_j |j\rangle\langle j|$ and $\sigma = \sum_j q_j |j\rangle\langle j|$. In this case, we can define the trace distance of quantum states similarly to equation (3.1). However, when the eigenstates of the two quantum states are different, we need to modify the definition.

Trace Norm

The core idea here is to use the ***trace norm*** of a matrix. For any matrix A with its singular value decomposition $A = \sum_j \lambda_j |s_j\rangle\langle v_j|$, the trace norm is defined as:

$$\|A\|_1 = \mathrm{Tr}[|A|] = \mathrm{Tr}[\sqrt{AA^\dagger}] = \sum_j |\lambda_j|. \tag{3.6}$$

Some important properties of the trace norm are as follows:

- ***Non-negativity***: $\|A\|_1 \geq 0$, with equality if and only if $A = 0$.
- ***Triangle inequality***: $\|A\|_1 + \|B\|_1 \geq \|A+B\|_1$.
- ***Invariance under isometries***: For isometric transformations U and V, we have $\|UAV\|_1 = \|A\|_1$.
- ***Convexity***: $\|\lambda A + (1-\lambda)B\|_1 \leq \lambda\|A\|_1 + (1-\lambda)\|B\|_1$.
- ***Variational characterization 1***: $\|A\|_1 = \max_{U:\text{unitary}} \mathrm{Tr}[UA]$.
- ***Variational characterization 2***: $\|A\|_1 = \max_{-\mathbb{I}\leq P\leq \mathbb{I}} \mathrm{Tr}[PA]$, where A is a Hermitian matrix.

Trace Distance

Based on the trace norm, we define the trace distance between two quantum states as

$$D(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \tag{3.7}$$

The trace distance between quantum states satisfies the following properties:

- ***Symmetry***: $D(\rho,\sigma) = D(\sigma,\rho)$.
- ***Bounds***: $D(\rho,\sigma) \in [0,1]$, where $D(\rho,\sigma) = 0$ if and only if $\rho = \sigma$, and $D(\rho,\sigma) = 1$ if and only if $\mathrm{Tr}[\rho\cdot\sigma] = 0$.

- ***Triangle inequality***: $D(\rho,\sigma)+D(\sigma,\gamma) \geq D(\rho,\gamma)$.
- ***Invariance under unitary transformations***: $D(\rho,\sigma) = D(U\rho U^\dagger, U\sigma U^\dagger)$.
- ***Tensor product property***: $D(\rho\otimes\gamma,\sigma\otimes\gamma) = D(\rho,\sigma)$.
- ***Variational form***: $D(\rho,\sigma) = \max_{0\leq P\leq \mathbb{I}} \mathrm{Tr}[P(\rho-\sigma)]$.

  Proof. Let $A=\rho-\sigma$, and consider its spectral decomposition $A=\sum_j \lambda_j |s_j\rangle\langle v_j|$. Define
  $$A_+ = \sum_{j:\lambda_j\geq 0} \lambda_j |s_j\rangle\langle s_j|, \quad A_- = \sum_{j:\lambda_j\leq 0} \lambda_j |s_j\rangle\langle s_j|, \tag{3.8}$$
  so that $A = A_+ + A_-$ and $|A| = A_+ - A_-$. Note that $0 = \mathrm{Tr}[A] = \mathrm{Tr}[A_+] + \mathrm{Tr}[A_-]$, so $D(\rho,\sigma) = \mathrm{Tr}[A_+] = -\mathrm{Tr}[A_-]$, and we have
  $$\mathrm{Tr}[P(\rho-\sigma)] = \mathrm{Tr}[PA] = \mathrm{Tr}[PA_+] + \mathrm{Tr}[PA_-] \leq \mathrm{Tr}[PA_+] \leq \mathrm{Tr}[A_+] = D(\rho,\sigma). \tag{3.9}$$
  Choosing $P = \sum_{j:\lambda_j\geq 0} |s_j\rangle\langle s_j|$ achieves equality. □

- ***Quantum state discrimination***: The trace distance $D(\rho,\sigma)$ also characterizes our ability to distinguish between two quantum states ρ and σ. Specifically, assume that we receive ρ or σ with uniform random probability and aim to distinguish the state. Suppose we perform a POVM measurement $\{\Lambda, \mathbb{I}-\Lambda\}$ on each state, and when the measurement result is Λ, we identify the state as ρ, and otherwise as σ. The average probability of successfully distinguishing between the quantum states ρ and σ is
  $$P_{\mathrm{succ}} = \frac{1}{2}\left(\mathrm{Tr}[\Lambda\rho] + \mathrm{Tr}[(\mathbb{I}-\Lambda)\sigma]\right) = \frac{1}{2}\left(1 + \mathrm{Tr}[\Lambda(\rho-\sigma)]\right). \tag{3.10}$$
  By optimizing $\Lambda \in [0,\mathbb{I}]$ to maximize P_{succ}, we obtain the maximum probability of quantum state discrimination as
  $$\max P_{\mathrm{succ}} = \frac{1+D(\rho,\sigma)}{2}. \tag{3.11}$$
  Thus, the trace distance between quantum states is directly related to the maximum success probability for distinguishing between them.

- ***Monotonicity under quantum channels***: $D(\rho,\sigma) \geq D(\mathcal{E}(\rho),\mathcal{E}(\sigma))$.

Proof. Any quantum channel can be realized by a unitary transformation on the system and an auxiliary system, i.e., $\mathcal{E}(\rho_A) = \mathrm{Tr}_E[U(\rho_A \otimes |0\rangle\langle 0|_E)U^\dagger]$. We first prove the monotonicity of the trace distance under partial trace, i.e.,

$$D(\rho_A, \sigma_A) \geq D(\rho_{AE}, \sigma_{AE}), \tag{3.12}$$

where $\rho_A = \mathrm{Tr}_E[\rho_{AE}]$ and $\sigma_A = \mathrm{Tr}_E[\sigma_{AE}]$. This can be shown using the variational form of the trace distance:

$$\begin{aligned} D(\rho_{AE}, \sigma_{AE}) &= \max_{0 \leq P_{AE} \leq \mathbb{I}_{AE}} \mathrm{Tr}[P_{AE}(\rho_{AE} - \sigma_{AE})], \\ &\geq \max_{0 \leq P_A \leq \mathbb{I}_A} \mathrm{Tr}[P_A \otimes \mathbb{I}_E(\rho_{AE} - \sigma_{AE})], \\ &= \max_{0 \leq P_A \leq \mathbb{I}_A} \mathrm{Tr}[P_A(\rho_A - \sigma_A)], \\ &= D(\rho_A, \sigma_A). \end{aligned} \tag{3.13}$$

Since the trace distance is invariant under unitary transformations, we can prove the monotonicity of the trace distance under general quantum channels:

$$\begin{aligned} D(\rho_A, \sigma_A) &= D(\rho_A \otimes |0\rangle\langle 0|_E, \sigma \otimes |0\rangle\langle 0|_E), \\ &= D(U(\rho_A \otimes |0\rangle\langle 0|_E)U^\dagger, U(\sigma \otimes |0\rangle\langle 0|_E)U^\dagger), \\ &\geq D(\mathrm{Tr}_E[U(\rho_A \otimes |0\rangle\langle 0|_E)U^\dagger], \mathrm{Tr}_E[U(\sigma \otimes |0\rangle\langle 0|_E)U^\dagger]), \\ &= D(\mathcal{E}(\rho), \mathcal{E}(\sigma)). \end{aligned} \tag{3.14}$$

□

Note that the trace distance characterizes the distinction between two quantum states. When performing the same operation on both quantum states, if the operation is reversible (i.e., unitary), the information is preserved and the difference between the states remains unchanged, so the trace distance is invariant. However, if the operation is irreversible, such as a partial trace, information is lost, and the distinction between the quantum states diminishes, thus reducing the trace distance. Therefore, the monotonicity of the trace distance under quantum channels describes how the trace distance evolves under quantum operations.

- ***Strong convexity***:

$$D\left(\sum_j p_j \rho_j, \sum_j q_j \sigma_j\right) \leq D(\mathbf{p}, \mathbf{q}) + \sum_j p_j D(\rho_j, \sigma_j), \tag{3.15}$$

where $\{\rho_j\}$ and $\{\sigma_j\}$ are quantum states, and $\mathbf{p}$ and $\mathbf{q}$ are probability distributions.

Proof. Introduce $\sum_j p_j\sigma_j$. By the triangle inequality, we have

$$\begin{aligned} D\left(\sum_j p_j\rho_j, \sum_j q_j\sigma_j\right) &\leq D\left(\sum_j p_j\sigma_j, \sum_j q_j\sigma_j\right) + D\left(\sum_j p_j\rho_j, \sum_j p_j\sigma_j\right), \\ &\leq \sum_j D\left(p_j\sigma_j, q_j\sigma_j\right) + \sum_j D\left(p_j\rho_j, p_j\sigma_j\right), \\ &\leq D(\mathbf{p},\mathbf{q}) + \sum_j p_j D(\rho_j,\sigma_j). \end{aligned} \tag{3.16}$$

□

- ***Relation to classical trace distance***: We can always perform measurements on a quantum state to transform it into a probability distribution. For example, consider a POVM $\{E_j\}$, then we have $p_j = \mathrm{Tr}[\rho E_j]$ and $q_j = \mathrm{Tr}[\sigma E_j]$. The trace distance between quantum states and the classical probability distributions is related by

$$D(\rho,\sigma) = \max_{\{E_j\}} D(\mathbf{p},\mathbf{q}). \tag{3.17}$$

Note that the measurement process can be described by a quantum channel:

$$\mathcal{M}(\cdot) = \sum_j \mathrm{Tr}[E_j\cdot]\,|j\rangle\langle j|. \tag{3.18}$$

The trace distance for classical probability distributions is $D(\mathbf{p},\mathbf{q}) = D(\mathcal{M}(\rho),\mathcal{N}(\sigma))$. Therefore, by the monotonicity of the trace distance, we have $D(\mathbf{p},\mathbf{q}) \leq D(\rho,\sigma)$. The equality holds when we construct the measurement using the spectral decomposition of $\rho-\sigma$.

3.1.2 Fidelity

Similar to the trace distance, fidelity measures the similarity between two quantum states. We first introduce the definition of classical probability distribution fidelity, and then define and discuss the properties of quantum fidelity.

Classical Case

Consider two probability distributions $\mathbf{p}$ and $\mathbf{q}$. Their fidelity is defined as[1]

$$F(\mathbf{p},\mathbf{q}) = \sum_j \sqrt{p_j q_j}. \tag{3.19}$$

Although fidelity F or its complement $1-F$ is not a strict distance for quantum states (it does not satisfy the triangle inequality), fidelity F indeed captures the similarity between two quantum states. Specifically, when $F=1$, we have $\mathbf{p}=\mathbf{q}$, and when $F=0$, the probability distributions $\mathbf{p}$ and $\mathbf{q}$ are perfectly distinguishable.

The classical trace distance and fidelity are closely related. On one hand, we have

$$\begin{aligned} D(\mathbf{p},\mathbf{q}) &= \frac{1}{2}\sum_j |p_j - q_j| \\ &= \frac{1}{2}\sum_j |\sqrt{p_j}-\sqrt{q_j}||\sqrt{p_j}+\sqrt{q_j}| \\ &\geq \frac{1}{2}\sum_j |\sqrt{p_j}-\sqrt{q_j}|^2 \\ &= \frac{1}{2}\sum_j p_j + q_j - 2\sqrt{p_j q_j} \\ &= 1 - F(\mathbf{p},\mathbf{q}). \end{aligned} \tag{3.20}$$

Thus, when the fidelity $F(\mathbf{p},\mathbf{q})$ is close to 1, it implies that the trace distance $D(\mathbf{p},\mathbf{q})$ is close to 0. On the other hand, we can also prove

$$D(\mathbf{p},\mathbf{q}) \leq \sqrt{1-F(\mathbf{p},\mathbf{q})^2}. \tag{3.21}$$

That is, a small trace distance also implies a large fidelity. We will later prove this inequality using quantum fidelity.

Quantum Fidelity

Pure States Case Consider two pure states of the same dimension, $|\psi\rangle = \sum_j \sqrt{p_j}\,|j\rangle$ and $|\phi\rangle = \sum_j \sqrt{q_j}\,|j\rangle$. We can similarly define their fidelity as

$$F(\psi,\phi) = |\langle\psi|\phi\rangle| = \sum_j \sqrt{p_j q_j}. \tag{3.22}$$

[1]The square root here is included to satisfy the normalization condition.

Using the density matrix representation, we have

$$F(\psi,\phi) = \mathrm{Tr}[\sqrt{\psi}\sqrt{\phi}], \tag{3.23}$$

where ψ and ϕ correspond to the density matrices of $|\psi\rangle$ and $|\phi\rangle$, respectively.

Special Mixed States Consider two special mixed states $\rho = \sum_j p_j |j\rangle\langle j|$ and $\sigma = \sum_j q_j |j\rangle\langle j|$, their fidelity can also be defined similarly as

$$F(\rho,\sigma) = \mathrm{Tr}[\sqrt{\rho}\sqrt{\sigma}] = \sum_j \sqrt{p_j q_j}. \tag{3.24}$$

General Case Based on the two special cases above, we can attempt to define the fidelity of quantum states as $F(\rho,\sigma) = \mathrm{Tr}[\sqrt{\rho}\sqrt{\sigma}]$. However, note that $\sqrt{\rho}\sqrt{\sigma}$ is not a Hermitian matrix, so its trace could be complex. To obtain a real-valued fidelity, we define the fidelity of quantum states as

$$F(\rho,\sigma) = \mathrm{Tr}[|\sqrt{\rho}\sqrt{\sigma}|] = \mathrm{Tr}\left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right]. \tag{3.25}$$

Some important properties of quantum state fidelity are as follows.

- First, the above definition is consistent with the definition for pure states and special mixed states. Specifically, when ρ or σ is a pure state, for example, $\rho = \psi$ being a pure state, the fidelity between quantum states is

  $$F(\psi,\sigma) = \sqrt{\langle\psi|\sigma|\psi\rangle}. \tag{3.26}$$

  Thus, the fidelity between quantum states ψ and σ can be understood as the square root of the expectation value of σ when measuring the quantum state ψ.

- ***Symmetry***: $F(\rho,\sigma) = F(\sigma,\rho)$.

- Special values: $F(\rho,\sigma) \in [0,1]$, and $F(\rho,\sigma) = 1$ if and only if $\rho = \sigma$; $F(\rho,\sigma) = 0$ if and only if $\mathrm{Tr}[\rho\cdot\sigma] = 0$.

- ***Unitary invariance***: $F(\rho,\sigma) = F(U\rho U^\dagger, U\sigma U^\dagger)$.

- Tensor product form: $F(\rho_1\otimes\rho_2,\sigma_1\otimes\sigma_2) = F(\rho_1,\sigma_1)F(\rho_2,\sigma_2)$.

- ***Triangle inequality***: Since F essentially characterizes the "inner product" between two states, it does not satisfy the triangle inequality. We can define $A = \arccos F$ to represent the angle between two quantum states, in which case the triangle inequality holds:

$$A(\rho,\sigma) + A(\sigma,\gamma) \geq A(\rho,\gamma). \tag{3.27}$$

  The proof of this can be derived from the triangle inequality for vector angles and Uhlmann's theorem.

- ***Uhlmann's Theorem***: The fidelity between two quantum states corresponds to the maximum fidelity of their purifications:

$$F(\rho,\sigma) = \max_{\psi,\phi} F(\psi,\phi), \tag{3.28}$$

  where ψ and ϕ are the purifications of ρ and σ (note that we assume the purifying systems have the same size).

  Proof. Let ρ and σ represent systems A, and the purified auxiliary system be E. Consider the maximally entangled state $|\Phi^+\rangle = \sum_j |jj\rangle$, and one special purification of ρ and σ can be given by $|\psi\rangle_{AE} = \sqrt{\rho_A} \otimes \mathbb{I}_E |\Phi^+\rangle_{AE}$ and $|\phi\rangle_{AE} = \sqrt{\sigma_A} \otimes \mathbb{I}_E |\Phi^+\rangle_{AE}$. Since different purifications can be transformed by isometric operations, any purification of ρ and σ can be written as $U_E |\psi\rangle_{AE}$ and $V_E |\phi\rangle_{AE}$, where U_E and V_E are isometric transformations. Thus, we have

$$\begin{aligned} F(\psi,\phi) &= |\langle \phi_{AE}|V_E U_E|\psi_{AE}\rangle|, \\ &= |\langle \Phi^+|_{AE} \sqrt{\sigma_A} V_E U_E |\sqrt{\rho_A}|\Phi^+\rangle_{AE}|, \\ &= |\mathrm{Tr}[\sqrt{\sigma_A}\sqrt{\rho_A} \cdot V_E U_E|]|. \end{aligned} \tag{3.29}$$

  Here, we use the identity $\mathrm{Tr}[A^\dagger B] = \langle \Phi^+|A \otimes B|\Phi^+\rangle$. Using the variational form of the trace norm, we have

$$\begin{aligned} \max_{\psi,\phi} F(\psi,\phi) &= \max_{U_E V_E} |\mathrm{Tr}[\sqrt{\sigma_A}\sqrt{\rho_A} \cdot V_E U_E^\dagger]|, \\ &= |\mathrm{Tr}[|\sqrt{\sigma_A}\sqrt{\rho_A}|]|, \\ &= F(\rho,\sigma). \end{aligned} \tag{3.30}$$

  □

Uhlmann's theorem can be viewed as a variational form of the quantum fidelity and establishes the relationship between the fidelity of complex mixed states and pure states.

- ***Monotonicity under Quantum Channels***: $F(\rho,\sigma) \leq F(\mathcal{E}(\rho),\mathcal{E}(\sigma))$, where $\mathcal{E}$ is a quantum channel.

  Proof. Similar to the monotonicity of the trace distance, we need only prove that the fidelity is monotonic under partial trace. This can be simply derived from Uhlmann's theorem. □

- ***Strong Concavity***: $F\left(\sum_j p_j\rho_j, \sum_j q_j\sigma_j\right) \geq \sum_j \sqrt{p_j q_j} F(\rho_j,\sigma_j)$, where $\{\rho_j\}$ and $\{\sigma_j\}$ are quantum states, and $\{p_j\}$ and $\{q_j\}$ are normalized probability distributions.

  Proof. Let ψ_j and ϕ_j be the purifications of ρ_j and σ_j, satisfying $F(\rho_j,\sigma_j) = F(\psi_j,\phi_j)$. Then, $|\psi\rangle = \sum_j \sqrt{p_j}\,|\psi_j\rangle\,|j\rangle$ and $|\phi\rangle = \sum_j \sqrt{q_j}\,|\phi_j\rangle\,|j\rangle$ are purifications of $\sum_j p_j\rho_j$ and $\sum_j q_j\sigma_j$, respectively. Therefore, we have

$$\begin{aligned} F\left(\sum_j p_j\rho_j, \sum_j q_j\sigma_j\right) &\geq F(\psi,\phi), \\ &= \sum_j \sqrt{p_j q_j}\left|\langle\psi_j|\phi_j\rangle\right|, \\ &= \sum_j \sqrt{p_j q_j} F(\rho_j,\sigma_j). \end{aligned} \tag{3.31}$$

  The first inequality here follows from Uhlmann's theorem. □

- ***Relationship with Classical Fidelity***: Similarly, we can measure quantum states to obtain classical probability distributions, and the fidelity between quantum states is closely related to the classical fidelity derived from measurement. Specifically, consider a POVM $\{E_j\}$ such that $p_j = \mathrm{Tr}[\rho E_j]$ and $q_j = \mathrm{Tr}[\sigma E_j]$, we have

$$F(\rho,\sigma) = \min_{\{E_j\}} F(\mathbf{p},\mathbf{q}). \tag{3.32}$$

The inequality $F(\mathbf{p},\mathbf{q}) \geq F(\rho,\sigma)$ can be proved similarly to the trace distance. The POVM that achieves equality corresponds to the Fuchs-Caves measurement, which is the operator

$$M = \rho^{-1/2}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\rho^{-1/2} \tag{3.33}$$

of eigenstates (assuming that ρ is invertible). Let the spectral decomposition of M be $M = \sum_x \lambda_x |x\rangle\langle x|$, then we have $M\rho M = \sigma$, and

$$\begin{aligned}
F(\mathbf{p},\mathbf{q}) &= \sum_x \sqrt{\langle x|\rho|x\rangle \cdot \langle x|\sigma|x\rangle}, \\
&= \sum_x \sqrt{\langle x|\rho|x\rangle \cdot \langle x|M\rho M|x\rangle}, \\
&= \sum_x \sqrt{\langle x|\rho|x\rangle \cdot \langle x|\lambda_x\rho\lambda_x|x\rangle}, \\
&= \sum_x \lambda_x \langle x|\rho|x\rangle, \\
&= \mathrm{Tr}[M\rho], \\
&= \mathrm{Tr}\left[\rho^{-1/2}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\rho^{-1/2}\rho\right], \\
&= F(\rho,\sigma).
\end{aligned} \tag{3.34}$$

Note that when ρ is not full rank, i.e., non-invertible, we can define Π_ρ as the projection onto the non-zero eigenvalue subspace of ρ, satisfying $\Pi_\rho^2 = \Pi_\rho$ and $\Pi_\rho\rho\Pi_\rho = \rho$. In this case, we can first prove that $F(\rho,\sigma) = F(\rho,\Pi_\rho\sigma\Pi_\rho)$. By defining the inverse of ρ as the inverse of its non-zero eigenvalues, we can use the above method to prove the general case.

Relationship between Trace Distance and Fidelity

When ψ and ϕ are pure states, the trace distance and fidelity are directly related by

$$D(\psi,\phi) = \sqrt{1 - F(\phi,\psi)^2}. \tag{3.35}$$

For general mixed states ρ and σ, we have

$$1 - F(\rho,\sigma) \leq D(\rho,\sigma) \leq \sqrt{1 - F(\rho,\sigma)^2}. \tag{3.36}$$

The first inequality can be proved by considering the degeneration of trace distance and fidelity to the case of classical probability distributions and applying equation (3.20). For the second inequality, we first consider the purifications of ρ and σ, denoted as ψ and ϕ, satisfying $F(\rho,\sigma) = F(\psi,\phi)$. Therefore, we have

$$D(\rho,\sigma) \leq D(\psi,\phi) = \sqrt{1 - F(\phi,\psi)^2} = \sqrt{1 - F(\rho,\sigma)^2}. \tag{3.37}$$

The first inequality follows from the monotonicity of the trace distance under partial trace. According to equation (3.36), the trace distance and fidelity are closely related. In particular, a small trace distance implies a large fidelity, and vice versa.

3.2 Tensor Network

Tensor networks are a powerful graphical framework for representing multi-dimensional tensors. By decomposing high-dimensional tensors into a network of lower-rank tensors, they provide an efficient way to represent quantum states and processes. Tensor networks also offer an intuitive visualization of the relationships between vectors, dual vectors, and matrices, facilitating the simplification of quantum information equalities and protocols.

In this section, we introduce the fundamentals of tensor networks, including their definition, contraction rules, and applications in quantum information processing. Additionally, we discuss a special class of tensor networks known as matrix product states (MPS) and explore their role in computational tasks.

3.2.1 Basics of Tensors

In quantum information, tensors can be understood as high-dimensional matrices with multiple indices. A general tensor representation of an operator M can be written as:

$$M = \sum_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}} M_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}} \left| i_0 i_1 \ldots i_{p-1} \right\rangle \left\langle j_0 j_1 \ldots j_{q-1} \right|. \tag{3.38}$$

Here, the indices $i_0 i_1 \ldots i_{p-1}$ correspond to row indices, while $j_0 j_1 \ldots j_{q-1}$ correspond to column indices. If the basis is fixed, then the tensor M is fully characterized by its components $M_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}}$.

Tensor Contractions In the ket-bra notation, vectors and dual vectors are explicitly distinguished using bras and kets. Tensor multiplication corresponds to contractions of tensor components. Consider another tensor:

$$N = \sum_{j_0 j_1 \ldots j_{q-1}, k_0 k_1 \ldots k_{s-1}} N_{j_0 j_1 \ldots j_{q-1}, k_0 k_1 \ldots k_{s-1}} \left| j_0 j_1 \ldots j_{q-1} \right\rangle \left\langle k_0 k_1 \ldots k_{s-1} \right|. \tag{3.39}$$

The contraction of M and N is then given by:

$$\begin{aligned} M \cdot N = & \sum_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}, k_0 k_1 \ldots k_{s-1}} M_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}} N_{j_0 j_1 \ldots j_{q-1}, k_0 k_1 \ldots k_{s-1}} \\ & \times \left| i_0 i_1 \ldots i_{p-1} \right\rangle \left\langle k_0 k_1 \ldots k_{s-1} \right|. \end{aligned} \tag{3.40}$$

Thus, the tensor component of $M \cdot N$ is computed as:

$$(M \cdot N)_{i_0 i_1 \ldots i_{p-1}, k_0 k_1 \ldots k_{s-1}} = \sum_{j_0 j_1 \ldots j_{q-1}} M_{i_0 i_1 \ldots i_{p-1}, j_0 j_1 \ldots j_{q-1}} N_{j_0 j_1 \ldots j_{q-1}, k_0 k_1 \ldots k_{s-1}}, \tag{3.41}$$

which follows the tensor contraction (matrix multiplication) rules. Thus, when using a fixed basis, tensor multiplication can be understood as contracting tensor components.

Index Notation and Einstein Summation When omitting the explicit basis, it becomes difficult to distinguish vectors from dual vectors. To address this, we adopt the convention that vectors are represented by upper indices, while dual vectors are represented by lower indices. The tensor component of M is then written as:

$$M^{i_0 i_1 \ldots i_{p-1}}_{j_0 j_1 \ldots j_{q-1}}. \tag{3.42}$$

For tensor contraction, we follow Einstein summation notation, which automatically sums over repeated upper and lower indices. This allows us to express Eq. (3.41) as:

$$(M \cdot N)^{i_0 i_1 \ldots i_{p-1}}_{k_0 k_1 \ldots k_{s-1}} = M^{i_0 i_1 \ldots i_{p-1}}_{j_0 j_1 \ldots j_{q-1}} N^{j_0 j_1 \ldots j_{q-1}}_{k_0 k_1 \ldots k_{s-1}}, \tag{3.43}$$

where summation over repeated indices $j_0 j_1 \ldots j_{q-1}$ is implied.

Problem 3.1. *Write down the tensor representations for pure quantum states, density matrices, and observables.*

Problem 3.2. *Express the tensor contraction for $\langle \psi | O | \psi \rangle$ in Einstein notation.*

With such a notation, we can always contract the same indices, which eventually corresponds to trace and tensor multiplications. For example, we can show that

$$M^{i_0 i_1 \dots i_{p-1}}_{j_0 j_1 \dots j_{q-1}} N^{j_0 j_1 \dots j_{q-1}}_{i_0 i_1 \dots i_{p-1}} \leftrightarrow \mathrm{Tr}[MN]. \tag{3.44}$$

While the distinction between upper and lower indices is useful for differentiating vectors from dual vectors and ensuring correct tensor contractions (where contractions occur only between an upper and a lower index), this distinction is not always necessary. In practical applications, indices can often be treated symmetrically as long as one carefully tracks the way they are contracted. In this note, we use upper and lower indices primarily for clarity and consistency, helping to visualize the flow of tensor operations and ensuring that contractions follow Einstein summation conventions. However, this notation is not mandatory in practice.

Operator-State Isomorphism A general tensor $M^{i_0 i_1 \dots i_{p-1}}_{j_0 j_1 \dots j_{q-1}}$ can be viewed as a multidimensional array indexed by $i_0 i_1 \dots i_{p-1} j_0 j_1 \dots j_{q-1}$. Even if we do not explicitly distinguish between upper and lower indices in computations, the tensor components remain unchanged. This becomes particularly useful when dealing with tensor contractions where contracted indices are swapped.

For example, we have:

$$M_{i,j} N^{i,j} = M^j_i N^i_j. \tag{3.45}$$

Although these equalities naturally hold in tensor contraction, they are less obvious in quantum information terminology. Specifically, $M_{i,j} N^{i,j}$ corresponds to the inner product of quantum states $|\psi\rangle = \sum_{i,j} N^{ij} |ij\rangle$ and $\langle\psi| = \sum_{i,j} M_{ij} \langle ij|$, whereas $M^j_i N^i_j$ represents the trace of the product of two operators:

$$\begin{aligned} M_{i,j} N^{i,j} &\leftrightarrow \langle\psi|\psi\rangle, \\ M^j_i N^i_j &\leftrightarrow \mathrm{Tr}[MN]. \end{aligned} \tag{3.46}$$

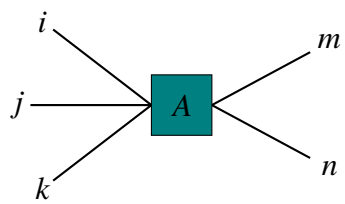
This illustrates the fundamental isomorphism between quantum states and operators, establishing a mathematical bridge between different physical processes.

Problem 3.3. *For the identity tensor $I = \sum_{i,j} I^i_j |i\rangle \langle j|$, where $I^i_j = \delta_{i,j}$, describe its isomorphic state representation.*

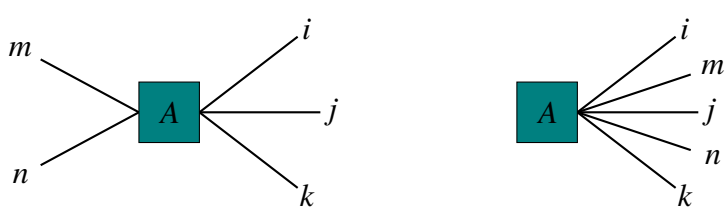
3.2.2 Graphical Representation of Tensors

Basics In the following, we will refer to tensor components simply as tensors. Here, we introduce a graphical method to represent tensors and their contractions, which is particularly useful in quantum information processing.

For instance, consider a tensor A^{mn}_{ijk}. We use the following graphical notation to represent it, where the rectangular node represents the tensor and lines pointing to the left (right) correspond to lower (upper) indices [2]:



The arrangement and direction of the lines are not crucial after contraction, as long as they are properly connected (see definitions of contraction below). In particular, the above tensor A^{mn}_{ijk} is isomorphic to A^{ijk}_{mn} and A^{ijkmn}, represented as follows:



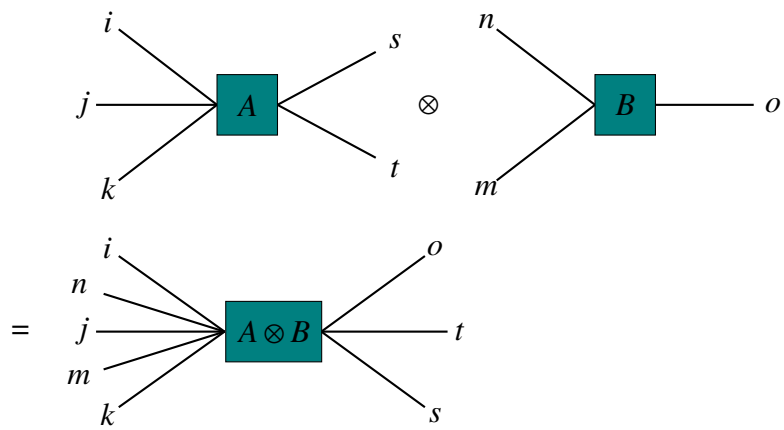
We note that while these three tensors represent different physics quantities, they are mathematically equivalent. Therefore, the tensor network provides a very useful tool to study these equivalences.

For the tensor product of two tensors,

$$(A \otimes B)^{sto}_{ijkmn} = A^{st}_{ijk} B^{o}_{mn}, \tag{3.47}$$

we can simply put the two graphs together by combining the indices:

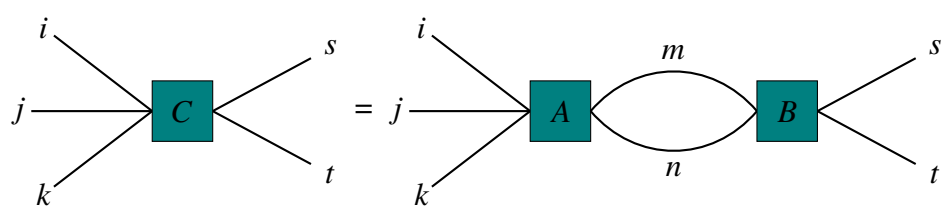
[2]This convention follows from the convention way draw a quantum circuit, and it is in an opposite direction for the ketbra notation. One can also use the ketbra notation convention where lines pointing to the left (right) correspond to vectors (dual vectors)



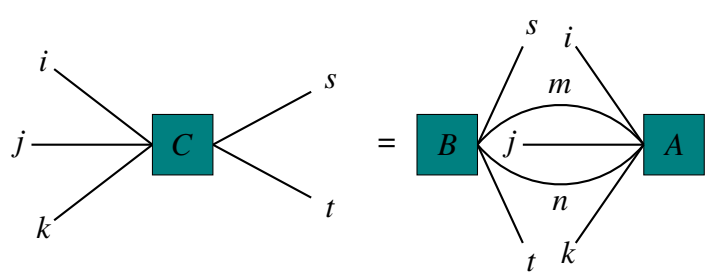
Tensor contraction describes the summation over repeated indices. Consider two tensors A^{mn}_{ijk} and B^{st}_{mn}, their contraction is defined as

$$C^{st}_{ijk} = \sum_{mn} A^{mn}_{ijk} B^{st}_{mn}. \tag{3.48}$$

Graphically, this contraction is represented as:



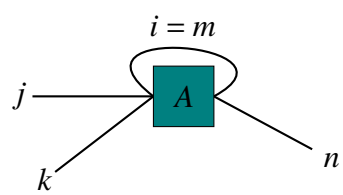
As we have noted above, we can freely move the contracted indices without affecting the result:



Similarly, we can contract two indices of the same tensor, analogous to taking a trace or a partial trace. For example, contracting the i and m indices gives:

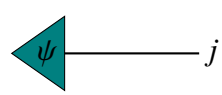
$$A^n_{jk} = \sum_{i=m} A^{mn}_{ijk} \tag{3.49}$$

Graphically, this is represented as:

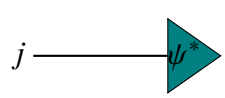


A tensor network is hence a graph of connected tensors. Since it clearly represents how the tensors are contracted, it is very useful to represent quantum information processes.

Representation of Quantum Information Objects With the above definitions, we can exploit tensor networks to describe quantum information components. In the computational basis, any quantum state $|\psi\rangle = \sum_j \psi_j |j\rangle$ defines a tensor ψ^j and denoted as

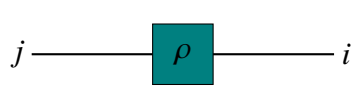


Consider the complex transpose of the state, it corresponds to $\langle\psi| = \psi^*_j \langle j|$ and hence a tensor ψ^*_j.



Note that, here we use triangles to represent quantum states just for clarification.

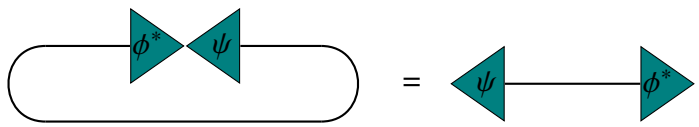
A density matrix $\rho = \sum_{ij} \rho^i_j |i\rangle \langle j|$ corresponds to a tensor ρ^i_j:



We can similarly represent unitary and observables. With the above definitions, we can use the tensor network rules to prove the following inequality

$$\mathrm{Tr}[|\psi\rangle\langle\phi|] = \langle\phi||\psi\rangle. \tag{3.50}$$

with the following tensor network:

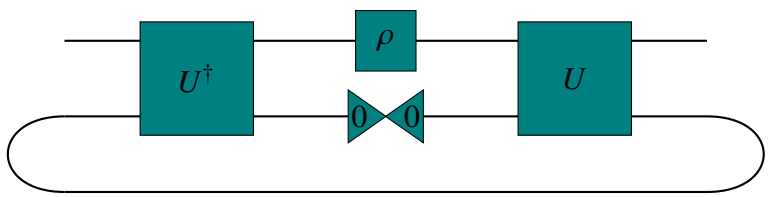


Problem 3.4. *Draw the graphical representation of* $\langle\psi|O|\phi\rangle$ *and* $\mathrm{Tr}[U\rho U^\dagger O]$.

We can also represent quantum channels as tensor networks. For example, consider the joint evolution description of a channel:

$$\mathcal{E}(\rho_A) = \mathrm{Tr}_E[U_{AE}(\rho_A \otimes |0\rangle\langle 0|_E)U_{AE}^\dagger]. \tag{3.51}$$

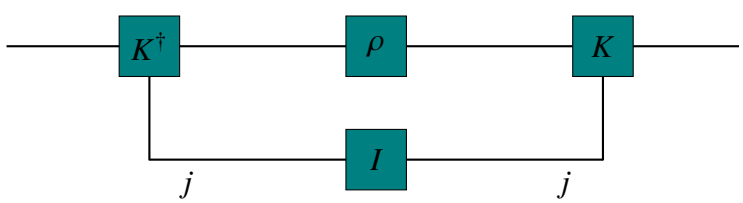
It can be represented as:



A quantum channel can also be represented in the Kraus form

$$\mathcal{E}_{A\to B}(\rho_A) = \sum_j K_j \rho_A K_j^\dagger. \tag{3.52}$$

Its corresponding tensor network representation is



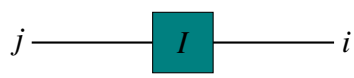
Here I is the identity tensor as we introduce below shortly.

Problem 3.5. *Prove that the two tensor network representations for quantum channels are equivalent.*

Some basic tensors and rules Tensor networks are not only useful for representing the relationship of tensors, but helpful for deriving and proving equalities after specifying the definitions of tensors. Here, we introduce several basic tensors, tensor operation, and their corresponding rules.

- ***Identity tensor and maximally entangled states*** — The most simple but intriguing tensor is the identity tensor and its isomorphic ones with swapped indices — the maximally entangled states.

  Specifically, the identity tensor is denoted by $I^i_j = \delta_{i,j}$ as

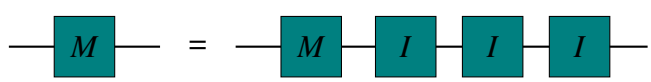


  Since the identity tensor corresponds to the identity matrix,

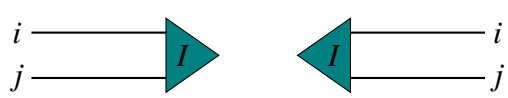
$$I = \sum_{i,j} I^i_j \left|i\right\rangle \left\langle j\right| = \sum_i \left|i\right\rangle \left\langle i\right| \tag{3.53}$$

  whose effect is nothing when acting on another matrix, we can generally omit the tenor node as

  Indeed, for any tensor, we can imagine that it is multiplied by the identity tensor for every index. For example, we have



  Now, let's swap the indices to left and right, which gives two new tensors



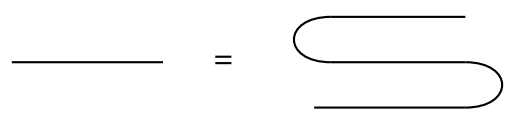
These tensors corresponds to the following vectors:

$$\begin{aligned} \langle I| &= \sum_{i,j} I_{ij} \langle ij| = \sum_i \langle ii|, \\ |I\rangle &= \sum_{i,j} I^{ij} |ij\rangle = \sum_i |ii\rangle, \end{aligned} \tag{3.54}$$

which is proportional to the maximally entangled state up to a normalization factor. In practice, we can explicitly multiply this factor besides the tenor network. Furthermore, since these two tenors are essentially equivalent to the identity matrix, we can also regard it as 'bended lines' and denote them as:

The major reason that we can treat them as 'bended lines' comes from the delta function property of the tensors. Therefore, we have shown the deep connection between the identity tensor and maximally entangled states. Such a connection combined with the tensor contraction rules, can be applied to prove many interesting rules and equalities. In the following, we consider from different perspectives to understand this.

- ***Snake equality*** — The snake equality is a very intriguing rule, whose graphical representation is

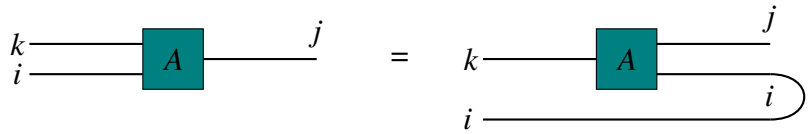
  That is, when combine two maximally entangled states in the above order, it simply reduces to the identity operation.

  Problem 3.6. *Prove the snake equality using tensor definitions.*

  Problem 3.7. *Prove the corresponding quantum information equality:*

  $$\langle \Phi^+|_{AB} |\Phi^+\rangle_{BC} = I_{BC}/2. \tag{3.55}$$

- ***Swapping indices*** — When acting the maximally entangled tensors on a general tensor, its effect is to swap upper and lower indices. For example, consider a tensor A_k^{ij} and we connect it with $I_{ii'} = \delta_{i,i'}$, it gives

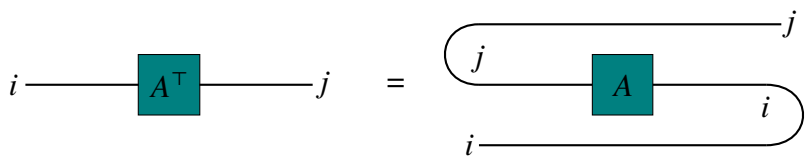


Here we have replaced $i' = i$ for clarity.

Problem 3.8. *Prove the above equality using tensors and quantum information language:*

$$\langle \Phi^+ |_{AA} \sum_{ijk} A_k^{ij} \, |ij\rangle_{AB} \, \langle k|_C = \frac{1}{\sqrt{2}} \sum_{ijk} A_{ik}^{j} \, |j\rangle_B \, \langle ik|_{AC} \, . \tag{3.56}$$

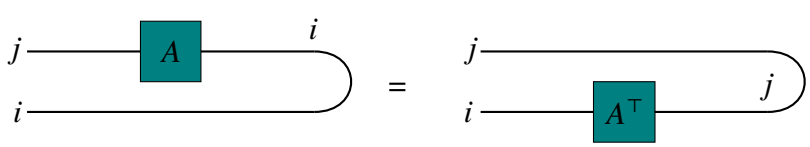
– ***Transpose*** — The transpose operation of a matrix basically corresponds to swapping the lower and upper indices, which can be easily obtained using the maximally entangled tensors:



Problem 3.9. *Prove the above equality using tensors and quantum information language:*

$$A_{AB}^{\top} = 2 \, \langle \Phi^+ |_{AA'} \, A_{AB}^{\top} \, |\Phi^+\rangle_{BB'} \, . \tag{3.57}$$

– ***Sliding matrices*** — When connect maximally entangled tensors to a matrix, we can effectively slide the matrix through the line. Nevertheless, we need to add the transpose operation on the matrix since it changes the indices. For example, for a matrix A^i_j, we can slide it through the line to obtain $A^j_i = (A^\top)^i_j$ as

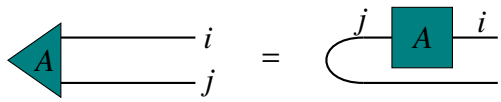


As we discuss below, such a relationship is also related to the vectorization of matrices. Furthermore, when combined with the snake equality, we can study how matrices slides on multiple bended lines.

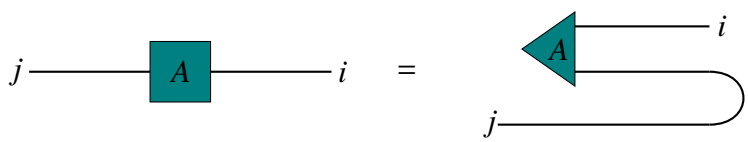
Problem 3.10. *Prove the above equality using tensors and quantum information language:*

$$A_A \left|\Phi^+\right\rangle_{AB} = A_B^\top \left|\Phi^+\right\rangle_{AB} . \tag{3.58}$$

– ***Matrix-vector duality*** — The above rule naturally defines the vectorization of matrices. For a matrix A^i_j, its vectorization can be represented as



Then, combining the above rules, we can easily prove the inverse vectorization as



Problem 3.11. *Prove the above equalities.*

Problem 3.12. *Prove matrix vectorization properties: consider matrices $X \in \mathcal{B}(B \to C)$ and $Y \in \mathcal{B}(A \to D)$, then we have*

$$(X \otimes Y)\mathrm{vec}(M) = \mathrm{vec}(XMY^\top) \tag{3.59}$$

– ***Trace and partial trace*** — The trace of a matrix can also be obtained using the maximally entangled tensors:

That is

$$\mathrm{Tr}[A] = 2\left\langle \Phi^+_{AB} \middle| A \middle| \Phi^+_{AB} \right\rangle . \tag{3.60}$$

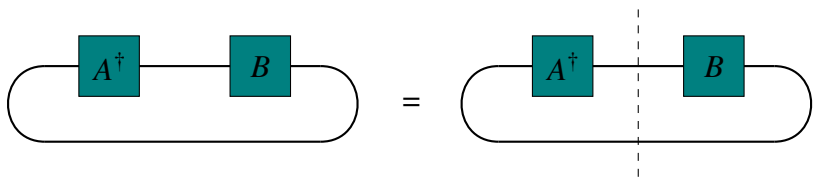
Similarly, one can obtain partial trace by connecting not all the indices.

Problem 3.13. *Use the tensor network approach to prove that the partial trace of the maximally entangled state is the maximally mixed state.*

Combining the matrix-vector duality, we can easily prove the norm of matrices using their vectorized forms:

$$\mathrm{Tr}[A^\dagger B] = \langle\langle A|B\rangle\rangle \tag{3.61}$$

with the following tensor network.



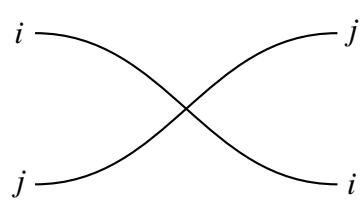
- ***SWAP tensor*** — Another very useful tensor is the SWAP tensor, which is defined as

$$\mathtt{SWAP}^{ij}_{kl} = \delta_{il}\delta_{jk}. \tag{3.62}$$

The corresponding SWAP gate is then defined as

$$\mathtt{SWAP} = \sum_{ij} |ij\rangle\langle ji|. \tag{3.63}$$

Since the effect of the SWAP tensor essentially swaps the indices according to the definition, it can be represented by simply swapping the lines:

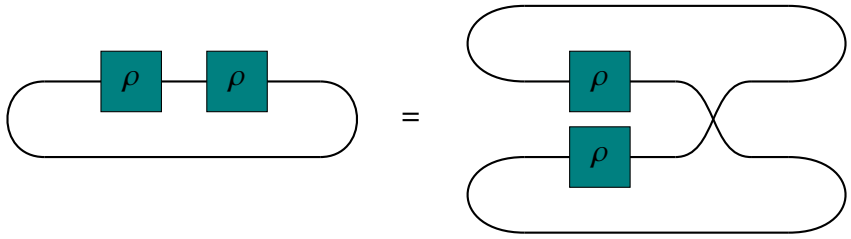


Problem 3.14. *Apply the SWAP tensor on states and matrices to show its effect.*

A very useful example of the SWAP tensor is the swap test, which gives a way to estimate the moments of density matrices by measuring the SWAP operator on copies of the density matrix:

$$\mathrm{Tr}[\rho^2] = \mathrm{Tr}[(\rho\otimes\rho)\mathtt{SWAP}] \tag{3.64}$$

Using the tensor network language, it can be simply proved as follows:



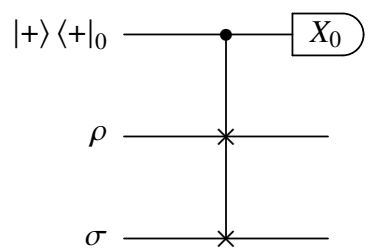
Problem 3.15. *Construct the swap test for measuring* $\mathrm{Tr}[\rho^k], \forall k = 3, 4, \ldots$ *and prove it using the tensor network language.*

- ***Other typical tensors*** — One can further define other tensors, such as ones for single and two qubit operations, and derive more rules for tensor contraction, such as ZX-calculus. We will leave its introduction to other textbooks.

  Here, we give one further example of one way to handle a special type of two qubit operations, i.e., controlled operations. Consider a controlled U, it can be written as

$$CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U, \tag{3.65}$$

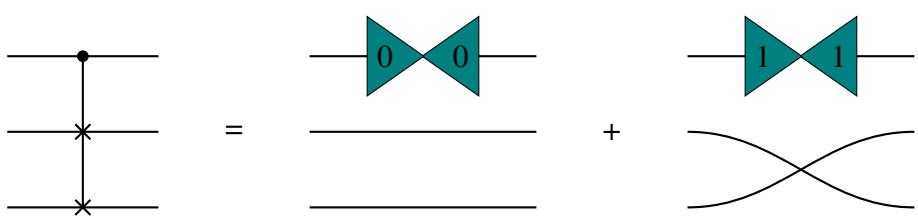
  which could be decomposed as the sum of two tensor product operations. Using such a property, we can verify the full SWAP test circuit:



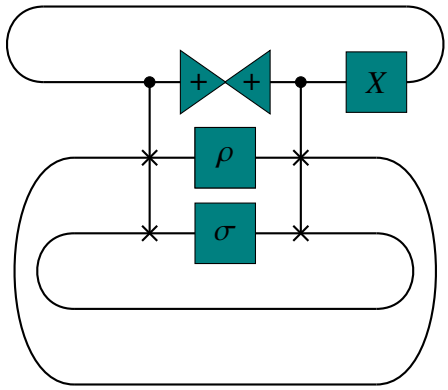
  where the expectation on the first qubit gives

$$\langle X_0 \rangle = \mathrm{Tr}[\rho\sigma]. \tag{3.66}$$

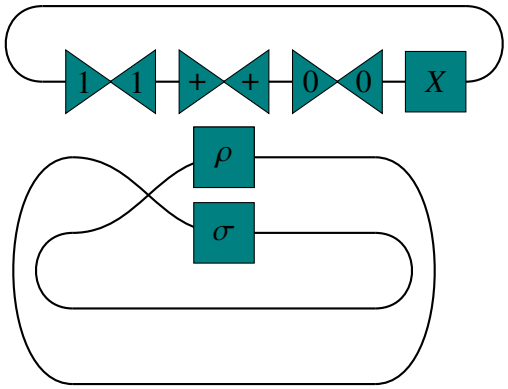
  To prove it using the tensor network language, we first decompose the controlled `SWAP` gate as



With such a decomposition, we can now represent the circuit as



Based on the decomposition of the `SWAP` gate, it can also be decomposed as four sub-graphs. However, note that the X gate is off-diagonal, it only remains two nonzero ones that are hermitian conjugate to each other. Therefore, the final result becomes 2 times the real part of the following graph:



One can then verify that it gives the Eq. (3.66).

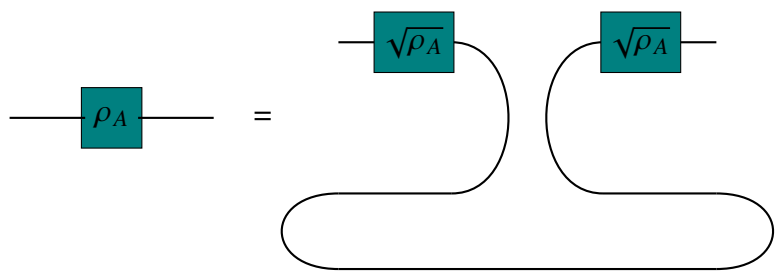
3.2.3 Applications

Here, we give some further examples of the tensor network method. We mainly focus on quantum information processing involving the maximally entangled state.

Purification Recall that a purification of a density matrix ρ_A is given by

$$|\psi\rangle_{AB} = \sqrt{\rho_A} \otimes \mathbb{I}_A |I\rangle_{AB}.$$

That is, we have $\rho_A = \mathrm{Tr}_B[\psi_{AB}]$. We can use tensor network to verify this equation:



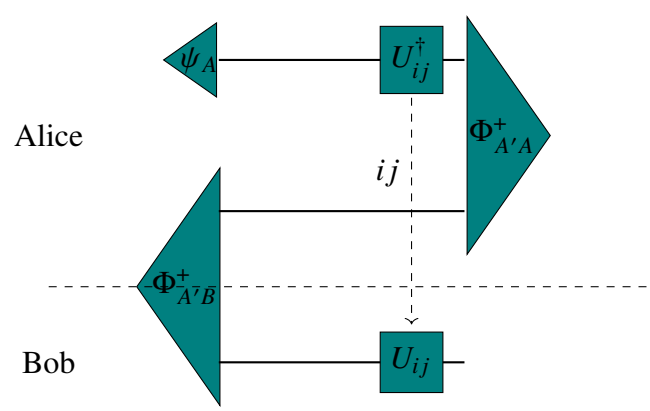
Problem 3.16. *Use the tensor network language to prove the Choi-Jamiolkowski Isomorphism. That is, given the Choi state,*

$$\Phi_{\mathcal{E}_{A\to B}} = \mathcal{E}_{A\to B}(|I\rangle_{A'A}), \tag{3.67}$$

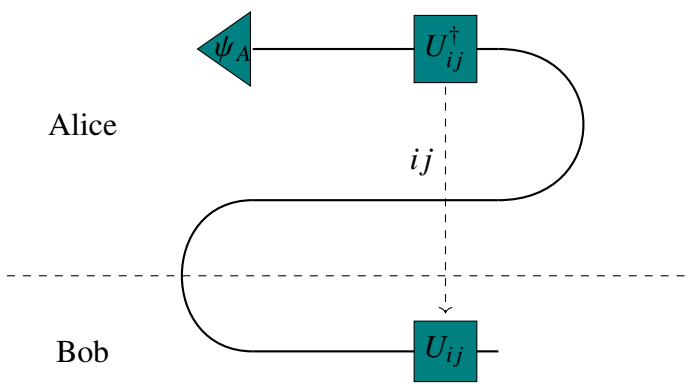
the inverse transform is

$$\mathcal{E}_{A\to B}(\rho_A) = \mathrm{Tr}_{A'}[\Phi_{A'B} \cdot \rho_{A'}^{\top}]. \tag{3.68}$$

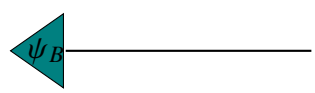
Quantum teleportation Using the tensor network language, we can represent the teleportation protocol graphically as follows.



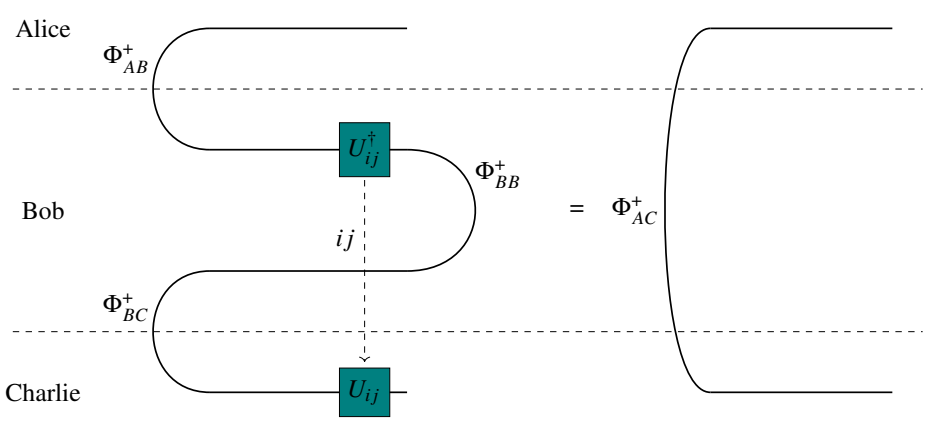
Here, we have denoted ij as the measurement outcome of Alice and use the arrow to show that the outcome is transmitted to Bob, which applies the inverted operation accordingly. The magic is that the tensor corresponding to Bell state and measurement is actually the same tensor for the identity matrix. Therefore, the above tensor network is linear proportional (up to a factor of 1/4) to the following one



which is equivalent to having the quantum state at Bob's site:



Entanglement swapping We can similarly prove the entanglement swapping protocol using the below tensor network. Again, the proof utilizes the 'bended line' feature of the maximal entangled state.



3.2.4 Matrix Product States

XXX

3.3 Semi-Definite Programming

Semidefinite Programming (SDP), a subfield of convex optimization, is a powerful optimization technique that deals with optimization problems involving semidefinite constraints. These constraints typically involve matrix variables, and since they are non-linear with respect to the matrix elements, SDP is particularly effective for solving complex and non-trivial problems. SDP has found widespread applications in various fields, including quantum information theory, where it is used to study quantum entanglement, quantum communication protocols, and quantum error correction.

From a theoretical standpoint, SDP and its duality are essential tools for analyzing complex quantum information problems. They provide insight into the interconnections between different quantum systems and help characterize the efficiency of quantum information protocols. By formulating quantum information tasks as SDP problems, one can leverage the powerful duality theory to derive bounds and optimal solutions.

On the computational side, there are several efficient algorithms available for solving SDP problems. Notable methods include the ellipsoid method, interior-point methods, and others, which enable the solution of convex optimization problems in polynomial time. Thanks to these algorithms, SDP has become an indispensable tool in both theoretical and applied optimization.

There are numerous modern SDP solvers that can efficiently handle such problems. For instance, CVXPY (Python) and CVX (Matlab) are popular solvers that provide convenient interfaces for solving SDPs. These solvers are often used in general-purpose convex optimization tasks. For very large-scale or sparse problems, solvers like SCS and DSDP offer improved efficiency by utilizing advanced techniques such as dual scaling and operator-splitting methods. These solvers are designed to exploit the structure of large-scale problems, making them particularly useful in fields like machine learning, control theory, and quantum computing. QPSolver is designed for solving quantum optimization problems and is particularly suited for problems involving quantum entanglement and quantum information theory.

In this section, we will provide a brief introduction to the definitions of SDP,

discuss the concepts of weak and strong duality, and highlight some practical examples of its applications.

3.3.1 Linear programming

Definition We first introduce ***linear programming (LP)***, a special case of SDP, as an appetizer. LP is defined as the optimization of linear functions of vectors subject to linear constraints. Specifically, for a d-dimensional real vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_d)$, a general LP can be written as

$$\begin{aligned} \text{maximize} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & A\boldsymbol{x} \leq \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0, \end{aligned} \tag{3.69}$$

where $\boldsymbol{b}, \boldsymbol{c}$ are also d-dimensional real vectors and A is a matrix with d columns and multiple rows, each corresponding to different linear functions of $\boldsymbol{x}$.

Now, we denote the constraints as

$$\mathcal{F} = \{\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}. \tag{3.70}$$

Then $\mathcal{F}$ is called the ***feasible set***, and $\boldsymbol{x}$ is called ***feasible*** when $\boldsymbol{x} \in \mathcal{F}$. We can prove that $\mathcal{F}$ is convex, so LP is a special case of convex optimization.

Problem 3.17. *Prove that $\mathcal{F}$ is convex, i.e., $p\boldsymbol{x} + (1-p)\boldsymbol{y} \in \mathcal{F}$, $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathcal{F}, p \in [0,1]$.*

Duality With the feasible set $\mathcal{F}$, we can rewrite the optimization as

$$\alpha = \max_{\boldsymbol{x} \in \mathcal{F}} \boldsymbol{a}^\top \boldsymbol{x}, \tag{3.71}$$

which is also called the ***primal*** form of LP. Its corresponding dual form is given by

$$\begin{aligned} \text{minimise} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & A^\top \boldsymbol{y} \geq \boldsymbol{c}, \\ & \boldsymbol{y} \geq 0. \end{aligned} \tag{3.72}$$

Denote the new constraints as

$$\tilde{\mathcal{F}} = \{\boldsymbol{y} : A^\top \boldsymbol{y} \geq \boldsymbol{c}, \boldsymbol{y} \geq 0\} \tag{3.73}$$

then the ***dual*** form is simplified as

$$\beta = \min_{\boldsymbol{y}\in\tilde{\mathcal{F}}} \boldsymbol{b}^\top \boldsymbol{y} \tag{3.74}$$

The primal and dual forms are deeply related with the help of Lagrangian multipliers. For the primal form, we consider the Lagrangian with Lagrangian multiplier $\boldsymbol{y}$:

$$\begin{aligned}\mathcal{L} &= \boldsymbol{c}^\top \boldsymbol{x} + (\boldsymbol{b} - A\boldsymbol{x})^\top \boldsymbol{y} \\ &= -\boldsymbol{x}^\top A^\top \boldsymbol{y} + \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{b}^\top \boldsymbol{y} \\ &= -\boldsymbol{x}^\top A^\top \boldsymbol{y} + \boldsymbol{x}^\top \boldsymbol{c} + \boldsymbol{b}^\top \boldsymbol{y} \\ &= \boldsymbol{b}^\top \boldsymbol{y} - \boldsymbol{x}^\top (A^\top \boldsymbol{y} - \boldsymbol{c})\end{aligned} \tag{3.75}$$

Therefore, $\boldsymbol{x}$ can be regarded as the Lagrangian multiplier for the dual form optimization. Furthermore, the primal and dual forms can be rewritten as the following optimizations of $\mathcal{L}$:

$$\begin{aligned}\alpha &= \max_{\boldsymbol{x}\geq 0} \min_{\boldsymbol{y}\geq 0} \mathcal{L}, \\ \beta &= \min_{\boldsymbol{y}\geq 0} \max_{\boldsymbol{x}\geq 0} \mathcal{L}.\end{aligned} \tag{3.76}$$

Problem 3.18. *Prove Eq.* (3.76).

Based on the max-min inequality, we thus derive the weak duality of LPs.

Theorem 3.1. ***[Weak duality]*** *For the primal and dual forms of LP defined in Eq.* (3.71) *and Eq.* (3.74)*, we have*

$$\alpha \leq \beta. \tag{3.77}$$

Problem 3.19. *Directly prove the weak duality.*

Interestingly, for most common LPs, there is a stronger version of the duality, where the primal and dual solutions coincide.

Definition 3.1. ***[Strong duality]*** *When certain conditions are satisfied, the primal and dual solutions of LP defined in Eq.* (3.71) *and Eq.* (3.74) *coincide:*

$$\alpha = \beta. \tag{3.78}$$

In particular, one can prove the following theorem.

Theorem 3.2. *Whenever the primal LP is feasible and bounded, strong duality holds with $\alpha = \beta$.*

Problem 3.20. *Write the primal and dual LPs for $\|\boldsymbol{x}\|_\infty$.*

3.3.2 Primal and Dual SDPs

SDPs are similar to LPs, with vectors $\boldsymbol{x}$, $\boldsymbol{c}$, and $\boldsymbol{b}$ replaced by matrices X, C, and B, respectively, the linear transformation of $A\boldsymbol{x}$ replaced by the linear transformation of matrices $\Phi(X)$, and the positivity of $\boldsymbol{x}$ replaced by the semi-definiteness of the matrix $X \geq 0$. Here, since we need $\Phi(X) \leq B$, we require $\Phi(X)$ to be Hermitian, and thus Φ must be Hermitian-preserving. With these replacements, we can define the primal and dual forms of SDPs as follows:

$$\begin{array}{lll@{\qquad}lll} \text{maximize} & \mathrm{Tr}[AX] & & \text{minimize} & \mathrm{Tr}[BY] & \\ \text{subject to} & \Phi(X) \leq B, & & \text{subject to} & \Phi^\dagger(Y) \geq A, & \\ & X \geq 0. & & & Y \geq 0. & \end{array} \tag{3.79}$$

Define the constraints:

$$\begin{aligned} \mathcal{F} &= \{X \geq 0 : \Phi(X) \leq B\}, \\ \tilde{\mathcal{F}} &= \{Y \geq 0 : \Phi^\dagger(Y) \geq A\}. \end{aligned} \tag{3.80}$$

The optimal solutions to the primal and dual SDPs are:

$$\begin{aligned} \alpha &= \sup_{X \in \mathcal{F}} \mathrm{Tr}[AX], \\ \beta &= \inf_{Y \in \tilde{\mathcal{F}}} \mathrm{Tr}[BY], \end{aligned} \tag{3.81}$$

We can see why these are dual to each other with the Lagrangian multiplier:

$$\begin{aligned} \mathcal{L} &= \mathrm{Tr}[AX] + \mathrm{Tr}[(B - \Phi(X))Y] \\ &= \mathrm{Tr}[AX] + \mathrm{Tr}[BY] - \mathrm{Tr}[X\Phi^\dagger(Y)] \\ &= \mathrm{Tr}[BY] + \mathrm{Tr}[X(A - \Phi^\dagger(Y))]. \end{aligned} \tag{3.82}$$

We can prove the following result:

$$\begin{aligned} \alpha &= \sup_{X \geq 0} \inf_{Y \geq 0} \mathcal{L}, \\ \beta &= \inf_{Y \geq 0} \sup_{X \geq 0} \mathcal{L}. \end{aligned} \tag{3.83}$$

Let's consider the primal case as an example. We can see that even though there is no explicit constraint on X in the above equation, the optimization over Y implicitly implies the constraint. Specifically, for any X that violates the constraint $\Phi(X) \leq B$, we can always choose Y such that $\inf_{Y \geq 0} \mathcal{L} = -\infty$. However, for any

feasible X, we have $\inf_{Y\geq 0}\mathrm{Tr}[(B-\Phi(X))Y]\geq 0$ and hence $\mathcal{L}\geq \mathrm{Tr}[AX]$. Therefore, we can restrict ourselves to feasible $X\in\mathcal{F}$, which consequently reduces to the constraints of the primal form.

Meanwhile, when there is no feasible X satisfying $\Phi(X)\leq B$, we can simply choose Y such that $\inf_{Y\geq 0}\mathcal{L}=-\infty$ for all X. Therefore, we have $\alpha=0$. Similarly, we have that $\beta=\infty$ when there is no feasible Y satisfying $\Phi^\dagger(Y)\geq A$.

Problem 3.21. *Prove Eq.* (3.83).

There are also different standard prime and dual forms of SDPs. For example, we can consider the equality constraints:

$$\begin{array}{lll} \text{maximize} & \mathrm{Tr}[AX] & \\ \text{subject to} & \Phi(X)=B, & \\ & X\geq 0. & \end{array} \qquad \begin{array}{lll} \text{minimize} & \mathrm{Tr}[BY] & \\ \text{subject to} & \Phi^\dagger(Y)\geq A, & \\ & Y=Y^\dagger. & \end{array} \tag{3.84}$$

Or even more general ones with both equality and inequality constraints:

$$\begin{array}{lll} \text{maximize} & \mathrm{Tr}[AX] & \\ \text{subject to} & \Phi_1(X)=B, & \\ & \Phi_2(X)\leq C. & \end{array} \qquad \begin{array}{lll} \text{minimize} & \mathrm{Tr}[BY]+\mathrm{Tr}[CZ] & \\ \text{subject to} & A-\Phi_1^\dagger(Y)-\Phi_2^\dagger(Z)=0, & \\ & Z\geq 0. & \end{array} \tag{3.85}$$

Problem 3.22. *Write the Lagrangian and prove the duality of the above SDPs. In particular, think about how the prime and dual constraints are related.*

Problem 3.23. *Show the equivalence between the above SDP forms by converting one to the other.*

Problem 3.24. *Given the prime SDP*

$$\begin{array}{ll} \textit{maximize} & \mathrm{Tr}[AX] \\ \textit{subject to} & \Phi_1(X)=B, \\ & \Phi_2(X)\leq C, \\ & X\geq 0. \end{array} \tag{3.86}$$

write the corresponding dual form.

3.3.3 Weak and Strong Duality

SDPs also satisfy weak duality.

Theorem 3.3. ***[Weak duality]*** *For the primal and dual forms of SDP defined in Eq.* (3.81)*, we have*

$$\alpha \leq \beta. \tag{3.87}$$

Here, we explicitly prove it.

Proof. For all feasible X and Y, we have the following inequalities:

$$\mathrm{Tr}[AX] \leq \mathrm{Tr}[\Phi^\dagger(Y)X] \leq \mathrm{Tr}[Y\Phi(X)] \leq \mathrm{Tr}[YB], \tag{3.88}$$

where the first and second inequalities come from the constraints in the dual and primal forms. If no feasible X or Y exists, these inequalities still hold by definition, since the infimum and supremum are defined over the feasible sets, which may be empty in such cases. □

While strong duality is not universally applicable to all SDPs, it holds for most general cases. A common sufficient condition for strong duality is given by Slater's theorem, which ensures the existence of a feasible solution in the interior of the feasible set.

Theorem 3.4. ***[Slater's theorem]*** *Strong duality holds when any of the following conditions are satisfied:*

- *If the primal SDP is strictly feasible and bounded, i.e.,* $\tilde{\mathcal{F}} \neq \emptyset$ *and there exists* X *such that* $\Phi(X) < B$*, then strong duality holds, i.e.,* $\alpha = \beta$*, and there exists a* $Y \in \tilde{\mathcal{F}}$ *such that* $\beta = \mathrm{Tr}[BY]$.
- *If the dual SDP is strictly feasible and bounded, i.e.,* $\mathcal{F} \neq \emptyset$ *and there exists a* Y *such that* $\Phi^\dagger(Y) > A$*, then strong duality holds, i.e.,* $\alpha = \beta$*, and there exists an* $X \in \mathcal{F}$ *such that* $\alpha = \mathrm{Tr}[AX]$.
- *If both the primal and dual problems are strictly feasible, strong duality holds as well.*

3.3.4 Examples

Here, we provide a few typical examples of SDPs for calculating nonlinear matrix norms and fidelity. We will discuss their application in general quantum information tasks, such as quantum entanglement, in later sections.

Matrix Norms We begin by showing how to calculate matrix norms. For a Hermitian matrix M, we consider the trace norm $\|M\|_1 = \mathrm{Tr}[|M|]$ and the spectral norm $\|M\|_\infty = \max_\psi \langle\psi|M|\psi\rangle$, which are both nonlinear functions of M.

For the trace norm, recall the derivation of trace distance, we note that there always exist positive matrices X_+ and X_- such that $M = X_+ - X_-$. We can first prove the inequality:

$$\|M\|_1 = \mathrm{Tr}[|M|] \leq \mathrm{Tr}[X_+] + \mathrm{Tr}[X_-], \tag{3.89}$$

and the equality holds when X_+ and X_- correspond to the positive and negative parts of M. Therefore, we can reformulate it as an SDP as follows:

$$\begin{aligned} \text{minimize} \quad & \mathrm{Tr}[X_+] + \mathrm{Tr}[X_-] \\ \text{subject to} \quad & X_+ - X_- = M, \\ & X_-, X_+ \geq 0. \end{aligned} \tag{3.90}$$

Problem 3.25. *Prove the following alternative SDP forms of the trace norm:*

$$\begin{aligned} \textit{maximize} \quad & \mathrm{Tr}[M(X-Y)] \\ \textit{subject to} \quad & 0 \leq X, Y \leq I. \end{aligned} \tag{3.91}$$

and

$$\begin{aligned} \textit{minimize} \quad & \mathrm{Tr}[X] \\ \textit{subject to} \quad & -X \leq M \leq X. \end{aligned} \tag{3.92}$$

and

$$\begin{aligned} \textit{maximize} \quad & \mathrm{Tr}[M(X-Y)] \\ \textit{subject to} \quad & X + Y = I, \\ & X, Y \geq 0. \end{aligned} \tag{3.93}$$

Therefore, even for the same task, the trace norm can be expressed in different SDP forms.

The above SDPs can be used to compute the trace distance between two density matrices.

For the spectral norm, it corresponds to the maximal absolute eigenvalue of M, and can be formulated as an SDP:

$$\begin{aligned} \text{minimize} \quad & \mathrm{Tr}[M\rho] \\ \text{subject to} \quad & \mathrm{Tr}[\rho] = 1, \\ & \rho \geq 0. \end{aligned} \tag{3.94}$$

Problem 3.26. *Write the dual form of the above SDP and prove strong duality.*

Problem 3.27. *Prove the following alternative SDP forms of the spectral norm:*

$$\begin{aligned} \textit{minimize} \quad & x \\ \textit{subject to} \quad & -x \leq M \leq x. \end{aligned} \tag{3.95}$$

and

$$\begin{aligned} \textit{maximize} \quad & \mathrm{Tr}[M(X-Y)] \\ \textit{subject to} \quad & \mathrm{Tr}[X+Y] = 1, \\ & X, Y \geq 0. \end{aligned} \tag{3.96}$$

Fidelity While matrix norms are relatively straightforward to convert into SDPs, even more nonlinear and complex matrix functions can be expressed as SDPs. The fidelity between quantum states serves as a perfect example of such a transformation.

Recall that the fidelity between two quantum states ρ and σ is defined as

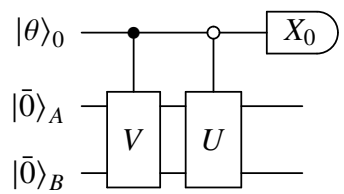
$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1, \tag{3.97}$$

where the trace norm $\|\cdot\|_1$ is a common SDP expression. However, the presence of the square roots $\sqrt{\rho}$ and $\sqrt{\sigma}$ complicates the calculation. To simplify this, we can utilize Uhlmann's theorem, which relates the fidelity of mixed states to the overlap of their purified states. Specifically, we have the following formulation:

$$\begin{aligned} F(\rho, \sigma) = \text{maximize} \quad & |\langle\psi|\phi\rangle_{AB}|, \\ \text{subject to} \quad & \mathrm{Tr}_B\left[|\psi\rangle\langle\psi|_{AB}\right] = \rho_A, \\ & \mathrm{Tr}_B\left[|\phi\rangle\langle\phi|_{AB}\right] = \sigma_A. \end{aligned} \tag{3.98}$$

In this formulation, the subscripts AB indicate the combined system of the auxiliary and original quantum systems, while the subscripts A and B are used to denote the respective parts of the state.

While the constraints in this reformulation are more straightforward, the objective function still involves a nonlinear optimization target. Fortunately, for certain nonlinear objectives or constraints, we can construct a larger matrix to convert it into a linear optimization problem. For example, in the case of the inner product $\langle\psi|\phi\rangle_{AB}$, we can use a method based on the Hadamard test circuit to estimate this quantity:



where we assumed $|\psi\rangle_{AB} = U|\bar{0}\rangle_A|\bar{0}\rangle_B$, $|\phi\rangle_{AB} = V|\bar{0}\rangle_A|\bar{0}\rangle_B$, and $|\theta\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$. Specifically, this circuit prepares the following state

$$|\Psi\rangle_{0AB} = \frac{1}{\sqrt{2}}(|0\rangle_0|\psi\rangle_{AB} + e^{i\theta}|1\rangle_0|\phi\rangle_{AB}), \tag{3.99}$$

with a density matrix

$$\Psi_{0AB} = \frac{1}{2}\begin{pmatrix} |\psi\rangle\langle\psi|_{AB} & e^{-i\theta}|\psi\rangle\langle\phi|_{AB} \\ e^{i\theta}|\phi\rangle\langle\psi|_{AB} & |\phi\rangle\langle\phi|_{AB} \end{pmatrix} \tag{3.100}$$

Suppose we choose θ such that $e^{-i\theta}\langle\phi|\psi\rangle_{AB} = |\langle\phi|\psi\rangle_{AB}|$, the expectation on the first qubit gives

$$\langle X_0\rangle = \mathrm{Tr}[\Psi_{0AB}X_0] = \Re\mathrm{Tr}[e^{-i\theta}|\psi\rangle\langle\phi|_{AB}] = |\langle\psi|\phi\rangle_{AB}|. \tag{3.101}$$

Therefore, the real part of the trace of the off-diagonal components corresponds to the target absolute state overlap.

Next, since the calculation does not involve B at all, we can trace out the purification system B, which gives a density matrix

$$\Psi_{0A} = \frac{1}{2}\begin{pmatrix} \rho_A & \mathrm{Tr}_B[e^{-i\theta}|\psi\rangle\langle\phi|_{AB}] \\ \mathrm{Tr}_B[e^{i\theta}|\pi\rangle\langle\psi|_{AB}] & \sigma_A. \end{pmatrix} \tag{3.102}$$

Then, we can relax the off-diagonal component to a general form with variable M_A as

$$\frac{1}{2}\begin{pmatrix} \rho_A & M_A \\ M_A^\dagger & \sigma_A \end{pmatrix} \geq 0. \tag{3.103}$$

Since $\mathrm{Tr}_B[e^{-i\theta}|\psi\rangle\langle\phi|_{AB}]$ is always a solution for the above constraint, we have

$$|\langle\psi|\phi\rangle_{AB}| \leq \Re\mathrm{Tr}[M_A] = \frac{1}{2}\left(\mathrm{Tr}[M_A] + \mathrm{Tr}[M_A^\dagger]\right). \tag{3.104}$$

Since the constraint and the optimization target are both linear, we thus obtained an upper bound of the fidelity $F(\rho,\sigma)$ with an SDP:

$$\begin{aligned} F(\rho,\sigma) \leq \text{maximize} \quad & \frac{1}{2}\left(\mathrm{Tr}[M_A] + \mathrm{Tr}[M_A^\dagger]\right) \\ \text{subject to} \quad & \frac{1}{2}\begin{pmatrix} \rho_A & M_A \\ M_A^\dagger & \sigma_A \end{pmatrix} \geq 0. \end{aligned} \tag{3.105}$$

Now, it remains to show that the upper bound is tight. To see why this is true intuitively, we can go back to Eq. (3.99) and find that the off-diagonal component cannot be 'larger' for pure states. However, since the off-diagonal component is a matrix, we still need to prove it in a rigorous way.

We first note that for any M_A satisfying the constraint, $\tilde{\rho}_{0A} = \frac{1}{2}\begin{pmatrix} \rho_A & M_A \\ M_A^\dagger & \sigma_A \end{pmatrix}$ is still a density matrix, which admits a purification $|\tilde{\Psi}\rangle_{0AB'}$ satisfying

$$\tilde{\rho}_{0A} = \mathrm{Tr}_{B'}[\tilde{\Psi}_{0AB'}]. \tag{3.106}$$

Then we can again expand $|\tilde{\Psi}\rangle_{0AB'}$ in a similar way to Eq. (3.99) as

$$|\Psi\rangle_{0AB'} = \frac{1}{\sqrt{2}}(|0\rangle_0 |\tilde{\psi}\rangle_{AB'} + |1\rangle_0 |\tilde{\phi}\rangle_{AB'}), \tag{3.107}$$

with $|\tilde{\psi}\rangle_{AB'}$ and $|\tilde{\phi}\rangle_{AB'}$ being the purifications of ρ_A and σ_A, respectively. Then, we have

$$\begin{aligned} \frac{1}{2}\left(\mathrm{Tr}[M_A] + \mathrm{Tr}[M_A^\dagger]\right) &= \mathrm{Tr}[|\Psi\rangle_{0AB'} X_0], \\ &= \Re\mathrm{Tr}[|\tilde{\psi}\rangle\langle\tilde{\phi}|_{AB'}], \\ &\leq |\,|\tilde{\psi}\rangle\langle\tilde{\phi}|_{AB'}\,|, \\ &\leq F(\rho_A,\sigma_A). \end{aligned} \tag{3.108}$$

Here, the last inequality uses Uhlmann's theorem. Therefore, we proved the other direction of the inequality. Together, we have

$$\begin{aligned} F(\rho,\sigma) = \text{maximize} \quad & \frac{1}{2}\left(\mathrm{Tr}[M] + \mathrm{Tr}[M^\dagger]\right) \\ \text{subject to} \quad & \begin{pmatrix} \rho & M \\ M^\dagger & \sigma \end{pmatrix} \geq 0. \end{aligned} \tag{3.109}$$

where the $1/2$ factor in the constraint is omitted without loss of generality. Note that the constraint might not be straightforward to be a linear constraint. However,

we can indeed prove that it can be written in the above standard forms. In general, we can combine known and unknown variables as a matrix, and the semidefinite constraint on this large matrix is still a valid SDP constraint.

Problem 3.28. *Prove the dual form of the SDP:*

$$\begin{aligned} \textit{minimize} \quad & \frac{1}{2}\mathrm{Tr}[\rho X] + \frac{1}{2}\mathrm{Tr}[\sigma Y] \\ \textit{subject to} \quad & \begin{pmatrix} X & -1 \\ -1 & Y \end{pmatrix} \geq 0 \\ & X = X^\dagger, Y = Y^\dagger \end{aligned} \tag{3.110}$$

Applications Beside the very intriguing derivation procedure of the fidelity as an SDP, one may wonder why we need to do it at all since we can easily calculate the fidelity given the two quantum states. Indeed, we generally do not need to use SDP to calculate the fidelity. However, it becomes very useful when another more complicated optimization problem involves some constraints based on the fidelity.

For example, consider an artificial problem where we aim to maximize some linear properties of quantum states that are ε-close to a given state ρ:

$$\begin{aligned} \text{maximize} \quad & \mathrm{Tr}[\sigma W] \\ \text{subject to} \quad & F(\rho, \sigma) \leq \varepsilon \end{aligned} \tag{3.111}$$

Without the SDP formulation of fidelity, it would be unclear how to represent this optimization problem as an SDP, and its solution would be complex due to the nonlinearity in the fidelity function. Fortunately, with the SDP form of the fidelity, we can rewrite this optimization as an SDP problem:

$$\begin{aligned} F(\rho, \sigma) = \text{maximize} \quad & \mathrm{Tr}[\sigma W] \\ \text{subject to} \quad & \frac{1}{2}\left(\mathrm{Tr}[M] + \mathrm{Tr}[M^\dagger]\right) \leq \varepsilon \\ & \begin{pmatrix} \rho & M \\ M^\dagger & \sigma \end{pmatrix} \geq 0. \end{aligned} \tag{3.112}$$

This reformulation leverages the SDP structure of fidelity to transform a non-trivial optimization problem into a convex optimization problem, making it solvable efficiently using known SDP techniques.

The SDP technique is widely used in quantum information science, with broad applications in the study of quantum states, quantum channels, and measurements.

For a more comprehensive introduction to the application of SDPs in quantum information science, we refer to https://doi.org/10.1088/978-0-7503-3343-6.

3.4 Random Matrix Theory

Part II

Quantum Information

Chapter 4

Quantum information theory

In this chapter, we introduce the basics of classical and quantum information theory.

4.1 Classical information theory

4.1.1 Uncertainty and entropy

What is information? Consider the password as an example. Suppose Alice has a password consists of 6 digits, does this password contain information? The answer is yes and no. On the one hand, if Alice does not know the password, such as she forget it, then knowing the password indeed gains information. However, if Alice already knows the password, telling her the password again does not increase any information. We can consider other similar examples, such as reading a book, listen to a lecture, download a file, etc. Therefore, intuitively speaking, information is defined with respect to unknownness or uncertainty. Denote an event by a random variable X with probability $p(X = x)$ or simply denoted as p_x. For the password example, it corresponds to the case of X being a 6-digit number. Then p_x is uniformly distributed and has maximal uncertainty if we have no information of the password, whereas it becomes a δ-function like distribution and has minimal uncertainty if we know the password. Then to quantify information, it is equivalent to quantify the uncertainty.

Shannon builds a beautiful theory and introduce the concept of entropy to quantify uncertainty or information. We first consider an abstract way of defining entropy. Suppose we can always use a real number/function I to quantify the

uncertainty of a random event $X = E$ happens with probability $p \in [0,1]$, it should satisfying the following conditions.

- The function $I(X)$ should only depend on the probability of E (instead of the value), i.e., $I(E) = I(p)$.
- The function is additive for independent events, that is $I(E \,\&\, F) = I(E) + I(F)$ or $I(pq) = I(p) + I(q)$ for two independent events E and F with probabilities $p \in [0,1]$ and $q \in [0,1]$, respectively. (We require the condition holds for any $p, q > 0$.)
- The function I is smooth.

We can show that the only function satisfying the above condition is $I(p) = k \log p$ for any constant k.

Proof. Since I is smooth, it has derivatives for all $p > 0$. Then

$$\begin{aligned} I'(p) &= \lim_{\varepsilon \to 0^+} \frac{I(p+\varepsilon) - I(p)}{\varepsilon}, \\ &= \lim_{\varepsilon \to 0^+} \frac{I(p(1+\varepsilon/p)) - I(p)}{\varepsilon}, \\ &= \lim_{\varepsilon \to 0^+} \frac{I(1+\varepsilon/p)}{\varepsilon}, \\ &= \frac{1}{p} \lim_{\varepsilon \to 0^+} \frac{I(1+\varepsilon)}{\varepsilon}. \end{aligned} \tag{4.1}$$

We note that $I(1) = 0$ and

$$I'(1) = \lim_{\varepsilon \to 0^+} \frac{I(1+\varepsilon) - I(1)}{\varepsilon} = \lim_{\varepsilon \to 0^+} \frac{I(1+\varepsilon)}{\varepsilon}. \tag{4.2}$$

Therefore we have

$$I'(p) = \frac{I'(1)}{p} \tag{4.3}$$

and hence

$$I(p) = \int_{x=1}^{p} I'(p) + I(1) = k \log p. \tag{4.4}$$

Here we have changed to the $\log_2$ basis and omitted 2. □

Since the uncertainty for each event E with probability p is $I(p) = k \log p$, the average uncertainty or ***entropy*** of the random variable X is

$$S(X) = S(\mathbf{p}) = \mathbb{E}(I(p)) = k \sum_x p_x \log p_x. \tag{4.5}$$

Now consider a binary distribution with probability $\mathbf{p} = (p, 1-p)$, the entropy is

$$S(\mathbf{p}) = k(p \log(p) + (1-p)\log(1-p)). \tag{4.6}$$

When we have a uniform distribution with $p = 1/2$, we have $S(\mathbf{p}) = -k$. Since a uniform distributed random variable has maximal uncertainty/entropy and we can regard a maximally random bit as a random unit, we set $S(\mathbf{p}) = -k = 1$ which leads to $k = -1$. We thus arrive at the definition of ***Shannon entropy***

$$S(\mathbf{p}) = -\sum_j p_x \log p_x. \tag{4.7}$$

Again, the entropy $S(\mathbf{p})$ measures how random or how uncertain the event is. Information is gained when we become to know the event so that the randomness or uncertainty is eliminated. Therefore, the entropy $S(\mathbf{p})$ also measures the information gain of the random variable.

An important property of entropy is concavity,

$$S(a\mathbf{p} + (1-a)\mathbf{q}) \geq aS(\mathbf{p}) + (1-a)S(\mathbf{q}). \tag{4.8}$$

One way to understand it is as follows. Consider a random coin with head and tail probability a and $1-a$, respectively. When we get head, we generate a random variable with probability $\mathbf{p}$, otherwise we generate it with probability $\mathbf{q}$. Then, the left hand size of the above equation quantifies the total randomness/uncertainty of the coin and the random variable. While on the right hand size, it only contains the randomness of the random variable given known coin outcome, which is no larger than the total randomness.

4.1.2 Shannon's noiseless coding theorem

What is the operational meaning of the entropy? It actually exactly measures the asymptotic average number of bits we need to know the random variable. Consider two parties Alice and Bob, where Alice aims to tell Bob some information that he does not know. Suppose the information is encoded into independent and

identically distributed (i.i.d.) random variables $X_1, X_2, \ldots, X_n$, where the probability of each X_i being 0 is p and being 1 is $1-p$. A naive strategy is of course to transfer all the n bits, which seems very inefficient if the $(p, 1-p)$ distribution is very biased. For example, if $p = 1/10$, we only have very rare random variables being 0s and we should be able to *compress* it quite well. There are various ways to construct different encoding/compression algorithm. Yet, Shannon's noiseless coding theorem tells us the limit of all encoding algorithms. Roughly speaking, Shannon's theorem says that ***we need at least $nH(X)$ bits to transfer the n bits, and there is a protocol that achieves it for $n \to \infty$ with a failure probability $\delta \to 0^+$.***

The key idea is to use the concentration lemma or the typical space. Consider the probability distribution of $X_1, X_2, \ldots, X_n$, it is called a typical sequence if we have np number of 0s and $n(1-p)$ number of 1s with probability

$$p(x_1, x_2, \ldots, x_n) = p^{np}(1-p)^{n(1-p)} = 2^{-nH(p)}. \tag{4.9}$$

Here we have denoted $H(X) = H(p) = -p\log(p) - (1-p)\log(1-p)$. According to the concentration lemma or large number theorem, the number of 0s N_0 is not far from np. Specifically, for arbitrary error $\varepsilon > 0$ and failure probability $\delta > 0$, we can choose a sufficiently large n so that $P(|N_0 - np| \leq n\varepsilon) \geq \delta$. Therefore, with a small failure probability δ, we have

$$2^{-n(H(p)+\tilde{\varepsilon})} \leq p(x_1, x_2, \ldots, x_n) \leq 2^{-n(H(p)-\tilde{\varepsilon})}, \tag{4.10}$$

where $\tilde{\varepsilon} = H'(p-\varepsilon)\varepsilon$. We thus call all sequences that satisfy the above equation ε-typical. We can see that when choosing sufficiently large n, all sequences are ε-typical except for an arbitrary small δ failure probability. Therefore, we only need to focus on these ε-typical sequences and encode them. Since there are at most $2^{n(H(p)+\tilde{\varepsilon})}$ different sequences, we only need $n(H(p)+\tilde{\varepsilon})$ bits or on average $H(p)+\tilde{\varepsilon}$ bit for each random variable. With the limit of $n \to \infty$, the optimal compression rate is $H(p)$.

4.1.3 Conditional entropy and mutual information

Suppose we have two random variables X and Y, we can quantify their joint entropy as

$$S(X,Y) = -\sum_{x,y} p(x,y)\log p(x,y). \tag{4.11}$$

We define the condition entropy as

$$S(X|Y) = S(X,Y) - S(Y), \tag{4.12}$$

which is the joint entropy of X and Y minus the entropy of Y. The mutual information is defined as

$$S(X:Y) = S(X) + S(Y) - S(X,Y) = S(X) - S(X|Y) = S(Y) - S(Y|X), \tag{4.13}$$

which measures information that is contained in both X and Y.

There are several important properties of the conditional entropy and mutual information, such as

- $S(X,Y) = S(Y,X)$, $S(X:Y) = S(Y:X)$.
- $S(X|Y)$, $S(Y|X)$, $S(X:Y) \geq 0$.
- Strong subadditivity: $S(X,Y,Z) + S(Y) \leq S(X,Y) + S(Y,Z)$ or equivalently $S(Z|X,Y) \leq S(Z|Y)$.

All these properties could be understood and remembered using the Venn diagram. Other important properties include the Chain rule and the data processing inequality (not required in this lecture). We refer to Nielsen & Chuang's book for more details and the proofs.

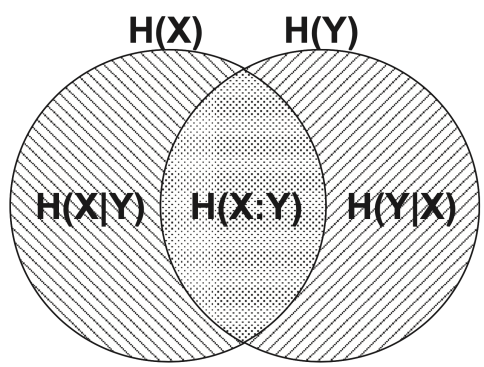


Figure 4.1: Venn diagram of the entropies. From Nielsen & Chuang's book.

4.1.4 Relative entropy/(K-L divergence)

The relative entropy is somehow the *most* important entropy definition. It aims to measure the difference between two distributions $\mathbf{p}$ and $\mathbf{q}$ as

$$S(\mathbf{p}\|\mathbf{q}) = \sum_x p(x)\log(p(x)/q(x)) = \sum_x p(x)[I(q(x)) - I(p(x))]. \tag{4.14}$$

(optional) How to understand this quantifier. Imagine that we have two random variables $X_{\mathbf{p}}$ and $\tilde{X}_{\mathbf{q}}$. Suppose we are given $X_{\mathbf{p}}$ which produces x with probability $p(x)$. When we do not have much samples of $X_{\mathbf{p}}$, we actually do not know the distribution, so as to the random variable. We may falsely thought that the random variable is $\tilde{X}_{\mathbf{q}}$ and, in this case, we falsely thought that we gained an amount of information $I(q(x))$. Therefore the relative entropy $S(\mathbf{p}\|\mathbf{q})$ roughly measures the average information discrepancy $I(q(x)) - I(p(x))$.

(optional) Another way to understand relative entropy is via hypothesis testing. Suppose we are given n samples of X, which are mostly ε-typical satisfying Eq. (4.10). Now we calculate the probability that those samples are produced by $\tilde{X}$, which is

$$p(\text{samples are } \tilde{X}|x_1, x_2, \ldots, x_n) = \binom{n}{N_0} q_0^{N_0}(1-q_0)^{N_1} \approx 2^{-nS(\mathbf{p}\|\mathbf{q})}. \tag{4.15}$$

Here we have used $N_0 \approx np_0$, $\binom{n}{N_0} \approx 2^{S(\mathbf{p})}$, and $q_0^{N_0}(1-q_0)^{N_1} \approx 2^{p_0\log q_0 + p_1 \log q_1}$. Therefore, $S(\mathbf{p}\|\mathbf{q})$ is related to the probability that $\mathbf{q}$ fakes $\mathbf{p}$.

(optional) Finally, the relative entropy measures the average extra information we need to transfer X if we used $\mathbf{q}$ for the distribution. For any event x happens with probability $p(x)$, we on average need $-\log p(x)$ bits to transfer it. Thus, if we used $\mathbf{q}$ for X, we need to transfer

$$-\sum_x p(x)\log q(x) = S(\mathbf{p}) + S(\mathbf{p}\|\mathbf{q}). \tag{4.16}$$

The power of the relative entropy is that it defines all other entropic measures. We will see this for the more general quantum case.

4.2 Quantum information theory

4.2.1 Quantum entropy measures

The Von Neumann entropy of ρ is defined as

$$S(\rho) = -\mathrm{Tr}[\rho\log\rho]. \tag{4.17}$$

Given a spectral decomposition of $\rho = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$, the Von Neumann entropy is

$$S(\rho) = -\sum_j \lambda_j \log \lambda_j. \tag{4.18}$$

Therefore, the quantum Von Neumann entropy is simply the Shannon entropy of the diagonal elements.

The conditional entropy and mutual information is defined the same. For a bipartite quantum state ρ^{AB},

$$S(A:B) = S(A) + S(B) - S(A,B) = S(A) - S(A|B) = S(B) - S(B|A). \tag{4.19}$$

Important properties of the above entropies are

- $S(\rho)$, $S(A:B)$ are non-negative, whereas $S(A|B)$ or $S(B|A)$ could be negative.
- For the Von Neumann entropy, we have
  - $S(\rho) \in [0, \log d]$ for qudit states.
  - $S(\rho \otimes \sigma) = S(\rho) + S(\sigma)$.
  - $S(A) = S(B)$ for pure states ψ^{AB}.
  - $S(\sum_j p_j |j\rangle\langle j| \otimes \rho_j) = S(\mathbf{p}) + \sum_j p_j S(\rho_j)$.
  - Mixture of states

    $$\sum_j p_j S(\rho_j) \le S(\sum_j p_j \rho_j) \le S(\mathbf{p}) + \sum_j p_j S(\rho_j), \tag{4.20}$$

    The first equal sign achieves when all ρ_j are the same. The second equality satisfies iff ρ_j are in orthogonal subspaces. We will shortly show its proof.
- Subadditivity: $|S(A) - S(B)| \le S(A,B) \le S(A) + S(B)$.
- Strong Subadditivity: $S(A,B,C) + S(B) \le S(A,B) + S(B,C)$ or equivalently $S(C|A,B) \le S(C|B)$. The proof of strong subadditivity is not required in this lecture.

4.2.2 Quantum relative entropy

Suppose $\mathrm{supp}(\sigma) \subseteq \mathrm{supp}(\rho)$[1], we define the quantum relative entropy as

$$S(\rho\|\sigma) = \mathrm{Tr}[\rho \log \rho] - \mathrm{Tr}[\rho \log \sigma], \tag{4.21}$$

otherwise $S(\rho\|\sigma) = \infty$.

Three important theorem of the quantum relative entropy are

Theorem 4.1. *The quantum relative entropy is always non-negative*

$$S(\rho\|\sigma) \geq 0 \tag{4.22}$$

with equal sign iff $\rho = \sigma$.

Theorem 4.2. *The quantum relative entropy is monotonic under quantum operations*

$$S(\rho\|\sigma) \geq S(\mathcal{E}(\rho)\|\mathcal{E}(\sigma)) \tag{4.23}$$

for any quantum channel $\mathcal{E}$.

Theorem 4.3. *The quantum relative entropy is jointly convex*

$$S(\sum_j p_j \rho \| \sum_j p_j \sigma) \leq \sum_j p_j S(\rho_j \| \sigma_j) \tag{4.24}$$

Interestingly, we can derive non-negativity and convexity with monotonicity. Likewise, we can derive monotonicity with convexity. To see Theorem 2 $\rightarrow$ 1, we only need to apply the trace map. To see Theorem 2 $\rightarrow$ 3, we first consider $\rho = \sum_j p_j |j\rangle\langle j| \otimes \rho_j$ and $\sigma = \sum_j p_j |j\rangle\langle j| \otimes \sigma_j$, and apply the partial trace map on the first system. To see Theorem 3 $\rightarrow$ 2, we only need to show the monotonicity under partial trace (since an arbitrary channel is equivalent to a joint unitary and a partial trace). We can consider a joint system and apply the twirling map, i.e., a random unitary, to effectively realize partial trace. Specifically, for state ρ^{AB}, we have

$$\mathrm{Tr}_A[\rho^{AB}] = \int dU_A \, U_A \rho^{AB} U_A^\dagger, \tag{4.25}$$

[1]The support of a hermitian matrix A is defined to be the vectors $|\psi\rangle$ satisfying $\langle v|A|v\rangle \neq 0$

here the integral is under a uniform distributed U_A with the Haar measure. Then we can show that

$$\begin{aligned} S(\rho^A\|\sigma^A) &= S\left(\int dU_A\ U_A\rho^{AB}U_A^\dagger \| \int dU_A\ U_A\sigma^{AB}U_A^\dagger\right), \\ &\leq \int dU_A S\left(U_A\rho^{AB}U_A^\dagger \| U_A\sigma^{AB}U_A^\dagger\right), \\ &\leq \int dU_A S\left(\rho^{AB}\|\sigma^{AB}\right), \\ &= S\left(\rho^{AB}\|\sigma^{AB}\right). \end{aligned} \tag{4.26}$$

Here the first inequality uses convexity and the second inequality uses unitary invariance of the relative entropy.

Quantum relative entropy is important since it defines all other entropies.

- Von Neumann entropy: consider $\sigma = \mathbb{I}_d/d$, we have $S(\rho\|\mathbb{I}_d/d) = \log d - S(\rho)$ or equivalently

$$S(\rho) = \log d - S(\rho\|\mathbb{I}_d/d).$$

- Conditional entropy: for ρ^{AB}, consider $\sigma^{AB} = \mathbb{I}_d^A/d \otimes \rho^B$, we have $S(\rho^{AB}\|\sigma^{AB}) = -\mathrm{Tr}[\rho^{AB}\log(\mathbb{I}_d^A/d\otimes\rho^B)] - S(\rho^{AB}) = \log d + S(\rho^B) - S(\rho^{AB}) = \log d - S(A|B)$ or equivalently

$$S(A|B) = \log d - S(\rho^{AB}\|\mathbb{I}_d^A/d\otimes\rho^B).$$

- Mutual information: for ρ^{AB}, consider $\sigma^{AB} = \rho^A\otimes\rho^B$, we have $S(\rho^{AB}\|\sigma^{AB}) = -\mathrm{Tr}[\rho^{AB}\log(\rho^A\otimes\rho^B)] - S(\rho^{AB}) = S(\rho^A) + S(\rho^B) - S(\rho^{AB}) = S(A:B)$ or equivalently

$$S(A:B) = S(\rho^{AB}\|\rho^A\otimes\rho^B).$$

Using the above definitions, we can now prove several interesting properties such as Eq. (4.20), subadditivity, and strong subadditivity.

First, the second inequality of subadditivity could be easily shown by noticing that $S(A:B) = S(A) + S(B) - S(A,B) = S(\rho^{AB}\|\rho^A\otimes\rho^B) \geq 0$. For the first one, we only need to consider a purification ψ^{ABC} of ρ^{AB}, apply $S(C) + S(A) \geq S(A,C)$, and notice $S(C) = S(A,B)$, $S(A,C) = S(B)$.

To prove the first inequality of Eq. (4.20), we first consider the state $\rho^{AB} = \sum_j p_j |j\rangle\langle j| \otimes \rho_j$, then $S(A) = S(\mathbf{p})$ and $S(B) = S(\sum_j p_j\rho_j)$, $S(A,B) = S(\mathbf{p}) +$

$\sum_j p_j S(\rho_j)$. Applying subadditivity, we have $S(A)+S(B)-S(A,B)=S(\sum_j p_j\rho_j)-\sum_j p_j S(\rho_j) \geq 0$. We can similarly prove it using the joint convexity of quantum relative entropy. The proof of the second inequality and strong subadditivity could be found from Nielsen & Chuang's book.

4.2.3 Schumacher's quantum noiseless channel coding theorem

The Von Neumann entropy plays a similar role for quantum state compression. Schumacher's quantum noiseless channel coding theorem guarantees that for any $\varepsilon, \delta > 0$, we can choose a sufficiently large n, an encoding channel $\mathcal{E}$ such that the dimension of $\mathcal{E}(\rho^{\otimes n})$ is no larger than $2^{n(S(\rho)+\varepsilon)}$, and a decoding channel $\mathcal{D}$ such that

$$F(\mathcal{D}\circ\mathcal{E}(\rho^{\otimes n}), \rho^{\otimes n}) \geq 1-\delta, \tag{4.27}$$

for any state ρ. We can thus compress n copies of ρ into $n(S(\rho)+\varepsilon)$ qubits and recover them with negligible error. The proof is very similar to the one for Shannon's noiseless coding theorem.

Chapter 5

Quantum communication

In this chapter, we introduce three of the most well-known quantum communication protocols — quantum teleportation, entanglement swapping, and quantum dense coding. All of these protocols rely on the manipulation of quantum entanglement.

5.1 Quantum Teleportation

Imagine a future where we have a quantum Internet, where each user possesses a quantum computer, and two users can communicate quantum information. A fundamental feature of a quantum Internet would be the ability to transfer a quantum state from one user to another. While the most straightforward approach would be simply to send the quantum state stored in the quantum computer, this would be practically challenging. Quantum computers may be realized using different solid-state technologies, such as trapped ions and superconducting qubits, and transmitting these solid states is difficult, at least with current technology. Even in the classical world, solid-state devices like laptop CPUs are typically local machines; while they can be moved, they are generally not used to transmit information to distant users. Instead, we usually employ light or photons to carry and transmit information, as they are inexpensive to produce and photons interact weakly with the environment. A similar situation can be expected in the quantum realm, where we will likely use solid-state quantum computers and transmit information using photons in most scenarios.

Now, imagine Alice has prepared a qubit state $|\psi\rangle = a|0\rangle + b|1\rangle$ in her local quantum computer. Can she send it to a distant party, Bob, using photons? There

are two related challenges:

- How can quantum information be transferred from solid-state qubits to photons? This seems to require complete knowledge of the quantum state, which in turn implies costly state tomography[1]. Is there a way to achieve this more efficiently, without requiring full knowledge of the state?

- When transmitting a photon, it still suffers from photon losses, phase shifts, and other types of noise. How can we mitigate these issues to ensure reliable quantum information transmission?

These two problems might initially seem insurmountable. However, surprisingly, they can be addressed through the quantum teleportation protocol. The central idea is to utilize quantum entanglement, allowing quantum information to be transmitted by sending classical information. In the following, we introduce the protocol, explain how it works for any qubit state, and discuss its extension to multi-qubit states.

5.1.1 Teleporting a Qubit State

Suppose Alice prepares a qubit $|\psi\rangle_A = a|0\rangle_A + b|1\rangle_A$, and Alice and Bob share an entangled state $|\Phi^+\rangle_{A'B} = \frac{1}{\sqrt{2}}(|00\rangle_{A'B} + |11\rangle_{A'B})$[2]. The quantum teleportation protocol proceeds as follows:

1. Alice performs a Bell state measurement (BSM) on qubits AA' and obtains a classical measurement outcome of two bits [3].

   A Bell state measurement is a projective measurement on one of the four Bell states:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \ |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \tag{5.1}$$

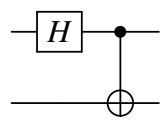
[1] State tomography is a procedure used to determine the density matrix of a quantum state. It involves preparing multiple copies of the state and measuring it in the complete Pauli basis. As the system size increases, state tomography becomes exponentially expensive, i.e., with the number of qubits.

[2] The entangled state is generally distributed using photons, which are then stored in solid-state quantum memories. For instance, Alice can generate a Bell pair and send one half to Bob. Both Alice and Bob then store the Bell pair in a quantum memory, which can input photon states and output them when needed.

[3] It is possible to interact solid-state systems and photons. For example, photons can excite the orbitals of atoms, and this process could be exploited to implement the interaction and thus the BSM.

It is easy to verify that $\{|\Phi^{\pm}\rangle, |\Psi^{\pm}\rangle\}$ forms an orthonormal basis for two-qubit states.

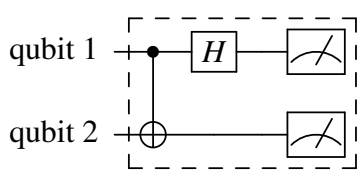
How can we realize a BSM? Recall that a normal projective measurement on the computational basis is a projection onto one of the four states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, which can be achieved by measuring each qubit in the $\{|0\rangle, |1\rangle\}$ basis. Additionally, we can always use a unitary transformation to rotate a basis to another one. In this case, we can rotate the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ to the Bell basis $\{|\Phi^{\pm}\rangle, |\Psi^{\pm}\rangle\}$ using the following circuit (refer to Chapter 10 for a detailed introduction).



Denote the circuit by U, then we have

$$U|00\rangle = |\Phi^{+}\rangle,\ U|01\rangle = |\Psi^{+}\rangle,\ U|10\rangle = |\Phi^{-}\rangle,\ U|11\rangle = |\Psi^{-}\rangle. \tag{5.2}$$

To implement the BSM, we can thus apply $U^{\dagger}$ first and measure in the computational basis, as follows.



Then the four computational basis measurement outcomes $00, 01, 10, 11$ corresponds to projection to $|\Phi^{+}\rangle, |\Psi^{+}\rangle, |\Phi^{-}\rangle, |\Psi^{-}\rangle$, respectively.

2. Alice send the measurement outcomes to Bob.

3. Bob applies I, σ_X, σ_Z, $\sigma_Z\sigma_X$ if the measurement outcome is respectively $00, 01, 10, 11$.

Let's see why the above protocol works. At the beginning, the initial state is

$$
\begin{aligned}
|\psi\rangle_{AA'B} &= \frac{1}{\sqrt{2}}(a|0\rangle_A + b|1\rangle_A)(|00\rangle_{A'B} + |11\rangle_{A'B}), \\
&= \frac{1}{\sqrt{2}}(a|000\rangle_{AA'B} + a|011\rangle_{AA'B} + b|100\rangle_{AA'B} + b|111\rangle_{AA'B}), \\
&= \frac{1}{2}\Big[a(|\Phi^+\rangle_{AA'} + |\Phi^-\rangle_{AA'})|0\rangle_B + a(|\Psi^+\rangle_{AA'} + |\Psi^-\rangle_{AA'})|1\rangle_B \\
&\quad + b(|\Psi^+\rangle_{AA'} - |\Psi^-\rangle_{AA'})|0\rangle_B + b(|\Phi^+\rangle_{AA'} - |\Phi^-\rangle_{AA'})|1\rangle_B\Big] \\
&= \frac{1}{2}\Big[|\Phi^+\rangle_{AA'}\big(a|0\rangle_B + b|1\rangle_B\big) + |\Phi^-\rangle_{AA'}\big(a|0\rangle_B - b|1\rangle_B\big) \\
&\quad + |\Psi^+\rangle_{AA'}\big(a|1\rangle_B + b|0\rangle_B\big) + |\Psi^-\rangle_{AA'}\big(a|1\rangle_B - b|0\rangle_B\big)\Big]
\end{aligned}
\tag{5.3}
$$

Here we have used $|00\rangle = \frac{1}{\sqrt{2}}(|\Phi^+\rangle + |\Phi^-\rangle)$, $|11\rangle = \frac{1}{\sqrt{2}}(|\Phi^+\rangle - |\Phi^-\rangle)$, $|01\rangle = \frac{1}{\sqrt{2}}(|\Psi^+\rangle + |\Psi^-\rangle)$, and $|10\rangle = \frac{1}{\sqrt{2}}(|\Psi^+\rangle - |\Psi^-\rangle)$.

After Alice's BSM measurement, the whole state collapse to one of the following four cases with equal $1/4$ probability

$$
\begin{aligned}
00: &\ |\Phi^+\rangle_{AA'}\big(a|0\rangle_B + b|1\rangle_B\big), \\
01: &\ |\Psi^+\rangle_{AA'}\big(a|1\rangle_B + b|0\rangle_B\big), \\
10: &\ |\Phi^-\rangle_{AA'}\big(a|0\rangle_B - b|1\rangle_B\big), \\
11: &\ |\Psi^-\rangle_{AA'}\big(a|1\rangle_B - b|0\rangle_B\big).
\end{aligned}
\tag{5.4}
$$

Note that the state on Bob's side is almost the initial state of Alice up to some local unitary transformation. After applying the corresponding unitary based on the measurement outcome of Alice, Bob recovers Alice's state with 100% probability.

We remark that the classical measurement outcome is crucial to recover Alice's state. Without it, one can easily verify that Bob's quantum state is a maximally mixed state.

Problem 5.1. *Prove that without Alice's outcome, Bob's quantum state is a maximally mixed state.*

Such a teleportation protocol partially addresses the two challenges mentioned earlier. For the first challenge, one can first create an entangled state between

solid-state qubits and photons, and then teleport the solid-state information onto the photons. For the second challenge, one can create an entangled state between the two users and then teleport the state from one user to the other. The reason we say it only partially solves these challenges is because the protocol still assumes the creation of perfect entangled states, which is generally not feasible in real experiments. For instance, in the second challenge, the state fidelity typically decays exponentially with the distance between the two users. As a result, the created entangled state, and thus the teleported state, will inevitably be imperfect.

However, fortunately, we can still address such imperfections in the entangled state. The key idea is to use entanglement distillation, a systematic approach that extracts high-fidelity entangled states from multiple copies of low-fidelity entangled states, as introduced in Chapter 7. Thus, even if we cannot create a perfect entangled state, we can still distill it into one that is nearly perfect. However, there are still challenges associated with entanglement distillation. First, distillation protocols generally rely on operations involving multiple copies of noisy entangled states, which requires quantum memory to store the photons. Second, entanglement distillation protocols are effective only when the fidelity of the noisy entangled state is above a certain threshold. As mentioned earlier, when transmitting a photon over a distance, its fidelity decays exponentially, which can easily breach the threshold over distances of hundreds of kilometers. Fortunately, as we will discuss below, quantum memory and entanglement swapping can help mitigate this issue. In conclusion, we can see that an efficient and robust protocol exists for transmitting quantum information between distant users.

Let's explore why quantum teleportation can address the two challenges mentioned earlier. When we think about transmitting quantum information, it is typically assumed that the information is encoded in a quantum state. This quantum state, and hence the information, is inherently susceptible to disturbances caused by noise during transmission. In the context of quantum teleportation, instead of directly sending the quantum information, we first create an entangled state, which itself does not carry any specific information. This crucial step allows us to mitigate the effects of noise: even if the entangled state is affected by noise, we can still apply entanglement distillation techniques to recover high-fidelity entanglement without worrying about losing the information. This is a key advantage over traditional quantum state transmission methods, where any noise directly impacts the information being carried by the state. In quantum teleportation, the quantum information is ultimately encoded in two classical bits of information, which are transmitted between Alice and Bob. Classical information is immune to quantum noise and can be cloned and transmitted without any degradation, as per the

no-cloning theorem. Therefore, even though the quantum state may be subject to noise, the transmission of classical information does not suffer from the same limitations, ensuring that the quantum information is preserved upon reaching its destination. This separation between the quantum state (which can be disturbed) and the classical information (which is robust to noise) allows quantum teleportation to offer a solution to the problem of transmitting quantum information over noisy channels. The ability to distill the entangled state further reduces the impact of noise, and the use of classical bits for encoding the quantum information ensures that the information can be reliably transferred.

Problem 5.2. *Consider another robust method for transmitting quantum information using quantum error correction. What strategies might be employed to correct errors during quantum information transmission, and how do they compare to the principles of quantum teleportation?*

5.1.2 Teleporting a Multi-Qubit State

A natural question arises: can we extend the quantum teleportation protocol to teleport an arbitrary n-qubit state, which is essential for practical applications? The answer is surprisingly straightforward — this can be done by applying quantum teleportation to each qubit of the multi-qubit state. The key insight is that quantum teleportation preserves the entanglement, which is crucial for transferring quantum information.

To see why this works, let us consider that Alice prepares a multi-qubit quantum state given by

$$|\psi_{A_1 A_{2\ldots n}}\rangle = a\,|0\rangle_{A_1}\,|\psi_0\rangle_{A_{2\ldots n}} + b\,|1\rangle_{A_1}\,|\psi_1\rangle_{A_{2\ldots n}}\,,$$

where $|\psi_0\rangle_{A2\ldots n}$ and $|\psi_1\rangle_{A2\ldots n}$ are arbitrary quantum states of the remaining qubits $A_{2\ldots n}$, and a and b are complex coefficients satisfying $|a|^2 + |b|^2 = 1$. Alice now wishes to teleport this entire n-qubit state to Bob. The strategy is to first entangle the qubits between Alice and Bob, and then teleport the quantum state of each qubit sequentially.

The initial combined state of Alice's system ($A_1 A_{2\ldots n}$) and the shared entan-

gled pairs ($A'B$) is given by:

$$
\begin{aligned}
|\psi\rangle_{A_1A_{2\ldots n}A'B} = \frac{1}{2}\Big[& |\Phi^+\rangle_{AA'}\left(a|0\rangle_B|\psi_0\rangle_{A_{2\ldots n}} + b|1\rangle_B|\psi_1\rangle_{A_{2\ldots n}}\right) \\
& + |\Phi^-\rangle_{AA'}\left(a|0\rangle_B|\psi_0\rangle_{A_{2\ldots n}} - b|1\rangle_B|\psi_1\rangle_{A_{2\ldots n}}\right) \\
& + |\Psi^+\rangle_{AA'}\left(a|1\rangle_B|\psi_0\rangle_{A_{2\ldots n}} + b|0\rangle_B|\psi_1\rangle_{A_{2\ldots n}}\right) \\
& + |\Psi^-\rangle_{AA'}\left(a|1\rangle_B|\psi_0\rangle_{A_{2\ldots n}} - b|0\rangle_B|\psi_1\rangle_{A_{2\ldots n}}\right)\Big].
\end{aligned}
\tag{5.5}
$$

Here, the Bell states $|\Phi^\pm\rangle$ and $|\Psi^\pm\rangle$ are the maximally entangled states shared between Alice and Bob. Alice performs a Bell state measurement (BSM) on her system (AA'), which yields a classical outcome. Based on this outcome, Alice sends two classical bits of information to Bob, and Bob applies a corresponding local unitary operation on his part of the system to recover the quantum state.

After the BSM and the corresponding local unitary operation based on the measurement outcome, the final state becomes:

$$
a|0\rangle_B|\psi_0\rangle_{A_{2\ldots n}} + b|1\rangle_B|\psi_1\rangle_{A_{2\ldots n}},
$$

which is remarkably similar to the original state $|\psi_{A_1A_{2\ldots n}}\rangle = a|0\rangle_{A_1}|\psi_0\rangle_{A_{2\ldots n}} + b|1\rangle_{A_1}|\psi_1\rangle_{A_{2\ldots n}}$. The only difference is that the A_1 system has been replaced by Bob's system B. Therefore, through this process, quantum teleportation effectively transfers the quantum state, preserving its entanglement with the remaining qubits.

To teleport the entire state $|\psi_{A_1A_{2\ldots n}}\rangle$, Alice and Bob need to sequentially apply quantum teleportation to each qubit system: A_1 and $A_{2\ldots n}$. Each qubit requires one pre-shared Bell pair and two classical bits for transmission. Thus, for an n-qubit state, the quantum teleportation protocol requires n pre-shared Bell pairs and $2n$ classical bits of information.

It is important to note that quantum teleportation does not violate the no-cloning theorem. This is because the protocol does not involve copying the quantum state; rather, it transfers the state from one party (Alice) to another (Bob). After the teleportation is completed, there is still only one copy of the quantum state. The state is merely moved, not duplicated, which ensures compliance with the no-cloning theorem.

5.1.3 Remarks

At last, let's try to understand why teleportation works. We first note that the four Bell states are related via local unitary

$$\begin{aligned}
|\Phi^+\rangle_{AA'} &= |\Phi^+\rangle_{AA'}, \\
|\Psi^+\rangle_{AA'} &= \sigma_X^A |\Phi^+\rangle_{AA'}, \\
|\Phi^-\rangle_{AA'} &= \sigma_Z^A |\Phi^+\rangle_{AA'}, \\
|\Psi^-\rangle_{AA'} &= \sigma_Z^A \sigma_X^A |\Phi^+\rangle_{AA'}.
\end{aligned} \tag{5.6}$$

One may found that these four unitary operators are actually just the four unitary operators Bob needs apply at the third step. We denote these unitary operators by U_{ij}, i.e., $U_{ij} = \mathbb{I}^A$, $U_{01} = \sigma_X^A$, $U_{10} = \sigma_Z^A$, $U_{11} = \sigma_Z^A \sigma_X^A$, and relabel the four Bell states as

$$|\Phi^+\rangle_{AA'} \equiv |\Phi_{00}\rangle_{AA'}, \ |\Psi^+\rangle_{AA'} \equiv |\Phi_{01}\rangle_{AA'}, \ |\Phi^-\rangle_{AA'} \equiv |\Phi_{10}\rangle_{AA'}, \ |\Psi^-\rangle_{AA'} \equiv |\Phi_{11}\rangle_{AA'}, \tag{5.7}$$

then we have

$$|\Phi_{ij}\rangle_{AA'} = U_{ij} |\Phi^+\rangle_{AA'}. \tag{5.8}$$

Now we consider the projective measurement of $|\Phi_{ij}\rangle_{AA'}$ on the initial state $|\psi\rangle_{AA'B} = \frac{1}{\sqrt{2}}(a|0\rangle_A + b|1\rangle_A)(|00\rangle_{A'B} + |11\rangle_{A'B})$. Ignoring the measurement probability, the output state of system B is

$$\langle\Phi_{ij}|_{AA'} |\psi\rangle_{AA'B} = \langle\Phi^+|_{AA'} U_{ij}^\dagger (a|0\rangle_A + b|1\rangle_A)(|00\rangle_{A'B} + |11\rangle_{A'B}). \tag{5.9}$$

We note that this equation is equivalent to the following one with a partial transpose on system A',

$$\begin{aligned}
\langle\Phi_{ij}|_{AA'} |\psi\rangle_{AA'B} &= \mathrm{Tr}_{AA'}[\langle\Phi_{ij}|_{AA'} |\psi\rangle_{AA'B}]^{T_{A'}}, \\
&= \mathrm{Tr}_{AA'}\left[\left(\langle\Phi^+|_{AA'}\right)^{T_{A'}} U_{ij}^\dagger (a|0\rangle_A + b|1\rangle_A)(|00\rangle_{A'B} + |11\rangle_{A'B})^{T_{A'}}\right], \\
&\propto \mathrm{Tr}_{AA'}\left[\mathbb{I}_{A'A} U_{ij}^\dagger (a|0\rangle_A + b|1\rangle_A)\mathbb{I}_{BA'}\right], \\
&= \mathrm{Tr}_A\left[\mathbb{I}_{BA} U_{ij}^\dagger (a|0\rangle_A + b|1\rangle_A)\right], \\
&= U_{ij}^\dagger (a|0\rangle_B + b|1\rangle_B).
\end{aligned} \tag{5.10}$$

Here the key observation is that $\langle\Phi^+|_{AA'}^{T_{A'}} = \mathbb{I}_{A'A}$. Therefore, the output state of system B is actually proportional to $U_{ij}^\dagger(a|0\rangle_B + b|1\rangle_B)$. We can thus apply U_{ij} to invert the unnecessary unitary and obtain the target state $a|0\rangle_B + b|1\rangle_B$.

Problem 5.3. *Prove that* $\langle\Phi^+|_{AA'}^{T_{A'}} = \mathbb{I}_{A'A}$.

5.2 Entanglement Swapping and Quantum Repeaters

As discussed earlier, when attempting to create entanglement between two distant parties, the state fidelity typically decays exponentially over large distances, making it challenging to directly apply entanglement distillation. To overcome this, we utilize ***quantum repeaters***, which rely on the entanglement swapping protocol to extend the range of entanglement between distant parties.

We begin by considering the ideal case. Suppose Alice and Bob share a Bell pair, denoted as $|\Phi^+\rangle_{AB}$, and Bob and Charlie share another Bell pair, $|\Phi^+\rangle_{BC}$. In this setup, we can apply the entanglement swapping protocol to create a Bell pair $|\Phi^+\rangle_{AC}$ between Alice and Charlie. The procedure involves using quantum teleportation to teleport Bob's state from $|\Phi^+\rangle_{AB}$ to Charlie, which consumes the Bell pair $|\Phi^+\rangle_{BC}$. Through this process, entanglement is effectively swapped between Alice and Charlie, without directly creating a Bell pair between them.

In the noisy scenario, where the fidelity of entanglement decays due to environmental disturbances or imperfections, we can first establish multiple noisy Bell pairs between Alice and Bob, and between Bob and Charlie. To improve the fidelity, we apply entanglement distillation separately to these pairs. This distillation process extracts high-fidelity Bell pairs from multiple copies of noisy entanglement. Since entanglement distillation requires several copies of noisy Bell pairs to be processed, quantum memory is necessary to store these pairs during the distillation process. Therefore, quantum repeaters require the ability to store and process entangled states over time.

In this context, Bob serves as an intermediate node between Alice and Charlie, facilitating the extension of quantum entanglement over long distances. This intermediate node is called a ***quantum repeater***, and it plays a crucial role in enabling long-range quantum communication by mitigating the effects of noise and distilling high-quality entanglement.

Problem 5.4. *Let's compare the two scenarios with and without quantum repeaters.*

- *The major error when transmitting a photon is photon loss. Specifically, the probability to detect the photon scales as $e^{-\alpha d}$, which decreases exponentially with the distance d with coefficient α, which typically ranges between 0.2 dB/km and 0.5 dB/km for standard single-mode fibers used in quantum communication.*
  - *Prove that when directly transmitting the signal, with failure probability δ, we have to repeat $O(\log(1/\delta)e^{ld})$ times to get one photon.*
  - *Calculate the loss for 100 kms and 1000 kms with $\alpha = 0.2dB/km$ and the average repeats to find one photon.*
- *Now, suppose that we divide the total distance into L equal segments. For each segments, we have $d' = d/L$ and the loss is greatly reduced to $e^{-\alpha d/L}$ with large L. Since we have a quantum repeater, we can then store the received photon between every two segments and run quantum swapping sequentially.*
  - *Prove that in this case, with failure probability δ, we have to repeat $O(\log(L/\delta)e^{\alpha d/L})$ times to get one Bell pair between all the segments.*
  - *Interestingly, when $L = O(d)$, we can thus reduce the exponential overhead from $O(\log(L/\delta)e^{ld})$ to $O(\log(d/\delta))$.*
  - *Calculate the average repeats to find one photon for 100 kms and 1000 kms with $\alpha = 0.2dB/km$.*

5.3 Dense Coding

Quantum teleportation uses two classical bits to transmit one quantum bit, while dense coding essentially serves the inverse purpose. This concept is connected to a fundamental question in quantum communication: can we send more information using quantum states, such as photons, compared to classical communication? The answer is yes. By sending a qubit state, we can effectively transmit two classical bits of information. The protocol works as follows:

1. Alice and Bob share an entangled state $|\Phi^+\rangle_{AB}$.
2. Alice applies one of the following operations to her system, depending on the classical information she wants to send:

- $\mathbb{I}$ for sending 00,
- σ_X for sending 01,
- σ_Z for sending 10,
- $\sigma_Z\sigma_X$ for sending 11.

3. Alice then sends her system to Bob.

4. Bob performs a Bell state measurement (BSM) on the received state and obtains the two classical bits based on the four possible BSM outcomes.

The proof of this protocol is straightforward. We only need to recall the transformation from the entangled state $|\Phi^+\rangle_{AB}$ to the other three Bell states as shown in Eq. (5.6). This transformation allows us to encode two classical bits into a single qubit, as Alice's operations correspond to projecting the state into one of four possible states, and Bob's BSM reveals the classical information.

Problem 5.5. *Prove the dense coding protocol.*

5.4 Advanced protocols

Based on the quantum teleportation and dense coding protocols, we are now in a position to study their connections. We note that there are three basic components in these two protocols:

- A quantum channel that transmits a qubit state from one party to another. We denote it as $[q \to q]$.
- A classical channel that transmits a classical state from one party to another. We denote it as $[c \to c]$.
- An entangled Bell state between the two parties, denoted as $[qq]$.

For these two processes, we denote $A \geq B$ when we can use A to simulate B without using extra resources.

Problem 5.6. *Prove the following inequality:*

$$[q \to q] \geq \{[c \to c], [qq]\}$$

Based on quantum teleportation and dense coding, we can derive the following inequalities:

$$\begin{aligned} 2[c \to c] + [qq] &\geq [q \to q], \\ [q \to q] + [qq] &\geq 2[c \to c]. \end{aligned} \tag{5.11}$$

When the entanglement $[qq]$ can be regarded as free, these two inequalities reduce to:

$$[q \to q] = 2[c \to c], \tag{5.12}$$

which means that a quantum bit channel is equivalent to two classical bits channels and the two protocols are reversible and optimal.

Problem 5.7. *Prove that the quantum teleportation and dense coding protocols cannot be more efficient. Specifically, when $\alpha < 2$, Eq. (5.11) cannot be satisfied.*

However, in many scenarios, quantum entanglement is not free. In this case, the two protocols work by consuming entanglement. Is there any way to modify the two protocols so that they become reversible? The answer is yes, with the help of ***coherent bit (cobit) channels*** [PhysRevLett.92.097902], which is defined as an isometry $V_{A\to AB}$:

$$V_{A\to AB} |x\rangle_A = |x\rangle_A |x\rangle_B, \forall x = 0, 1. \tag{5.13}$$

We denote a cobit as $[q \to qq]$. It can be understood as a 'cloning' procedure, which coherently copies A's classical information into B. When A and B are in the same location, it could be realized with the CNOT gate.

Problem 5.8. *Prove the following relation:*

$$[q \to q] \geq [q \to qq] \geq \{[c \to c], [qq]\}$$

Therefore, a cobit can be regarded as sending classical information while also generating entanglement.

The quantum teleportation and dense coding protocols can be generalized with cobits, as illustrated in the following two problems.

Problem 5.9. ***Generalized quantum dense coding.***

- *Prove the equality*

$$(H \otimes I)\mathit{CNOT}(Z^i X^j \otimes I)|\Phi^+\rangle = |i\rangle|j\rangle.$$

- *The generalized teleportation protocol works as follows:*
  - *Consider any unknown two-qubit state $|\psi\rangle_{A_1A_2}$ on Alice's side and the maximally entangled state $|\Phi^+\rangle_{A'B}$ between Alice and Bob.*
  - *Apply controlled Z with control qubit A_1 and target A', and controlled X with control qubit A_2 and target A'.*
  - *Send A' to Bob and apply $(H\otimes I)\texttt{CNOT}$ on $A'B$ on Bob's side.*

  Prove that the whole procedure implements two cobit channels, i.e.,

$$[q\to q]+[qq]\geq 2[q\to qq]. \tag{5.14}$$

  In this case, the generated resources are two cobits, which are stronger than two classical bits.

Problem 5.10. ***Generalized quantum teleportation.*** *The protocol works as follows:*

- *Consider an unknown qubit state $|\psi\rangle_A$ and a shared entangled state $|\Phi^+\rangle_{A'B}$.*
- *Apply $\texttt{CNOT}_{AB}$ and $\texttt{H}_A$ to the system AA'.*
- *Apply two cobit channels sequentially on A and A' to 'copy' the state onto B_1 and B_2, respectively.*
- *Apply the coherent correction on Bob's side, i.e.,*

$$|00\rangle\langle 00|_{B_1B_2}+|01\rangle\langle 01|_{B_1B_2}\otimes X_B+|10\rangle\langle 10|_{B_1B_2}\otimes Z_B+|11\rangle\langle 11|_{B_1B_2}\otimes Z_BX_B$$

Then, prove that it generates one qubit channel and two Bell pairs, i.e.,

$$2[q\to qq]+[qq]\geq [q\to q]+2[qq]. \tag{5.15}$$

Note that, when regarding $[qq]$ as a catalyst and allowing catalysts for the $\geq$ sign, the above equation reduces to:

$$2[q\to qq]\geq [q\to q]+[qq]. \tag{5.16}$$

Therefore, we find that

$$[q\to q]+[qq]=2[q\to qq]. \tag{5.17}$$

Hence, the generalized teleportation and superdense coding protocols become reversible when extending classical bit channels to coherent bit channels.

Chapter 6

Quantum key distribution

In this chapter, we introduce the basic concepts about quantum key distribution, including the BB84, B92, and E91, and the idea of the security proof. We also introduce a related topic about quantum random number generation.

6.1 Why we need key distribution

In many scenarios, we need to privately communicate without worrying about eavesdropping. For example, we want privacy when we chat with each other; we want our account and password protected when inputting online; bank, hospital, government, and many sensitive organizations want their communication secret; secure communication is critical in military.

In order to secretly communicate n-bit information, the most secure way is to first share n-bit of private keys between the two parties and then apply one-time-pad. There is currently no information theoretical provable way to share private keys classically[1], and most existing key distribution or sharing or extension protocols rely on certain assumptions, such as RSA assuming the hardness of factoring. Quantum key distribution can solve this part, but let's wait a bit and study what is one-time-pad first. Imagine Alice wants to privately send

$$x_1, x_2, \ldots x_n$$

[1]This is not exact, since there is always one way which requires the two parties to meet first and then exchange keys face-to-face. But of course, this is not realistic or efficient when considering online communication or when they cannot meet physically. Even though, this kind of idea has already been applied in many cases in reality.

to Bob. Of course, if she directly sends it (or apply some easy function to it), the bad guy Eve can easily learn the information. The idea of one-time-pad is to add the keys

$$k_1, k_2, \ldots, k_n$$

securely shared by Alice and Bob to the x string,

$$y_1 = x_1 \oplus k_1,\ y_2 = x_2 \oplus k_2,\ \ldots y_n = x_n \oplus k_n$$

and send the y string instead. Then Bob can decode the information by adding the keys again

$$x_1 = y_1 \oplus k_1,\ x_2 = y_2 \oplus k_2,\ \ldots x_n = y_n \oplus k_n$$

Since Eve has no information about the keys k, she has no information about y either. By saying this, we simply mean that Eve's ability to guess the value of y is not better than random guessing. Meanwhile, using more keys cannot increase the security, since then Eve can simply randomly guess the value of y. While having less keys will leak certain information to Eve. This is because Eve now only need to guess the keys instead and gain a certain amount of information.

Another important point to remark is that the key in one-time-pad can only be used once. Imagine that we used the same keys to send $\boldsymbol{a}$ and $\boldsymbol{b}$, that is the sent information is

$$\boldsymbol{a} \oplus \boldsymbol{k} \quad \text{and} \quad \boldsymbol{b} \oplus \boldsymbol{k}$$

Then Eve can obtain n-bit of information of $\boldsymbol{a} \oplus \boldsymbol{b}$ by bitwise plussing $\boldsymbol{a} \oplus \boldsymbol{k}$ and $\boldsymbol{b} \oplus \boldsymbol{k}$. This implies that we have to consume the same length of keys to send that length of information.

6.2 BB84, B92, E91

6.2.1 BB84

The first quantum key distribution protocol was proposed by Charles Bennett and Gilles Brassard in 1984, so the protocol is also called BB84. The protocol works as follows.

1. State preparation and measurement. Repeat the step (1a), (1b), (1d), and (1e) for n rounds.

   (a) For the ith round, Alice generate a random bit $a_i \in \{X, Z\}$ to decide the basis of the state to prepare.

(b) Alice generate another random bit $x_i \in \{0,1\}$. If $a_i = X$, she prepares $|+\rangle$ or $|-\rangle$ if $x_i = 0$ or $x_1 = 1$; If $a_i = Z$, she prepares $|0\rangle$ or $|1\rangle$ if $x_i = 0$ or $x_1 = 1$.

(c) Alice sends the n quantum states to Bob.

(d) For each state Bob receives, he randomly generate a bit $b_i \in \{X, Z\}$ to decide the basis of the state to measure.

(e) He measures in the b_i basis and obtain a random outcome y_i.

2. Basis reconciliation. After the procedure, they obtain n_X bits with $a_i = b_i = X$ and n_Z bits with $a_i = b_i = Z$.

   (a) Alice and Bob publicly announces the basis information of $\boldsymbol{a}$ and $\boldsymbol{b}$.

   (b) They discard the events when $a_i \neq b_i$.

3. Error estimation. Alice and Bob randomly select a part of the keys to estimate the bit and phase flip errors e_b and e_p, respectively. The bit (phase) error is defined by the probability when $a_i = b_i = Z$ ($a_i = b_i = X$) whereas $x_i \neq y_i$. Suppose the keys are defined by the outcomes with $a_i = b_i = Z$. A non-zero phase error implies nonzero information leakage.

4. Privacy amplification. Alice and Bob apply a privacy amplification protocol to extract the leaked information, which consumes about $S(e_p)$ bits of keys.

5. Error correction. Alice and Bob run an error correction procedure to correct bit flip errors, which consumes about $S(e_b)$ bits of keys.

Finally, Alice and Bob obtain $(n_X + n_Z)\big(1 - S(e_p) - S(e_b)\big)$ bits of private keys with a key rate of $r = 1 - S(e_p) - S(e_b)$.

A rigorous proof of the BB84 protocol is actually quite complicated. Here, we only show its security for some special cases. We will also see some insights of the security of BB84 after we introduce E91.

- First, we consider the noiseless case where $e_b = e_p = 0$. In this case, Eve has no information of the quantum state. This is because she has to disturb the quantum state to extract any information, which leads to nonzero phase or bit flip errors.

  Here we prove the intuition of information requires disturbance. Consider two different quantum states $|\psi_0\rangle$ and $|\psi_1\rangle$ denoting the bit of 0 and 1, respectively. Suppose we can apply a unitary U to extract certain information

without disturbing the quantum state. Specifically, we have

$$U|\psi_0\rangle|0\rangle = |\psi_0\rangle|v_0\rangle, \quad U|\psi_1\rangle|0\rangle = |\psi_1\rangle|v_1\rangle.$$

Calculating the inner product of the two states, we have

$$\langle\psi_1|\langle 0|U^\dagger U|\psi_0\rangle|0\rangle = \langle\psi_1|\psi_0\rangle\langle v_1|v_0\rangle,$$

which is equivalent to

$$\langle\psi_1|\psi_0\rangle = \langle\psi_1|\psi_0\rangle\langle v_1|v_0\rangle.$$

This equation holds either when $\langle\psi_1|\psi_0\rangle = 0$ or $\langle v_1|v_0\rangle = 1$, corresponding to the exactly distinguishable and no information extracted cases, respectively.

- Next, we consider the noisy case when $e_b = e_p \neq 0$. We consider several strategy of Eve and show the security.
  - Suppose Eve randomly guesses the measurement basis and measure the quantum state. Naively, it seems that Even has a 1/2 probability to correctly guess the basis and hence extract 1/2 of the bits. This is not true, since it introduces very large bit and phase errors. Consider that Alice and Bob are both in the X basis. Suppose Alice prepares $|+\rangle$. Since Eve does not know the basis, with probability 1/2, she randomly measures it in the X basis to correctly get 0 without disturbing the state. With probability 1/2, she measures in the Z basis to get an incorrect outcome and also disturbs the state to either $|0\rangle$ or $|1\rangle$. Then we can evaluate $e_p = 1/4$ and similarly get $e_b = 1/4$. Then the key rate $r = 1 - 2S(0.25) < 0$.

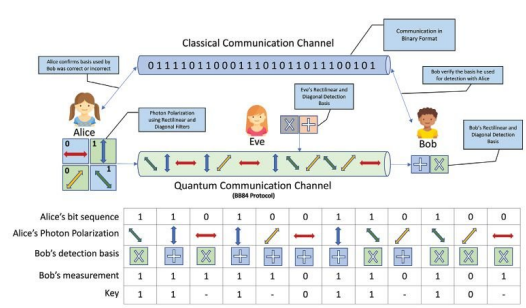


Figure 6.1: Schematic of the BB84 protocol.

- Of course, we can consider a more generate strategy of Eve, who randomly measures in the X and Z bases with probability p and $1-p$. We can similarly estimate $e_p = (1-p)/2$ and $e_b = p/2$ with a key rate $r = 1 - S((1-p)/2) - S(p/2) \leq 0$. The equal sign is achieved when $p = 0$ or 1.
- One may argue that this is not Eve's optimal strategy yet. Eve cannot extract information without disturbance simply because she does not know the state basis. However, Alice and Bob will announce the basis at a later stage, Eve can then store the state and measure it until the basis information is revealed. A rigorous proof of the security against such attacks is out of the scope, so here we only show intuitively why it works. The basic idea is actually still the information - disturbance relationship. Since Eve does not know the basis information, whenever Eve tries to get any information, either before or after the basis announcement, she has to apply some disturbance to the quantum state at the state transmission stage. Otherwise, the state she has will be independent of the state Alice sends and she cannot obtain any information even if she knows the basis. Then if she applies any disturbance to the transmitted state, Alice and Bob will know it from the phase or bit error, and those information will be extracted out at the privacy amplification and error correction stage.

6.2.2 B92

In the B92 protocol, by Charles Bennett in 1992, Alice only prepares $|0\rangle$ and $|+\rangle$ and Bob measures in the X and Z basis. When Bob measures $|-\rangle$, he knows that the state cannot be $|+\rangle$; similarly when Bob measures $|1\rangle$, he knows that the state cannot be $|0\rangle$. Then Bob only tells Alice which state he is sure about without announcing the measurement basis and they key is shared by setting $|0\rangle$ to be 0 and $|+\rangle$ to be 1.

The B92 protocol only requires to prepare two states and does not need the basis reconciliation part. However, it is secure only when there is no loss. We consider the so-called intercept and resend attack. Specifically, Eve intercepts the state, measures it randomly in the X and Z basis. There are several casese.

- Suppose she measures in the X basis.
  - When she obtains $|+\rangle$, she does not know whether the state is $|0\rangle$ or $|+\rangle$ and she just sends nothing to Bob.

  - When she obtains $|-\rangle$, she knows that the state must be $|0\rangle$. Then she also prepares $|0\rangle$ and send it to Bob.

- Suppose she measures in the Z basis.
  - When she obtains $|0\rangle$, she does not know whether the state is $|0\rangle$ or $|+\rangle$ and she just sends nothing to Bob.
  - When she obtains $|1\rangle$, she knows that the state must be $|+\rangle$. Then she also prepares $|+\rangle$ and send it to Bob.

In this case, the sent quantum state is lost with 50% probability and Eve has perfect information of Alice's prepared state.

6.2.3 E91

The E91 protocol, proposed by Artur Ekert in 1991, uses Bell nonlocality and quantum entanglement. The basic idea is to observe nonlocality by measuring an entangled state in the $\{Z_0, Z_{\pi/8}, Z_{\pi/4}\}$ and $\{Z_0, Z_{\pi/8}, Z_{-\pi/8}\}$ for Alice and Bob respectively. They can detect nonlocality using the $\{Z_0, Z_{\pi/4}\}$ and $\{Z_{\pi/8}, Z_{-\pi/8}\}$ basis and generate keys in the (Z_0, Z_0) and $(Z_{\pi/8}, Z_{-\pi/8},)$ bases. While the E91 protocol has a lot of debate when it was initially introduced and it takes more than 20 years until we found rigorous proofs for it.

Without realizing Bell nonlocality, we can actually show that a modified E91 protocol is equivalent to BB84. The protocol works as follows.

- Alice prepares a Bell pair $|\Phi^+\rangle_{AB} = (|00\rangle_{AB} + |11\rangle_{AB})/\sqrt{2}$ and send the B part to Bob.
- Alice and Bob randomly select from the $\{X, Z\}$ basis and measure the state.
- Alice and Bob announce the basis they select and discards the cases where the basis choices are different.
- Alice and Bob estimate the bit (Z basis) and phase (X basis) flip errors.
- Alice and Bob apply error correction and privacy amplification.

To see why this is equivalent to BB84, we can simply think that Alice measures her state before she sends the B part. Using the modified E91 protocol, we can actually more easily see why it and BB84 work. In the ideal case, we have

$|\Phi^+\rangle_{AB} = (|00\rangle_{AB} + |11\rangle_{AB})/\sqrt{2} = (|{+}{+}\rangle_{AB} + |{-}{-}\rangle_{AB})/\sqrt{2}$. Then if Alice and Bob both measure in the Z basis, they will get either 00 or 11 randomly with uniform probability. Since whether it is 00 or 11 is uniformly random, Eve does not have any information than random guessing. The analysis is similar for the X basis measurement. For the noisy case where the state shared by Alice and Bob is ρ_{AB}, we can imagine that they run an entanglement distillation protocol which transforms $\rho_{AB}^{\otimes n}$ into $m \leq n$ copies of Φ_{AB}^+. Then we can measure the ideal Bell pair to generate secure keys. Note that we can design the entanglement distillation protocol such that it commutes with the XX or ZZ basis measurements. The basic idea is that the projectors $\Pi_{XX=\pm 1} = (\mathbb{I} \pm XX)/2$ and $\Pi_{ZZ=\pm 1} = (\mathbb{I} \pm ZZ)/2$ can detect the error syndrome and we can use the CSS code to correct errors.

6.3 Quantum random number generation

We have mentioned that when we prepared a Bell state $|\Phi^+\rangle_{AB} = (|00\rangle_{AB} + |11\rangle_{AB})/\sqrt{2}$ and measure system A and B in the ZZ basis, we will obtain a full random outcome. To see this, we can focus on the randomness generation process and consider a qubit state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. When we measure the state in the Z basis, it will randomly collapse to $|0\rangle$ or $|1\rangle$ with probability $1/2$ and $1/2$. No one can predict the measurement outcome unless the basic rule of quantum mechanics fails. Mathematically, suppose the state $|+\rangle_A$ is held by Alice and consider an adversary Eve who holds a state $|e\rangle_E$. Then Alice measures the state collapsing it into $(|0\rangle\langle 0|_A + |1\rangle\langle 1|_A) \otimes |e\rangle\langle e|_E$. The tensor product structure maintains since the measurement is performed locally and independently on Eve's system. Then we can evaluate the conditional entropy to have $S(A|E) = 1$, indicating randomness of a single bit.

Next, we consider a noisy state ρ_A and study how much randomness can we obtain if we measure it in the Z basis. Suppose Eve's state is $|e\rangle_E$, then the state after measurement is $(|0\rangle\langle 0|_A \rho_A |0\rangle\langle 0|_A + |1\rangle\langle 1|_A \rho_A |1\rangle\langle 1|_A) \otimes |e\rangle\langle e|_E$. Suppose $p_0 = \langle 0|_A \rho_A |0\rangle_A$ and $p_1 = \langle 1|_A \rho_A |1\rangle_A$, the randomness we have is $S(A|E) = S(p_0)$. This analysis seems to indicate that Eve's state is not useful since $S(A|E) = S(A)$. However, there is a huge flaw in the analysis. Consider the maximally mixed state $\rho_A = |0\rangle\langle 0|_A + |1\rangle\langle 1|_A$, the measurement outcome is indeed uniformly random if we measure in the Z basis. However, imagine that Eve holds a purification of ρ_A, that is, Alice and Eve's state is jointly described as $|\Phi^+\rangle_{AE} = (|00\rangle_{AB} + |11\rangle_{AB})/\sqrt{2}$. Eve can always perfectly learn Alice's measurement outcome by simply measuring her system in the Z basis. Therefore, the

randomness of Alice's outcome conditioned on Eve's prediction is actually 0.

To have the correct proof, we need to consider a purification of ρ_A as ψ_{AE} satisfying $\rho_A = \mathrm{Tr}_E[\psi_{AE}]$. After we measure A in the Z basis, it becomes $\tilde{\psi}_{AE} = \sum_j |j\rangle\langle j|_A \otimes \langle j|_A \psi_{AE} |j\rangle_A$. Denote the normalization of $\langle j|_A \psi_{AE} |j\rangle_A$ as $p_j = \mathrm{Tr}[\langle j|_A \psi_{AE} |j\rangle_A]$, and the normalized pure state as $\psi_E^j = \langle j|_A \psi_{AE} |j\rangle_A / p_j$. Then we have $\tilde{\psi}_{AE} = \sum_j p_j |j\rangle\langle j|_A \otimes \psi_E^j$. Now we calculate the conditional entropy of the measured state, we obtain

$$\begin{aligned} S(A|E)_{\tilde{\psi}_{AE}} &= S(AE) - S(E), \\ &= S(p_j) + \sum_j p_j S(\psi_E^j) - S(\rho_A). \end{aligned} \tag{6.1}$$

Here we have used $S(AE) = S(p_j) + \sum_j p_j S(\psi_E^j)$, $S(\psi_E^j) = 0$, and $S(E) = S(\rho_A)$. Denote $\Delta_Z(\rho_A) = (|0\rangle\langle 0|_A \rho_A |0\rangle\langle 0|_A + |1\rangle\langle 1|_A \rho_A |1\rangle\langle 1|_A) \otimes |e\rangle\langle e|_E$, we have $S(p_j) = S(\Delta_Z(\rho_A))$. Then the randomness conditioned on Eve's system is

$$\mathcal{R}(\rho_A) = S(\Delta(\rho_A)) - S(\rho_A). \tag{6.2}$$

For example, when ρ_A is a pure state, we have $S(\rho_A) = 0$ and $\mathcal{R}(\rho_A) = S(\Delta(\rho_A))$. When ρ_A is the maximally mixed state, we have $S(\Delta(\rho_A)) = S(\rho_A) = 1$, and thus $\mathcal{R}(\rho_A) = 0$.

While the above analysis seems unrelated to the analysis of QKD, we can actually see their relationship by carefully designing a workable randomness generation protocol. We note that the randomness formula of Eq. (6.2) requires to evaluate $S(\rho_A)$ and hence assumes the knowledge of ρ_A. In practice, we may not know the state and only have partial information of it. For example, imagine we measure the state in the X basis and obtain a state $\Delta_X(\rho_A)$, we can prove an uncertainty relation

$$\mathcal{R}(\rho_A) + S(\Delta_X(\rho_A)) \geq 1. \tag{6.3}$$

We can thus lower bound the randomness as

$$\mathcal{R}(\rho_A) \geq 1 - S(\Delta_X(\rho_A)). \tag{6.4}$$

Note that we can treat $S(\Delta_X(\rho_A))$ as the phase error in QKD. In particular, we have $p_- = \langle -|\rho|-\rangle$ and $S(\Delta_X(\rho_A)) = S(p_-)$. That is, suppose we aim to prepare and measure the $|+\rangle$ state. When a phase error happens to the state, it transforms it to $|-\rangle$ and p_- thus represents the error probability.

Chapter 7

Quantum entanglement (by Xiao Yuan, Wenhao Zhang, and Long Xiong)

In this chapter, we introduce the basic concepts of quantum entanglement.

7.1 Quantum Entanglement

7.1.1 Definitions

To study entanglement in quantum states, we need to first clarify what entanglement means. Intuitively, if we consider a composite system which consists of two single qubits A and B, a tensor product state $|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$ is supposed to contain no entanglement, while a maximally entangled state denotes

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

for A, B being single-qubit systems. For mixed states, we would regard entanglement as kind of a "resource", and then we might follow the methodology of "Resource theory":

- Define resource-free objects;
- Define free operations.

These steps suggest we first define "entanglement-free" states, and then define "free operations", i.e. operations that do not introduce entanglement into an "entanglement-free" state.
For pure states, we define the tensor product states

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

to be free of entanglement. Those pure states are also called "separable states". For mixed states,

$$\rho_{AB} = \rho_A \otimes \rho_B$$

is obviously free of entanglement, or "separable". But we may also consider a state of the following form:

$$\rho_{AB} = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)},$$

with $\sum_i p_i = 1$. Although this state seemingly contains some kind of correlation, it is worth noticing that one may prepare this state via the following process:

- randomly pick i according to probability distribution p_i;
- independently prepare $\rho_A^{(i)}$ in system A and $\rho_B^{(i)}$ in system B.

We realize that during the preparation of the states in this ensemble, no quantum entanglement is introduced. Thus, this kind of state is "classically correlated" instead of "quantumly entangled". We will refer to all those entanglement-free states as *separable states*.
In the process of preparing ρ_{AB}, we have locally prepared two states in different sub-systems, and we have also communicated classically to distribute the information of i. We define the free operations to be

Local Operation + Classical Communication,

referred to as "LOCC". Those free operations would not introduce entanglement into separable states.

7.1.2 Common Entangled States

Bell States (EPR pairs)

The Bell states, also called EPR pairs, are maximally entangled two-qubit pure states. There are four orthonormal Bell states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{7.1}$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{7.2}$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{7.3}$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{7.4}$$

These states form a complete basis for the two-qubit Hilbert space and are widely used in quantum communication protocols such as quantum teleportation and superdense coding.

Derivation of Bell States: Start with $|00\rangle$ and apply the Hadamard gate H on the first qubit:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \tag{7.5}$$

The combined state becomes

$$(H \otimes I)|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle). \tag{7.6}$$

Apply the CNOT gate with the first qubit as control and the second as target:

$$\text{CNOT}\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{7.7}$$

which yields the $|\Phi^+\rangle$ Bell state.

GHZ State

The Greenberger-Horne-Zeilinger (GHZ) state is a three-qubit maximally entangled state:

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle). \tag{7.8}$$

The GHZ state exhibits **monogamy of entanglement**, meaning that entanglement is shared globally, not between pairs. Tracing out one qubit leaves the remaining two qubits in a separable (non-entangled) state.

Reduced Density Matrix of GHZ: The reduced density matrix of two qubits, say AB, when tracing out qubit C is:

$$\rho_{AB} = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{7.9}$$

This matrix is diagonal and represents classical correlation only.

W State

The W state is another three-qubit entangled state with different entanglement properties:

$$|W\rangle = \frac{1}{\sqrt{3}}(|100\rangle + |010\rangle + |001\rangle). \tag{7.10}$$

Unlike the GHZ state, the W state retains entanglement even if one qubit is lost. This robustness makes W states useful in scenarios where particle loss is likely.

Cluster States

Cluster states are multi-qubit entangled states used as a resource in measurement-based quantum computation. For a one-dimensional three-qubit cluster state:

$$|\text{Cluster}\rangle = \frac{1}{2}(|000\rangle + |011\rangle + |101\rangle - |110\rangle). \tag{7.11}$$

Cluster states are prepared by applying controlled-phase gates between neighboring qubits initialized in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Summary of Properties

- Bell states: maximal bipartite entanglement, perfect correlation.
- GHZ state: global tripartite entanglement, exhibits monogamy.

- W state: robust against qubit loss, retains bipartite entanglement.
- Cluster states: entangled resource for measurement-based quantum computation.

These states are foundational for understanding entanglement in quantum information theory.

7.1.3 Examples of Stabilizer Entangled States

The stabilizer formalism provides a convenient way to describe a large class of quantum states, including many important entangled states used in quantum information processing. A stabilizer group $\mathcal{S}$ is an abelian subgroup of the n-qubit Pauli group P_n, which does not contain $-I$. A stabilizer state $|\psi\rangle$ is the unique common +1 eigenstate of all elements in $\mathcal{S}$.

For an n-qubit system, if $\mathcal{S}$ has n independent generators $\{g_1, g_2, \ldots, g_n\}$, the stabilizer state is uniquely determined by the equations

$$g_j |\psi\rangle = |\psi\rangle, \quad \text{for all } j = 1, \ldots, n. \tag{7.12}$$

Example: Bell State (EPR Pair)

Consider the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle). \tag{7.13}$$

The stabilizer generators for this state are

$$g_1 = Z \otimes Z, \quad g_2 = X \otimes X. \tag{7.14}$$

One can verify that

$$(Z \otimes Z) |\Phi^+\rangle = |\Phi^+\rangle, \quad (X \otimes X) |\Phi^+\rangle = |\Phi^+\rangle. \tag{7.15}$$

The Bell state is the unique +1 eigenstate of these two commuting Pauli operators.

Example: Three-Qubit GHZ State

The three-qubit GHZ state is given by

$$|\mathrm{GHZ}_3\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle). \tag{7.16}$$

The stabilizer generators are

$$g_1 = X \otimes X \otimes X, \quad g_2 = Z \otimes Z \otimes I, \quad g_3 = I \otimes Z \otimes Z. \tag{7.17}$$

These operators commute and stabilize $|\mathrm{GHZ}_3\rangle$:

$$g_j |\mathrm{GHZ}_3\rangle = |\mathrm{GHZ}_3\rangle, \quad \text{for } j = 1, 2, 3. \tag{7.18}$$

Example: Four-Qubit Cluster State

The four-qubit cluster state on a linear chain is defined as

$$|C_4\rangle = \prod_{(j,k)\in E} CZ_{j,k} |+\rangle^{\otimes 4}, \tag{7.19}$$

where $CZ_{j,k}$ is the controlled-Z gate acting on qubits j and k, $E = \{(1,2), (2,3), (3,4)\}$ is the set of edges of the linear graph, and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

The stabilizer generators are

$$g_1 = X_1 Z_2, \quad g_2 = Z_1 X_2 Z_3, \quad g_3 = Z_2 X_3 Z_4, \quad g_4 = Z_3 X_4. \tag{7.20}$$

Each g_j stabilizes $|C_4\rangle$:

$$g_j |C_4\rangle = |C_4\rangle, \quad \text{for } j = 1, 2, 3, 4. \tag{7.21}$$

Example: Five-Qubit Code State

The five-qubit code is the smallest possible quantum error-correcting code that can correct arbitrary single-qubit errors. The logical zero state $|0_L\rangle$ of the code is stabilized by the following set of operators:

$$\begin{aligned} g_1 &= XZZXI, \\ g_2 &= IXZZX, \\ g_3 &= XIXZZ, \\ g_4 &= ZXIXZ. \end{aligned} \tag{7.22}$$

All these generators mutually commute and stabilize the logical subspace.

Summary

Stabilizer states provide a systematic way to construct and describe a wide variety of entangled states used in quantum information, including Bell states, GHZ states, cluster states, and quantum error-correcting code states. Their mathematical structure allows efficient classical simulation via the Gottesman-Knill theorem and underpins many applications in quantum computing and quantum error correction.

7.2 Entanglement quantification

For any given bipartite quantum state ρ_{AB}, can we quantify its entanglement using a real-valued function $E(\rho_{AB})$? A general resource quantifier should satisfy the following two requirements.

- Non-negativity and normalization. Since separable states have no entanglement, we should have $E(\sigma_{AB}) = 0$. Since the Bell states have maximal entanglement, we can normalize it to have $E(\Phi^+_{AB}) = 1$.
- Monotonicity. The entanglement cannot be increased under any LOCC operation Λ, i.e., $E(\rho_{AB}) \geq E(\Lambda(\rho_{AB}))$.

A widely used quantifier of entanglement is defined as

$$E(\rho) = \min_{\sigma_{AB} \in \mathrm{SEPP}} S(\rho_{AB} \| \sigma_{AB}), \tag{7.23}$$

where $S(\rho\|\sigma)$ is the relative entropy. We can easily show non-negativity and normalization for it, and here we prove its monotonicity.

Proof. Note that any LOCC can only transform a separable state to a separable state, thus

$$\begin{aligned} E(\rho_{AB}) &= \min_{\sigma_{AB} \in \mathrm{SEPP}} S(\rho_{AB} \| \sigma_{AB}), \\ &\geq \min_{\sigma_{AB} \in \mathrm{SEPP}} S(\Lambda(\rho_{AB}) \| \Lambda(\sigma_{AB})), \\ &\geq \min_{\sigma_{AB} \in \mathrm{SEPP}} S(\Lambda(\rho_{AB}) \| \sigma_{AB}), \\ &= E(\Lambda(\rho_{AB})) \end{aligned} \tag{7.24}$$

Here the first inequality used the monotonicity of relative entropy under physical operations and the second inequality used $\Lambda(\sigma_{AB}) \in \mathrm{SEPP}$. □

While $E(\rho_{AB})$ is a valid and useful entanglement quantifier, calculating it is extremely complicated for a general mixed state due to the minimization. Fortunately, we have a simple formula of $E(\psi_{AB})$ for pure states ψ_{AB},

$$E(\psi_{AB}) = S(\rho_A) = S(\rho_B), \tag{7.25}$$

where ρ_A and ρ_B are the reduced density matrices on system A and B, respectively. One can show that the measure vanishes when ψ_{AB} is a product state. In the next section, we further show its operational meaning.

7.2.1 Other Entanglement Measures and Quantification

Schmidt Decomposition and Entropy

Given the Schmidt decomposition

$$|\psi\rangle_{AB} = \sum_k \lambda_k |u_k\rangle_A |v_k\rangle_B, \tag{7.26}$$

where $\{|u_k\rangle_A\}$ and $\{|v_k\rangle_B\}$ are orthonormal bases in A and B. The **entanglement entropy** of this state is the von Neumann entropy of the reduced density matrix. The reduced density matrix for subsystem A is

$$\rho_A = \mathrm{Tr}_B\left(|\psi\rangle_{AB}\langle\psi|\right) = \sum_k \lambda_k^2 |u_k\rangle\langle u_k|. \tag{7.27}$$

The entanglement entropy is defined as

$$S(\rho_A) = -\sum_k \lambda_k^2 \log_2 \lambda_k^2. \tag{7.28}$$

For the Bell state, $\lambda_k^2 = 1/2$, so

$$S(\rho_A) = -2 \cdot \frac{1}{2} \log_2 \frac{1}{2} = 1. \tag{7.29}$$

The entropy achieves its maximum when the Schmidt coefficients are all equal.

Concurrence

For a general 2×2 mixed state ρ, concurrence provides a direct measure of entanglement. Define the spin-flipped state

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{7.30}$$

where ρ^* is the complex conjugate of ρ in the computational basis and σ_y is the Pauli-Y matrix:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{7.31}$$

Then, define the matrix

$$R = \rho\tilde{\rho}. \tag{7.32}$$

Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ be the eigenvalues of R, sorted in descending order. The concurrence is

$$C(\rho) = \max\left(0, \sqrt{\lambda_1} - \sqrt{\lambda_2} - \sqrt{\lambda_3} - \sqrt{\lambda_4}\right). \tag{7.33}$$

For example, for the Bell state $|\Phi^+\rangle$, one finds $C = 1$, indicating maximal entanglement. For separable states, $C = 0$.

Negativity and Logarithmic Negativity

The partial transpose of ρ with respect to subsystem B is given by

$$\langle ij|\rho^{T_B}|kl\rangle = \langle il|\rho|kj\rangle. \tag{7.34}$$

Negativity is defined as the absolute value of the sum of the negative eigenvalues of ρ^{T_B}:

$$N(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}, \tag{7.35}$$

where $\|\rho^{T_B}\|_1$ is the trace norm, the sum of absolute eigenvalues.

The logarithmic negativity is

$$E_N(\rho) = \log_2 \|\rho^{T_B}\|_1. \tag{7.36}$$

Negativity detects entanglement whenever the partial transpose has negative eigenvalues. However, there exist entangled states with positive partial transpose (PPT), known as bound entangled states.

Entanglement of Formation

The entanglement of formation $E_F(\rho)$ quantifies the minimum average entanglement needed to prepare a mixed state ρ:

$$E_F(\rho) = \min_{\{p_i, |\psi_i\rangle\}} \sum_i p_i S(\rho_A^i), \tag{7.37}$$

where the minimization is over all pure state decompositions $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ and $\rho_A^i = \mathrm{Tr}_B |\psi_i\rangle\langle\psi_i|$.

For 2×2 systems, $E_F(\rho)$ is related to the concurrence by the formula:

$$E_F(\rho) = h\left(\frac{1+\sqrt{1-C(\rho)^2}}{2}\right), \tag{7.38}$$

where

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x). \tag{7.39}$$

Distillable Entanglement

The distillable entanglement $E_D(\rho)$ is the maximal asymptotic rate at which Bell pairs can be extracted from ρ using LOCC:

$$E_D(\rho) = \sup\left\{r : \lim_{n\to\infty} \left\|\Lambda_n(\rho^{\otimes n}) - \Phi_{2^r}^{\otimes n}\right\| \to 0\right\}, \tag{7.40}$$

where Λ_n denotes LOCC operations and Φ_{2^r} is the maximally entangled state of rank 2^r.

It holds that

$$E_D(\rho) \le E_F(\rho). \tag{7.41}$$

For pure states, $E_D = E_F$, but for mixed states they may differ. For bound entangled states, $E_F > 0$ while $E_D = 0$.

These entanglement measures together offer a comprehensive toolkit for analyzing and quantifying bipartite quantum entanglement from various perspectives.

7.2.2 Monogamy of Entanglement

In classical physics, correlations between multiple parties can be shared freely among many systems. However, in quantum mechanics, entanglement exhibits a restriction known as the **monogamy of entanglement**, which states that if two

systems are maximally entangled, they cannot be entangled with a third system at the same time. This property is essential in quantum cryptography and quantum networks, as it guarantees that entanglement cannot be freely distributed.

Consider three parties, labeled A, B, and C. Suppose A and B share a quantum state ρ_{AB}, and A and C share ρ_{AC}. Monogamy imposes constraints on how much entanglement A can share with B and C simultaneously.

The concept can be formalized using **concurrence**, an entanglement measure defined for two-qubit systems. For a three-qubit pure state $|\psi_{ABC}\rangle$, the monogamy inequality, first introduced by Coffman, Kundu, and Wootters (CKW), is given by:

$$C_{A|BC}^2 \geq C_{AB}^2 + C_{AC}^2, \tag{7.42}$$

where $C_{A|BC}$ is the concurrence between qubit A and the joint system BC, while C_{AB} and C_{AC} are the concurrences between A and B, and A and C, respectively.

The square of the concurrence C_{AB}^2 quantifies the bipartite entanglement between A and B, and similarly for C_{AC}^2.

To understand the physical significance of the inequality

$$C_{A|BC}^2 - C_{AB}^2 - C_{AC}^2 \geq 0, \tag{7.43}$$

note that the difference is called the **residual tangle**, or simply the **3-tangle**:

$$\tau_{ABC} = C_{A|BC}^2 - C_{AB}^2 - C_{AC}^2. \tag{7.44}$$

The 3-tangle τ_{ABC} captures the genuine tripartite entanglement that cannot be reduced to pairwise entanglement between any two qubits. For the three-qubit GHZ state

$$|\mathrm{GHZ}_3\rangle = \frac{1}{\sqrt{2}}\left(|000\rangle + |111\rangle\right), \tag{7.45}$$

the pairwise concurrences C_{AB}, C_{AC}, and C_{BC} all vanish, while the 3-tangle $\tau_{ABC} = 1$, indicating that all the entanglement is genuinely tripartite.

In contrast, for the W state

$$|W\rangle = \frac{1}{\sqrt{3}}\left(|100\rangle + |010\rangle + |001\rangle\right), \tag{7.46}$$

the 3-tangle $\tau_{ABC} = 0$, but the pairwise concurrences between qubits are nonzero:

$$C_{AB} = C_{AC} = C_{BC} = \frac{2}{3}. \tag{7.47}$$

This illustrates that the W state contains only bipartite entanglement shared among the pairs, and no genuine three-way entanglement.

Generalization to Multi-Qubit Systems

The CKW inequality has been generalized to n-qubit systems. For a given qubit A entangled with $n-1$ other qubits $B_1, B_2, \ldots, B_{n-1}$, the generalized monogamy relation is

$$C^2_{A|B_1B_2\ldots B_{n-1}} \geq \sum_{i=1}^{n-1} C^2_{AB_i}. \tag{7.48}$$

This inequality implies that the total amount of bipartite entanglement that A can share with all other qubits is limited by the total entanglement between A and the group of the remaining qubits as a whole.

Monogamy for Other Entanglement Measures

While concurrence satisfies monogamy inequalities, not all entanglement measures do. Negativity, for example, does not generally obey the monogamy inequality. Other measures like entanglement of formation may also fail to satisfy strict monogamy relations in higher dimensions.

However, for squared concurrence and related quantities such as the squared negativity under certain conditions, monogamy holds reliably in the qubit case.

7.3 Entanglement conversion via LOCC

Generally, specifying the conditions of transforming one mixed state to another state via LOCC is hard. However, we could provide a sufficient and necessary condition for two pure states $|\psi\rangle$ and $|\phi\rangle$ to be transformable via LOCC. We give the conclusion here:

Theorem 7.1. *Let $|\psi\rangle$ and $|\phi\rangle$ be two normalized pure states in the composite system AB, with the dimension of the systems $d_A = d_B = d$. Suppose we have their*

Schmidt Decomposition as

$$|\psi\rangle = \sum_i \sqrt{p_i}|\psi_i^A\rangle|\psi_i^B\rangle,$$
$$|\phi\rangle = \sum_i \sqrt{q_i}|\phi_i^A\rangle|\phi_i^B\rangle,$$

with $\sqrt{p_i}$ and $\sqrt{q_i}$ being the d-dim Schmidt coefficients of $|\psi\rangle$ and $|\phi\rangle$, respectively. Then we can transform $|\phi\rangle$ to $|\psi\rangle$ using LOCC operations alone, if and only if

$$\boldsymbol{p} \succ \boldsymbol{q}.$$

Here the symbol $\succ$ reads "majorize". $\boldsymbol{p} \succ \boldsymbol{q}$ means if we sort the entries of $\boldsymbol{p}$ and $\boldsymbol{q}$ in descending order, i.e.

$$p_{(1)} \geqslant p_{(2)} \geqslant \cdots p_{(d)} \geqslant 0,$$
$$q_{(1)} \geqslant q_{(2)} \geqslant \cdots q_{(d)} \geqslant 0,$$

and for the partial sum $S_p(k) = \sum_{i=1}^k p_{(i)}$, $S_q(k) = \sum_{i=1}^k q_{(i)}$,

$$S_p(k) \geqslant S_q(k), \quad \forall k = 1, 2, \cdots, d-1;$$
$$S_p(d) = S_q(d).$$

Graphically, the relationship of majorization could be illustrated in a so called *Lorenz Curve*. See Fig.7.1. Next we will introduce two lemmas.

Lemma 7.1. *For vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ of the same dimension, $\boldsymbol{p} \succ \boldsymbol{q}$ if and only if there exists a doubly stochastic matrix D, such that*

$$\boldsymbol{q} = D \cdot \boldsymbol{p}.$$

A *doubly stochastic matrix D* is a square matrix whose all entries are non-negative, and each row and each column sums to 1. The next lemma writes:

Lemma 7.2. *Each $d \times d$ doubly stochastic matrix D is a convex combination of permutation matrices of dimension d, i.e.*

$$D = \sum_\pi \lambda_\pi \pi,$$

with $\lambda_\pi \geqslant 0$ and $\sum_\pi \lambda_\pi = 1$.

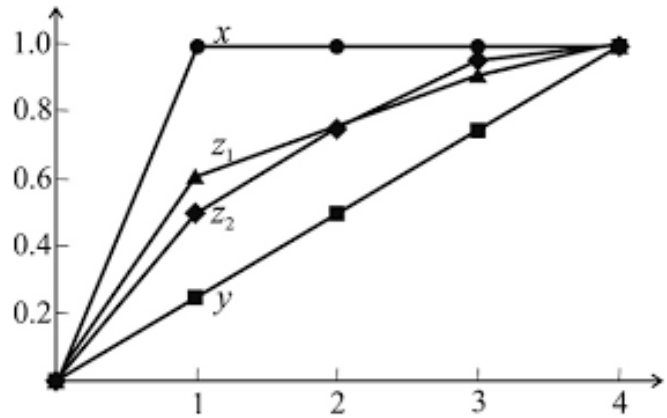


Figure 7.1: Lorenz curve. Horizontal axis represents k and vertical axis represents $S(k)$ for different states. $\boldsymbol{p} \succ \boldsymbol{q}$ if and only if the Lorenz curve of $\boldsymbol{p}$ lies fully above that of $\boldsymbol{q}$. Note the "largest" vector with respect to majorization corresponds to the curve x illustrated, whose entries look like $(1,0,\cdots,0)$; and the "smallest" vector corresponds to the curve y, whose entries look like $(1/d,1/d,\cdots,1/d)$. Intuitively, "larger" vectors are hence less entangled. Figure credited: G.Bellomo and G. M. Bosyk, *Quantum Worlds, Perspectives on the Ontology of Quantum Mechanics*.

Combining those results, we have the following useful conclusion:

Corollary 7.1. $\boldsymbol{p} \succ \boldsymbol{q}$ *if and only if*

$$\boldsymbol{q} = \sum_{\pi} \lambda_\pi \pi(\boldsymbol{p}),$$

with $\lambda_\pi \geqslant 0$, $\sum_\pi \lambda_\pi = 1$, *and the summation goes over all permutations* π *on the entries of* $\boldsymbol{p}$.

As an example, one may consider the "largest" vector with respect to majorization, $\boldsymbol{p} = (1,0,\cdots,0)$, hence $\boldsymbol{p}$ majorizes any $\boldsymbol{q} = (q_1,q_2,\cdots,q_d)$. It is obvious that we can permute entries of $\boldsymbol{p}$ and then convexly combine them to obtain $\boldsymbol{q}$.

Proof of Theorem 1

Since we are free to apply local operations on $|\psi\rangle$ and $|\phi\rangle$, we could apply reversible local unitary operations on them to shift their Schmidt Decomposition

onto the same basis. Without loss of generality, we will assume the states $|\psi\rangle$ and $|\phi\rangle$ possess the following form:

$$|\psi\rangle = \sum_i \sqrt{p_i}|i\rangle^A|i\rangle^B,$$
$$|\phi\rangle = \sum_i \sqrt{q_i}|i\rangle^A|i\rangle^B.$$

The proof goes as follows:

i). We first prove this condition is *sufficient*. Suppose $\boldsymbol{p} \succ \boldsymbol{q}$, then we can find coefficients $\lambda_\pi \geqslant 0, \sum_\pi \lambda_\pi = 1$, such that $\boldsymbol{q} = \sum_\pi \lambda_\pi \pi(\boldsymbol{p})$. Now that $\pi(\boldsymbol{p})$ is a permuted vector of $\boldsymbol{p}$, we may define a function $\tilde{\pi} : (1, 2, \cdots, d) \to (1, 2, \cdots, d)$ corresponding to π, such that $[\pi(\boldsymbol{p})]_i = p_{\tilde{\pi}(i)}$, i.e. the $i-th$ component of $\pi(\boldsymbol{p})$ is the $\tilde{\pi}(i)-th$ component of $\boldsymbol{p}$. Hence

$$q_i = \sum_\pi \lambda_\pi p_{\tilde{\pi}(i)}.$$

We can then define as following a set of *Kraus operators*:

$$K_\pi = \sum_i \sqrt{\lambda_\pi}\sqrt{\frac{p_{\tilde{\pi}(i)}}{q_i}}|\tilde{\pi}(i)\rangle\langle i|.$$

To verify that $\{K_\pi\}$ is indeed a set of Kraus operators, we need to show $\sum_\pi K_\pi^\dagger K_\pi = I$.

$$\begin{aligned}
\sum_\pi K_\pi^\dagger K_\pi &= \sum_\pi \sum_{i,j} (\sqrt{\lambda_\pi})^2 \sqrt{\frac{p_{\tilde{\pi}(j)}}{q_j}}\sqrt{\frac{p_{\tilde{\pi}(i)}}{q_i}}|j\rangle\langle\tilde{\pi}(j)|\tilde{\pi}(i)\rangle\langle i| \\
&= \sum_\pi \sum_i \lambda_\pi \frac{p_{\tilde{\pi}(i)}}{q_i}|i\rangle\langle i| \\
&= \sum_i \frac{1}{q_i}\left(\sum_\pi \lambda_\pi p_{\tilde{\pi}(i)}\right)|i\rangle\langle i| \\
&= \sum_i |i\rangle\langle i| = I.
\end{aligned}$$

Hence $\{K_\pi\}$ characterize an (irreversible) quantum channel. Now we apply local

operation $K_\pi^A \otimes \tilde{\pi}^B$ to the pure state $|\phi\rangle$ (Here we define $\tilde{\pi}|i\rangle := |\tilde{\pi}(i)\rangle$):

$$\begin{aligned}
K_\pi^A \otimes \tilde{\pi}^B|\phi\rangle &= \sum_i \sqrt{\lambda_\pi}\sqrt{\frac{p_{\tilde{\pi}(i)}}{q_i}}|\tilde{\pi}(i)\rangle\langle i|^A \otimes \tilde{\pi}^B \sum_j \sqrt{q_j}|j\rangle^A|j\rangle^B \\
&= \sum_i \sqrt{\lambda_\pi}\sqrt{\frac{p_{\tilde{\pi}(i)}}{q_i}}\sqrt{q_i}|\tilde{\pi}(i)\rangle^A|\tilde{\pi}(i)\rangle^B \\
&= \sqrt{\lambda_\pi}\sum_{i'}|i'\rangle^A|i'\rangle^B \\
&= \sqrt{\lambda_\pi}|\psi\rangle.
\end{aligned}$$

Here we relabel the states using $i' = \tilde{\pi}(i)$. We may verify $\sum_\pi K_\pi^A \otimes \tilde{\pi}^B|\phi\rangle\langle\phi|(K_\pi^A \otimes \tilde{\pi}^B)^\dagger = \sum_\pi \lambda_\pi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi|$. Thus, this channel transforms $|\phi\rangle$ to $|\psi\rangle$.

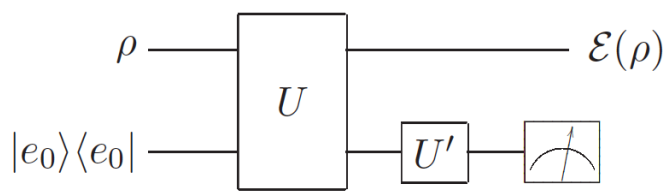


Figure 7.2: Implementation of a Kraus channel. The measurement on the ancillae is necessary to obtain information of which Kraus operation is being applied. Figure credited: *Nielson, Chuang, Quantum Computation and Quantum Information*.

To see that this channel is indeed an LOCC, we need to check the implementation of Kraus operations on system A. Generally, one need to introduce ancilla qubits and apply a global unitary to the bipartite system and then trace out the ancilla part. See Fig.7.2. However, to obtain the information of "which Kraus operation is being applied" one need to measure the ancilla qubit, and the outcome indicates the particular π-index of the Kraus operation. This information should then be transferred classically to system B, so as to apply proper local operation $\tilde{\pi}^B$ on system B. We realize that both local operation and classical communication are indispensable for transforming between two states $|\psi\rangle$ and $|\phi\rangle$.

ii). We then prove this condition is *necessary*. We will focus on a weaker version: Suppose Alice applies a Kraus channel $\{K_m^A\}$ to $|\phi\rangle^{AB}$ on system A, and sends the information of m to Bob (via measurement on the ancilla qubits), then Bob correspondingly applies quantum channel $\mathcal{E}_m^B$ to the state on system B. And

the resulting state becomes $|\psi\rangle$. Mathematically, we have

$$|\psi\rangle\langle\psi| = \sum_m \mathcal{E}_m^B \left(K_m^A |\phi\rangle\langle\phi| K_m^{A^\dagger} \right).$$

Here we notice that the left hand side is a pure state, while right hand side is a summation of multiple states. Since we can only obtain a mixed state if we sum up different states, it can be concluded that each term on the right hand side must be proportional to the pure state $|\psi\rangle\langle\psi|$. Hence

$$\begin{aligned}
\rho_\psi^A = \mathrm{Tr}_B[|\psi\rangle\langle\psi|] \propto \mathrm{Tr}_B \left[\mathcal{E}_m^B \left(K_m^A |\phi\rangle\langle\phi| K_m^{A^\dagger} \right) \right] \\
&= \mathrm{Tr}_B \left(K_m^A |\phi\rangle\langle\phi| K_m^{A^\dagger} \right) \\
&= K_m^A \left(\mathrm{Tr}_B |\phi\rangle\langle\phi| \right) K_m^{A^\dagger} \\
&= K_m^A \rho_\phi^A K_m^{A^\dagger}.
\end{aligned}$$

(From now on we will omit the superscript A on K_m.) So if we define a normalization coefficient $p_m = \mathrm{Tr}\left(K_m \rho_\phi^A K_m^\dagger \right)$, then

$$\rho_\psi^A = \frac{1}{p_m} K_m \rho_\phi^A K_m^\dagger.$$

On the other side, notice that $\sqrt{K_m \rho_\phi^A K_m^\dagger}$ is the positive operator that appears in the polar decomposition of $K_m \sqrt{\rho_\phi^A}$, i.e.

$$K_m \sqrt{\rho_\phi^A} = \sqrt{K_m \rho_\phi^A K_m^\dagger} U_m = \sqrt{p_m \rho_\psi^A}\, U_m,$$

for some unitary U_m. Hence we could manipulate the state a little bit:

$$\begin{aligned}
\rho_\phi^A &= \sqrt{\rho_\phi^A} \left(\sum_m K_m^\dagger K_m \right) \sqrt{\rho_\phi^A} \\
&= \sum_m U_m^\dagger \sqrt{p_m \rho_\psi^A} \sqrt{p_m \rho_\psi^A} U_m \\
&= \sum_m p_m U_m^\dagger \rho_\psi^A U_m.
\end{aligned}$$

Further notice that

$$\rho_\psi^A = \mathrm{Tr}_B[|\psi\rangle\langle\psi|] \quad = \mathrm{Tr}_B\left[\sum_{i,j}\sqrt{p_i p_j}|i\rangle\langle j|^A \otimes |i\rangle\langle j|^B\right] = \sum_i p_i |i\rangle\langle i|^A,$$

$$\rho_\phi^A = \mathrm{Tr}_B[|\phi\rangle\langle\phi|] \quad = \mathrm{Tr}_B\left[\sum_{i,j}\sqrt{q_i q_j}|i\rangle\langle j|^A \otimes |i\rangle\langle j|^B\right] = \sum_i q_i |i\rangle\langle i|^A,$$

So we have

$$\begin{aligned} q_i &= \langle i|\rho_\phi^A|i\rangle^A \\ &= \langle i|\sum_m p_m U_m^\dagger \left(\sum_j p_j |j\rangle\langle j|^A\right) U_m |i\rangle^A \\ &= \sum_j \sum_m p_m |\langle i|U_m^\dagger|j\rangle|^2 p_j \\ &= \sum_j D_{ij} p_j. \end{aligned}$$

Here we have defined $D_{ij} \equiv \sum_m p_m |\langle i|U_m^\dagger|j\rangle|^2$. We can easily verify D is a convex combination of doubly stochastic matrices $|\langle i|U_m^\dagger|j\rangle|^2$, and hence is itself a doubly stochastic matrix. Using Lemma 1, we have thus secured

$$\boldsymbol{p} \succ \boldsymbol{q}.$$

7.3.1 Probabilistic Conversion (Vidal's Formula)

If the majorization condition is not satisfied, conversion may still be possible with some probability p. The optimal success probability of converting $|\psi\rangle$ to $|\phi\rangle$ is given by

$$p_{\max} = \min_k \frac{\sum_{i=1}^k p_i}{\sum_{i=1}^k q_i}. \tag{7.49}$$

7.4 Entanglement Distillation

Consider a two-qubit entangled state

$$|\psi\rangle = \sum_i \sqrt{p_i}|i\rangle|i\rangle,$$

it is maximally entangled if $p_0 = p_1 = 1/2$. However, according to the previous section, for a general state $|\psi\rangle$ with biased distribution of p_0 and p_1, we cannot extract a maximally entangled state $|\Phi^+\rangle$ from it. Yet it is possible to "distill" $|\Phi^+\rangle$ from multiple copies of $|\psi\rangle$. Moreover, we may generate m copies of $|\Phi^+\rangle$ using n copies of $|\psi\rangle$. We now denote $|0\rangle|0\rangle$ and $|1\rangle|1\rangle$ by $|\overline{0}\rangle$ and $|\overline{1}\rangle$, respectively. Consider the full state of n copies of $|\psi\rangle$:

$$\begin{aligned}
|\psi\rangle^{\otimes n} &= (\sum_i \sqrt{p_i}|i\rangle|i\rangle)^{\otimes n} \\
&= (\sqrt{p_0}|\overline{0}\rangle + \sqrt{p_1}|\overline{1}\rangle)^{\otimes n} \\
&= \sqrt{p_1^n}|\overline{1}\overline{1}\cdots\overline{1}\rangle + \left(\sqrt{p_0 p_1^{n-1}}(|\overline{0}\overline{1}\cdots\overline{1}\rangle + |\overline{1}\overline{0}\cdots\overline{1}\rangle + \cdots + |\overline{1}\overline{1}\cdots\overline{0}\rangle)\right) + \cdots \\
&= \sum_{k=0}^{n} \sqrt{p_0^k \cdot p_1^{n-k}}(C_n^k \text{ terms, each with } k\ |\overline{0}\rangle\text{'s and } (n-k)\ |\overline{1}\rangle\text{'s}).
\end{aligned}$$

Note that for each k, all corresponding terms possess the same amplitude, suggesting that these C_n^k terms form a maximally entangled state of $\lfloor \log C_n^k \rfloor$ qubits. This result comes from the fact that $|\overline{0}\rangle$ and $|\overline{0}\rangle$ stand for $|00\rangle^{AB}$ and $|11\rangle^{AB}$, so we may write each state $|\bar{i}_1\bar{i}_2\cdots\bar{i}_n\rangle$ as $|i_1 i_2 \cdots i_n\rangle^A |i_1 i_2 \cdots i_n\rangle^B$, and a maximally entangled state over m qubits could be obtained by relabeling the superposition of 2^m states of the form $|\boldsymbol{i}\rangle^A|\boldsymbol{i}\rangle^B$.

We could implement on the full state the following projective measurement:

$$\mathbf{\Pi} = \{\Pi_0, \Pi_1, \cdots, \Pi_n\},$$

where each operator Π_k projects the state onto the subspace with k $|\overline{0}\rangle$'s. This measurement could be done locally, since it is equivalent to projecting onto the subspace with k $|0\rangle^A$'s on system A. With probability $p(k) = C_n^k \cdot p_0^k p_1^{n-k}$ we will have the system projected onto the subspace k, and hence generate a maximally entangled state of $m \sim \log C_n^k$ qubits. So the expectation value of m is

$$\mathbb{E}(m) = \sum_k C_n^k p_0^k p_1^{n-k} \log C_n^k.$$

When n is sufficiently large, the previous expectation value would be dominated by those terms in the typical space, or $k \sim np_0$. Hence we may replace $\log C_n^k$ by $\log 2^{nS(p_0)} = nS(p_0)$. Furthermore we have

$$\lim_{n\to\infty} \frac{\mathbb{E}(m)}{n} = S(p_0),$$

where $S(P_0)$ is the entropy of the probability distribution $(p_0, 1-p_0)$.

7.5 Entanglement witness

We can of course use an entanglement measure to see whether the state is entangled or not. However, in practice, calculating the entanglement measure generally requires full state tomography, which is costly for large systems. Suppose we can only measure partial information of the state, can we still detect entanglement?

Here, we introduce the concept of entanglement witness, a tool widely used in experiment to verify entanglement. An operator W is called an entanglement witness if it satisfy

$$\mathrm{Tr}[W\sigma] \geq 0, \forall \sigma \in \mathrm{SEPP}, \tag{7.50}$$

and for some entangled state ρ, we have

$$\mathrm{Tr}[W\rho] < 0. \tag{7.51}$$

Therefore, whenever we observe a negative witness value, we can conclude the existence of entanglement.

Here is an example entanglement witness,

$$W = \frac{1}{2} - \Phi^+_{AB}. \tag{7.52}$$

For any separable state σ, we have[1] $\mathrm{Tr}[W\sigma] = \frac{1}{2} - \mathrm{Tr}[\Phi^+_{AB}\sigma] \geq 0$. Consider the noisy maximally entangled state mixed with white noise,

$$\rho_p = (1-p)\Phi^+ + p\mathbb{I}/4, \tag{7.53}$$

we have

$$\mathrm{Tr}[\rho_p W] = \frac{3p}{4} - \frac{1}{2}. \tag{7.54}$$

We can see that the state is entangled whenever $p \leq 2/3$.

7.5.1 Examples of Entanglement Detection: PPT Criterion

Peres Criterion: Positive Partial Transpose (PPT)

For a bipartite quantum state ρ_{AB} acting on $\mathcal{H}_A \otimes \mathcal{H}_B$, the partial transpose with respect to subsystem B is defined by its matrix elements:

$$\langle ij|\rho^{T_B}|kl\rangle = \langle il|\rho|kj\rangle. \tag{7.55}$$

[1] Here we have used $\mathrm{Tr}[\Phi^+_{AB}\sigma] \leq 1/2$ for any separable state. To prove it, one first notice that we only need to consider pure product states and then use the Schmidt decomposition of Φ^+_{AB}.

The Peres-Horodecki criterion states that if ρ_{AB} is separable, then ρ^{T_B} is positive semidefinite:

$$\rho^{T_B} \geq 0. \tag{7.56}$$

Therefore, if ρ^{T_B} has at least one negative eigenvalue, the state ρ_{AB} must be entangled.

This criterion is both necessary and sufficient for separability when the dimensions of the subsystems satisfy 2×2 or 2×3. In higher dimensions, PPT positivity is only necessary but not sufficient, leading to the existence of bound entangled states (PPT entangled states).

Example: Consider the Werner state in $\mathbb{C}^2 \otimes \mathbb{C}^2$:

$$\rho_W(p) = p|\Psi^-\rangle\langle\Psi^-| + (1-p)\frac{I}{4}, \tag{7.57}$$

where $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ is the singlet Bell state and $0 \leq p \leq 1$.

The partial transpose of $\rho_W(p)$ is

$$\rho_W(p)^{T_B} = \frac{1}{4}\begin{pmatrix} 1-p & 0 & 0 & -p \\ 0 & 1+p & 0 & 0 \\ 0 & 0 & 1+p & 0 \\ -p & 0 & 0 & 1-p \end{pmatrix}. \tag{7.58}$$

The eigenvalues of $\rho_W(p)^{T_B}$ are:

$$\frac{1+p}{4} \text{ (double degenerate)}, \quad \frac{1-3p}{4}, \quad \frac{1+p}{4}. \tag{7.59}$$

The smallest eigenvalue becomes negative when $p > \frac{1}{3}$, so the Werner state is entangled for $p > \frac{1}{3}$.

7.5.2 CHSH Inequality as an Entanglement Witness

The CHSH inequality provides a method to detect entanglement using Bell nonlocality. Define the Bell operator:

$$\mathcal{B} = A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1, \tag{7.60}$$

where A_0, A_1 are observables for Alice, and B_0, B_1 are observables for Bob, each with eigenvalues ± 1.

For any local hidden variable model (LHVM), the CHSH inequality holds:

$$|\langle \mathcal{B} \rangle| \leq 2. \tag{7.61}$$

Quantum mechanics allows a maximal violation up to Tsirelson's bound:

$$|\langle \mathcal{B} \rangle| \leq 2\sqrt{2}. \tag{7.62}$$

Example: For the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, choosing optimal measurement settings

$$A_0 = \sigma_z, \quad A_1 = \sigma_x, \quad B_0 = \frac{\sigma_z + \sigma_x}{\sqrt{2}}, \quad B_1 = \frac{\sigma_z - \sigma_x}{\sqrt{2}}, \tag{7.63}$$

the expectation value of $\mathcal{B}$ becomes:

$$\langle \mathcal{B} \rangle = 2\sqrt{2}. \tag{7.64}$$

This violates the classical CHSH bound and certifies the presence of entanglement.

Relation to Entanglement Witness The Bell operator $\mathcal{B}$ can be rewritten as an entanglement witness:

$$W_{\text{CHSH}} = 2I - \mathcal{B}. \tag{7.65}$$

For separable states, $\langle W_{\text{CHSH}} \rangle \geq 0$. A negative expectation value signals entanglement.

The CHSH inequality provides a direct operational method to detect entanglement without full state tomography, which is particularly useful in experimental scenarios.

These two examples, PPT criterion and CHSH inequality, are fundamental tools in entanglement detection, illustrating different aspects of quantum correlations: the algebraic structure of density matrices versus measurement correlations violating local realism.

Chapter 8

Nonlocality

In this chapter, we introduce Bell inequalities, the concept of quantum nonlocality, and its applications.

8.1 Framework of nonlocal games

We focus on the nonlocal game scenario, where two distantly separated parties, Alice and Bob, are asked to play a game. In particular, as shown in Fig. 8.1, they receive random inputs x and y from a referee with probability $q(x,y)$, and give outputs a and b, respectively. Consider a payoff function $\beta(a,b,x,y)$ and denote the strategy of Alice and Bob by $p(a,b|x,y)$, the average payoff is

$$I = \sum_{a,b,x,y} \beta(a,b,x,y)p(a,b|x,y)q(x,y). \tag{8.1}$$

For simplicity, we consider binary inputs and outputs and assume that the inputs are generated uniformly random with $q(x,y) = 1/4$. Then we have

We will show that the average payoff obtained from any classical strategy is upper bounded, and such a bound is called a Bell inequality, i.e.,

$$I = \frac{1}{4} \sum_{a,b,x,y} \beta(a,b,x,y)p(a,b|x,y) \leq u_{\mathrm{C}} \tag{8.2}$$

Yet, this upper bound could be violated when we use quantum strategies. Therefore, the violation of Bell's inequality indicated the existence of quantumness. We discuss that this feature could be exploited to design device independent or self-testing quantum information protocols.

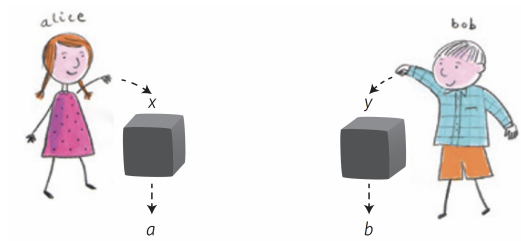


Figure 8.1: Bell Nonlocality. Figure from Nature Physics volume 10, pages 264–270 (2014).

8.2 Classical, quantum, no-signalling correlations

8.2.1 Classical

We first consider what is a classical strategy. In the Bell nonlocality scenario, we assume that Alice and Bob cannot communicate once they have started the game. In this case, Alice and Bob have to use independent strategies $p(a|x)$ and $p(b|y)$, respectively, which defines the joint strategy $p(a|x)p(b|y)$. Yet, since Alice and Bob can communicate before the game, their strategies can indeed depend on some predetermined rules, say λ. Therefore, any classical strategy could be described by

$$p_{\mathrm{C}}(a,b|x,y) = \sum_{\lambda} q(\lambda) p_A(a|x,\lambda) p_B(b|y,\lambda) \tag{8.3}$$

We note that this also includes the cases where the strategy could also vary with different rounds of the game. The theory described by $p_{\mathrm{C}}(a,b|x,y)$ is also called a local hidden variable model (LHVM), since the correlation[1] is actually generated by the local hidden variable λ.

Can we generate any correlation using a classical LHVM? The answer is no and a simple counter example is as follows,

$$p_{\mathrm{SWAP}}(a,b|x,y) = \begin{cases} 1 & a = y \\ 1 & b = x \end{cases} \tag{8.4}$$

which effectively output the swapped inputs of Alice and Bob. Such a probability distribution is equivalent to the swap gate. Obviously it cannot be realized by

[1]For two random variables X and Y, we say they are correlated simply mean $p(X,Y) \neq p(X)p(Y)$.

LHVM since it is equivalent to sending Alice and Bob's inputs to the other party and LHVM is forbid to communicate.

8.2.2 NS

Can we formalize the requirement of no-communication or no-signaling (NS)? We first define the partial probability

$$p(a|x,y) = \sum_b p(a,b|x,y), \quad p(b|x,y) = \sum_a p(a,b|x,y). \tag{8.5}$$

Since the output a (b) is determined by Alice (Bob) while the input y (x) is sent to Bob, a no-signaling strategy thus requires

$$p_{\mathrm{NS}}(a|x,y) = p(a|x,y'), \quad p_{\mathrm{NS}}(b|x,y) = p(b|x',y), \tag{8.6}$$

and such a condition is also called the no-signaling condition. It is easy to verify that LHVM satisfy NS.

Can LHVM generate any NS correlation? Interestingly, the answer is still no and an explicit counter example is the PR-box, defined as

$$p_{\mathrm{PR}}(a,b|x,y) = \begin{cases} 1/2 & a \oplus b = xy \\ 0 & \text{otherwise} \end{cases} \tag{8.7}$$

At this point, it is not obvious why LHVM cannot generate the PR-box. But we will see this soon using the CHSH inequality.

8.2.3 Quantum

Another important family of strategy is using quantum mechanics. Specifically, Alice and Bob can share some entangled state ρ_{AB} and measure the state with POVM $\{M^A_{a|x} \geq 0\}$ and $\{M^B_{b|y} \geq 0\}$ satisfying $\sum_a M^A_{a|x} = \mathbb{I}_A$ for all x and $\sum_b M^B_{b|y} = \mathbb{I}_B$ for all y. The correlation generated in this way is

$$p_{\mathrm{Q}}(a,b|x,y) = \mathrm{Tr}[\rho_{AB}(M^A_{a|x} \otimes M^B_{b|y})] \tag{8.8}$$

Again, can you generate any quantum correlation using LHVM? The answer is still no and we will prove it using the tool of Bell inequalities. In particular, we will introduce a Bell inequality such that the payoff I_{C} using LHVM is upper bounded by 2, yet the maximal payoff I_{Q} and I_{NS} using quantum and NS strategies reach $2\sqrt{2}$ and 4. The relationship between NS, quantum correlation, and classical LHVMs can be summarized as in Fig. 8.2.

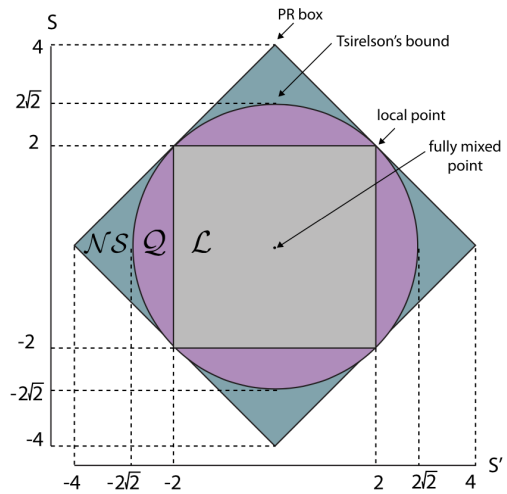


Figure 8.2: No-signalling, quantum, and classical correlations. Here the value is related to the CHSH inequality by $I' = 8I - 4$. Figure from Rev. Mod. Phys. 86, 419 (2014).

8.3 CHSH inequality

8.3.1 Definition

The CHSH inequality is different from the original inequality proposed by Bell, but it is much easier to understand and is more widely studied now. The payoff function of the CHSH inequality is defined by

$$\beta(a,b|x,y) = \begin{cases} 1 & a \oplus b = xy \\ 0 & \text{otherwise} \end{cases} \tag{8.9}$$

The payoff table with nonzero payoff β is given in Table 8.1.

Table 8.1: Payoff table for the CHSH inequality.

x	y	a	b	β
0	0	0	0	1
0	0	1	1	1
0	1	0	0	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1

8.3.2 NS

First we can prove that the average payoff I is upper bounded by 1. That is

$$\begin{aligned} I &= \frac{1}{4}\sum_{a,b,x,y}\beta(a,b,x,y)p(a,b|x,y), \\ &= \frac{1}{4}\sum_{a\oplus b=xy} p(a,b|x,y), \\ &\leq \frac{1}{4}\sum_{xy} 1, \\ &= 1. \end{aligned} \tag{8.10}$$

Here we have used the normalization condition $\sum_{ab} p(a,b|x,y) = 1$. It is not hard to see that the PR-box achieves this upper bound.

8.3.3 LHVMs

Next, we prove that the average payoff I_{C} using LHVMs is upper bounded by $3/4$. First, we have

$$\begin{aligned} I_{\mathrm{C}} &= \frac{1}{4}\sum_{a\oplus b=xy} p_{\mathrm{C}}(a,b|x,y), \\ &= \frac{1}{4}\sum_{a\oplus b=xy}\sum_{\lambda} q(\lambda)p_A(a|x,\lambda)p_B(b|y,\lambda), \\ &\leq \frac{1}{4}\max_{p_A(a|x,\lambda),p_B(b|y,\lambda)}\sum_{a\oplus b=xy} p_A(a|x,\lambda)p_B(b|y,\lambda), \end{aligned} \tag{8.11}$$

so that we can focus on $\sum_{a\oplus b=xy} p_A(a|x,\lambda)p_B(b|y,\lambda)$. Denote $p_A(a|x,\lambda)$ and $p_B(b|y,\lambda)$ by $p_x^A(a)$ and $p_y^B(b)$, respectively, we have

$$\begin{aligned} &\sum_{a\oplus b=xy} p_A(a|x,\lambda)p_B(b|y,\lambda) \\ &= p_0^A(0)p_0^B(0) + (1-p_0^A(0))(1-p_0^B(0)) + p_0^A(0)p_1^B(0) + (1-p_0^A(0))(1-p_1^B(0)), \\ &+ p_1^A(0)p_0^B(0) + (1-p_1^A(0))(1-p_0^B(0)) + p_1^A(0)(1-p_1^B(0)) + (1-p_1^A(0))p_1^B(0), \\ &= \frac{1}{2}\left[2p_0^A(0)-1\right]\left[2p_0^B(0)-1\right] + \frac{1}{2}\left[2p_0^A(0)-1\right]\left[2p_1^B(0)-1\right] \\ &+ \frac{1}{2}\left[2p_1^A(0)-1\right]\left[2p_0^B(0)-1\right] - \frac{1}{2}\left[2p_0^A(0)-1\right]\left[2p_1^B(0)-1\right] + 2. \end{aligned} \tag{8.12}$$

Here we have used $p_0^A(0)p_0^B(0)+(1-p_0^A(0))(1-p_0^B(0)) = \frac{1}{2}\left[2p_0^A(0)-1\right]\left[2p_0^B(0)-1\right] + \frac{1}{2}$. Denote $O_i^A = 2p_i^A(0)-1$ and $O_i^B = 2p_i^B(0)-1$, which satisfy $|O_i^A|, |O_i^B| \in [-1,1]$, we have[2]

$$\begin{aligned} \sum_{a\oplus b=xy} p_A(a|x,\lambda)p_B(b|y,\lambda) &= \frac{1}{2}[O_0^A O_0^B + O_0^A O_1^B + O_1^A O_0^B - O_1^A O_1^B] + 2, \\ &\leq \frac{1}{2}\left[|O_0^A||O_0^B + O_1^B| + |O_1^A||O_0^B - O_1^B|\right] + 2, \\ &\leq 3. \end{aligned} \tag{8.13}$$

Putting everything back to Eq. (8.11), we thus have

$$I_{\mathrm{C}} \leq 3/4. \tag{8.14}$$

[2]Here the last line could be proved as follow. Let $x = O_0^B + O_1^B$ and $y = O_0^B - O_1^B$, we have $|x+y| \leq 2$ and $|x-y| \leq 2$. Then we can prove it by linear programming.

This inequality is usually called the Bell (here CHSH) inequality.

8.3.4 Quantum

We can then give an explicit quantum strategy that can beat the LHVM bound. Define a projective measurement as $\{\Pi_0(\theta), \Pi_1(\theta)\}$ with $\Pi_0(\theta) = |\psi_0(\theta)\rangle\langle\psi_0(\theta)|$ and $\Pi_1(\theta) = |\psi_1(\theta)\rangle\langle\psi_1(\theta)|$, and

$$\begin{aligned} |\psi_0(\theta)\rangle &= \cos\theta\,|0\rangle + \sin\theta\,|1\rangle\,, \\ |\psi_1(\theta)\rangle &= -\sin\theta\,|0\rangle + \cos\theta\,|1\rangle\,. \end{aligned} \tag{8.15}$$

The state Alice and Bob share is the Bell state $|\Phi^+\rangle_{AB} = (|00\rangle_{AB} + |11\rangle_{AB})/\sqrt{2}$ and the measurement basis is chosen as

$$M^A_{i|0} = \Pi_i(0), \quad M^A_{i|1} = \Pi_i(\pi/2), \quad M^B_{i|0} = \Pi_i(\pi/8), \quad M^B_{i|1} = \Pi_i(-\pi/8), \tag{8.16}$$

We can then calculate the average payoff using this strategy as $I = \cos(\pi/8)^2 \approx 0.85 \geq I_{\mathrm{C}}$, which show the violation of the Bell inequality using LHVMs.

Finally, we can show that the above strategy is optimal. Similar to the proof of LHVMs, we first define O^A_i and O^B_i as follows

$$O^A_i = M^A_{0|i} - M^A_{1|i}, \quad O^B_i = M^B_{0|i} - M^B_{1|i}. \tag{8.17}$$

Then we can show that

$$\begin{aligned} I_{\mathrm{Q}} &= \frac{1}{4}\sum_{a\oplus b=xy} \mathrm{Tr}[\rho_{AB}(M^A_{a|x}\otimes M^B_{b|y})], \\ &= \frac{1}{8}\mathrm{Tr}\left[\rho_{AB}\left(O^A_0\otimes O^B_0 + O^A_0\otimes O^B_1 + O^A_1\otimes O^B_0 - O^A_1\otimes O^B_1\right)\right] + \frac{1}{2}. \end{aligned} \tag{8.18}$$

Without loss of generality, we can consider a pure state $\rho_{AB} = |\psi\rangle\langle\psi|_{AB}$ and we show

$$\begin{aligned} \langle\psi|_{AB}\, O^A_0\otimes O^B_0 + O^A_0\otimes O^B_1 + O^A_1\otimes O^B_0 - O^A_1\otimes O^B_1\,|\psi\rangle_{AB} \\ &= \|\left(O^A_0\otimes O^B_0 + O^A_0\otimes O^B_1 + O^A_1\otimes O^B_0 - O^A_1\otimes O^B_1\right)|\psi\rangle_{AB}\|, \\ &\leq \|\left(O^A_0\otimes O^B_0 + O^A_0\otimes O^B_1\right)|\psi\rangle_{AB}\| + \|\left(O^A_1\otimes O^B_0 - O^A_1\otimes O^B_1\right)|\psi\rangle_{AB}\|, \\ &\leq \|\left(O^B_0 + O^B_1\right)|\psi\rangle_{AB}\| + \|\left(O^B_0 - O^B_1\right)|\psi\rangle_{AB}\|. \end{aligned} \tag{8.19}$$

Denote $|\psi_i\rangle = O_i^B |\psi\rangle_{AB}$, we have

$$\begin{aligned}\| \left(O_0^B + O_1^B\right) |\psi\rangle_{AB} \| + \| \left(O_0^B - O_1^B\right) |\psi\rangle_{AB} \| &= \| |\psi_0\rangle + |\psi_1\rangle \| + \| |\psi_0\rangle - |\psi_1\rangle \|, \\ &= \sqrt{2 + 2\Re \langle\psi_0|\psi_1\rangle} + \sqrt{2 - 2\Re \langle\psi_0|\psi_1\rangle}, \\ &\leq 2\sqrt{2}.\end{aligned} \tag{8.20}$$

Finally, we have

$$I_{\mathrm{Q}} \leq \frac{2+\sqrt{2}}{4}. \tag{8.21}$$

8.4 The Mermin inequality and the Greenberger-Horne-Zeilinger (GHZ) paradox

We can consider multipartite nonlocality and nonlocal games. For instance, a tripartite nonlocal game is given by

$$I = \sum_{a,b,c,x,y,z} \beta(a,b,c,x,y,z) p(a,b,c|x,y,z) q(x,y,z), \tag{8.22}$$

where $p(a,b,c|x,y,z)$ is the probability distribution of the outputs given the inputs. We can similarly consider observables as

$$\left\langle A_x B_y C_z \right\rangle = \sum_{a,b,c=\pm 1} abc p(a,b,c \mid x,y,z).$$

The Mermin inequality is defined by

$$|\langle A_1 B_2 C_2\rangle + \langle A_2 B_1 C_2\rangle + \langle A_2 B_2 C_1\rangle - \langle A_1 B_1 C_1\rangle| \leq 2. \tag{8.23}$$

Using a quantum strategy, we can achieve the maximal bound 4. Specifically, we just need to share the GHZ state $|\mathrm{GHZ}\rangle = (1/\sqrt{2})(|000\rangle + |111\rangle)$ and measure σ_x, σ_y, then we have

$$\begin{aligned} A_1 B_1 C_1 = +1, A_1 B_2 C_2 = -1 \\ A_2 B_1 C_2 = -1, A_2 B_2 C_1 = -1 \end{aligned} \tag{8.24}$$

This equality constraints are also called the GHZ paradox.

8.5 Applications

The Bell inequality has wide applications for designing self-testing or device independent quantum information protocols. Recall that a violation of the Bell inequality implies the existence of quantum entanglement. Therefore, the measurement outcomes must have genuine/unpredictable randomness. Such a feature could thus be used to generate randomness. Since the randomness is guaranteed by the violation of the Bell inequality without assuming the realization, it is thus robust to device implementation errors, and hence called device independent. Bell inequality can also be used for quantum key distribution, blind quantum computing, etc. One can read Rev. Mod. Phys. 86, 419 (2014) for more detailed discussions.

Chapter 9

Quantum sensing (by Long Xiong)

9.1 Quantum Sensing

Quantum sensing is a rapidly advancing area within quantum technologies that leverages the unique properties of quantum systems—such as superposition, entanglement, and coherence—to achieve extraordinarily high precision in measuring physical quantities. Unlike quantum computing, which aims to solve complex computational problems, or quantum communication, which ensures secure information transfer, quantum sensing addresses a different but equally significant challenge: extracting information about the environment with ultimate sensitivity [**?**, **?**].

One of the key motivations for quantum sensing lies in the inherent sensitivity of quantum systems to external perturbations. While this sensitivity is often a drawback in quantum computing due to decoherence, it becomes a powerful advantage in sensing applications. A quantum sensor typically operates by initializing a quantum system into a well-defined state, allowing it to interact with an unknown external stimulus—often modeled as a perturbation to the system's Hamiltonian—and then measuring the final state after evolution. The acquired measurement outcomes encode information about the stimulus. This general sensing framework includes techniques such as Ramsey and Rabi interferometry, which are foundational in extracting phase and frequency information from quantum systems.

Compared to their classical counterparts, quantum sensors offer numerous advantages. They can access extremely small spatial resolutions, achieve higher sensitivities, reduce power and size requirements, and offer better traceability to fun-

damental physical constants. Quantum principles have already been implemented in traditional devices like atomic clocks, superconducting quantum interference devices (SQUIDs), and vapor-based magnetometers. However, modern quantum sensing distinguishes itself by exploiting quantum coherence and entanglement in engineered ways to surpass classical performance limits.

A particularly significant enhancement arises from the use of entanglement. By entangling multiple sensing qubits, it is possible to approach the so-called Heisenberg limit, where the measurement precision scales as $1/N$, in contrast to the standard quantum limit (SQL), which scales as $1/\sqrt{N}$. This reflects not just a quantitative improvement in measurement outcomes, but also a transition to a fundamentally new regime of sensing performance.

The versatility of quantum sensing is also evident in the wide range of physical platforms it can employ. These include solid-state systems such as nitrogen-vacancy (NV) centers in diamond, neutral atoms, trapped ions, Rydberg atoms, superconducting qubits, and even elementary particles like muons and neutrons. Each of these systems offers unique advantages suited to different application domains. For instance, NV centers are highly suitable for nanoscale magnetic imaging, while atomic vapor cells have proven effective in biomedical magnetometry.

In practical terms, quantum sensors are already making an impact in diverse areas such as medical imaging, inertial navigation, geophysics, and materials characterization. Single-spin and ensemble-spin sensors have demonstrated capabilities such as detecting weak magnetic fields from neural activity and imaging atomic-scale magnetic structures. Even in high-energy physics and cosmology, quantum sensing is being explored as a tool to detect previously inaccessible signals.

Finally, quantum sensing is not only about achieving high sensitivity, but also about robustness and adaptability. Ongoing developments include adaptive sensing protocols that expand dynamic range, quantum error correction techniques that enhance stability, and the integration of quantum memories to preserve coherence during measurements. As the field continues to evolve, quantum sensing is expected to play a transformative role in both applied technologies and our fundamental understanding of the physical world.

Although quantum sensing and quantum metrology both use quantum effects to improve measurement performance, they differ in several ways:

(1) Different goals: Quantum sensing focuses on detecting specific signals in real-world environments with high sensitivity. Quantum metrology is more concerned with reaching the ultimate limits of precision under ideal or controlled conditions.

(2) Different application areas: Quantum sensing is widely used in practical applications such as biomedical imaging, navigation, and materials analysis. Quantum metrology is mainly used in fundamental science and in setting measurement standards, such as in timekeeping and fundamental physics experiments.

(3) Different technical approaches: Quantum sensing often uses the response of a single quantum system to measure a signal. Quantum metrology typically involves more advanced techniques like multiparticle entanglement and quantum interference to maximize precision.

In short, quantum sensing can be viewed as a practical branch of quantum metrology, focusing on real-world measurements that benefit from quantum-enhanced sensitivity.

9.1.1 Mathematical description

The following provides a detailed mathematical description of this process.

The system is first prepared in the ground state $|0\rangle$. A unitary operator U_{Init} is applied to create the initial state

$$|\psi_{\text{Init}}\rangle = a|0\rangle + b|1\rangle, \tag{9.1}$$

where $a, b \in \mathbb{C}$ satisfy $|a|^2 + |b|^2 = 1$. A common choice is the Hadamard gate

$$U_{\text{Had}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{9.2}$$

Applying U_{Had} to $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ yields

$$|\psi_{\text{Init}}\rangle = U_{\text{Had}}|0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{9.3}$$

This creates an equal superposition of $|0\rangle$ and $|1\rangle$.

The system then interacts with an external stimulus, modeled by a Hamiltonian

$$\hat{H} = \left(E_0 - \frac{1}{2}\gamma V_0\right)|0\rangle\langle 0| + \left(E_1 + \frac{1}{2}\gamma V_0\right)|1\rangle\langle 1|, \tag{9.4}$$

where E_0, E_1 are the energy levels, γ is a coupling constant, and V_0 is the stimulus strength.

The time-evolution operator is

$$\hat{U}_H(0,t) = e^{-i\hat{H}t}. \tag{9.5}$$

The state at time t becomes

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{-i(E_0 - \frac{1}{2}\gamma V_0)t} |0\rangle + \frac{1}{\sqrt{2}} e^{-i(E_1 + \frac{1}{2}\gamma V_0)t} |1\rangle. \tag{9.6}$$

Factoring out the global phase $e^{-i(E_0 - \frac{1}{2}\gamma V_0)t}$ gives

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{-i(E_0 - \frac{1}{2}\gamma V_0)t} \left(|0\rangle + e^{-i(E_1 - E_0 + \gamma V_0)t} |1\rangle \right). \tag{9.7}$$

Before measurement, a readout operation is applied, typically the Hadamard gate again:

$$|\psi_{\text{Final}}\rangle = U_{\text{Had}} |\psi(t)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} e^{-i(E_0 - \frac{1}{2}\gamma V_0)t} \begin{pmatrix} 1 \\ e^{-i(E_1 - E_0 + \gamma V_0)t} \end{pmatrix}. \tag{9.8}$$

Carrying out the matrix multiplication:

$$|\psi_{\text{Final}}\rangle = \frac{1}{2} e^{-i(E_0 - \frac{1}{2}\gamma V_0)t} \begin{pmatrix} 1 + e^{-i(E_1 - E_0 + \gamma V_0)t} \\ 1 - e^{-i(E_1 - E_0 + \gamma V_0)t} \end{pmatrix}. \tag{9.9}$$

The probability of measuring $|1\rangle$ is

$$p_{|0\rangle \to |1\rangle} = |\langle 1 | \psi_{\text{Final}} \rangle|^2 = \left| \frac{1}{2} \left(1 - e^{-i(E_1 - E_0 + \gamma V_0)t} \right) \right|^2. \tag{9.10}$$

Define

$$\phi = (E_1 - E_0 + \gamma V_0)t. \tag{9.11}$$

Then

$$|1 - e^{-i\phi}|^2 = 4 \sin^2 \left(\frac{\phi}{2} \right). \tag{9.12}$$

Thus, the transition probability becomes

$$p_{|0\rangle \to |1\rangle} = \sin^2 \left(\frac{\phi}{2} \right) = \frac{1 - \cos\phi}{2}. \tag{9.13}$$

Substituting back ϕ gives

$$p_{|0\rangle\to|1\rangle} = \frac{1-\cos\left((E_1 - E_0 + \gamma V_0)t\right)}{2}. \tag{9.14}$$

This probability depends on the unknown V_0. Repeating the measurement many times allows precise estimation of $p_{|0\rangle\to|1\rangle}$, from which γV_0 can be extracted by inverting the cosine relationship.

This method, known as Ramsey interferometry, is a fundamental technique in quantum sensing. It enables high-precision measurement of weak external fields by mapping the environmental interaction onto phase accumulation and measuring interference outcomes. The sensitivity can be further improved by optimizing the interrogation time t and using entangled probe states.

9.1.2 Quantum sensing protocol

The quantum sensing protocol uses a quantum sensor to measure a signal by three basic steps: initialization, interaction with the signal, and readout. The goal is to learn about the signal $V(t)$ by observing how it affects the quantum sensor [**?**].

The total Hamiltonian of the sensor is

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_V(t) + \hat{H}_{\text{control}}(t), \tag{9.15}$$

where $\hat{H}_0$ is the sensor's own Hamiltonian, $\hat{H}_V(t)$ is from the signal, and $\hat{H}_{\text{control}}(t)$ is used to control the sensor.

The internal Hamiltonian $\hat{H}_0$ gives the energy levels of the sensor without any signal. It is usually written as

$$\hat{H}_0 = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1|, \tag{9.16}$$

where $|0\rangle$ and $|1\rangle$ are the energy states. The energy gap is $\omega_0 = E_1 - E_0$.

The signal Hamiltonian $\hat{H}_V(t)$ shows how the signal $V(t)$ couples to the sensor. It can be split into two parts:

$$\hat{H}_V(t) = \hat{H}_{V_\parallel}(t) + \hat{H}_{V_\perp}(t), \tag{9.17}$$

with

$$\hat{H}_{V_\parallel}(t) = \frac{1}{2}\gamma V_\parallel(t)\left(|1\rangle\langle 1| - |0\rangle\langle 0|\right), \tag{9.18}$$

$$\hat{H}_{V_\perp}(t) = \frac{1}{2}\gamma\left(V_\perp(t)|1\rangle\langle 0| + V_\perp^\dagger(t)|0\rangle\langle 1|\right), \tag{9.19}$$

where γ is the strength of coupling. $V_{\parallel}(t)$ affects the energy levels, $V_{\perp}(t)$ can flip the states.

For signals like magnetic or electric fields, which are vectors $\boldsymbol{V}(t) = \{V_x(t), V_y(t), V_z(t)\}$, the Hamiltonian becomes

$$\hat{H}_V(t) = \gamma \boldsymbol{V}(t) \cdot \hat{\boldsymbol{\sigma}}, \tag{9.20}$$

where $\hat{\boldsymbol{\sigma}} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$ are the Pauli matrices. Here

$$V_{\parallel}(t) = V_z(t), \quad V_{\perp}(t) = V_x(t) + iV_y(t). \tag{9.21}$$

The control Hamiltonian $\hat{H}_{\text{control}}(t)$ applies quantum gates to the sensor. These gates help prepare the initial state, drive the evolution, or rotate the sensor for measurement.

The sensing protocol follows these steps:

First, the sensor is initialized to the state $|0\rangle$. A control operation $\hat{U}_a$ is applied to prepare the sensing state

$$|\psi_0\rangle = \hat{U}_a|0\rangle. \tag{9.22}$$

The sensor then evolves under the total Hamiltonian for time t. The state after time t is

$$|\psi(t)\rangle = \hat{U}_H(0,t)|\psi_0\rangle = c_0|\psi_0\rangle + c_1|\psi_1\rangle, \tag{9.23}$$

where $|\psi_1\rangle$ is orthogonal to $|\psi_0\rangle$.

Before readout, another control operation $\hat{U}_b$ is applied. Often $\hat{U}_b = \hat{U}_a^{\dagger}$. The final state becomes

$$|\alpha\rangle = \hat{U}_b|\psi(t)\rangle = c_0|0\rangle + c_1|1\rangle. \tag{9.24}$$

The measurement gives ”0” with probability $1-p$ and ”1” with probability

$$p = |c_1|^2. \tag{9.25}$$

Because one measurement gives only one binary result, the process is repeated N times to estimate p accurately.

In many cases, p is measured as a function of time t or some control parameter. The collected data $\{p_k\}$ is then used to find the signal $V(t)$.

This basic protocol works for many quantum sensing experiments and helps measure weak signals with high precision.

9.1.3 Example

The Ramsey measurement is a basic example of using a quantum sensor to measure the static energy splitting ω_0, or a static perturbation $V_{\|}$. In this method, the sensor is first prepared in the state $|0\rangle$. A $\pi/2$ pulse is applied to rotate the state into a superposition [**?**]. The resulting state is

$$|\psi_0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \tag{9.26}$$

This superposition state evolves under the internal Hamiltonian $\hat{H}_0$ for a time t. During this time, the two basis states accumulate a relative phase difference. The Hamiltonian is

$$\hat{H}_0 = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1|, \tag{9.27}$$

which gives the energy difference $\omega_0 = E_1 - E_0$. The time evolution of the state is governed by

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{-i\omega_0 t}|1\rangle\right), \tag{9.28}$$

up to an overall phase factor which does not affect measurement outcomes.

After the evolution, a second $\pi/2$ pulse is applied. This operation transforms the state into

$$|\alpha\rangle = \frac{1}{2}(1 + e^{-i\omega_0 t})|0\rangle + \frac{1}{2}(1 - e^{-i\omega_0 t})|1\rangle. \tag{9.29}$$

The measurement is then performed in the original basis $\{|0\rangle, |1\rangle\}$. The probability of finding the sensor in state $|1\rangle$ is given by

$$\begin{aligned} p &= 1 - |\langle 0|\alpha\rangle|^2 \\ &= 1 - \left|\frac{1}{2}(1 + e^{-i\omega_0 t})\right|^2 \\ &= \sin^2\left(\frac{\omega_0 t}{2}\right) \\ &= \frac{1}{2}\left[1 - \cos(\omega_0 t)\right]. \end{aligned} \tag{9.30}$$

By measuring p as a function of time t, an oscillating pattern called Ramsey fringes is observed. The frequency of these fringes directly gives the value of ω_0.

Another basic example is the Rabi measurement, which is used to measure the transition strength $|V_\perp|$. In this method, the sensor is again initialized in the state

$$|\psi_0\rangle = |0\rangle. \tag{9.31}$$

In the simplest case where $\hat{H}_0 = 0$, the evolution is driven only by the transverse signal through the Hamiltonian

$$\hat{H}_{V_\perp} = \frac{1}{2}\gamma V_\perp \sigma_x = \omega_1 \sigma_x, \tag{9.32}$$

where $\omega_1 = \frac{1}{2}\gamma V_\perp$ is the Rabi frequency. The time evolution under this Hamiltonian gives the state

$$|\psi(t)\rangle = |\alpha\rangle = \frac{1}{2}(1+e^{-i\omega_1 t})|0\rangle + \frac{1}{2}(1-e^{-i\omega_1 t})|1\rangle. \tag{9.33}$$

The transition probability is calculated as

$$p = 1 - |\langle 0|\alpha\rangle|^2 = \sin^2\left(\frac{\omega_1 t}{2}\right). \tag{9.34}$$

In a more general case where $\hat{H}_0 \neq 0$, the total Hamiltonian includes both $\hat{H}_0$ and $\hat{H}_{V_\perp}$. The transition probability then follows the solution of Rabi's problem:

$$p = \frac{\omega_1^2}{\omega_1^2+\omega_0^2}\sin^2\left(\sqrt{\omega_1^2+\omega_0^2}\,t\right). \tag{9.35}$$

This result shows that only signals with frequency close to ω_0 can efficiently drive transitions, a condition known as resonance. Therefore, the Rabi measurement not only gives the coupling strength $V_\perp$ but also provides information about the frequency of the signal.

9.2 Metrology: From Classical to Quantum

Metrology is the science of measurement. Although it is sometimes confused with meteorology (which deals with weather), metrology is fundamentally about how we make and improve measurements. It plays a crucial role in science and technology, since testing a scientific theory almost always requires comparing its predictions to experimental results [**?**].

As our measurement tools become more precise, we are able to test more detailed predictions. For instance, in 2016, the LIGO experiment detected gravitational waves, confirming an important prediction of Einstein's theory of general relativity. Later in 2021, the Fermilab experiment measured the magnetic moment of the muon with remarkable precision, showing a small deviation from the Standard Model, sparking discussions about new physics.

High-precision measurements are not limited to fundamental science. They are also essential in engineering and manufacturing. A machine part designed with an accuracy of 0.1 mm is significantly more reliable than one made to the nearest centimeter.

Not all quantities can be measured directly. In many cases, we need to estimate an unknown value based on related measurements. For example, in the LIGO experiment, the gravitational wave caused a tiny change in the phase of a laser beam. This phase shift was not directly observable, but it was inferred from the interference pattern of the light.

This brings us to estimation theory, a branch of statistics that helps us figure out how to estimate unknown parameters using data. There are two main approaches in estimation theory. The Bayesian approach treats the parameter as a random variable and incorporates prior knowledge, while the frequentist approach treats the parameter as fixed but unknown. In this note, we focus on the frequentist view. Here, the quality of an estimate is often evaluated by the mean-squared error (MSE), which measures how far off the estimate is on average.

The Cramér–Rao bound is a key result in frequentist estimation. It tells us the smallest possible MSE for an unbiased estimator. This bound depends on the Fisher information, which measures how much information the data carries about the unknown parameter.

In the second half of the 20th century, researchers such as Helstrom and Holevo extended estimation theory to quantum systems. This led to the development of quantum metrology. In quantum metrology, an unknown parameter (like a phase or frequency) is encoded in a quantum system that we can prepare and measure. Because of quantum properties like entanglement, these systems can sometimes estimate parameters more precisely than any classical method allows.

Quantum metrology has been demonstrated using different kinds of quantum systems, including photons, atomic ensembles, and superconducting qubits.

A central example in quantum metrology is phase estimation. One typical setup works as follows: we prepare a set of n qubits in an entangled state, such as a Greenberger-Horne-Zeilinger (GHZ) state. We let the unknown phase act on the system, and then we perform measurements to estimate the phase. Using

entangled states allows the mean-squared error to scale like $1/n^2$, which is known as the Heisenberg limit—the best possible scaling allowed by quantum mechanics. In contrast, if we do not use entanglement, the best scaling we can achieve is $1/n$, which is called the standard quantum limit (SQL), also known as the shot-noise limit.

Quantum metrology has a wide range of applications. For example, it can be used to detect extremely weak magnetic fields (magnetometry), measure temperature very accurately (thermometry), identify materials through their energy levels (spectroscopy), or improve the resolution of imaging systems using fewer photons (quantum imaging). In biomedicine, where excessive measurement may damage delicate samples, quantum-enhanced measurements help reduce the number of interactions needed while preserving high accuracy.

To summarize, classical metrology has enabled remarkable progress in science and engineering. Quantum metrology builds upon this foundation by using quantum effects to further improve precision. Concepts such as quantum Fisher information and quantum Cramér–Rao bounds help us understand the fundamental limits of measurement in the quantum world.

9.2.1 Estimation theory

What is Estimation?

Estimation is the process of using observed data to guess unknown values, called **parameters**. For example, to estimate the average height of people in a city, we can measure the heights of n people and use these measurements to guess the true average.

The **parameter** θ is the unknown value we want to estimate, such as the true average height.

The **estimator** $\hat{\theta}(X)$ is a function of the observed data X, used to estimate θ. For example, if $X = (X_1, X_2, \ldots, X_n)$ represents the measured heights of n people, the sample mean

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{9.36}$$

is an estimator of the true average height θ.

How Good is an Estimator?

The quality of an estimator is measured by several important properties:

Unbiasedness: An estimator $\hat{\theta}$ is unbiased if its expected value equals the true parameter:

$$E[\hat{\theta}] = \theta. \tag{9.37}$$

This means that, on average, the estimator correctly guesses the true value.

Variance: The variance of an estimator, denoted $\mathrm{Var}(\hat{\theta})$, measures how much the estimator fluctuates when the experiment is repeated with different data samples. Smaller variance implies higher precision. For example, if we estimate the probability p of a coin landing heads by flipping the coin n times and observing X heads, then the estimator

$$\hat{p} = \frac{X}{n} \tag{9.38}$$

is unbiased, since

$$E[\hat{p}] = p, \tag{9.39}$$

and its variance is

$$\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}. \tag{9.40}$$

The variance decreases as the number of trials n increases.

Mean Squared Error (MSE): The mean squared error combines both the bias and the variance of the estimator. For an estimator $\hat{\theta}$ of θ, the MSE is defined as

$$\mathrm{MSE}(\hat{\theta}) = E\left[(\hat{\theta}-\theta)^2\right]. \tag{9.41}$$

If the estimator is unbiased, the MSE reduces to the variance:

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}). \tag{9.42}$$

The Cramér-Rao Bound

There is a fundamental lower limit on the variance of any unbiased estimator, known as the **Cramér-Rao Bound (CRB)**. This bound is determined by the **Fisher Information** $I(\theta)$, which quantifies how much information the data provides about the parameter θ.

The Fisher Information is defined as

$$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta}\ln p(X \mid \theta)\right)^2\right], \tag{9.43}$$

where $p(X \mid \theta)$ is the probability density (or mass) function of the data X, and the term

$$\frac{\partial}{\partial \theta} \ln p(X \mid \theta) \tag{9.44}$$

is called the **score function**. The Fisher Information can also be written, under regularity conditions, as

$$I(\theta) = -E\left[\frac{\partial^2}{\partial \theta^2} \ln p(X \mid \theta)\right]. \tag{9.45}$$

The Cramér-Rao Bound states that for any unbiased estimator $\hat{\theta}$,

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}. \tag{9.46}$$

If an estimator reaches this lower bound, it is called **efficient**. An efficient estimator has the smallest possible variance among all unbiased estimators, meaning it is as precise as possible under these conditions.

Classical estimation theory provides the foundation for understanding how accurately we can estimate parameters from data. The Cramér-Rao Bound establishes a theoretical limit on precision, which plays a crucial role not only in classical statistics but also in quantum metrology, where similar principles apply to the estimation of parameters in quantum systems.

To understand why the Cramér-Rao Bound holds, we now derive it step by step.

Start with the **score function**, defined as

$$S(\theta) = \frac{\partial}{\partial \theta} \ln p(X \mid \theta), \tag{9.47}$$

where $p(X \mid \theta)$ is the probability density (or mass) function of the observed data X. The score function describes how sensitive the log-likelihood is to changes in θ.

The expected value of the score function is zero:

$$E[S(\theta)] = 0. \tag{9.48}$$

To verify this, notice that since $p(X \mid \theta)$ is a probability distribution, it satisfies

$$\int p(X \mid \theta)\, dX = 1. \tag{9.49}$$

Differentiating both sides with respect to θ gives

$$\frac{\partial}{\partial \theta} \int p(X \mid \theta)\, dX = \int \frac{\partial}{\partial \theta} p(X \mid \theta)\, dX = 0. \tag{9.50}$$

Using the identity

$$\frac{\partial}{\partial \theta} \ln p(X \mid \theta) = \frac{1}{p(X \mid \theta)} \frac{\partial}{\partial \theta} p(X \mid \theta), \tag{9.51}$$

we substitute into the expectation of the score:

$$E[S(\theta)] = \int \left(\frac{\partial}{\partial \theta} \ln p(X \mid \theta) \right) p(X \mid \theta)\, dX = \int \frac{\partial}{\partial \theta} p(X \mid \theta)\, dX = 0. \tag{9.52}$$

The **variance of the score function** is given by

$$\mathrm{Var}(S(\theta)) = E[S(\theta)^2] - (E[S(\theta)])^2 = E[S(\theta)^2], \tag{9.53}$$

since $E[S(\theta)] = 0$. This variance is called the **Fisher Information**, denoted by $I(\theta)$:

$$I(\theta) = E[S(\theta)^2] = E\left[\left(\frac{\partial}{\partial \theta} \ln p(X \mid \theta) \right)^2 \right]. \tag{9.54}$$

Alternatively, under suitable regularity conditions, the Fisher Information can also be written as

$$I(\theta) = -E\left[\frac{\partial^2}{\partial \theta^2} \ln p(X \mid \theta) \right]. \tag{9.55}$$

Assume that $\hat{\theta}$ is an **unbiased estimator** of θ, meaning

$$E[\hat{\theta}] = \theta. \tag{9.56}$$

Consider the covariance between $\hat{\theta}$ and $S(\theta)$:

$$\mathrm{Cov}(\hat{\theta}, S(\theta)) = E[\hat{\theta} S(\theta)] - E[\hat{\theta}] E[S(\theta)]. \tag{9.57}$$

Since $E[\hat{\theta}] = \theta$ and $E[S(\theta)] = 0$, this simplifies to

$$\mathrm{Cov}(\hat{\theta}, S(\theta)) = E[\hat{\theta} S(\theta)]. \tag{9.58}$$

To compute $E[\hat{\theta} S(\theta)]$, use the expression for the score:

$$E[\hat{\theta} S(\theta)] = \int \hat{\theta}(X) \left(\frac{\partial}{\partial \theta} \ln p(X \mid \theta) \right) p(X \mid \theta)\, dX. \tag{9.59}$$

Since $E[\hat{\theta}] = \theta$, differentiate both sides of

$$E[\hat{\theta}] = \int \hat{\theta}(X) p(X \mid \theta)\, dX = \theta \tag{9.60}$$

with respect to θ:

$$\frac{\partial}{\partial \theta} E[\hat{\theta}] = \int \hat{\theta}(X) \frac{\partial}{\partial \theta} p(X \mid \theta)\, dX = 1. \tag{9.61}$$

From the identity

$$\frac{\partial}{\partial \theta} p(X \mid \theta) = p(X \mid \theta) \cdot \frac{\partial}{\partial \theta} \ln p(X \mid \theta), \tag{9.62}$$

we substitute into the integral:

$$\int \hat{\theta}(X) \cdot p(X \mid \theta) \cdot \frac{\partial}{\partial \theta} \ln p(X \mid \theta)\, dX = 1. \tag{9.63}$$

This shows that

$$E[\hat{\theta} S(\theta)] = 1. \tag{9.64}$$

Therefore, the covariance becomes

$$\mathrm{Cov}(\hat{\theta}, S(\theta)) = 1. \tag{9.65}$$

Now apply the **Cauchy-Schwarz inequality** for random variables A and B:

$$|\mathrm{Cov}(A, B)| \leq \sqrt{\mathrm{Var}(A) \cdot \mathrm{Var}(B)}. \tag{9.66}$$

Substituting $A = \hat{\theta}$ and $B = S(\theta)$ gives

$$1 \leq \sqrt{\mathrm{Var}(\hat{\theta}) \cdot \mathrm{Var}(S(\theta))}. \tag{9.67}$$

Since $\mathrm{Var}(S(\theta)) = I(\theta)$, this becomes

$$1 \leq \sqrt{\mathrm{Var}(\hat{\theta}) \cdot I(\theta)}. \tag{9.68}$$

Squaring both sides yields

$$1 \leq \mathrm{Var}(\hat{\theta}) \cdot I(\theta), \tag{9.69}$$

which leads to the **Cramér-Rao Bound**:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}. \tag{9.70}$$

This inequality shows that the variance of any unbiased estimator $\hat{\theta}$ cannot be smaller than the inverse of the Fisher Information. If the equality holds, the estimator is called **efficient**.

9.2.2 Example 1

Consider estimating the mean μ of a normal distribution $\mathcal{N}(\mu, \sigma^2)$ where the variance σ^2 is known. Assume we have n independent and identically distributed (i.i.d.) samples $X_1, X_2, \ldots, X_n$ drawn from this distribution.

The likelihood function is

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right). \tag{9.71}$$

Taking the logarithm of the likelihood (the log-likelihood):

$$\ln L(\mu) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (X_i - \mu)^2. \tag{9.72}$$

The score function is the derivative of the log-likelihood with respect to μ:

$$S(\mu) = \frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \mu). \tag{9.73}$$

The expectation of the score is zero, because

$$E[S(\mu)] = \frac{1}{\sigma^2} \sum_{i=1}^{n} E[X_i - \mu] = \frac{1}{\sigma^2} \cdot n \cdot 0 = 0. \tag{9.74}$$

The Fisher Information is the variance of the score function:

$$I(\mu) = E\left[S(\mu)^2\right] = E\left[\left(\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)\right)^2\right]. \tag{9.75}$$

Because the X_i are independent and each has variance σ^2, we have

$$\mathrm{Var}\left(\sum_{i=1}^{n}(X_i - \mu)\right) = \sum_{i=1}^{n}\mathrm{Var}(X_i - \mu) = n\sigma^2. \tag{9.76}$$

Substituting into the expression for $I(\mu)$:

$$I(\mu) = \frac{1}{\sigma^4}\cdot n\sigma^2 = \frac{n}{\sigma^2}. \tag{9.77}$$

The Cramér-Rao Bound states that the variance of any unbiased estimator $\hat{\mu}$ satisfies

$$\mathrm{Var}(\hat{\mu}) \geq \frac{1}{I(\mu)} = \frac{\sigma^2}{n}. \tag{9.78}$$

The sample mean

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{9.79}$$

is an unbiased estimator of μ, with variance

$$\mathrm{Var}(\hat{\mu}) = \frac{\sigma^2}{n}. \tag{9.80}$$

Since the variance of the sample mean exactly reaches the Cramér-Rao Bound, the sample mean is an **efficient** estimator of μ in this case.

The method used to find such estimators is often **maximum likelihood estimation (MLE)**. The likelihood function $L(\theta) = p(X \mid \theta)$ measures how likely the observed data X is, given the parameter θ. The MLE chooses the value of θ that maximizes this likelihood.

Because maximizing $L(\theta)$ is equivalent to maximizing $\ln L(\theta)$, we often work with the log-likelihood for simplicity.

9.2.3 Example 2

Consider a simple example: estimating the probability p of heads in a biased coin flip. Flip the coin n times and observe k heads. The likelihood function is

$$L(p) = p^k (1-p)^{n-k}, \tag{9.81}$$

which comes from the binomial distribution. The log-likelihood is

$$\ln L(p) = k \ln p + (n-k) \ln(1-p). \tag{9.82}$$

To maximize this, take the derivative with respect to p:

$$\frac{d}{dp} \ln L(p) = \frac{k}{p} - \frac{n-k}{1-p}. \tag{9.83}$$

Set this derivative equal to zero:

$$\frac{k}{p} = \frac{n-k}{1-p}. \tag{9.84}$$

Cross-multiply to solve for p:

$$k(1-p) = (n-k)p, \tag{9.85}$$

$$k - kp = np - kp, \tag{9.86}$$

$$k = np, \tag{9.87}$$

$$p = \frac{k}{n}. \tag{9.88}$$

This gives the maximum likelihood estimator

$$\hat{p} = \frac{k}{n}. \tag{9.89}$$

For example, if the coin is flipped 10 times and 7 heads are observed, the estimate is $\hat{p} = 0.7$. This suggests that the probability of heads is 70% according to the observed data.

9.3 Quantum estimation

In quantum estimation, we use quantum systems to estimate unknown parameters more precisely than classical methods. For example, entangled particles can help measure a magnetic field with higher accuracy [**?**, **?**, **?**].

To evaluate how well we can estimate a parameter θ, we use **Fisher information**. In quantum systems, the maximum possible Fisher information over all measurements is called the **quantum Fisher information** (QFI), denoted by $F_Q(\theta)$. This tells us how much information the quantum state carries about θ.

Consider a pure quantum state $|\psi(\theta)\rangle$ that depends on the parameter θ. If the state evolves under a unitary operator generated by a Hamiltonian H:

$$|\psi(\theta)\rangle = e^{-iH\theta}|\psi\rangle, \tag{9.90}$$

then the QFI is

$$F_Q(\theta) = 4\,\mathrm{Var}(H), \tag{9.91}$$

where $\mathrm{Var}(H)$ is the variance of the Hamiltonian in the initial state $|\psi\rangle$, defined by

$$\mathrm{Var}(H) = \langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2. \tag{9.92}$$

If $|\psi\rangle$ is an eigenstate of H, the variance is zero, so $F_Q(\theta) = 0$. In this case, we gain no information about θ. However, if $|\psi\rangle$ is a superposition of eigenstates of H, the variance can be nonzero, allowing better estimation of θ.

Consider the example of a single qubit. Let $H = \frac{\sigma_z}{2}$, where σ_z is the Pauli-Z operator with eigenvalues ± 1.

First, if $|\psi\rangle = |0\rangle$, which is an eigenstate of σ_z with eigenvalue +1, then

$$H|\psi\rangle = \frac{\sigma_z}{2}|0\rangle = \frac{1}{2}|0\rangle. \tag{9.93}$$

Compute the expectation value of H:

$$\langle\psi|H|\psi\rangle = \langle 0|\frac{1}{2}|0\rangle = \frac{1}{2}. \tag{9.94}$$

Next, compute $\langle H^2\rangle$:

$$\langle\psi|H^2|\psi\rangle = \langle 0|\left(\frac{\sigma_z}{2}\right)^2|0\rangle = \frac{1}{4}\langle 0|I|0\rangle = \frac{1}{4}. \tag{9.95}$$

The variance is then

$$\mathrm{Var}(H) = \frac{1}{4} - \left(\frac{1}{2}\right)^2 = 0. \tag{9.96}$$

Thus, the QFI is

$$F_Q(\theta) = 4 \cdot 0 = 0. \tag{9.97}$$

This confirms that if the initial state is an eigenstate of H, no information about θ can be extracted.

Now consider $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, a superposition state.

Apply H to $|\psi\rangle$:

$$H|\psi\rangle = \frac{\sigma_z}{2} \cdot \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2} \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \tag{9.98}$$

Calculate the expectation value of H:

$$\langle\psi|H|\psi\rangle = \frac{\langle 0| + \langle 1|}{\sqrt{2}} \cdot \frac{1}{2} \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{1}{4}(1 - 0 + 0 - 1) = 0. \tag{9.99}$$

Next, compute $\langle H^2\rangle$:

$$\langle H^2\rangle = \langle\psi|\left(\frac{\sigma_z}{2}\right)^2|\psi\rangle = \frac{1}{4}\langle\psi|I|\psi\rangle = \frac{1}{4}. \tag{9.100}$$

The variance becomes

$$\mathrm{Var}(H) = \frac{1}{4} - 0^2 = \frac{1}{4}. \tag{9.101}$$

The QFI is then

$$F_Q(\theta) = 4 \cdot \frac{1}{4} = 1. \tag{9.102}$$

This shows that the superposition state allows nonzero Fisher information, giving us information about θ.

9.3.1 Standard Quantum Limit and Heisenberg Limit

In quantum metrology, the goal is to estimate an unknown parameter θ with the highest possible precision. Suppose θ is encoded into a quantum system through a unitary transformation

$$U_\theta = e^{-i\theta H}, \tag{9.103}$$

where H is the Hamiltonian of the system. The precision of any unbiased estimator $\hat{\theta}$ is bounded by the quantum Cramér-Rao bound (QCRB):

$$\Delta^2\hat{\theta} \geq \frac{1}{\nu\mathcal{Q}}, \tag{9.104}$$

where ν is the number of experimental repetitions and $\mathcal{Q}$ is the quantum Fisher information (QFI). For a pure state $|\psi\rangle$, the QFI under unitary evolution is given by

$$\mathcal{Q} = 4\left(\langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2\right) = 4\Delta^2 H, \tag{9.105}$$

where $\Delta^2 H$ is the variance of the Hamiltonian in the state $|\psi\rangle$.

The QCRB can be written as

$$\Delta^2\hat{\theta} \geq \frac{1}{4\Delta^2 H}, \tag{9.106}$$

which rearranges to

$$\Delta^2\hat{\theta}\cdot\Delta^2 H \geq \frac{1}{4}. \tag{9.107}$$

This inequality shows that increasing the variance of H improves the estimation precision.

Consider an n-qubit system with local Hamiltonians $H_j = \frac{\sigma_z^{(j)}}{2}$ for each qubit. The total Hamiltonian is

$$H = \sum_{j=1}^{n} H_j. \tag{9.108}$$

If the initial state is the separable product state

$$|\psi\rangle = \bigotimes_{j=1}^{n} \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \tag{9.109}$$

then the qubits are independent, and the total variance adds up:

$$\mathrm{Var}(H) = \sum_{j=1}^{n} \mathrm{Var}(H_j) = n \cdot \frac{1}{4} = \frac{n}{4}. \tag{9.110}$$

The QFI becomes

$$\mathcal{Q} = 4 \cdot \frac{n}{4} = n. \tag{9.111}$$

This scaling is known as the **standard quantum limit (SQL)**, where the precision scales as

$$\Delta^2 \hat{\theta}_{\mathrm{SQL}} = \frac{1}{\nu n}. \tag{9.112}$$

Now consider an **entangled state**, such as the GHZ state:

$$|\psi_{\mathrm{GHZ}}\rangle = \frac{1}{\sqrt{2}} \left(|00\cdots0\rangle + |11\cdots1\rangle \right). \tag{9.113}$$

The expectation value of H in this state is

$$\langle \psi_{\mathrm{GHZ}} | H | \psi_{\mathrm{GHZ}} \rangle = 0, \tag{9.114}$$

since for each j,

$$\sigma_z^{(j)} |00\cdots0\rangle = |00\cdots0\rangle, \quad \sigma_z^{(j)} |11\cdots1\rangle = -|11\cdots1\rangle, \tag{9.115}$$

leading to

$$\langle \psi_{\mathrm{GHZ}} | \sigma_z^{(j)} | \psi_{\mathrm{GHZ}} \rangle = 0. \tag{9.116}$$

The second moment $\langle H^2 \rangle$ is

$$\langle H^2 \rangle = \sum_{j,k=1}^{n} \langle \psi_{\mathrm{GHZ}} | H_j H_k | \psi_{\mathrm{GHZ}} \rangle. \tag{9.117}$$

For $j = k$,

$$H_j^2 = \frac{1}{4} I, \quad \langle \psi_{\mathrm{GHZ}} | H_j^2 | \psi_{\mathrm{GHZ}} \rangle = \frac{1}{4}. \tag{9.118}$$

For $j \neq k$, using the symmetry of the GHZ state,

$$\langle\psi_{\text{GHZ}}|H_j H_k|\psi_{\text{GHZ}}\rangle = \frac{1}{4}. \tag{9.119}$$

There are n diagonal terms and $n(n-1)$ off-diagonal terms, so

$$\langle H^2\rangle = n\cdot\frac{1}{4} + n(n-1)\cdot\frac{1}{4} = \frac{n^2}{4}. \tag{9.120}$$

The variance of H is then

$$\text{Var}(H) = \frac{n^2}{4}. \tag{9.121}$$

The QFI becomes

$$\mathcal{Q} = 4\cdot\frac{n^2}{4} = n^2. \tag{9.122}$$

The corresponding precision limit is

$$\Delta^2\hat{\theta}_{\text{HL}} = \frac{1}{\nu n^2}, \tag{9.123}$$

which is known as the **Heisenberg limit**. This represents a quadratic improvement over the SQL.

The Heisenberg limit illustrates how entanglement can enhance precision in parameter estimation. However, not all entangled states achieve this limit—specific states like the GHZ state are required, and the result depends on the structure of the Hamiltonian H.

9.3.2 Quantum fisher information

Quantum estimation theory studies how to estimate an unknown parameter θ that is encoded in a quantum state ρ_θ. By measuring this state, we obtain data to construct an estimator $\hat{\theta}$ for θ. This process is similar to classical estimation but uses quantum features like superposition and entanglement, which can improve precision beyond classical limits, sometimes reaching the **Heisenberg limit**.

The estimation process involves two steps. In the **quantum step**, we prepare the quantum state, encode θ into it, and measure the state to obtain classical data. In the **classical step**, we process this data using statistical methods to estimate θ. The encoding may use a unitary operation U_θ such as

$$U_\theta = e^{-i\theta H}, \tag{9.124}$$

where H is a Hamiltonian operator, or more generally, a completely positive trace-preserving map Λ_θ. The measurement can be a projective measurement or a more general positive operator-valued measure (POVM) $\mathcal{M} = \{M_m\}$.

The estimation precision is quantified by the **mean squared error (MSE)** of the estimator:

$$\Delta^2\hat{\theta} = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right]. \tag{9.125}$$

In classical estimation, the Cramér-Rao bound provides a lower limit on the MSE using the Fisher information. In quantum mechanics, we have the **quantum Cramér-Rao bound (QCRB)**, which uses the **quantum Fisher information (QFI)**. If the experiment is repeated ν times, the QCRB states

$$\Delta^2\hat{\theta} \geq \frac{1}{\nu\mathcal{I}(\rho_\theta, \mathcal{M})} \geq \frac{1}{\nu\mathcal{Q}(\rho_\theta)}, \tag{9.126}$$

where $\mathcal{I}(\rho_\theta, \mathcal{M})$ is the Fisher information for a specific measurement $\mathcal{M}$, and $\mathcal{Q}(\rho_\theta)$ is the QFI, defined as the maximum Fisher information over all possible measurements:

$$\mathcal{Q}(\rho_\theta) = \max_{\mathcal{M}} \mathcal{I}(\rho_\theta, \mathcal{M}). \tag{9.127}$$

For a given measurement $\mathcal{M} = \{M_m\}$, the probability of outcome m is

$$p(m \mid \theta) = \mathrm{Tr}(\rho_\theta M_m), \tag{9.128}$$

and the Fisher information for this measurement is

$$\mathcal{I}(\rho_\theta, \mathcal{M}) = \sum_m \frac{1}{p(m \mid \theta)} \left(\frac{\partial p(m \mid \theta)}{\partial \theta}\right)^2, \tag{9.129}$$

where the sum is over outcomes with $p(m \mid \theta) \neq 0$.

The QFI sets the ultimate precision limit. In many cases, calculating $\mathcal{Q}(\rho_\theta)$ requires finding the symmetric logarithmic derivative (SLD) L_θ, which satisfies

$$\frac{\partial \rho_\theta}{\partial \theta} = \frac{1}{2}\left(L_\theta \rho_\theta + \rho_\theta L_\theta\right). \tag{9.130}$$

The QFI is then given by

$$\mathcal{Q}(\rho_\theta) = \mathrm{Tr}\left(\rho_\theta L_\theta^2\right). \tag{9.131}$$

For pure states $\rho_\theta = |\psi(\theta)\rangle\langle\psi(\theta)|$, this simplifies to

$$\mathcal{Q}(\rho_\theta) = 4\left[\langle\partial_\theta\psi(\theta)|\partial_\theta\psi(\theta)\rangle - |\langle\psi(\theta)|\partial_\theta\psi(\theta)\rangle|^2\right]. \tag{9.132}$$

Consider phase estimation as an example. Let θ be a phase shift applied to each of n qubits using

$$U_\theta = e^{-i\theta H}, \quad H = \frac{\sigma_z}{2}. \tag{9.133}$$

If the initial state is a product state $|0\rangle^{\otimes n}$, after applying $U_\theta^{\otimes n}$ and measuring each qubit, the QFI scales as n:

$$\mathcal{Q}(\rho_\theta) = n. \tag{9.134}$$

This leads to the standard quantum limit:

$$\Delta^2\hat{\theta} \geq \frac{1}{\nu n}. \tag{9.135}$$

If instead we use the **GHZ state**

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right), \tag{9.136}$$

the QFI becomes

$$\mathcal{Q}(\rho_\theta) = n^2. \tag{9.137}$$

This gives the Heisenberg limit:

$$\Delta^2\hat{\theta} \geq \frac{1}{\nu n^2}. \tag{9.138}$$

This result shows how entanglement can significantly enhance precision compared to classical or unentangled strategies.

9.3.3 Error propagation formula for quantum parameter estimation

To estimate the parameter θ, one practical method is to measure an observable O and use its expectation value $\langle O \rangle = \mathrm{Tr}(O\rho_\theta)$, which we write as a function $f(\theta)$. We estimate $f(\theta)$ by measuring O, then invert this relation to find θ.

Assume O has eigenvalues $\{\lambda_j\}$ and corresponding eigenvectors $\{|\phi_j\rangle\}$, so the spectral decomposition of O is

$$O = \sum_j \lambda_j |\phi_j\rangle\langle\phi_j|. \tag{9.139}$$

Measuring the state ρ_θ in this basis yields outcome λ_j with probability

$$p_j = \langle\phi_j|\rho_\theta|\phi_j\rangle. \tag{9.140}$$

Suppose we repeat this measurement ν times, obtaining outcomes $m_1, m_2, \ldots, m_\nu$, where each $m_k = \lambda_j$ for some j. The estimator for $f(\theta)$ is the sample mean

$$\hat{f} = \frac{1}{\nu}\sum_{k=1}^{\nu} m_k. \tag{9.141}$$

This estimator is unbiased, because

$$\mathbb{E}[\hat{f}] = \frac{1}{\nu}\sum_{k=1}^{\nu}\mathbb{E}[m_k] = \mathbb{E}[m_1] = \sum_j \lambda_j p_j = \mathrm{Tr}(O\rho_\theta) = f(\theta). \tag{9.142}$$

Next, we calculate the variance of $\hat{f}$. The variance of a single measurement outcome is

$$\mathrm{Var}(m_1) = \mathbb{E}[m_1^2] - (\mathbb{E}[m_1])^2 = \sum_j \lambda_j^2 p_j - \left(\sum_j \lambda_j p_j\right)^2 = \mathrm{Tr}(O^2\rho_\theta) - [\mathrm{Tr}(O\rho_\theta)]^2 = \Delta^2 O, \tag{9.143}$$

where $\Delta^2 O$ is the variance of the observable O in the state ρ_θ.

Since the measurement outcomes $m_1, \ldots, m_\nu$ are independent and identically distributed, the variance of $\hat{f}$ is

$$\Delta^2 \hat{f} = \frac{1}{\nu^2} \sum_{k=1}^{\nu} \mathrm{Var}(m_k) = \frac{\mathrm{Var}(m_1)}{\nu} = \frac{\Delta^2 O}{\nu}. \tag{9.144}$$

To estimate θ, we use the inverse function $\hat{\theta} = f^{-1}(\hat{f})$. If $f(\theta)$ is not globally invertible, we assume θ is near a known value where $f(\theta)$ is locally invertible. Using a Taylor expansion around $f(\theta)$, we write

$$\hat{\theta} = f^{-1}(\hat{f}) \approx f^{-1}(f(\theta)) + \left(\frac{\partial f}{\partial \theta}\right)^{-1} (\hat{f} - f(\theta)) = \theta + \frac{\hat{f} - f(\theta)}{\frac{\partial \langle O \rangle}{\partial \theta}}, \tag{9.145}$$

since

$$\frac{\partial f^{-1}}{\partial \hat{f}} = \left(\frac{\partial f}{\partial \theta}\right)^{-1}. \tag{9.146}$$

The estimation error is approximately

$$\hat{\theta} - \theta \approx \frac{\hat{f} - f(\theta)}{\frac{\partial \langle O \rangle}{\partial \theta}}. \tag{9.147}$$

The mean squared error (MSE) of the estimator $\hat{\theta}$ is

$$\Delta^2 \hat{\theta} = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] \approx \mathbb{E}\left[\left(\frac{\hat{f} - f(\theta)}{\frac{\partial \langle O \rangle}{\partial \theta}}\right)^2\right] = \frac{\mathbb{E}\left[(\hat{f} - f(\theta))^2\right]}{\left(\frac{\partial \langle O \rangle}{\partial \theta}\right)^2} = \frac{\Delta^2 \hat{f}}{\left(\frac{\partial \langle O \rangle}{\partial \theta}\right)^2}. \tag{9.148}$$

Substituting the result for $\Delta^2 \hat{f}$ gives

$$\Delta^2 \hat{\theta} = \frac{\Delta^2 O}{\nu \left|\frac{\partial \langle O \rangle}{\partial \theta}\right|^2}. \tag{9.149}$$

This is the **error propagation formula** for quantum parameter estimation. It shows that the precision of estimating θ depends on both the variance $\Delta^2 O$ of the observable and the sensitivity of the expectation value $\langle O \rangle$ with respect to θ. A larger derivative $\frac{\partial \langle O \rangle}{\partial \theta}$ improves precision, as small changes in θ produce larger changes in the measurement outcome.

If needed, this approach can be combined with choosing optimal observables O to maximize the sensitivity and minimize the estimation error.

9.3.4 QFI for pure state and mixed state

The quantum Fisher information (QFI) quantifies how much information a quantum state ρ_θ carries about an unknown parameter θ. It sets the ultimate limit on how precisely we can estimate θ.

The starting point is the classical Fisher information. In the quantum setting, consider a state ρ_θ and a measurement described by a POVM $\mathcal{M} = \{E_m\}$. The Fisher information for this measurement is

$$\mathcal{I}(\rho_\theta, \mathcal{M}) = \int \frac{(\mathrm{Tr}(E_m \dot{\rho}_\theta))^2}{\mathrm{Tr}(E_m \rho_\theta)} \mathrm{d}m, \tag{9.150}$$

where $\dot{\rho}_\theta = \frac{\partial \rho_\theta}{\partial \theta}$ is the derivative of the state with respect to θ, $\mathrm{Tr}(E_m \rho_\theta)$ is the probability of outcome m, and $\mathrm{Tr}(E_m \dot{\rho}_\theta)$ shows how this probability changes with θ.

The QFI is the maximum Fisher information over all possible measurements:

$$\mathcal{Q}(\rho_\theta) = \max_{\mathcal{M}} \mathcal{I}(\rho_\theta, \mathcal{M}). \tag{9.151}$$

To compute this maximum, we introduce the **symmetric logarithmic derivative (SLD)**, defined through the equation

$$\frac{\partial \rho_\theta}{\partial \theta} = \frac{1}{2} \left(L_\theta \rho_\theta + \rho_\theta L_\theta \right). \tag{9.152}$$

The QFI can be expressed as

$$\mathcal{Q}(\rho_\theta) = \mathrm{Tr}\left(\rho_\theta L_\theta^2 \right). \tag{9.153}$$

To find L_θ explicitly, we use the spectral decomposition of ρ_θ:

$$\rho_\theta = \sum_j \lambda_j |j\rangle\langle j|, \tag{9.154}$$

where λ_j are eigenvalues and $|j\rangle$ are eigenvectors. The derivative $\dot{\rho}_\theta$ is

$$\dot{\rho}_\theta = \sum_j \dot{\lambda}_j |j\rangle\langle j| + \sum_j \lambda_j \left(|\dot{j}\rangle\langle j| + |j\rangle\langle \dot{j}| \right), \tag{9.155}$$

where $\dot{\lambda}_j = \frac{\partial \lambda_j}{\partial \theta}$ and $|\dot{j}\rangle = \frac{\partial}{\partial \theta}|j\rangle$.

Substituting this into the definition of the SLD and solving for the matrix elements of L_θ gives

$$\langle j|L_\theta|k\rangle = \frac{2\langle j|\dot{\rho}_\theta|k\rangle}{\lambda_j+\lambda_k}, \quad \text{for } \lambda_j+\lambda_k \neq 0. \tag{9.156}$$

The QFI then becomes

$$\mathcal{Q}(\rho_\theta) = \sum_j \frac{\dot{\lambda}_j^2}{\lambda_j} + 2\sum_{j\neq k} \frac{(\lambda_j-\lambda_k)^2}{\lambda_j+\lambda_k} |\langle j|\dot{k}\rangle|^2. \tag{9.157}$$

The first term represents classical contributions from changes in eigenvalues (probabilities), while the second term represents quantum contributions from changes in eigenvectors (superposition effects).

For a pure state $\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|$, where $\rho_\theta^2 = \rho_\theta$, differentiating $\rho_\theta^2 = \rho_\theta$ gives

$$\dot{\rho}_\theta = \dot{\psi}_\theta\langle\psi_\theta| + |\psi_\theta\rangle\langle\dot{\psi}_\theta|. \tag{9.158}$$

In this case, the SLD simplifies to $L_\theta = 2\dot{\rho}_\theta$, and the QFI reduces to

$$\mathcal{Q}(|\psi_\theta\rangle) = 4\left(\langle\dot{\psi}_\theta|\dot{\psi}_\theta\rangle - |\langle\dot{\psi}_\theta|\psi_\theta\rangle|^2\right). \tag{9.159}$$

This expression directly measures how much the state $|\psi_\theta\rangle$ changes with respect to θ.

For mixed states, the QFI can also be obtained by minimizing over all purifications $|\Psi_\theta\rangle$ of ρ_θ, leading to

$$\mathcal{Q}(\rho_\theta) = \min_{|\Psi_\theta\rangle} 4\left(\langle\dot{\Psi}_\theta|\dot{\Psi}_\theta\rangle - |\langle\dot{\Psi}_\theta|\Psi_\theta\rangle|^2\right). \tag{9.160}$$

This shows that QFI unifies classical probability changes and quantum coherence effects into a single formula that determines the best achievable precision in parameter estimation.

9.3.5 Geometric perspective of QFI

In quantum mechanics, a quantum state can be seen either as a vector in a Hilbert space or as a point in a geometric space called complex projective space $\mathbb{CP}^n$. For a single qubit, this is the Bloch sphere: pure states lie on the surface, mixed states inside. This space has a natural distance measure, called the **Fubini-Study metric** for pure states, which generalizes to the **Bures metric** for mixed states. These distances help quantify how much two quantum states differ, especially when they are close.

Consider a quantum state ρ_θ that depends smoothly on a parameter θ. The infinitesimal distance ds between ρ_θ and $\rho_{\theta+\delta\theta}$ is given by

$$ds^2 = \frac{1}{4}\mathcal{Q}(\rho_\theta)\,\delta\theta^2, \tag{9.161}$$

where $\mathcal{Q}(\rho_\theta)$ is the quantum Fisher information (QFI). This expression shows that QFI measures how sensitive the state is to small changes in θ, similar to classical Fisher information for probability distributions.

Another way to compare quantum states is via the **Bures angle** between two density matrices ρ_1 and ρ_2, defined as

$$d_A(\rho_1,\rho_2) = \arccos\sqrt{\mathscr{F}(\rho_1,\rho_2)}, \tag{9.162}$$

where the fidelity $\mathscr{F}(\rho_1,\rho_2)$ measures how close the two states are.

For two nearby states ρ_θ and $\rho_{\theta+\delta\theta}$, the fidelity approaches 1 as $\delta\theta \to 0$. Using a Taylor expansion, we approximate the Bures angle:

$$d_A(\rho_\theta,\rho_{\theta+\delta\theta}) \approx \sqrt{2-2\sqrt{\mathscr{F}(\rho_\theta,\rho_{\theta+\delta\theta})}} + \mathcal{O}(\delta\theta^2). \tag{9.163}$$

From the geometry of the Bures metric, the same distance can be written as

$$d_A(\rho_\theta,\rho_{\theta+\delta\theta}) \approx ds = \frac{1}{2}\sqrt{\mathcal{Q}(\rho_\theta)}\,\delta\theta + \mathcal{O}(\delta\theta^2). \tag{9.164}$$

Equating these two expressions gives

$$\sqrt{2-2\sqrt{\mathscr{F}(\rho_\theta,\rho_{\theta+\delta\theta})}} \approx \frac{1}{2}\sqrt{\mathcal{Q}(\rho_\theta)}\,\delta\theta. \tag{9.165}$$

Squaring both sides yields

$$2-2\sqrt{\mathscr{F}(\rho_\theta,\rho_{\theta+\delta\theta})} \approx \frac{1}{4}\mathcal{Q}(\rho_\theta)\,\delta\theta^2, \tag{9.166}$$

which simplifies to

$$1-\sqrt{\mathscr{F}(\rho_\theta,\rho_{\theta+\delta\theta})} \approx \frac{1}{8}\mathcal{Q}(\rho_\theta)\,\delta\theta^2. \tag{9.167}$$

Taking the limit $\delta\theta \to 0$ gives an exact expression for the QFI in terms of fidelity:

$$\mathcal{Q}(\rho_\theta) = \lim_{\delta\theta\to 0} \frac{8\left[1-\sqrt{\mathscr{F}(\rho_\theta,\rho_{\theta+\delta\theta})}\right]}{\delta\theta^2}. \tag{9.168}$$

This provides a geometric interpretation of QFI through the Bures distance and fidelity.

QFI also satisfies a monotonicity property under any completely positive trace-preserving (CPTP) map $\mathcal{E}$:

$$\mathcal{Q}(\mathcal{E}(\rho_\theta)) \leq \mathcal{Q}(\rho_\theta). \tag{9.169}$$

This inequality holds because fidelity increases under CPTP maps:

$$\mathscr{F}(\mathcal{E}(\rho_1), \mathcal{E}(\rho_2)) \geq \mathscr{F}(\rho_1, \rho_2). \tag{9.170}$$

This means that interaction with the environment can degrade the amount of information we can extract about θ.

The QFI determines the precision bound through the quantum Cramér-Rao bound (QCRB):

$$\Delta^2\hat{\theta} \geq \frac{1}{\nu\mathcal{Q}(\rho_\theta)}, \tag{9.171}$$

where ν is the number of repetitions of the experiment.

This geometric approach links QFI to distances between quantum states, while the Heisenberg limit describes the best precision achievable through quantum resources and optimal state preparation.

9.3.6 Multi-parameter quantum estimation

Consider a quantum state $\rho_{\boldsymbol{\theta}}$ that depends on several parameters $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_m)$. The goal is to estimate these parameters by performing measurements on the state and processing the outcomes to obtain estimators $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_m)$.

Assume the estimators are unbiased, meaning

$$\mathbb{E}[\hat{\boldsymbol{\theta}}] = \boldsymbol{\theta}. \tag{9.172}$$

The precision of the estimation is characterized by the **covariance matrix**

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}})_{i,j} = \mathbb{E}\left[(\hat{\theta}_i - \theta_i)(\hat{\theta}_j - \theta_j)\right]. \tag{9.173}$$

The diagonal elements represent the variances of the estimators, while the off-diagonal elements describe correlations between the estimators of different parameters.

The **quantum Cramér-Rao bound (QCRB)** for multi-parameter estimation generalizes the single-parameter case and takes the matrix form

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \geq \frac{1}{\nu}\mathcal{I}^{-1}(\boldsymbol{\theta}) \geq \frac{1}{\nu}\mathcal{Q}^{-1}(\boldsymbol{\theta}), \tag{9.174}$$

where ν is the number of independent repetitions of the experiment, $\mathcal{I}(\boldsymbol{\theta})$ is the classical Fisher information matrix, and $\mathcal{Q}(\boldsymbol{\theta})$ is the quantum Fisher information (QFI) matrix.

The classical Fisher information matrix is given by

$$\mathcal{I}(\boldsymbol{\theta})_{i,j} = \int p(\mathbf{x} \mid \boldsymbol{\theta}) \left(\frac{\partial \ln p(\mathbf{x} \mid \boldsymbol{\theta})}{\partial \theta_i}\right)\left(\frac{\partial \ln p(\mathbf{x} \mid \boldsymbol{\theta})}{\partial \theta_j}\right) \mathrm{d}\mathbf{x}, \tag{9.175}$$

where $p(\mathbf{x} \mid \boldsymbol{\theta})$ is the probability of measurement outcome $\mathbf{x}$ given the parameters $\boldsymbol{\theta}$.

The quantum Fisher information matrix, which provides the ultimate precision limit over all possible measurements, is defined as

$$\mathcal{Q}(\boldsymbol{\theta})_{i,j} = \mathrm{Tr}\left(\mathcal{R}_{\rho_\theta}^{-1}\left(\frac{\partial \rho_{\boldsymbol{\theta}}}{\partial \theta_i}\right)\rho_{\boldsymbol{\theta}}\mathcal{R}_{\rho_\theta}^{-1}\left(\frac{\partial \rho_{\boldsymbol{\theta}}}{\partial \theta_j}\right)\right), \tag{9.176}$$

where $\mathcal{R}_{\rho_\theta}^{-1}$ is the inverse of the superoperator

$$\mathcal{R}_{\rho_\theta}(O) = \frac{1}{2}\left(\rho_{\boldsymbol{\theta}} O + O \rho_{\boldsymbol{\theta}}\right). \tag{9.177}$$

The derivative $\frac{\partial \rho_{\boldsymbol{\theta}}}{\partial \theta_i}$ describes how the quantum state changes with respect to θ_i. The superoperator $\mathcal{R}_{\rho_\theta}^{-1}$ ensures that this change is properly weighted by the structure of the quantum state.

The QCRB states that the covariance matrix of any unbiased estimator is bounded from below by the inverse of the QFI matrix. The inequality is understood in the sense of positive semi-definite matrices.

When $\mathcal{Q}(\boldsymbol{\theta})$ is diagonal, the parameters are statistically independent, and the bound simplifies to

$$\mathrm{Var}(\hat{\theta}_i) \geq \frac{1}{\nu \mathcal{Q}(\theta_i)}, \tag{9.178}$$

recovering the single-parameter result. If the off-diagonal elements of $\mathcal{Q}(\boldsymbol{\theta})$ are nonzero, this indicates coupling between parameters, and the estimation of one parameter may affect the precision of others.

The inverse of a positive semi-definite matrix satisfies the inequality

$$M_{i,i} \geq \frac{1}{(M^{-1})_{i,i}}, \tag{9.179}$$

with equality if M is diagonal. This explains why parameter correlations (encoded in the off-diagonal terms) can reduce the achievable precision for individual parameters.

In quantum systems, achieving the bound requires measuring in the eigenbasis of the symmetric logarithmic derivative operators associated with each parameter. However, for multiple parameters, these measurements may not commute, meaning that the optimal measurements for different parameters may be incompatible.

This fundamental incompatibility leads to trade-offs in precision: estimating several parameters simultaneously may require compromises in measurement strategies. Such trade-offs are significant in tasks like quantum sensing with multiple sensors, multiparameter phase estimation, or characterizing complex quantum channels.

When the generators of the parameter encoding commute, the parameters can be estimated independently without loss of precision. In contrast, for non-commuting generators, entanglement and joint measurement strategies may be required to optimize precision across all parameters.

The quantum Cramér-Rao bound in the multi-parameter setting thus generalizes the classical concept, revealing how quantum correlations, measurement compatibility, and entanglement affect the fundamental limits of precision in parameter estimation.

9.3.7 Example of QFI

Consider a quantum system with N energy levels in thermal equilibrium at temperature T. The state of the system is given by the Gibbs state

$$\rho_T = \frac{1}{Z} e^{-\frac{H}{T}}, \tag{9.180}$$

where H is the Hamiltonian of the system, and $Z = \mathrm{Tr}\, e^{-\frac{H}{T}}$ is the partition function. The Hamiltonian can be expressed as

$$H = \sum_{k=1}^{N} \epsilon_k |\epsilon_k\rangle\langle\epsilon_k|, \tag{9.181}$$

with eigenvalues ϵ_k and eigenvectors $|\epsilon_k\rangle$. Throughout, we set the Boltzmann constant $k_B = 1$ for simplicity.

The goal is to estimate the temperature T by performing measurements on ρ_T. The quantum Fisher information (QFI) quantifies the ultimate precision limit for estimating T.

To calculate the QFI, we first compute the derivative of ρ_T with respect to T. Since

$$\rho_T = \frac{1}{Z} e^{-\frac{H}{T}}, \tag{9.182}$$

the derivative of the exponential term is

$$\frac{\partial}{\partial T} e^{-\frac{H}{T}} = e^{-\frac{H}{T}} \cdot \frac{H}{T^2}. \tag{9.183}$$

For the partition function Z, we have

$$\frac{\partial Z}{\partial T} = \mathrm{Tr}\left(e^{-\frac{H}{T}} \cdot \frac{H}{T^2}\right) = \frac{1}{T^2}\langle H\rangle, \tag{9.184}$$

where $\langle H\rangle = \mathrm{Tr}(H\rho_T)$ is the average energy.

Applying the product rule to $\rho_T = \frac{1}{Z} e^{-\frac{H}{T}}$, the derivative $\dot{\rho}_T = \frac{\partial \rho_T}{\partial T}$ becomes

$$\dot{\rho}_T = -\frac{1}{Z^2}\frac{\partial Z}{\partial T} e^{-\frac{H}{T}} + \frac{1}{Z} e^{-\frac{H}{T}} \cdot \frac{H}{T^2}. \tag{9.185}$$

Substituting the expression for $\frac{\partial Z}{\partial T}$ and using $\rho_T = \frac{1}{Z} e^{-\frac{H}{T}}$, this simplifies to

$$\dot{\rho}_T = -\frac{\langle H\rangle}{T^2}\rho_T + \frac{1}{T^2}\rho_T H. \tag{9.186}$$

Because H and ρ_T commute, $\rho_T H = H\rho_T$, so this can be rewritten as

$$\dot{\rho}_T = \frac{1}{T^2}(H - \langle H\rangle)\rho_T. \tag{9.187}$$

To symmetrize this expression, note that

$$\dot{\rho}_T = \frac{1}{2T^2}\left((H - \langle H\rangle)\rho_T + \rho_T(H - \langle H\rangle)\right). \tag{9.188}$$

This matches the general form of the symmetric logarithmic derivative (SLD) L, defined through

$$\dot{\rho}_T = \frac{1}{2}(L\rho_T + \rho_T L). \tag{9.189}$$

Comparing terms, we identify the SLD as

$$L = \frac{1}{T^2}(H - \langle H \rangle). \tag{9.190}$$

The quantum Fisher information is then given by

$$\mathcal{Q}(\rho_T) = \mathrm{Tr}(L\rho_T L). \tag{9.191}$$

Substituting L into this expression yields

$$\mathcal{Q}(\rho_T) = \mathrm{Tr}\left(\frac{1}{T^4}(H - \langle H \rangle)\rho_T(H - \langle H \rangle)\right), \tag{9.192}$$

which simplifies to

$$\mathcal{Q}(\rho_T) = \frac{1}{T^4}\Delta^2 H, \tag{9.193}$$

where the variance of the Hamiltonian is

$$\Delta^2 H = \mathrm{Tr}(H^2\rho_T) - \langle H \rangle^2. \tag{9.194}$$

The quantum Cramér-Rao bound (QCRB) relates the QFI to the precision of an unbiased estimator $\hat{T}$ measured ν times:

$$\Delta^2 \hat{T} \geq \frac{1}{\nu \mathcal{Q}(\rho_T)} = \frac{T^4}{\nu \Delta^2 H}. \tag{9.195}$$

This result connects quantum estimation theory to thermodynamics. The heat capacity C is defined by

$$C = \frac{\partial \langle H \rangle}{\partial T}. \tag{9.196}$$

For the Gibbs state, there is a thermodynamic identity relating heat capacity and energy fluctuations:

$$\Delta^2 H = T^2 C. \tag{9.197}$$

Substituting this into the expression for QFI gives

$$\mathcal{Q}(\rho_T) = \frac{C}{T^2}. \tag{9.198}$$

The QCRB then becomes

$$\Delta^2\hat{T} \geq \frac{T^2}{\nu C}. \tag{9.199}$$

Alternatively, using the identity $C = \frac{\Delta^2 H}{T^2}$, the QCRB can also be written as

$$\Delta^2\hat{T} \geq \frac{\Delta^2 H}{\nu\left(\frac{\partial\langle H\rangle}{\partial T}\right)^2}, \tag{9.200}$$

which is consistent because $\frac{\partial\langle H\rangle}{\partial T} = C$.

The optimal measurement to achieve this bound is the projective measurement of the energy H. Measuring H yields outcomes ϵ_k with probabilities

$$p_k = \frac{e^{-\frac{\epsilon_k}{T}}}{Z}. \tag{9.201}$$

From repeated measurements, we can estimate $\langle H\rangle$, and through the functional dependence of $\langle H\rangle$ on T, infer T. The error propagation formula in this case matches the QCRB exactly, confirming that energy measurements saturate the bound.

In certain cases, such as two-level systems with one low-energy state and multiple degenerate high-energy states, the relative error $\Delta^2\hat{T}/T^2$ can scale favorably as $\mathcal{O}(1/\log N)$, where N is the system size, making these systems efficient quantum thermometers.

This demonstrates how quantum thermometry leverages thermal fluctuations and energy measurements to achieve optimal precision in temperature estimation, directly linking quantum statistics to fundamental thermodynamic quantities.

Part III

Quantum Computing

Chapter 10

Quantum computing model

One of the main models of quantum computing is the quantum circuit, which consists of single-qubit and multi-qubit gates. In this chapter, we will briefly introduce the quantum computing circuit model and explain the universality of this model.

10.1 Single-Qubit Gates

There are many types of gate operations for a single qubit, with the most representative being the Pauli gates σ_x, σ_y, σ_z, σ_I, represented as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \ \sigma_I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{10.1}$$

Where σ_x is the bit flip, σ_z is the phase flip, σ_y flips both the bit and the phase, and σ_I is the identity matrix. σ_x, σ_y, σ_z are generally also referred to as Pauli-X, Y, Z gates.

Problem 10.1. *For the single-qubit state $|\psi\rangle = a|0\rangle + b|1\rangle$, discuss the effect of the Pauli gates on this quantum state.*

Using the single-qubit Pauli gates as Hamiltonians and selecting a time $t = \theta/2$, the corresponding unitary operations are

$$\mathtt{R}_X(\theta) = \exp(-i\sigma_X\theta/2) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix},$$
$$\mathtt{R}_Y(\theta) = \exp(-i\sigma_Y\theta/2) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix},$$
$$\mathtt{R}_Z(\theta) = \exp(-i\sigma_Z\theta/2) = \begin{pmatrix} \exp(-i\theta/2) & 0 \\ 0 & \exp(i\theta/2) \end{pmatrix}.$$

Where $\mathtt{R}_Z(\theta)$ is also referred to as the phase gate, denoted as $\mathtt{P}(\alpha)$. Meanwhile, for a more general Hamiltonian $\boldsymbol{n}\cdot\boldsymbol{\sigma} = n_x\sigma_X + n_y\sigma_Y + n_z\sigma_Z$, the corresponding quantum state operation is:

$$\mathtt{R}_{\boldsymbol{n}}(\theta) = \exp(-i\boldsymbol{n}\cdot\boldsymbol{\sigma}\theta/2) = \cos\left(\frac{\theta}{2}\right)\sigma_I - i\sin\left(\frac{\theta}{2}\right)\left(n_x\sigma_X + n_y\sigma_Y + n_z\sigma_Z\right). \quad (10.2)$$

As we have mentioned earlier, single-qubit pure states can be uniquely mapped to unit vectors on the Bloch sphere, so for the three-dimensional sphere rotation operation $R_{\boldsymbol{n}}(\theta)$, we have

$$\mathtt{R}_{\boldsymbol{n}}(\theta)\cdot\boldsymbol{\sigma}\cdot\mathtt{R}_{\boldsymbol{n}}(\theta)^\dagger = R_{\boldsymbol{n}}(\theta)\cdot\boldsymbol{\sigma}. \quad (10.3)$$

Thus, the operation on the quantum state $|\psi(\boldsymbol{x})\rangle$ can be viewed as an operation on the coordinates $\boldsymbol{x}$ on the Bloch sphere:

$$\mathtt{R}_{\boldsymbol{n}}(\theta)\,|\psi(\boldsymbol{x})\rangle = |\psi(R_{\boldsymbol{n}}(\theta)\cdot\boldsymbol{x})\rangle. \quad (10.4)$$

Problem 10.2. *Prove the universality of single-qubit rotation gates.*

- *Prove that any single-qubit unitary operation can be written as*

$$U = \exp(i\alpha)R_{\boldsymbol{n}}(\theta). \quad (10.5)$$

  Note that since the global phase $\exp(i\alpha)$ *has no effect on quantum state measurements, any rotation around an arbitrary axis* $\boldsymbol{n}$ *can represent any single-qubit quantum gate.*

- *Based on the relationship between* $\mathtt{R}_{\boldsymbol{n}}(\theta)$ *and* $R_{\boldsymbol{n}}(\theta)$*, prove that any single-qubit unitary operation can be written as*

$$U = \exp(i\alpha)R_a(\theta_1)R_b(\theta_2)R_a(\theta_3), \quad (10.6)$$

  where $a \neq b \in \{X, Y, Z\}$.

Although single-qubit rotation gates are universal, in quantum circuit and algorithm design, we often use some more specialized gates:

$$\mathtt{H} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix},\ \mathtt{S} = \begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix},\ \mathtt{T} = \begin{pmatrix}1 & 0\\ 0 & e^{i\pi/4}\end{pmatrix}. \tag{10.7}$$

The $\mathtt{H}$ gate is called the Hadamard gate, which satisfies $\mathtt{H}^\dagger = \mathtt{H}$ and swaps the basis states of σ_x and σ_z, i.e.,

$$H|0\rangle = |+\rangle, H|1\rangle = |-\rangle, H|+\rangle = |0\rangle, H|-\rangle = |1\rangle.$$

In the computational basis, since the $\mathtt{H}$ gate can create superpositions of 0 and 1, it is widely used in quantum computing, especially for initializing quantum circuits. The $\mathtt{S} = \sqrt{\mathtt{Z}}$ gate is a phase gate, which is non-Hermitian, and swaps the basis states of σ_x and σ_y, i.e.,

$$S|\pm\rangle = |\pm i\rangle,\ S^\dagger|\pm i\rangle = |\pm\rangle.$$

We can see that the Pauli gates σ_x, σ_y, σ_z and the gates $\mathtt{H}, \mathtt{S}$ are essentially transformations in the Pauli basis, so their actions are discrete and limited. Thus, we need to introduce a new gate to implement transformations beyond the Pauli basis, which is the $\mathtt{T}$ gate. The $\mathtt{T}$ gate is also a phase gate, non-Hermitian, and can be understood as half of the $\mathtt{S}$ gate, i.e.,

$$\mathtt{T} = \sqrt{\mathtt{S}} = \sqrt[4]{\mathtt{Z}}.$$

It is easy to prove that when the $\mathtt{T}$ gate acts on the Pauli basis, the resulting quantum states are no longer Pauli states. Therefore, the $\mathtt{T}$ gate, along with the $\mathtt{H}$ and $\mathtt{S}$ gates, is crucial for achieving universal quantum computation.

Problem 10.3. *Prove that* $S = \exp(i\pi/4)R_Z(\pi/2)$, $T = \exp(i\pi/8)R_Z(\pi/4)$.

Problem 10.4. *Use* H *and* S *to construct transformations for the* σ_y *and* σ_z *basis states.*

Problem 10.5. *Express the* H *gate using rotation gates.*

Problem 10.6. *Prove the following properties of the* H *gate:*

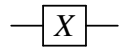
$$HXH = Z,\ HYH = -Y,\ HZH = X. \tag{10.8}$$

Problem 10.7. *Use* H *and* T *gates to express the* X, Y, Z, S *gates.*

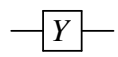
In the following, we will further discuss how to use the H and T gates to approximate any single-qubit quantum gate to arbitrary precision. Therefore, the H and T gates form a complete set of single-qubit gates.

In quantum circuits, a single qubit is typically represented by a line, and a gate operation is represented by a box. Here, we summarize the single-qubit gates introduced above.

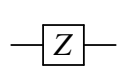
$$\text{Pauli-X} = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



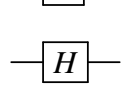
$$\text{Pauli-Y} = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$



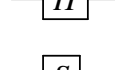
$$\text{Pauli-Z} = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



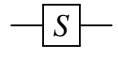
$$\mathtt{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



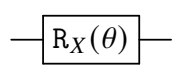
$$\mathtt{S} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$



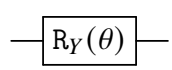
$$\mathtt{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$



$$\mathtt{R}_X(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$



$$\mathtt{R}_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$



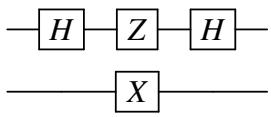
$$\mathtt{R}_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

10.2 Multi-Qubit Gates

Next, we introduce multi-qubit gate operations. In general, for multiple qubits, each qubit is represented by a parallel horizontal line, and the corresponding multi-qubit gate operation is represented by a box acting on the corresponding qubit. We will introduce two-qubit and multi-qubit gates separately.

10.2.1 Two-Qubit Controlled Gates

For two qubits, if the gate operation acts on each qubit individually, we still treat it as a single-qubit operation, for example:



A two-qubit gate is only considered when the operation acts on both qubits simultaneously. A commonly used two-qubit gate in quantum computing is the controlled gate, which has the general form:

Here, the first qubit is the control qubit, and the second qubit is the target qubit. The left gate is a 1-controlled gate, and the right gate is a 0-controlled gate, represented as:

$$CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U, \qquad \bar{CU} = |0\rangle\langle 0| \otimes U + |1\rangle\langle 1| \otimes I \tag{10.9}$$

The matrix representations of the two gate operations are:

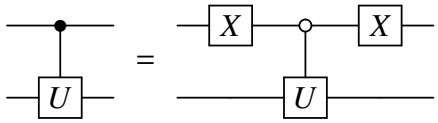
$$CU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{pmatrix}, \qquad \bar{CU} = \begin{pmatrix} u_{00} & u_{01} & 0 & 0 \\ u_{10} & u_{11} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{10.10}$$

For the computational basis $|c\rangle|t\rangle$, we have:

$$CU|c\rangle|t\rangle = |c\rangle U^c|t\rangle, \; \bar{CU}|c\rangle|t\rangle = |c\rangle U^{c\oplus 1}|t\rangle, \tag{10.11}$$

Problem 10.8. *Write the specific results of CU and $\bar{CU}$ when acting on the computational basis.*

Problem 10.9. *Prove that $CU = X \otimes I \cdot \bar{CU} \cdot X \otimes I$, i.e.,*



Problem 10.10. *For $U = e^{-iHt}$ (e.g., H is a sum of Pauli operators), prove that $CU = e^{-i|1\rangle\langle 1|\otimes H} = e^{-i(I-Z)\otimes H/2}$. Therefore, when we have the Hamiltonian of U, we also have the corresponding Hamiltonian for the implementation of CU.*

When U is the Pauli X and Z matrices, the corresponding (1) controlled operations are denoted as CX and CZ, represented as:

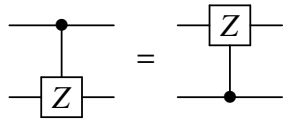
The matrix representations of CX and CZ are:

$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad \bar{CU} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{10.12}$$

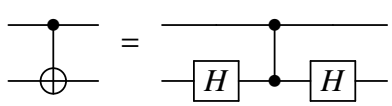
For the computational basis $|c\rangle\,|t\rangle$, we have:

$$CX\,|c\rangle\,|t\rangle = |c\rangle\,|c \oplus t\rangle\,,\ CZ\,|c\rangle\,|t\rangle = (-1)^{ct}\,|c\rangle\,|t\rangle\,. \tag{10.13}$$

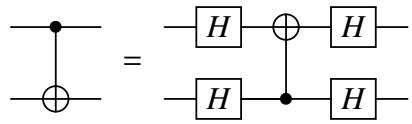
The action of the CX gate is to apply the flip operation X on the target qubit based on the control qubit. The flip operation corresponds to the classical NOT operation, so the CX gate is also called the CNOT gate. The CZ gate applies a phase flip operation Z to the target qubit based on the control qubit. Since the phase flip operation is applied to the target qubit only when the control qubit is 1, the effect of the CX gate is to apply a phase flip when both the control and target qubits are 1. Therefore, the CX gate is symmetric in that the control and target qubits are interchangeable, and swapping the two qubits results in the same CX operation, i.e.,



Note that X and Z can be converted into each other via the H gate, so we have:



Problem 10.11. *Prove*

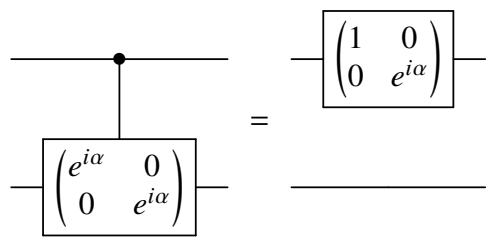


Thus, CNOT acts as if it swaps the control and target qubits when applied to the computational basis and the $|\pm\rangle$ basis after the H gate rotation.

As mentioned earlier, the global phase gate $U = e^{i\alpha}$ for a single qubit has no physical significance and can be ignored. However, when we consider controlling the global phase operation, we cannot neglect it. Specifically, when $U = e^{i\alpha}$, we have

$$CU\,|\psi\rangle_{ct} = \mathtt{R}_Z^c(\alpha)\,|\psi\rangle_{ct}\,, \tag{10.14}$$

which corresponds to a phase operation controlled by the control qubit, essentially a single-qubit gate:



At the same time, we need to note that when $U = \mathtt{R}_Z(\alpha)$, the controlled phase gate operation is

$$\mathtt{CR}_z(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix} \tag{10.15}$$

Unlike the controlled global phase operation mentioned above, this is essentially a two-qubit gate.

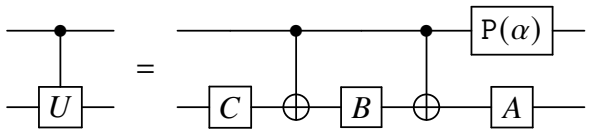
Finally, if we treat the CNOT gate as a basic two-qubit operation, an important question is whether we can use only CNOT and single-qubit operations to realize any two-qubit gate. Since the fidelity of two-qubit gates is generally lower than that of single-qubit gates, we aim to use as few two-qubit gates as possible. Here, we first introduce how to implement a general controlled gate operation. For this purpose, we need a special decomposition method for single-qubit gates:

$$U = e^{i\alpha} AXBXC, \tag{10.16}$$

where $ABC = I$.

Problem 10.12. *Prove Equation* (10.16).

Therefore, for $U = e^{i\alpha} AXBXC$, any CU can be decomposed into the following form, using only two CNOT operations and four single-qubit operations:



The derivation of this formula can be done using the ket-bra representation of CU, i.e., $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{i\alpha} AXBXC$. The right-hand side then becomes:

$$|0\rangle\langle 0| \otimes ABC + |1\rangle\langle 1| \otimes e^{i\alpha} AXBXC = CU. \tag{10.17}$$

Thus, we can see that $ABC = I$ is the key to making this expression valid, otherwise, we would need to separately implement control-A, control-B, control-C, and two CNOTs.

Problem 10.13. *Derive two CNOT implementations for the CU gate.*

10.2.2 Other Two-Qubit Gates

In addition to controlled single-qubit gates, there are other common two-qubit gates, such as the SWAP gate and two-qubit Hamiltonian rotation gates.

The SWAP gate, as the name suggests, swaps two quantum states, i.e.,

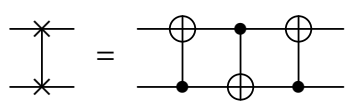
$$\texttt{SWAP}\,|\psi\rangle\,|\phi\rangle = |\phi\rangle\,|\psi\rangle\,, \tag{10.18}$$

Its matrix form is

$$\texttt{CX} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{10.19}$$

The circuit diagram for the SWAP gate is

The SWAP gate satisfies commutativity, Hermiticity, and can also be implemented using CNOT gates:



Another important property of the SWAP gate is

$$\mathrm{Tr}[\rho \otimes \sigma \cdot \texttt{SWAP}] = \mathrm{Tr}[\rho \cdot \sigma]. \tag{10.20}$$

This means that the SWAP operation can be applied to $\rho \otimes \sigma$ and allows us to measure the overlap between ρ and σ. Note that the SWAP gate should be viewed more as a measurement operator rather than a gate. Later, we will discuss how to implement this operation, which is the SWAP test method.

In addition to the SWAP gate, we can also consider more general Hamiltonian evolution (rotation) operations. Similar to single-qubit rotation operations, we consider a Hamiltonian H and an evolution time $t = \theta/2$, which corresponds to the evolution:

$$\mathtt{R}_H(\theta) = \exp(-iH\theta/2). \tag{10.21}$$

Some common choices of Hamiltonians are $H = XX, YY, ZZ, (XX+YY)/2$.

$$\mathtt{R}_{XX}(\theta) = \exp(-iXX\theta/2) = \begin{pmatrix} \cos\left(\frac{\phi}{2}\right) & 0 & 0 & -i\sin\left(\frac{\phi}{2}\right) \\ 0 & \cos\left(\frac{\phi}{2}\right) & -i\sin\left(\frac{\phi}{2}\right) & 0 \\ 0 & -i\sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) & 0 \\ -i\sin\left(\frac{\phi}{2}\right) & 0 & 0 & \cos\left(\frac{\phi}{2}\right) \end{pmatrix} \tag{10.22}$$

$$\mathtt{R}_{YY}(\theta) = \exp(-iYY\theta/2) = \begin{pmatrix} \cos\left(\frac{\phi}{2}\right) & 0 & 0 & i\sin\left(\frac{\phi}{2}\right) \\ 0 & \cos\left(\frac{\phi}{2}\right) & -i\sin\left(\frac{\phi}{2}\right) & 0 \\ 0 & -i\sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) & 0 \\ i\sin\left(\frac{\phi}{2}\right) & 0 & 0 & \cos\left(\frac{\phi}{2}\right) \end{pmatrix} \tag{10.23}$$

$$\mathtt{R}_{ZZ}(\theta) = \exp(-iZZ\theta/2) = \begin{bmatrix} e^{-i\phi/2} & 0 & 0 & 0 \\ 0 & e^{i\phi/2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} & 0 \\ 0 & 0 & 0 & e^{-i\phi/2} \end{pmatrix} \tag{10.24}$$

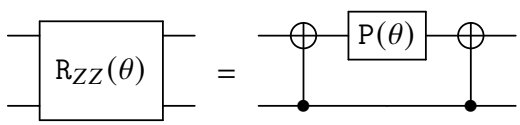
$$\mathtt{R}_{XY}(\theta) = \exp(-i(XX+YY)\theta/4) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\phi/2) & -i\sin(\phi/2) & 0 \\ 0 & -i\sin(\phi/2) & \cos(\phi/2) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{10.25}$$

In experiments, we often implement Hamiltonian evolution operations, and we can use these operations to first implement basic two-qubit gates.

Problem 10.14. *Prove that the CNOT gate can be implemented by the following Hamiltonian evolution:*

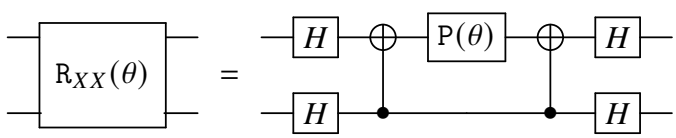
$$\text{CNOT} = e^{-i\frac{\pi}{4}} R_{Y_1}(-\pi/2) R_{Y_1}(-\pi/2) R_{X_2}(-\pi/2) R_{XX}(\pi/2) R_{Y_1}(\pi/2). \tag{10.26}$$

On the other hand, assuming that basic gate operations can be implemented by other hardware methods, an important issue in quantum computing is how to compile Hamiltonian evolution operations into basic gate operations. For $\mathtt{R}_{ZZ}(\theta)$, we can implement it using two CNOT gates and single-qubit gates as follows:



Additionally, since $\mathtt{R}_{ZZ}(\theta)$ is symmetric for two qubits, the above circuit also holds when the two qubits are swapped.

When the Hamiltonian is not ZZ, we can first use single-qubit rotations to transform it into ZZ, and then use the above circuit. For example, for XX, we have:



Problem 10.15. *Write the CNOT circuit decomposition for the Hamiltonian evolution operator corresponding to any general Pauli matrix for a multi-qubit system.*

For $\mathtt{R}_{XY}(\theta)$, we first note that XX and YY commute, so we have:

$$\mathtt{R}_{XY}(\theta) = \mathtt{R}_{XX}(\theta/2)\mathtt{R}_{YY}(\theta/2), \tag{10.27}$$

which means it can be implemented using 4 CNOT gates. So, is it possible to use fewer CNOT gates to implement $\mathtt{R}_{XY}(\theta)$ or more general two-qubit gates? The following exercise provides a general answer.

Problem 10.16. *Prove that any two-qubit gate operation can be implemented using at most three CNOT gates. (Reference: arXiv:quant-ph/0308006)*

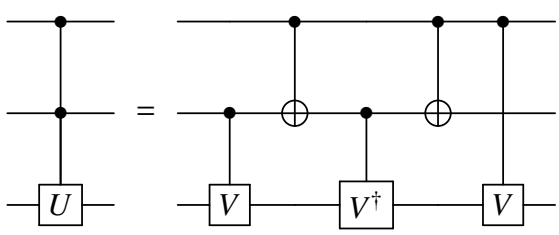
10.2.3 Multi-Qubit Gates

Similar to two-qubit gates, we first consider multi-qubit controlled gate operations. For the case of one control qubit and multiple target qubits, suppose we know the specific implementation of the control operation $U = U_n U_{n-1} \dots U_1$, where each U_i is a two-qubit gate. The control operation CU can be implemented by applying each control U_i sequentially, i.e.,

$$CU = CU_n \cdot CU_{n-1} \cdot \dots \cdot CU_1. \tag{10.28}$$

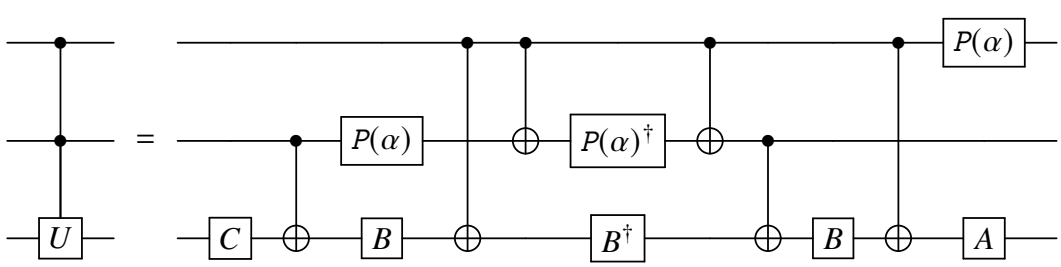
At the same time, we know that each two-qubit gate can be decomposed into single-qubit gates and CNOT gates. Therefore, the key here is how to implement controlled CNOT gates, specifically the CCNOT gate or Toffoli gate.

Here, we consider a more general case with two control qubits and one target qubit, i.e., C^2U. Note that for a general U, we cannot implement it using only two-qubit operations and CU. To address this, we need to introduce a new gate $V = \sqrt{U}$, and then we can implement C^2U using the following circuit:

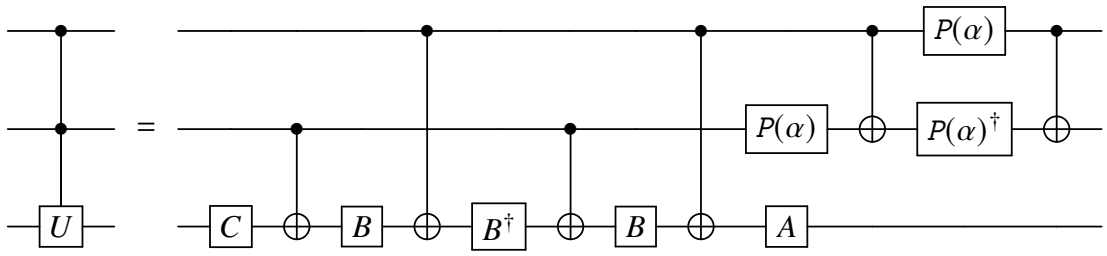


Problem 10.17. *Prove the implementation of the C^2U circuit.*

Problem 10.18. *Assume that $V = e^{i\alpha}AXBXC$. Prove that C^2U can be further simplified as:*



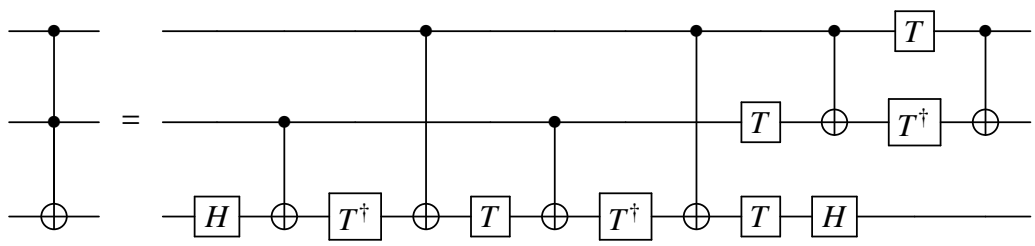
Further, prove that this circuit is equivalent to the following circuit:



Specifically, when $U = X$, the corresponding C^2U is the Toffoli gate, also known as CCX, which is key to implementing classical reversible computation. It is represented as

$$\texttt{CCX}\,|c_1\rangle\,|c_2\rangle\,|t\rangle = |c_1\rangle\,|c_2\rangle\,|t \oplus c_1c_2\rangle\,. \tag{10.29}$$

The corresponding gate is $V = \sqrt{X} = e^{i\alpha}AXBXC$, with $\alpha = \pi/4$, $A = HT$, $B = T^\dagger$, and $C = H$. Therefore, the corresponding circuit is:



It can be proven that the Toffoli gate with two control qubits requires at least 6 CNOT gates to implement. Therefore, the above circuit is the optimal one.

Problem 10.19. *Think about how to implement any two-qubit gate using the* CCX *gate and single-qubit gates.*

Another commonly used controlled two-qubit gate is the controlled swap gate, also called the Fredkin gate, denoted as CSWAP, which is represented as

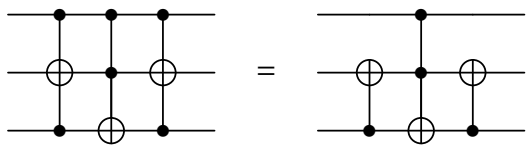
$$\texttt{CSWAP}\,|c\rangle\,|\psi\rangle\,|\phi\rangle = \begin{cases} |c\rangle\,|\psi\rangle\,|\phi\rangle & c = 0 \\ |c\rangle\,|\phi\rangle\,|\psi\rangle & c = 1 \end{cases} \tag{10.30}$$

or

$$\texttt{CSWAP}\,|c\rangle\,|t_1\rangle\,|t_2\rangle = |c\rangle\,|t_1 \oplus c_1(t_1 \oplus t_2)\rangle\,|t_2 \oplus c_1(t_1 \oplus t_2)\rangle\,. \tag{10.31}$$

If we decompose the SWAP gate into three CNOT gates and then apply the Toffoli decomposition, it requires a total of 18 CNOT gates to implement SWAP. However, we can notice the following phenomenon that reduces the number of CNOT gates to 8.

Problem 10.20. *Prove that the following circuit requires only 8 CNOT gates:*



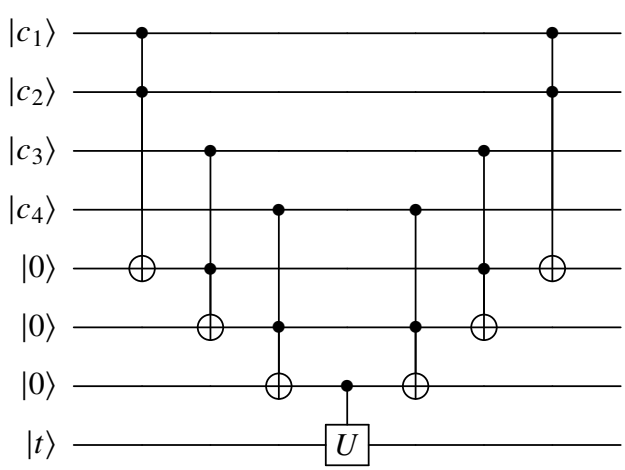
Think about how to further simplify this circuit.

Using the SWAP gate, we can implement the SWAP test to measure the overlap of quantum states. Details will be discussed in later chapters.

In practical computations, we also need to consider the case with multiple control qubits. For instance, consider the control operation $C^m U$ with m control qubits, defined as:

$$C^m U |c_1\rangle |c_2\rangle \ldots |c_m\rangle |t\rangle = |c_1\rangle |c_2\rangle \ldots |c_m\rangle |t \oplus c_1 c_2 \ldots c_m\rangle . \tag{10.32}$$

Therefore, the key here is how to compute $c_1 c_2 \ldots c_m$. We note that the Toffoli gate with two control qubits essentially implements the $c_1 c_2$ calculation for two control qubits, so we can use multiple Toffoli gates to compute $c_1 c_2 \ldots c_m$. Specifically, we can introduce $m-1$ auxiliary qubits and sequentially compute $c_1 c_2$, $c_1 c_2 c_3$, $\ldots$, $c_1 c_2 \ldots c_m$. For example, the case when $m = 4$ is as follows:



Problem 10.21. *Think about how to implement a multi-control qubit control gate without using auxiliary qubits.*

In addition to controlled gates, a common multi-qubit gate in Hamiltonian simulation is multi-qubit Hamiltonian evolution. For example, consider a Hamiltonian that is a multi-qubit Pauli operator $\boldsymbol{\sigma} = \sigma_1 \otimes \sigma_2 \otimes \cdots \otimes \sigma_n$, and we need to implement

$$U(\boldsymbol{\sigma}, t) = e^{-i\sigma t} = e^{-i\sigma_1 \otimes \sigma_2 \otimes \cdots \otimes \sigma_n t}. \tag{10.33}$$

Note that U is a global operation. Here, we will discuss how to implement this operation using a linear number of CNOT gates. First, we can transform the Hamiltonian into the standard form $Z_1 \otimes Z_2 \otimes \cdots \otimes Z_n$ by using local single-qubit operations, leading to $U(\boldsymbol{Z}, t)$. Next, we focus on how to implement

$$U(\boldsymbol{Z}, t) = e^{-iZ_1 \otimes Z_2 \otimes \cdots \otimes Z_n t}. \tag{10.34}$$

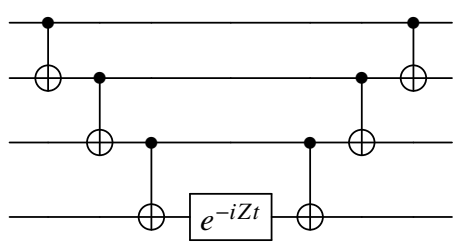
Note that $U(\boldsymbol{Z}, t)$ is essentially a global phase operator, and its action can be represented as:

$$U(\boldsymbol{Z}, t) |x_1\rangle |x_2\rangle \ldots |x_n\rangle = \begin{cases} e^{-it} |x_1\rangle |x_2\rangle \ldots |x_n\rangle & x_1 \oplus x_2 \oplus \cdots \oplus x_n = 0 \\ e^{it} |x_1\rangle |x_2\rangle \ldots |x_n\rangle & x_1 \oplus x_2 \oplus \cdots \oplus x_n = 1 \end{cases} \tag{10.35}$$

Problem 10.22. *Prove Equation* (10.35)*. Specifically, prove that*

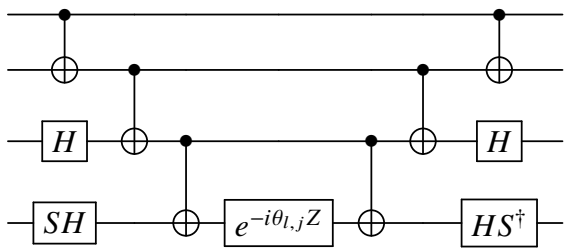
$$\langle x_1, x_2, \ldots, x_n | Z_1 \otimes Z_2 \otimes \cdots \otimes Z_n | x_1, x_2, \ldots, x_n \rangle = 1 - 2x_1 \oplus x_2 \oplus \cdots \oplus x_n. \tag{10.36}$$

Therefore, we can first compute $x_1 \oplus x_2 \oplus \cdots \oplus x_n$, and then apply different phases to the entire quantum state depending on the value of this sum. We know that the `CNOT` gate can perform modulo-2 addition of quantum bits, so we can use the following quantum circuit to implement $U(\boldsymbol{Z}, t)$:



Here we consider the example with four qubits. The `CNOT` sequence on the left calculates $x_1 \oplus x_2$, $x_1 \oplus x_2 \oplus x_3$, and $x_1 \oplus x_2 \oplus x_3 \oplus x_4$ for the second, third, and fourth qubits. The single-qubit gates in the middle apply the corresponding phase operations, while the `CNOT` sequence on the right restores the computation results for $x_1 \oplus x_2$, $x_1 \oplus x_2 \oplus x_3$, and $x_1 \oplus x_2 \oplus x_3 \oplus x_4$ back to the original states of x_2, x_3, x_4.

Problem 10.23. *Consider $Z_1Z_2X_3Y_4$, and prove that it can be decomposed into the following circuit:*



For general Hamiltonian operators, we can generally decompose them into a sum of Pauli operators, and then use methods such as Trotter decomposition to implement them. A detailed introduction to this will be given in subsequent chapters.

10.3 Universality of Quantum Gates

We have previously introduced single-qubit gates, two-qubit gates, and some special multi-qubit gates. So, can we use these gates to implement general multi-qubit operations?

To answer this question, we first consider how to use arbitrary single-qubit gates and CNOT gates to exactly implement any unitary operation. Next, we consider how to approximate any unitary operation using a limited set of quantum gates. Note that here we are only concerned with the universality of quantum gates, and there are no restrictions on the number of gates used for implementation. For unitary matrices with special structures, there might be more efficient implementation methods. Of course, we will eventually explain that any general multi-qubit gate generally requires an exponential number of single-qubit and two-qubit gates to be synthesized.

10.3.1 Exact Implementation: Single-Qubit Gates + `CNOT`

For an n-qubit unitary operation U, we can treat it as a $d \times d$ matrix, where $d = 2^n$. To implement this matrix exactly, we first need the following result:

Any unitary matrix can be decomposed into a product of two-dimensional unitary matrices.

Specifically, we can find $U_1, U_2, \ldots, U_{d(d-1)/2}$ such that

$$U = U_1 U_2 \ldots U_{d(d-1)/2}, \tag{10.37}$$

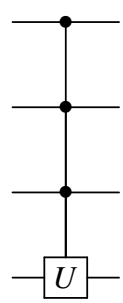
where each U_i acts only on two basis vectors. The construction of each U_i is similar to the elimination method in matrix diagonalization, applying each U_i to eliminate the off-diagonal elements of the matrix U, thus satisfying

$$U_{d(d-1)/2} \ldots U_2^\dagger U_1^\dagger U = I. \tag{10.38}$$

Problem 10.24. *Write the decomposition of the following 3-dimensional unitary matrix into a product of two-dimensional unitary matrices:*

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}. \tag{10.39}$$

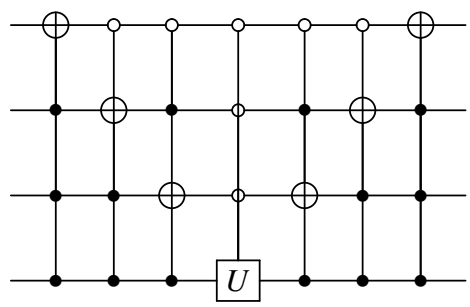
The next question is how to use single-qubit gates and `CNOT` gates to implement any 2-dimensional unitary matrix. For example, consider a general 2-dimensional unitary matrix $U = \sum_{x,x'} U(x,x') \left|x\right\rangle \left\langle x'\right|$. If $x = x_1 x_2 \ldots x_n$ and $x' = y_1 y_2 \ldots y_n$ differ by only one bit (e.g., $x = 1111$ and $y = 1110$), we can use a multi-controlled U gate to implement it, i.e.,



Of course, if the bits where x and y differ are not 1, we can still use local X operations to turn them into 0 controls. For general x and y, we would first need to transform them into a form where they differ by just one bit. We can use Gray code, which modifies x bit by bit, changing at most $n-1$ bits to make it differ from y by just one bit. For example, consider $x = 1111$ and $y = 0000$, we can consider the following sequence:

$$\begin{matrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix} \tag{10.40}$$

Thus, the first row is x, the last row is y, and each pair of consecutive rows differs by only one bit. For each pair of rows, we can perform the conversion using multi-controlled Toffoli gates. Specifically, we can implement this 2-dimensional unitary transformation using the following circuit:



Here, the Toffoli gates on the left perform the sequence transformation, and the Toffoli gates on the right perform the inverse operation of that transformation.

Problem 10.25. *Prove that any n-qubit unitary matrix can be implemented using $\mathcal{O}(n^2 4^n)$ single-qubit gates and* `CNOT` *gates.*

10.3.2 Approximate Implementation

Although single-qubit gates and `CNOT` gates can implement any unitary transformation, the set of gates that can be implemented in experiments (quantum error correction codes) is typically discrete and special. Therefore, we discuss how to approximate any unitary matrix using a discrete set of gates. Specifically, we consider a discrete set of gates $\mathbb{S}$ and find a sequence of gates $U_1, U_2, \ldots U_m \in \mathbb{S}$ that satisfies

$$\|U - U_1 U_2 \ldots U_m\| \leq \varepsilon, \tag{10.41}$$

where $\varepsilon \geq 0$ is the chosen approximation accuracy and $\|A\|$ denotes the spectral norm of the matrix A.

Based on the conclusions from the previous section, we can construct this sequence using single-qubit gates and `CNOT` gates. When the set $\mathbb{S}$ includes `CNOT`, we can transform this approximation problem into how to implement any single-qubit gate using a discrete set of single-qubit gates. As mentioned earlier, when the set $\mathbb{S}$ only includes $X, Y, Z, \mathtt{H}, \mathtt{S}$, we can only implement a limited set of single-qubit gates (the single-qubit Clifford gates), so we need to introduce another gate, namely the `T` gate. According to the Solovay–Kitaev theorem, we can efficiently

approximate any single-qubit gate using a combination of single-qubit Clifford gates and the T gate.

Solovay–Kitaev Theorem. Any single-qubit gate can be approximated to an accuracy of ε_1 using $\log^c(1/\varepsilon_1)$ Clifford + T gates, where $c \approx 2$.

Thus, combining the above results, we can select $m = n^2 4^n$, $\varepsilon_1 = \varepsilon/m$, to ensure the overall accuracy is ε, as in equation (10.41). Therefore, the total number of gates required is

$$\mathcal{O}(n^2 4^n \log^c(n^2 4^n/\varepsilon)). \tag{10.42}$$

Note that here we only need H (a Clifford gate) + T gates to implement all the Clifford gates, so the following set forms a universal quantum gate set:

$$\mathbb{S} = \{\mathtt{H}, \mathtt{CNOT}, \mathtt{T}\}. \tag{10.43}$$

Problem 10.26. *Prove that the following sets also form universal quantum gate sets:*

$$\begin{aligned} \mathbb{S} &= \{S, CNOT, R_Y(\pi/4)\}, \\ \mathbb{S} &= \{Toffoli, H\}. \end{aligned} \tag{10.44}$$

Problem 10.27. *Prove that any n-qubit unitary transformation requires at least* $\mathcal{O}(2^n \log(1/\varepsilon)/\log(n))$ *single-qubit and two-qubit gates to achieve an accuracy of* ε.

Chapter 11

Quantum error correction

In this chapter, we study quantum error correction, including the Shor code, the stabilizer formalism, and fault-tolerance.

11.1 The Shor code

11.1.1 Classical repetition code

Consider a classical bit $x \in \{0,1\}$ and suppose bit flip error, i.e., $x \to x \oplus 1$, happens with probability $p < 0.5$. Can we protect x? The answer is yes and a simple strategy is to use redundant information. In particular, we can use $2n+1$ bits to encode x as

$$0_L := 00\ldots0, \quad 1_L := 11\ldots1. \tag{11.1}$$

Then we can use majority vote to decide whether it is 0_L or 1_L. Suppose that error happens independently, then the failure probability is

$$p_{\text{fail}} = \sum_{k=n+1}^{2n+1} \binom{2n+1}{k} p^k (1-p)^{2n+1-k} \leq \sum_{k=n+1}^{2n+1} \left(\frac{e(2n+1)p}{n+1}\right)^k = \left(\frac{e(2n+1)p}{n+1}\right)^{n+1} \frac{1-\left(\frac{e(2n+1)p}{n+1}\right)^{n+1}}{1-\frac{e(2n+1)p}{n+1}}. \tag{11.2}$$

Here we used $\binom{2n+1}{k} \leq \left(\frac{e(2n+1)}{n+1}\right)^k$ for $k \geq n+1$. Therefore, as long as $\frac{e(2n+1)p}{n+1} \leq 1$ or equivalently $p \leq \frac{n+1}{e(2n+1)}$, we can exponentially suppress the failure probability[1].

[1] In fact, the bound here is quite loose. We only need $p < 1/2$ to have exponentially small error.

11.1.2 Quantum repetition code — bit flip error

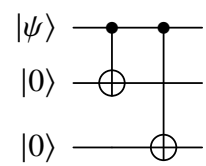
There are several challenges to construct a quantum error correcting code.

- Quantum states are continuous and cannot be cloned — a quantum state is $|\psi\rangle = a|0\rangle + b|1\rangle$ with continuous a and b, and we cannot clone it to have $|\psi\rangle^{\otimes n}$.
- Errors are also continuous — a general error channel is $\mathcal{E}(\rho) = \sum_j K_j \rho K_j^\dagger$ where each $K_j = a_0 I + a_1 X + a_2 Y + a_3 Z$ with continuous a_i.
- Measurement destroys quantum states — the state collapses if we measure the state or extract information.

For quantum state $|\psi\rangle = a|0\rangle + b|1\rangle$, the quantum repetition code is defined as

$$|\psi\rangle_L = a|0\rangle_L + b|1\rangle_L = a|000\rangle + b|111\rangle. \tag{11.3}$$

We can realize it with the following quantum circuit.



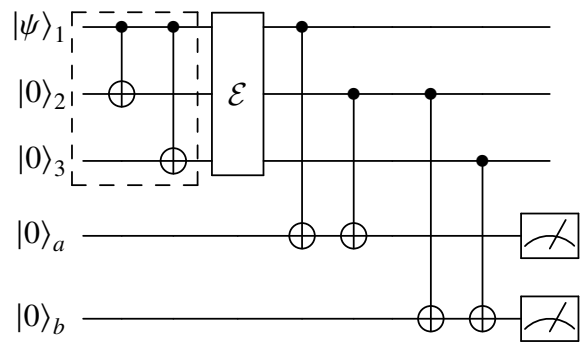
What can we do with this code? Suppose independent bit flip error happens to every qubit, i.e., $\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$, then

$$\mathcal{E}_1 \circ \mathcal{E}_2 \circ \mathcal{E}_3(\psi_L) = (1-p)^3\psi_L + p(1-p)^2\left[X_1\psi_L X_1 + X_2\psi_L X_2 + X_3\psi_L X_3\right] + \mathcal{O}(p^2). \tag{11.4}$$

Now the leading order errors are $X_1\psi_L X_1 + X_2\psi_L X_2 + X_3\psi_L X_3$, which are independent bit flip errors on each qubit. We can detect these errors by majority vote, or using the nondestructive measurement

$$\begin{aligned} P_0 &= |000\rangle\langle 000| + |111\rangle\langle 111|, \\ P_1 &= |100\rangle\langle 100| + |011\rangle\langle 011|, \\ P_2 &= |010\rangle\langle 010| + |101\rangle\langle 101|, \\ P_3 &= |001\rangle\langle 001| + |110\rangle\langle 110|, \end{aligned} \tag{11.5}$$

which can be realize using the following circuit



Specifically, the ancillary qubit a is 0 iff qubit 1 and 2 are the same and it is 1 otherwise, the ancillary qubit b is 0 iff qubit 2 and 3 are the same and it is 1 otherwise. Note that the two ancillary qubits effectively measures an eigenstate of Z_1Z_2 and Z_2Z_3. Specifically, when $a = 0$ (1) we have $Z_1Z_2 = 1$ (-1).

Therefore, the there are four cases of the two ancillary qubit measurement corresponds to four eigenstates of Z_1Z_2 and Z_2Z_3, that is the four nondestructive measurement

$$\begin{aligned}
&a = 0, b = 0 \leftrightarrow Z_1Z_2 = 1, Z_2Z_3 = 1 \leftrightarrow P_0,\\
&a = 0, b = 1 \leftrightarrow Z_1Z_2 = 1, Z_2Z_3 = -1 \leftrightarrow P_3,\\
&a = 1, b = 0 \leftrightarrow Z_1Z_2 = -1, Z_2Z_3 = 1 \leftrightarrow P_1,\\
&a = 1, b = 1 \leftrightarrow Z_1Z_2 = -1, Z_2Z_3 = -1 \leftrightarrow P_2,
\end{aligned} \tag{11.6}$$

The ZZ or P measurements are called the syndrome measurements.

To see the equivalence between the ZZ operators and the projectors P, we first define the projector of $Z_iZ_k = \pm 1$ as $\Pi_{Z_iZ_k=\pm 1} = (\mathbb{I} \pm Z_iZ_k)/2$. For example, $\Pi_{Z_1Z_2=1} = (\mathbb{I} + Z_1Z_2)/2 = (|00\rangle\langle 00| + |11\rangle\langle 11|)_{12} \otimes \mathbb{I}_3$ and $\Pi_{Z_2Z_3=-1} = (\mathbb{I} - Z_2Z_3)/2 = \mathbb{I}_1 \otimes (|01\rangle\langle 01| + |10\rangle\langle 10|)_{23}$. Then the projector with both $Z_1Z_2 = 1$ and $Z_2Z_3 = -1$ is $\Pi_{Z_1Z_2=1\&Z_2Z_3=-1} = \Pi_{Z_1Z_2=1}\Pi_{Z_2Z_3=-1} = |001\rangle\langle 001| + |110\rangle\langle 110| = P_3$.

Using the four measurements in Eq. (11.5), we can therefore detect which error happens. That is, P_i corresponds to the case of bit flip error on the ith qubit ($i \in [1,3]$). Then we can apply a recovery operation to correct single qubit errors. The above discussion ignores the case with more than one errors and it is not hard to see that we cannot detect or correct more those cases. Now we study whether

error correction is useful at all. Without error correction, the state is $\mathcal{E}(\psi)$ whose fidelity to ψ is

$$F(\mathcal{E}(\psi),\psi) = \sqrt{\langle\psi|\mathcal{E}(\psi)|\psi\rangle} = \sqrt{(1-p)+p\,\langle\psi|X|\psi\rangle\,\langle\psi|X|\psi\rangle}, \tag{11.7}$$

with a minimal value of $\sqrt{1-p}$. With error correction, we have the ideal encoded state $|\psi\rangle_L$ and the corrected noisy state $\rho = [(1-p)^3+3p(1-p)^2]\psi_L+\ldots$. The fidelity of the corrected state is

$$F(\rho,\psi_L) = \sqrt{\mathrm{Tr}[\psi_L\rho]} \geq \sqrt{(1-p)^3+3p(1-p)^2} = \sqrt{1-3p^2+2p^3}. \tag{11.8}$$

To have $F(\rho,\psi_L) > F(\mathcal{E}(\psi),\psi)$, we need $1-3p^2+2p^3 \geq 1-p$ or equivalently $p < 1/2$. Therefore, as long as $p < 1/2$, we can decrease the error or increase the state fidelity.

11.1.3 Quantum repetition code — phase flip error

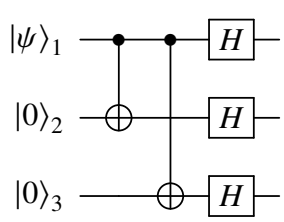
Now suppose the error channel is $\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z$, can we still detect and correct those errors? Lets see how phase flip affects the state. For state $|\psi\rangle = a|0\rangle + b|1\rangle$, we have

$$Z|\psi\rangle = a|0\rangle - b|1\rangle, \tag{11.9}$$

which flips the phase instead of the value, just as the name of the error indicates. While this error seems quite different from the bit flip error, they are actually very related. Remember the transformation between X and Z with $Z = HXH$, a phase flip error could thus be understood as a bit flip error in a different basis. Remember that a transformation of operator is equivalent to a corresponding transformation of the state, i.e., $\langle\psi|[U^\dagger OU]|\psi\rangle = \langle\tilde{\psi}|O|\tilde{\psi}\rangle$ with $|\tilde{\psi}\rangle = U|\psi\rangle$. Therefore, we can apply the $H_L = H^{\otimes 3}$ on the encoded state to have

$$|\psi\rangle_L = H_L(a|0\rangle_L + b|1\rangle_L) = a|+\rangle_L + b|-\rangle_L = a|+++\rangle + b|---\rangle, \tag{11.10}$$

with the circuit



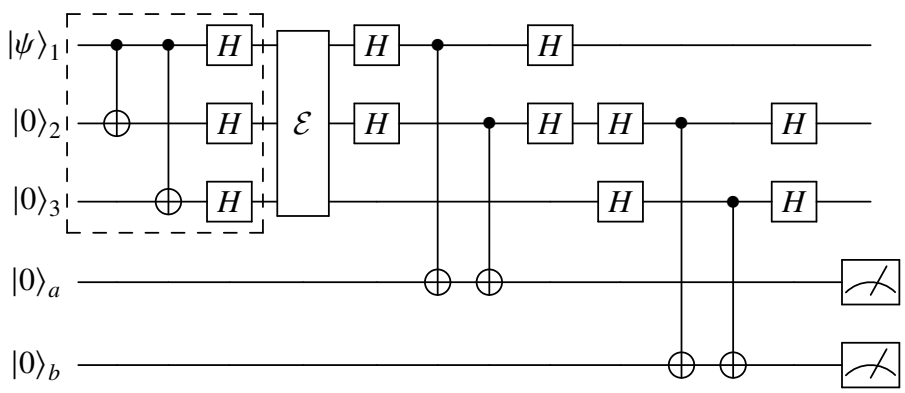
Now suppose Z_1 error happens on qubit 1, we have

$$Z_1 |\psi\rangle_L = a|-++\rangle + b|+--\rangle, \tag{11.11}$$

which is analog to the effect of a X_1 error on $a|0\rangle_L + b|1\rangle_L$. Focusing on the cases with zero and one error, we can apply a transformed nondestructive measurements of Eq. (11.5),

$$\{H_L P_i H_L\}, \tag{11.12}$$

to detect phase errors. Specifically, the measurement $H_L P_i H_L$ detects the error Z_i on the ith qubit with $i \in \{1,2,3\}$. The circuit to detect errors is



The relation between the a, b measurement outcomes and the nondestructive measurements $H_L P_i H_L$ is

$$\begin{aligned}
&a=0, b=0 \leftrightarrow X_1X_2=1, X_2X_3=1 \leftrightarrow H_L P_0 H_L,\\
&a=0, b=1 \leftrightarrow X_1X_2=1, X_2X_3=-1 \leftrightarrow H_L P_3 H_L,\\
&a=1, b=0 \leftrightarrow X_1X_2=-1, X_2X_3=1 \leftrightarrow H_L P_1 H_L,\\
&a=1, b=1 \leftrightarrow X_1X_2=-1, X_2X_3=-1 \leftrightarrow H_L P_2 H_L,
\end{aligned} \tag{11.13}$$

Here we can similarly understand the a and b measurement as X_1X_2 and X_2X_3, respectively.

11.1.4 The Shor code

Since the encoding circuits are different for bit and phase flip errors, can we combine them to correct both of them? The answer is yes with the Shor code. The

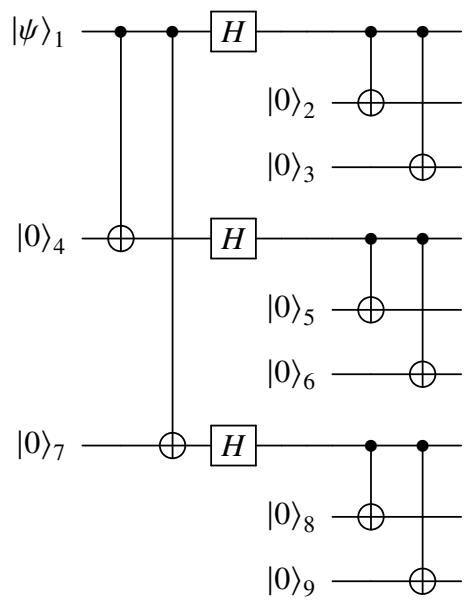
basic idea is to use code concatenation. We first apply the phase flip code to have

$$|0\rangle_L = |+++\rangle = \frac{(|0\rangle + |1\rangle)^{\otimes 3}}{2\sqrt{2}}, |1\rangle_L = |---\rangle = \frac{(|0\rangle - |1\rangle)^{\otimes 3}}{2\sqrt{2}}. \tag{11.14}$$

Then for each of $|0\rangle$ and $|1\rangle$, we apply the bit flip code with $|0\rangle \to |000\rangle$ and $|1\rangle \to |111\rangle$. Then we have the Shor code

$$|0\rangle_L = \frac{(|000\rangle + |111\rangle)^{\otimes 3}}{2\sqrt{2}}, |1\rangle_L = \frac{(|000\rangle - |111\rangle)^{\otimes 3}}{2\sqrt{2}}. \tag{11.15}$$

The encoding circuit is



We can now detect any single qubit phase or bit flip error. For example, suppose a bit flip error happens to the first qubit, we can then measure Z_1Z_2 and Z_2Z_3 to detect it. Any other bit flip error could be detected similarly. On the other hand, suppose a phase flip error happen to the first qubit, then we can measure $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$ to detect it. To summarize, we can measure the following observables

$$\begin{aligned} &Z_1Z_2,\ Z_2Z_3,\ Z_4Z_5,\ Z_5Z_6,\ Z_7Z_8,\ Z_8Z_9, \\ &X_1X_2X_3X_4X_5X_6,\ X_4X_5X_6X_7X_8X_9, \end{aligned} \tag{11.16}$$

and any single qubit bit or phase flip error could be detected using those measurement outcomes.

Can we do more than this? Suppose both bit and phase flip errors happen to the first qubit, we can actually still detect it using the measurement results of Z_1Z_2, Z_2Z_3, $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$. This works for simultaneous bit and phase flip errors on any qubit.

However, a practical error may be like $\mathcal{E}(\rho) = \sum_j K_j \rho K_j^\dagger$ with $K_j = a_0 I + a_1 X + a_2 Y + a_3 Z$, which is a superposition of the bit, phase, bit+phase errors. Can we still correct them. For simplicity, we assume that an error $K_j \rho K_j^\dagger$ happens to the first qubit of state $|\psi\rangle_L$. Then we have

$$K_j |\psi\rangle_L = a_0 |\psi\rangle_L + a_1 X_1 |\psi\rangle_L + a_2 Y_1 |\psi\rangle_L + a_3 Z_1 |\psi\rangle_L . \tag{11.17}$$

Now we measure Z_1Z_2, Z_2Z_3, $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$, or equivalently applying one of the 16 projectors $\Pi_{Z_1Z_2=\pm1}$, $\Pi_{Z_2Z_3=\pm1}$, $\Pi_{X_1X_2X_3X_4X_5X_6=\pm1}$, and $\Pi_{X_4X_5X_6X_7X_8X_9=\pm1}$. Suppose we have obtained $Z_1Z_2 = -1$, $Z_2Z_3 = -1$, $X_1X_2X_3X_4X_5X_6 = -1$ and $X_4X_5X_6X_7X_8X_9 = -1$, then the state is projected to

$$\Pi_{Z_1Z_2=-1}\Pi_{Z_2Z_3=-1}\Pi_{X_1X_2X_3X_4X_5X_6=-1}\Pi_{X_4X_5X_6X_7X_8X_9=-1} K_j |\psi\rangle_L = a_2 Y_1 |\psi\rangle_L . \tag{11.18}$$

Therefore, when we apply the non-destructive syndrome measurements, we also project the state to either $|\psi\rangle_L$, $X_1 |\psi\rangle_L$, $Y_1 |\psi\rangle_L$, or $Z_1 |\psi\rangle_L$, i.e., one of the state with single qubit I, X, Y, Z errors. So as long as we can correct discrete single qubit I, X, Y, Z errors, we can also correct continuous errors!

Now we analyze whether the state fidelity is improved with the Shor code. Consider the depolarizing channel $\mathcal{E}(\rho) = (1-p)\rho + p/3(X\rho X + Y\rho Y + Z\rho Z)$ as a special case, the fidelity without error correction is

$$F(\psi, \mathcal{E}(\psi)) = \sqrt{\mathrm{Tr}[\psi \mathcal{E} \psi]} = \sqrt{\mathrm{Tr}[\psi[(1-4p/3)\psi + 2p/3]]} = \sqrt{1-2p/3} = 1 - p/3 + \mathcal{O}(p^2). \tag{11.19}$$

On the other hand, suppose we encode the state using the Shor code with logical state ψ_L and independent depolarizing channel happens to every qubit. Since we can correct all single qubit errors, the corrected state is $\rho = \left[(1-p)^9 + 9p(1-p)^8\right]\psi_L + \ldots$, and the fidelity is

$$F(\psi_L, \rho) \geq \sqrt{(1-p)^8(1+8p)} = 1 - 18p^2 + \mathcal{O}(p^3) \tag{11.20}$$

Consider infidelity as a measure of error, the infidelity is then quadratically improved from $p/3$ to $18p^2$. In practice, we can apply code concatenation to further

decrease errors.

Revisiting the aforementioned three challenges, we overcome them using the following ideas.

- Quantum states are continuous and cannot be cloned — we can encode the state into a subspace of a larger entangled state.
- Errors are also continuous — errors become effectively discrete when we apply syndrome measurements.
- Measurement destroys quantum states — we apply non-destructive syndrome measurements.

11.2 A general theory of QEC

The theory of error correction could be understood abstractly as follows. We first encode the state, say a qubit, into a multi-qubit entangled state $|\psi_L\rangle = a|0\rangle_L + b|1\rangle_L$. The logical state space $\{|0\rangle_L, |1\rangle_L\}$ is only a subspace of the physical state space and we denote the projection from the physical space to the logical space as Π. For example, the projection of the Shor code is

$$\Pi = \Pi_{Z_1Z_2=1}\Pi_{Z_2Z_3=1}\Pi_{Z_4Z_5=1}\Pi_{Z_5Z_6=1}\Pi_{Z_7Z_8=1}\Pi_{Z_8Z_9=1}\Pi_{X_1X_2X_3X_4X_5X_6=1}\Pi_{X_4X_5X_6X_7X_8X_9=1}. \tag{11.21}$$

That is, for any 9-qubit state $|\psi\rangle$, applying the projection $\Pi|\psi\rangle$ projects it into the code space. Now, suppose an error channel $\mathcal{E}(\psi_L) = \sum_j K_j\psi_L K_j^\dagger$ happens to the logical state ψ_L, it now becomes a mixture of states $\rho_j = K_j\psi_L K_j^\dagger$. We denote the state space with K_j error as $\mathcal{S}_j = \{K_j|\psi_L\rangle, \forall\psi_L\}$. Then error correction works if S_j forms orthogonal subspaces so that there exist a projective measurement $\{\Pi_j\}$ to distinguish them, as shown in Fig. 11.1(b). Specifically, we need

$$\langle\psi|_L K_j^\dagger K_k|\psi\rangle_L = d_{jk},\ \forall|\psi\rangle_L, \tag{11.22}$$

for diagonal d_{jk}. Note that $|\psi\rangle_L = \Pi|\psi\rangle_L$, the above equation is equivalent to

$$\langle\psi|_L \Pi K_j^\dagger K_k \Pi|\psi\rangle_L = d_{jk},\ \forall|\psi\rangle_L, \tag{11.23}$$

or

$$\Pi K_j^\dagger K_k \Pi = d_{jk}\Pi. \tag{11.24}$$

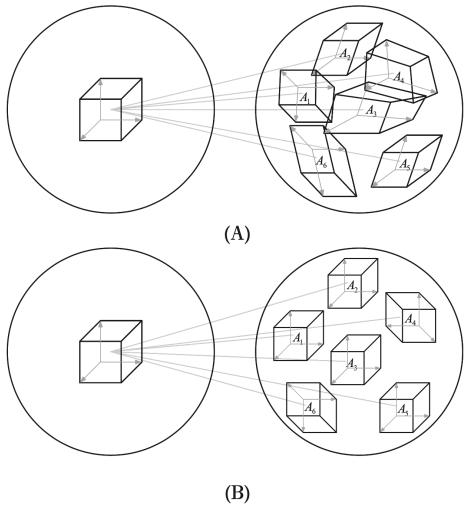


Figure 11.1: Illustration of quantum error correction. From Nielsen & Chuang's book.

Here we thus proved the necessary condition of error correction (not strictly, see discussion below). We can further prove that the above equation is sufficient for error correction. When we have $\Pi K_j^\dagger K_k \Pi = d_{jk}\Pi$, we can explicitly construct the projective measurement $\{\Pi_j\}$ to distinguish between S_j. Consider the polar decomposition of $K_k\Pi = U_k\sqrt{\Pi K_k^\dagger K_k \Pi} = \sqrt{d_{kk}}U_k\Pi$ with some unitary U_k, the effect of K_k on the logical space is thus equivalent to a unitary U_k. Now define projectors $\{\Pi_k = U_k\Pi U_k^\dagger\}$, we have

$$\Pi_k K_k |\psi\rangle_L = U_k \Pi U_k^\dagger K_k |\psi\rangle_L = K_k |\psi\rangle_L, \tag{11.25}$$

and $\Pi_k\Pi_j = \delta_{k,j}$. Therefore, we can apply a projective measurement $\{\Pi_k\}$, which distinguishes S_j or K_j. Then we can apply the inverse map of U_j to rotate the space S_j or state $K_j|\psi\rangle_L$ back to $\{|\psi\rangle_L\}$.

In the above analysis, we assumed that we need to perfectly distinguish the errors K_j. However, we have seen that even if K_j is a sum of Pauli errors (in this case, we cannot distinguish them), we can still correct them. Actually, we can prove that

Theorem 11.1. *If error correction works for channel with Kraus operators $\{E_j\}$, it also works for channels with Kraus operators $\{K_k = \sum_k c_{jk}E_j\}$, which is a linear combination of E_j.*

The idea is very similar to our analysis of the Shor code. Basically, since $\{\Pi_k\}$ project the state to the subspace $\mathcal{S}_j = \{K_j|\psi_L\rangle, \forall\psi_L\}$, applying the projection $\{\Pi_k\}$ also project $\{K_k = \sum_k c_{jk}E_j\}$ into a mixture of $\{E_j\}$, which becomes correctable.

Taking the above theorem into account, we thus arrive at the final result for the condition of error correction.

Theorem 11.2. *Suppose the logical state is defined by a projector P and consider errors with Kraus operator $\{K_j\}$, a necessary and sufficient condition for the existence of an error-correction protocol is*

$$PK_j^\dagger K_k P = \alpha_{jk} P, \tag{11.26}$$

for any hermitian matrix α_{jk}.

We refer to Nielsen & Chuang's book for the detailed and rigourous proof.